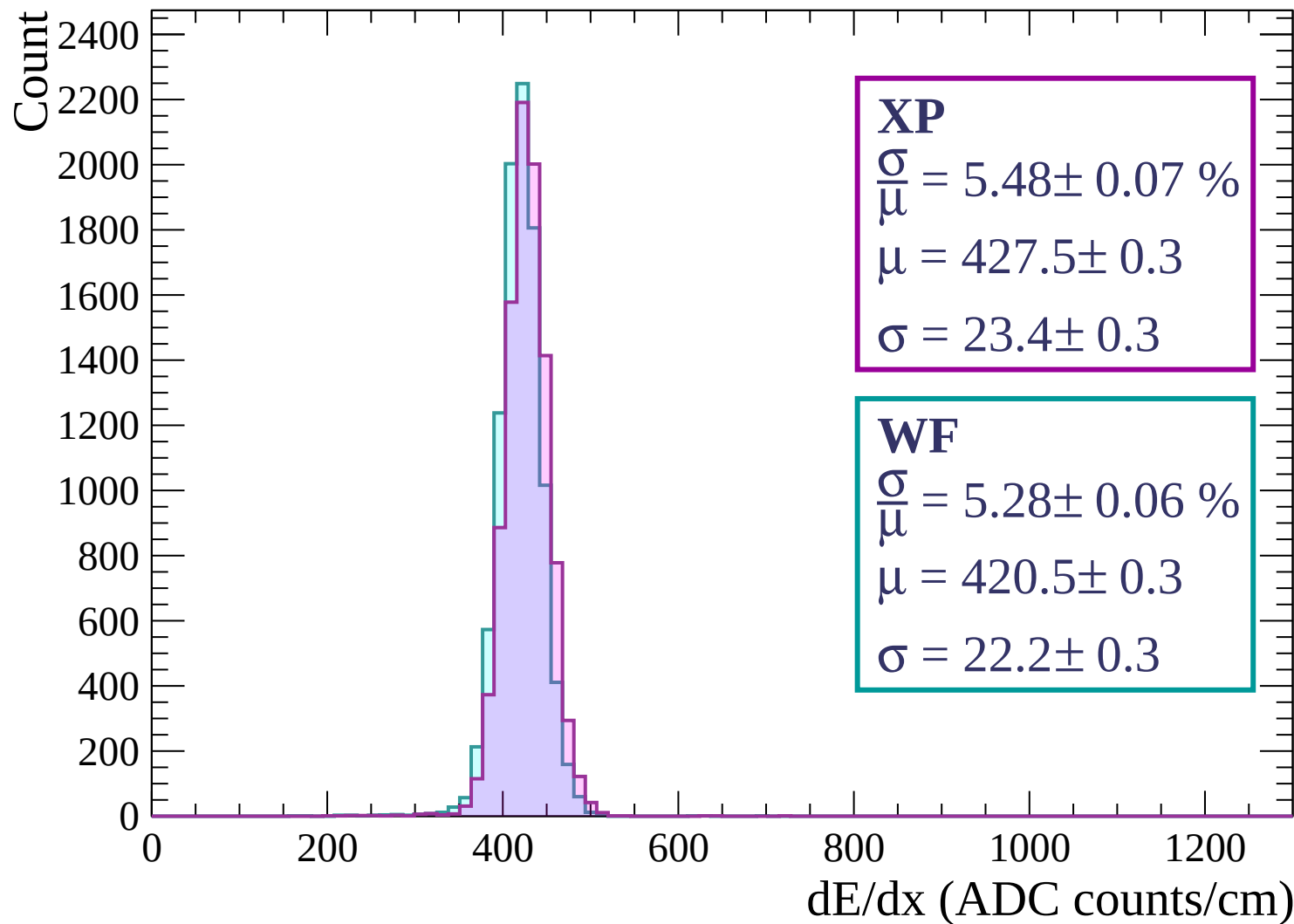
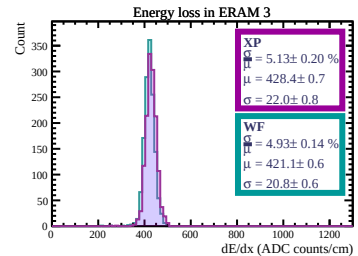
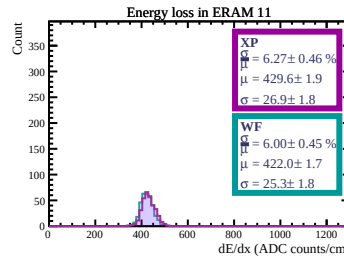
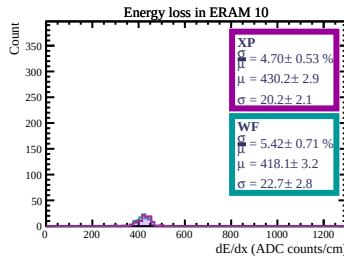
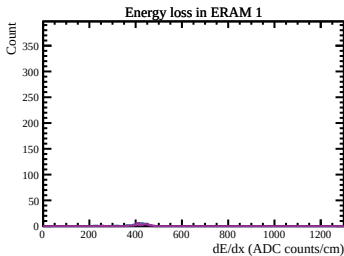
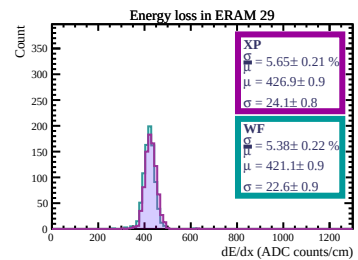
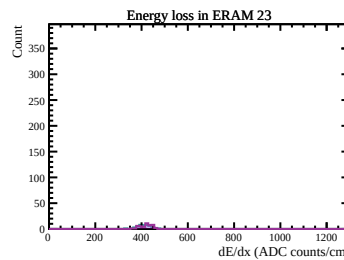
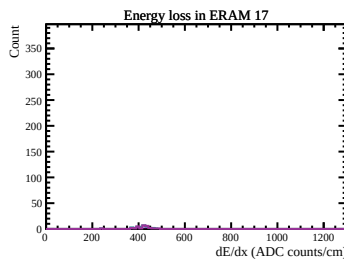
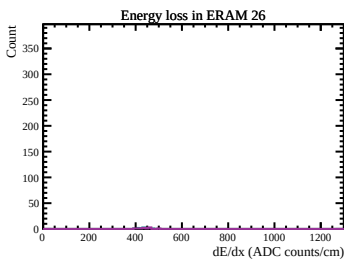
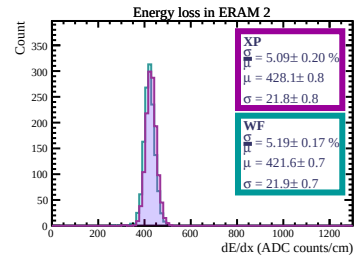
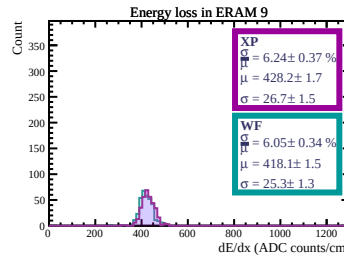
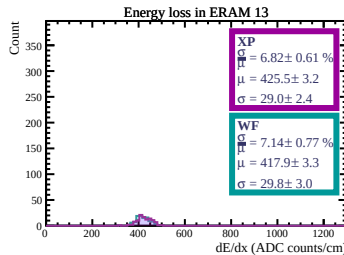
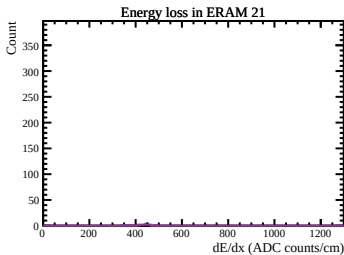
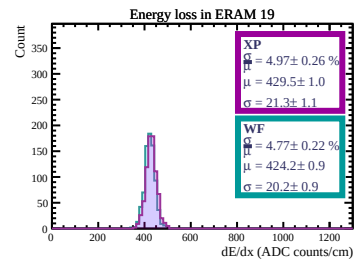
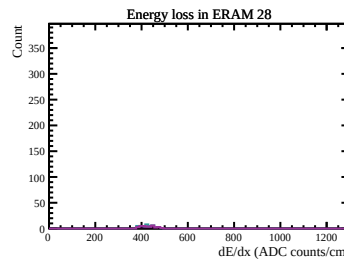
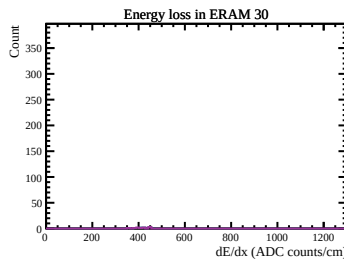
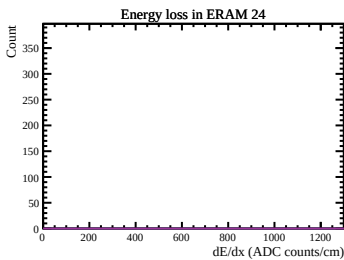
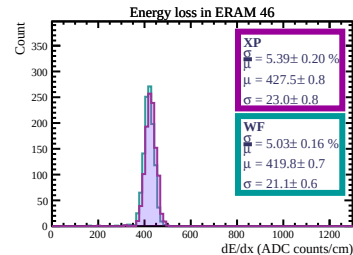
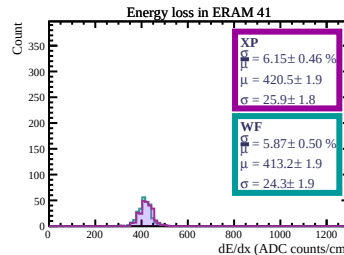
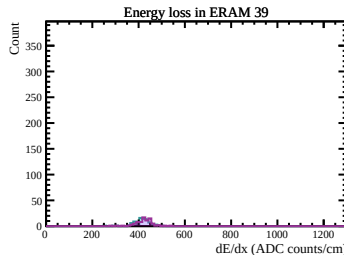
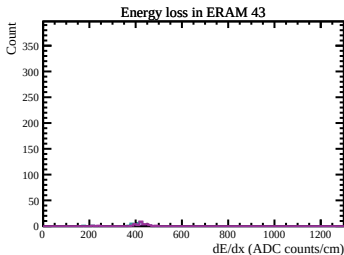
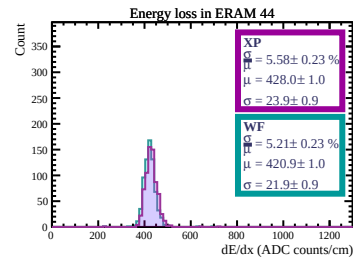
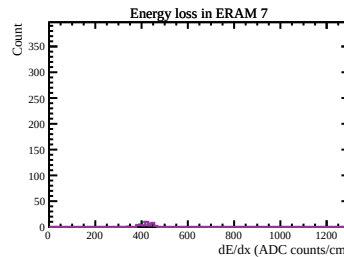
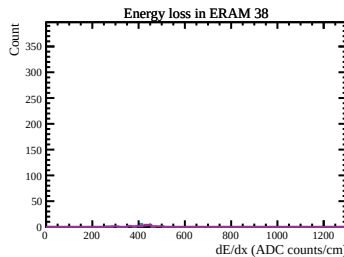
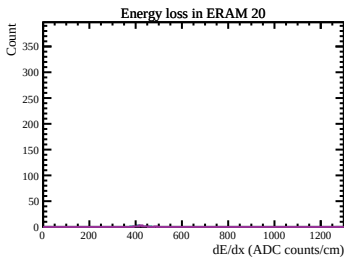
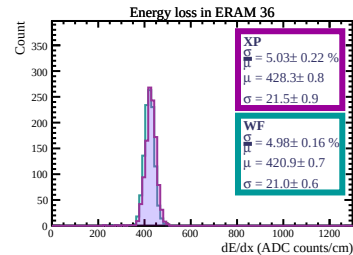
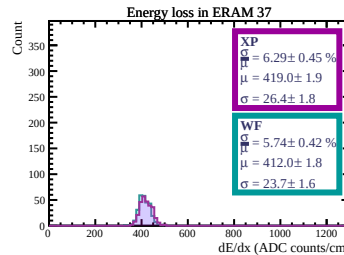
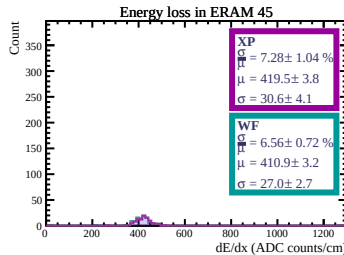
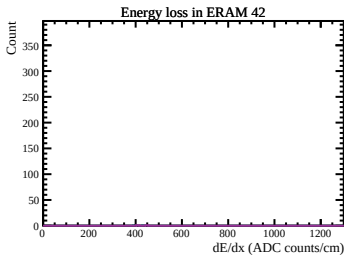
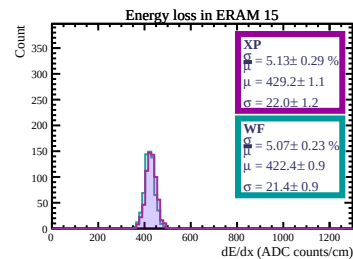
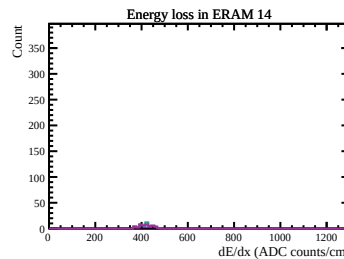
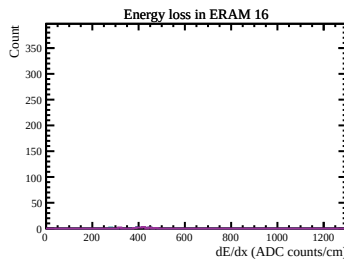
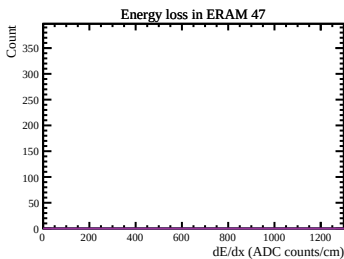
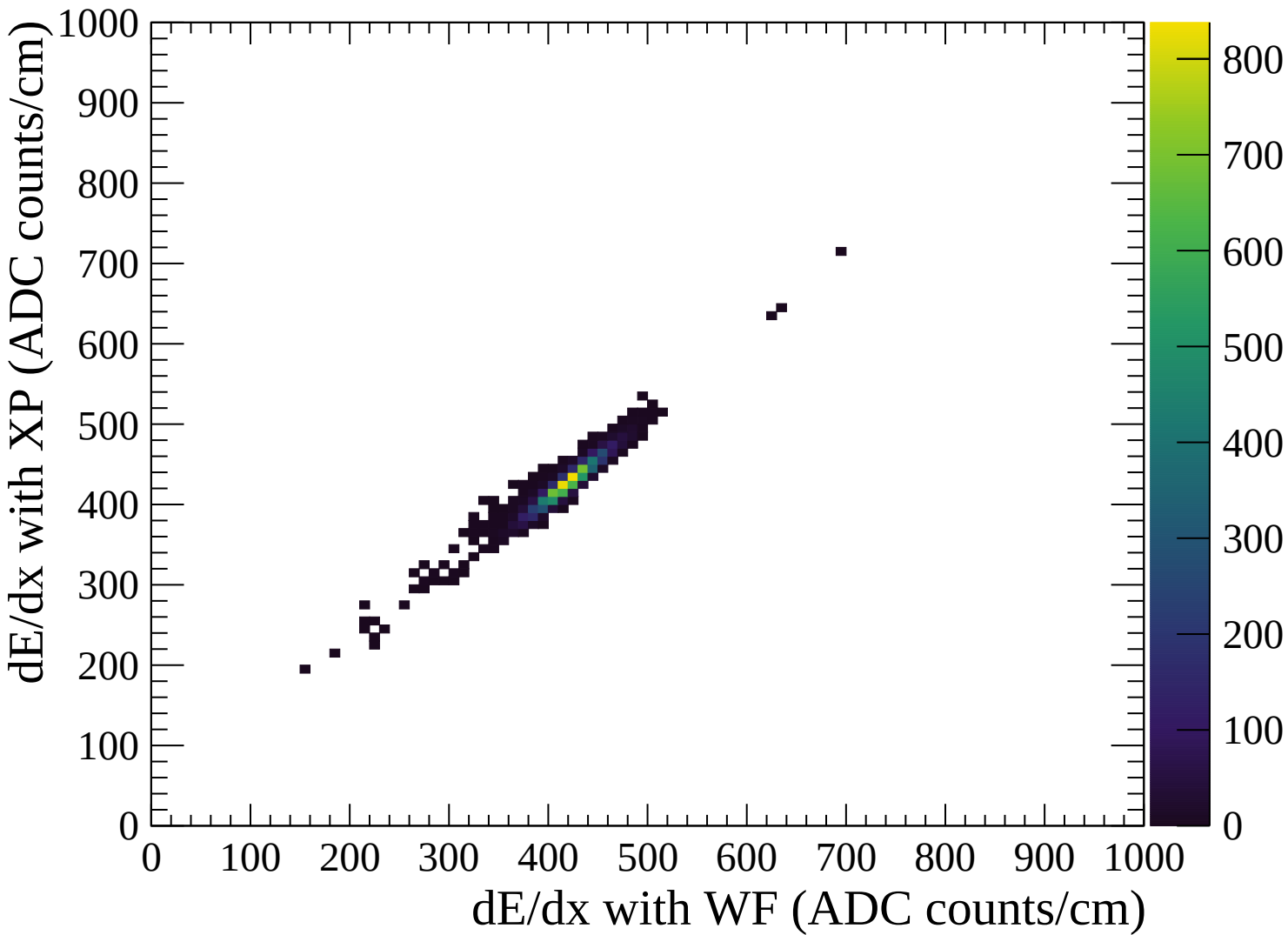
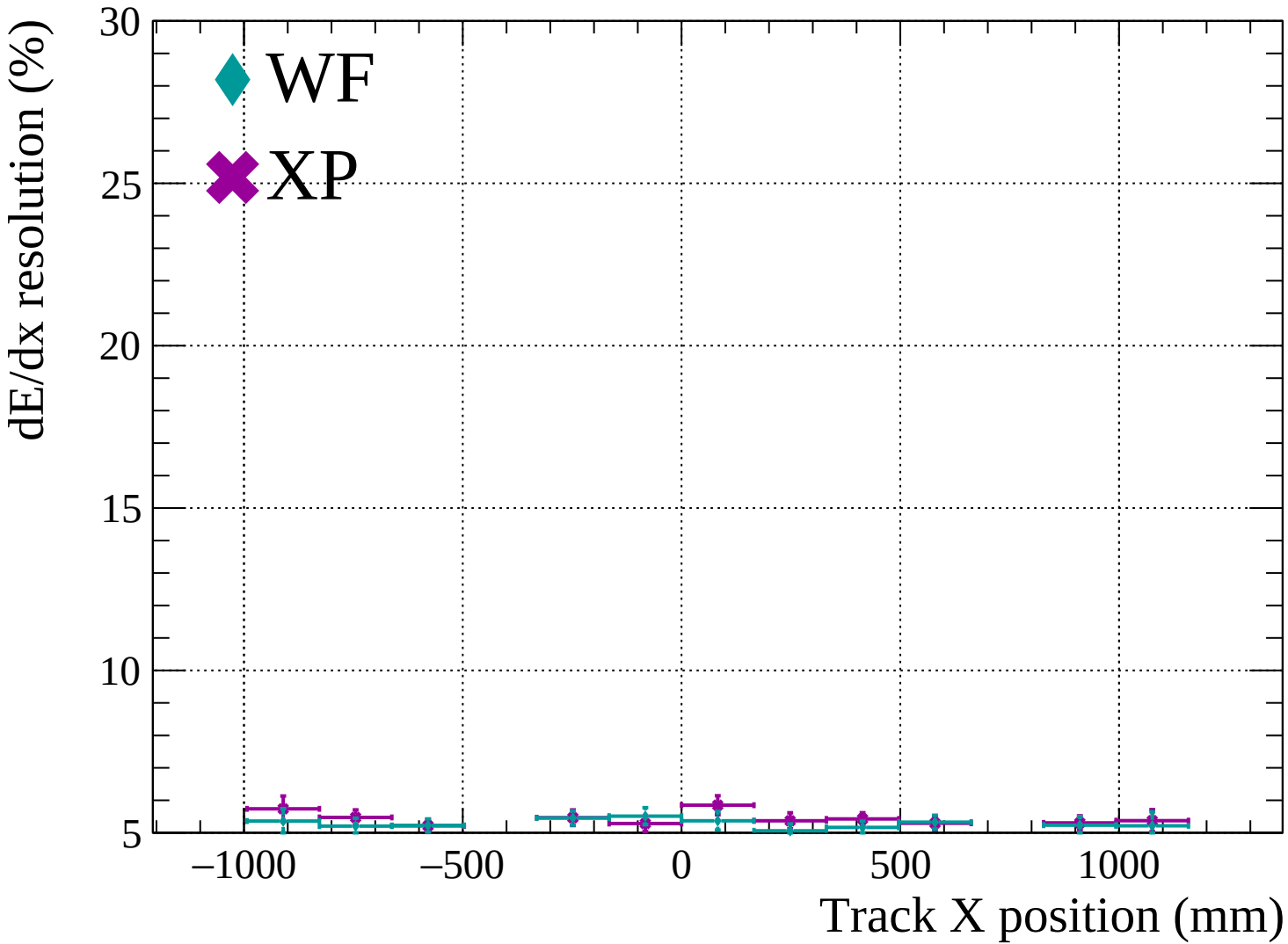


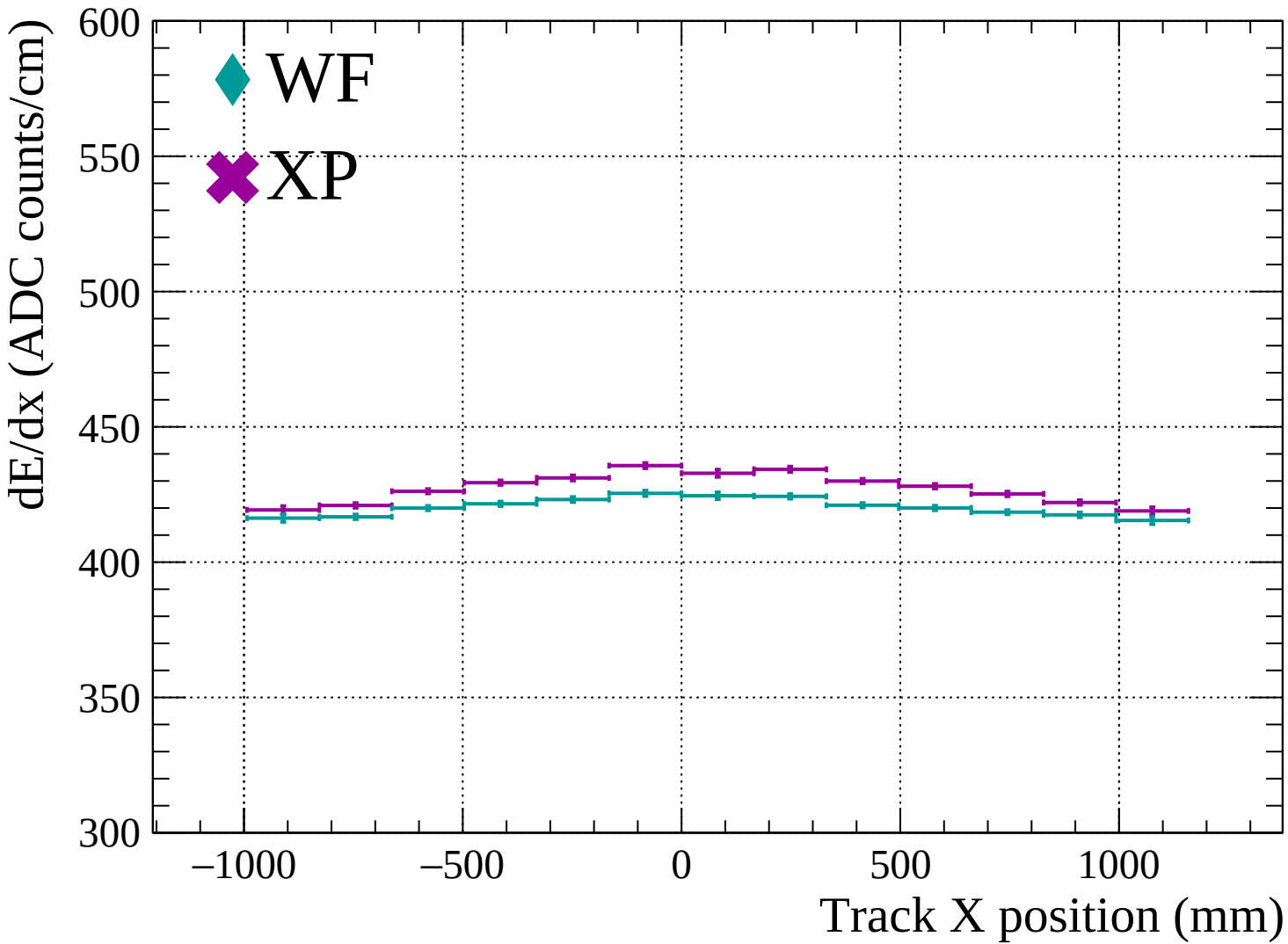


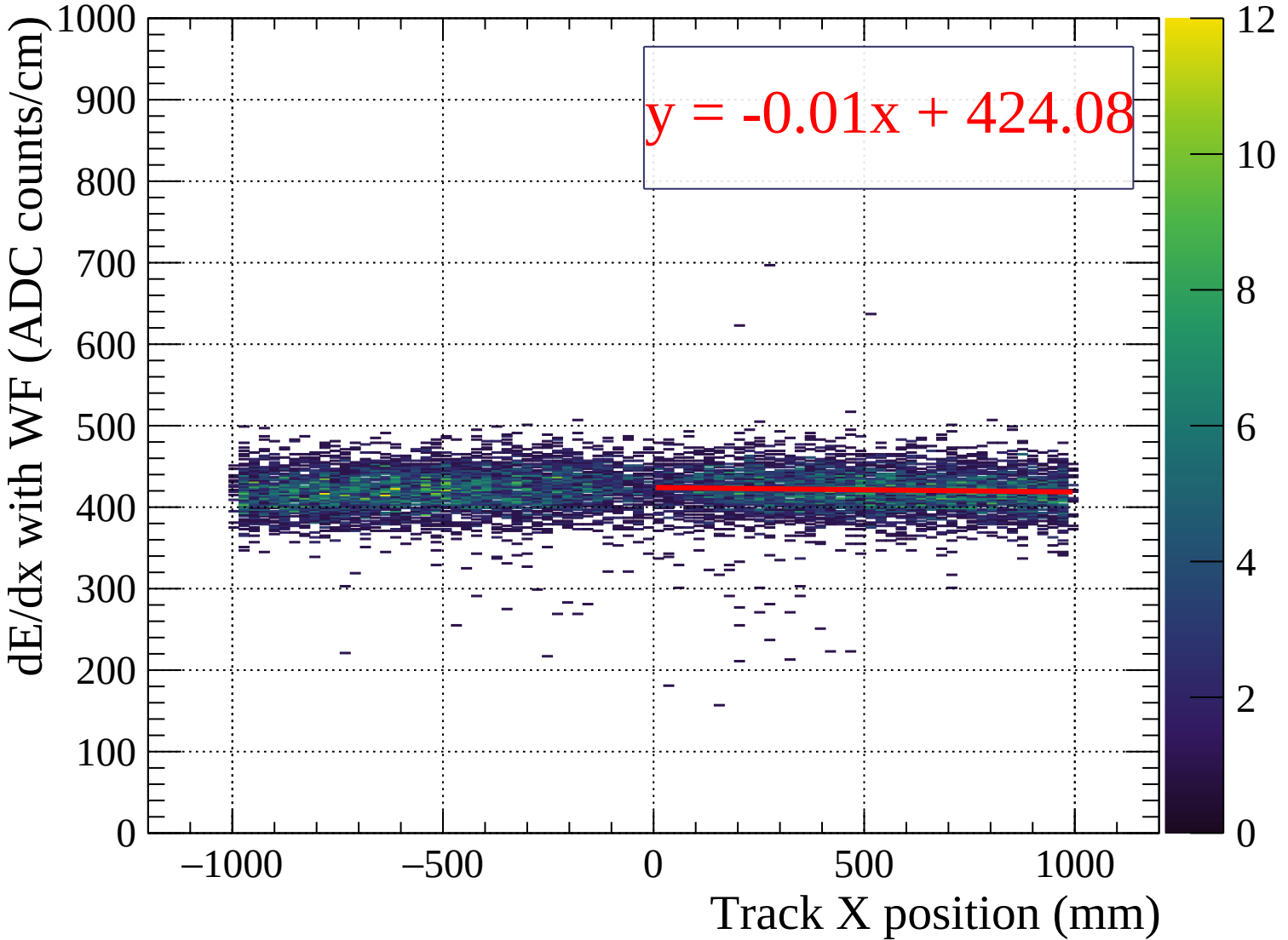


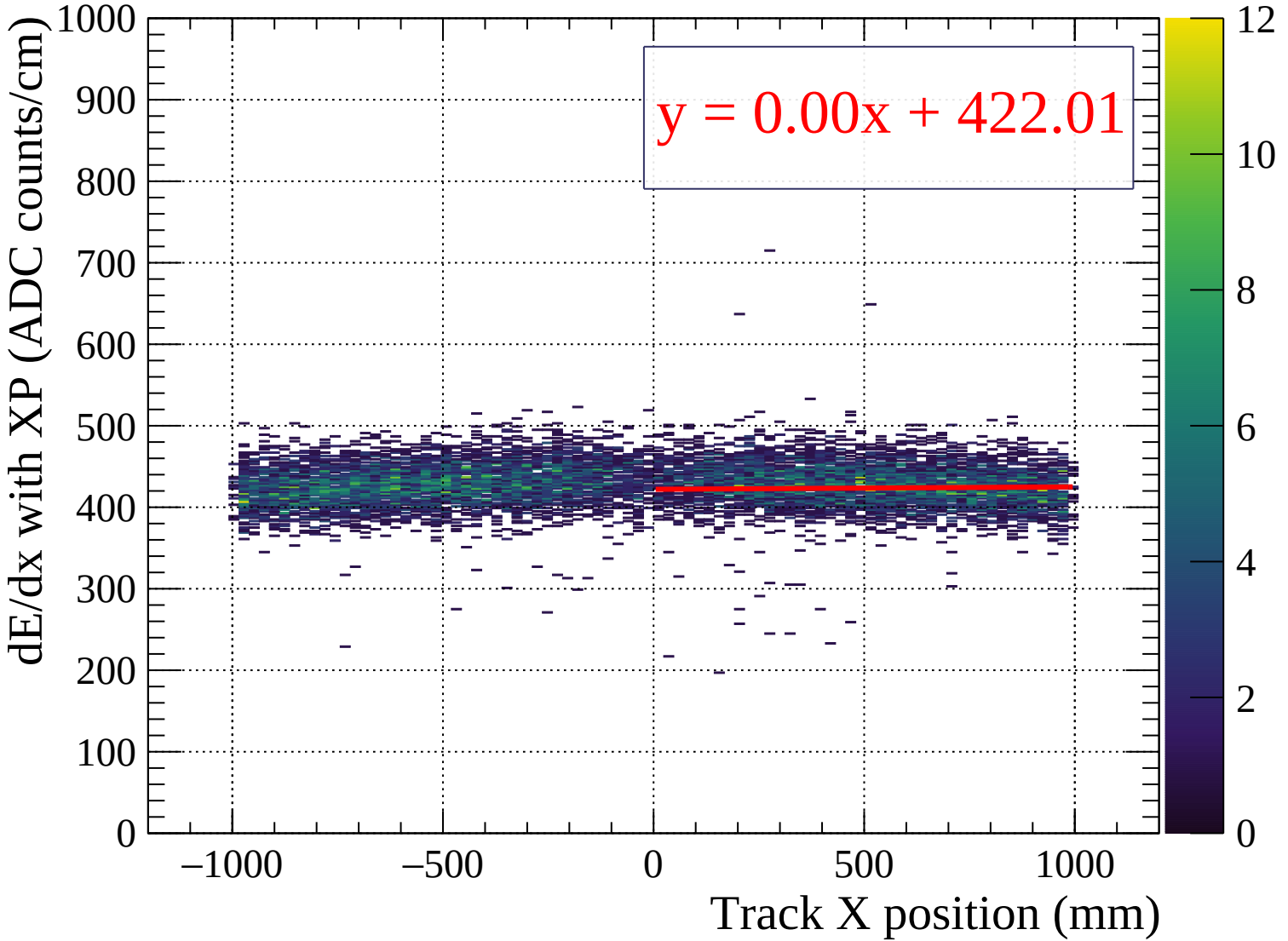


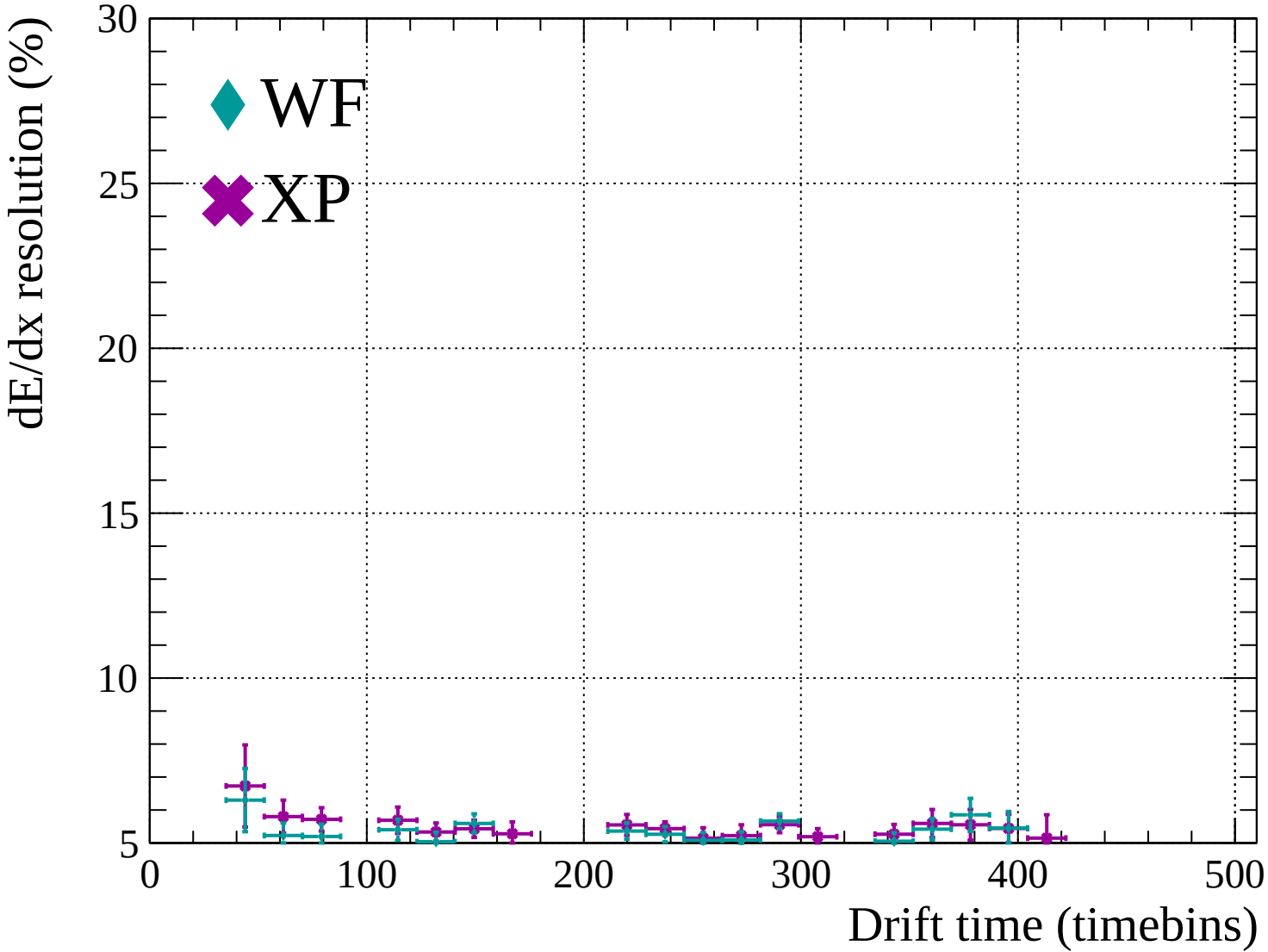


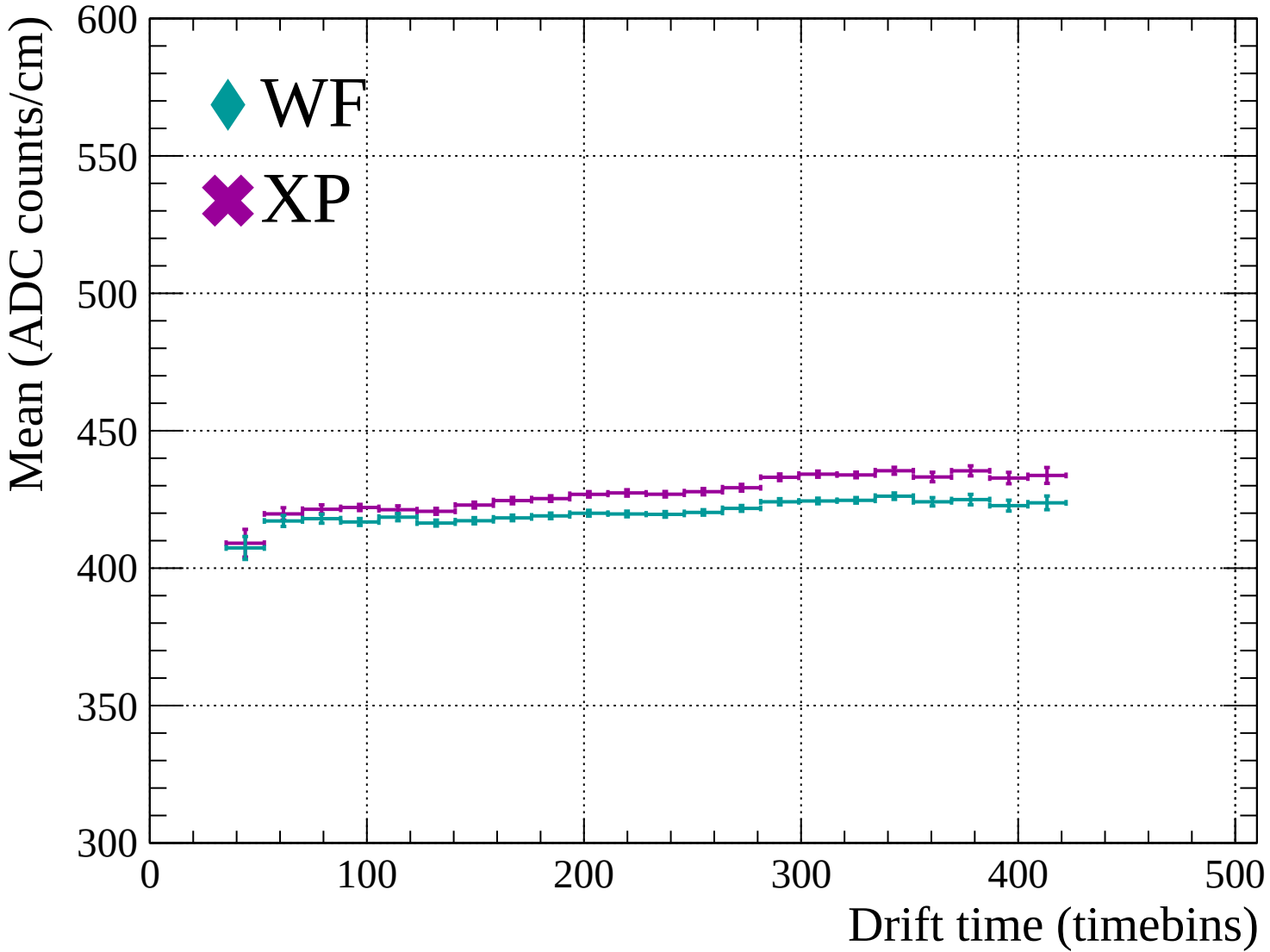


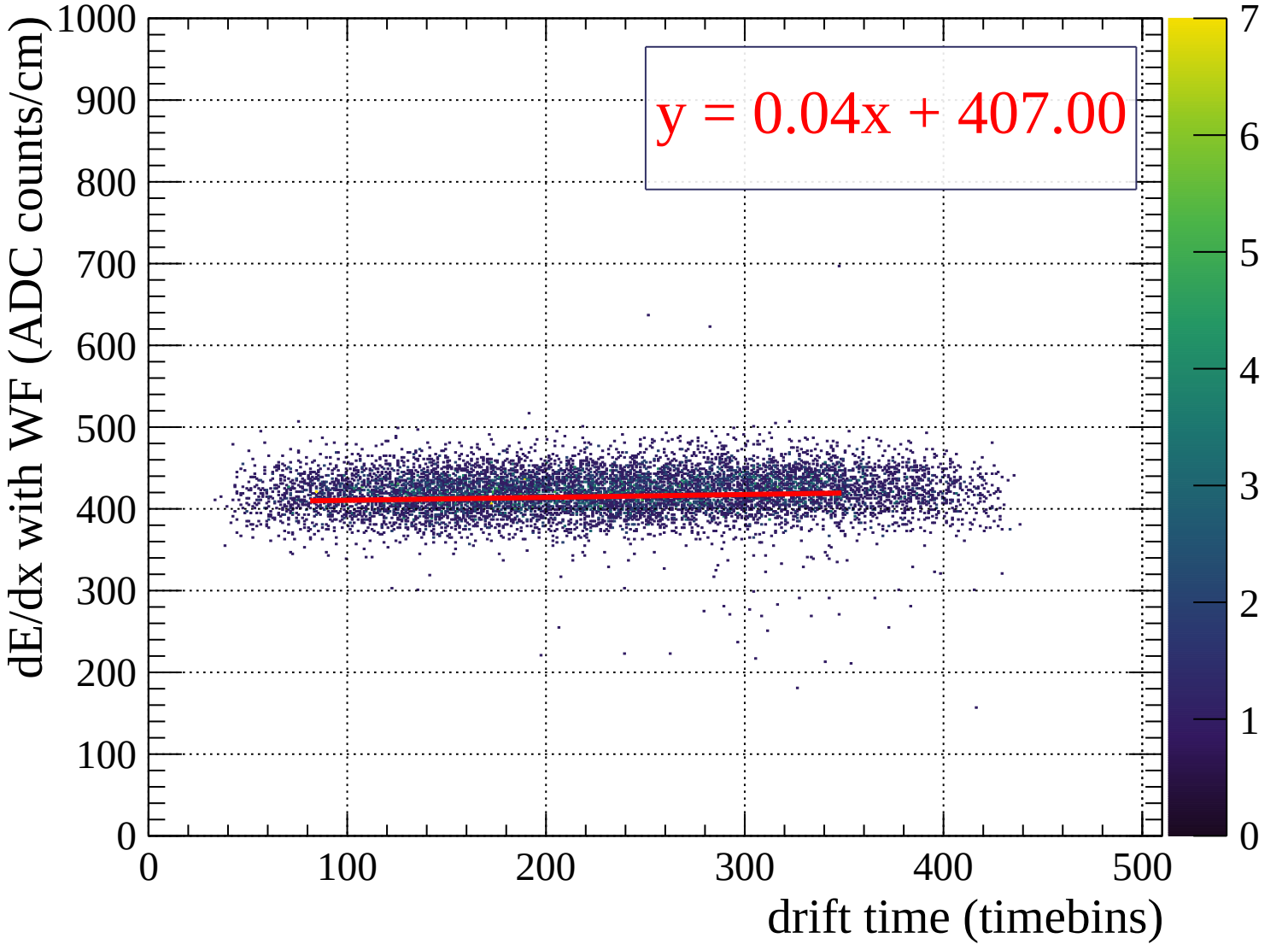


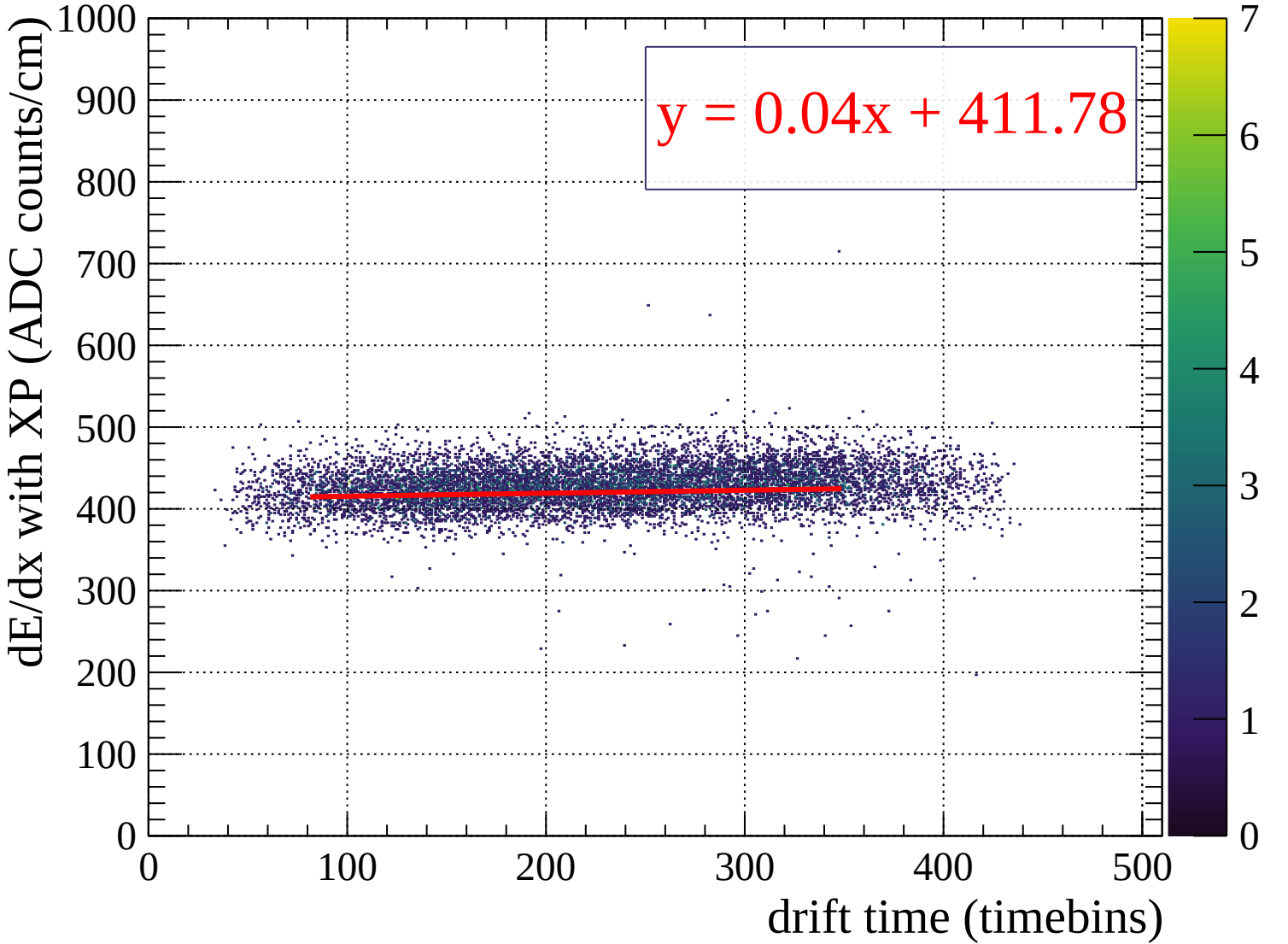


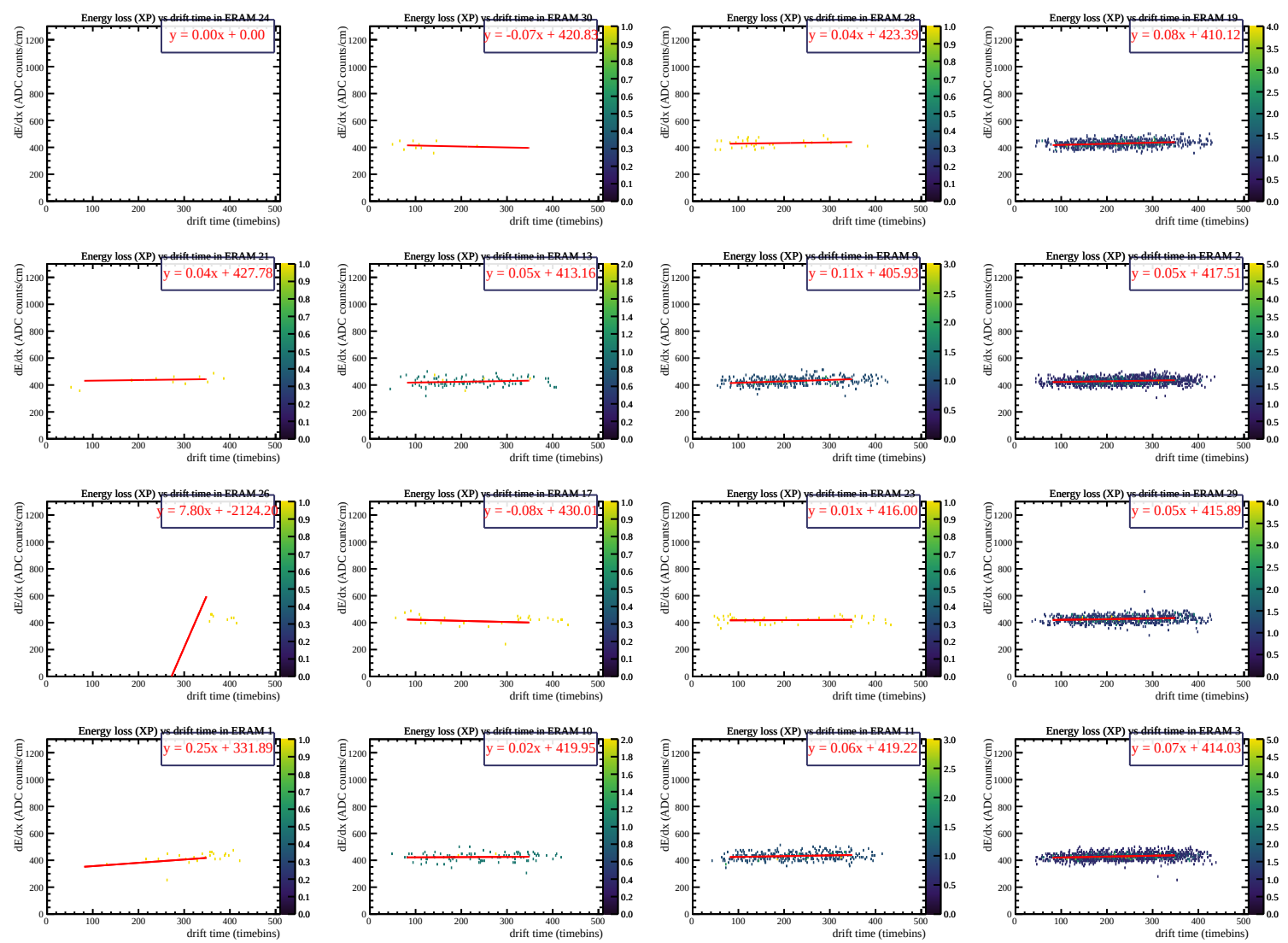


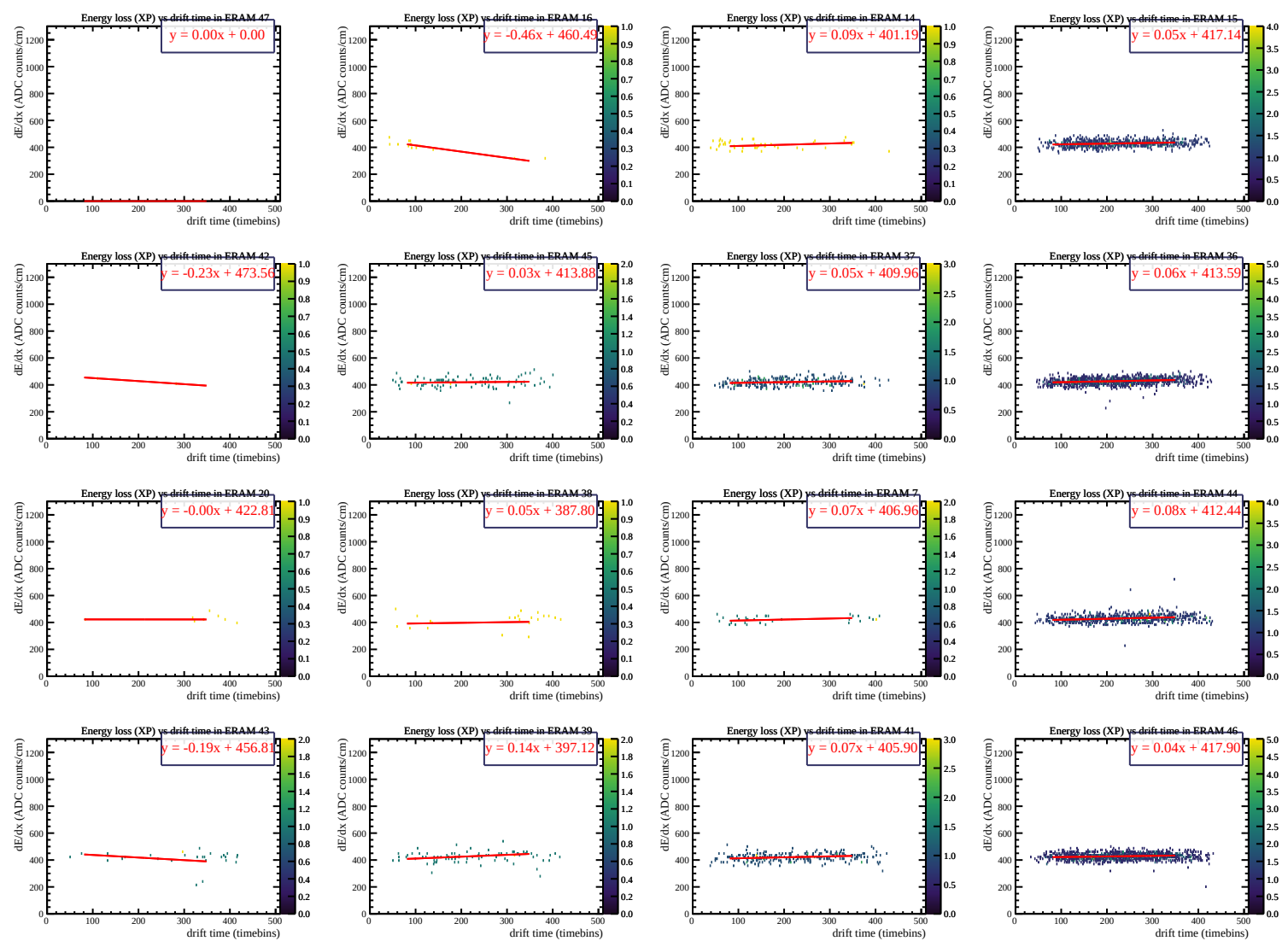


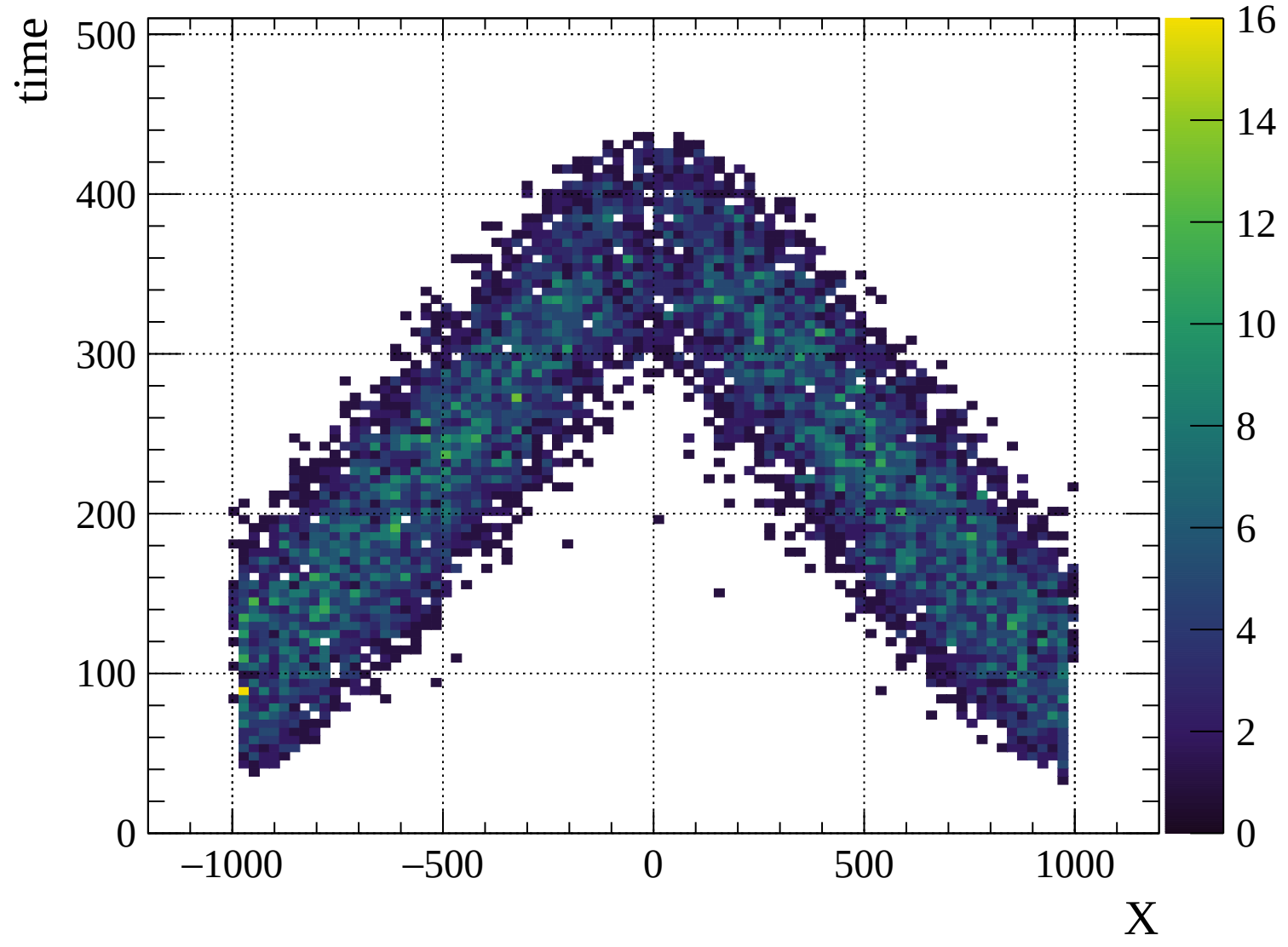


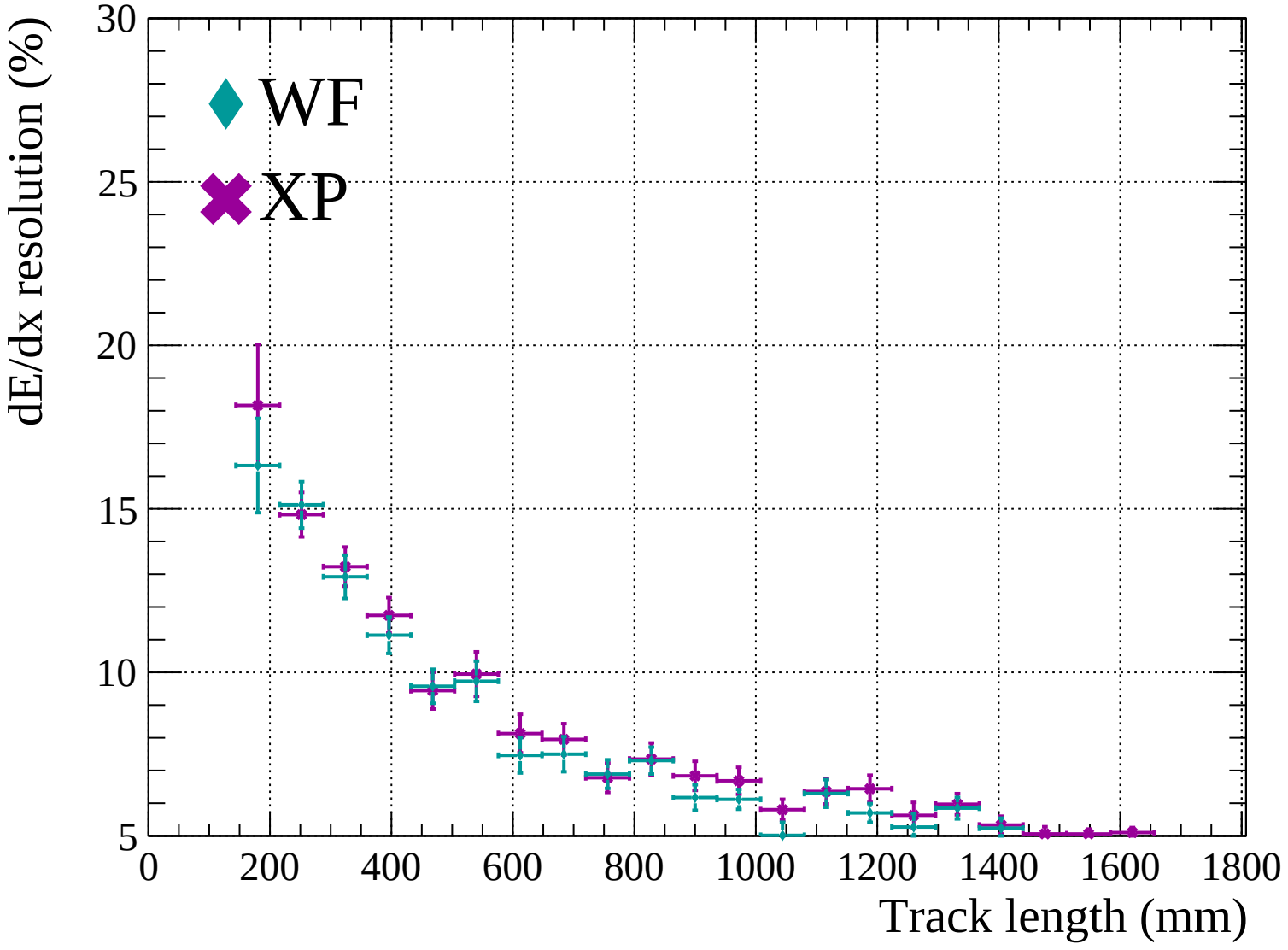


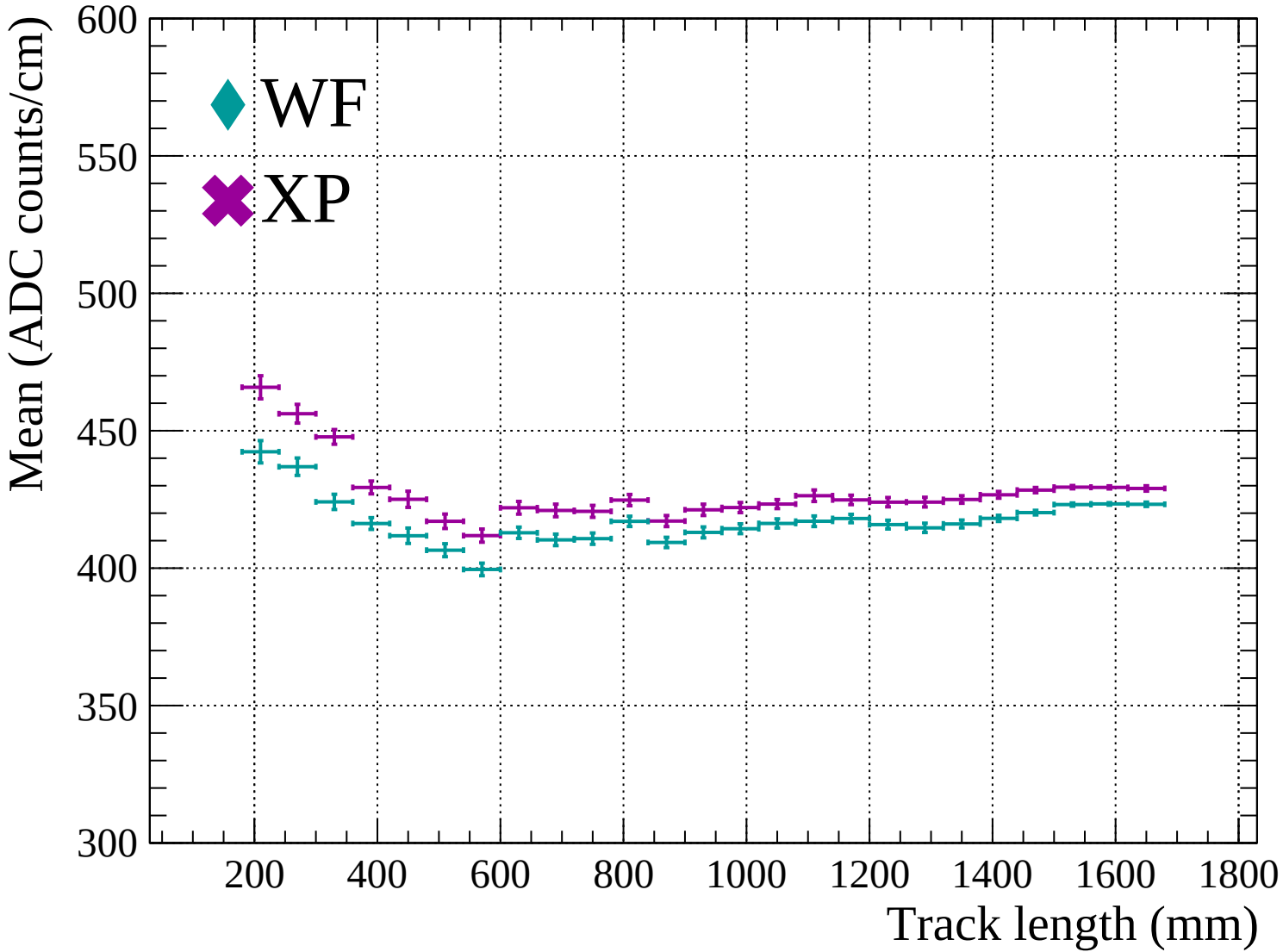


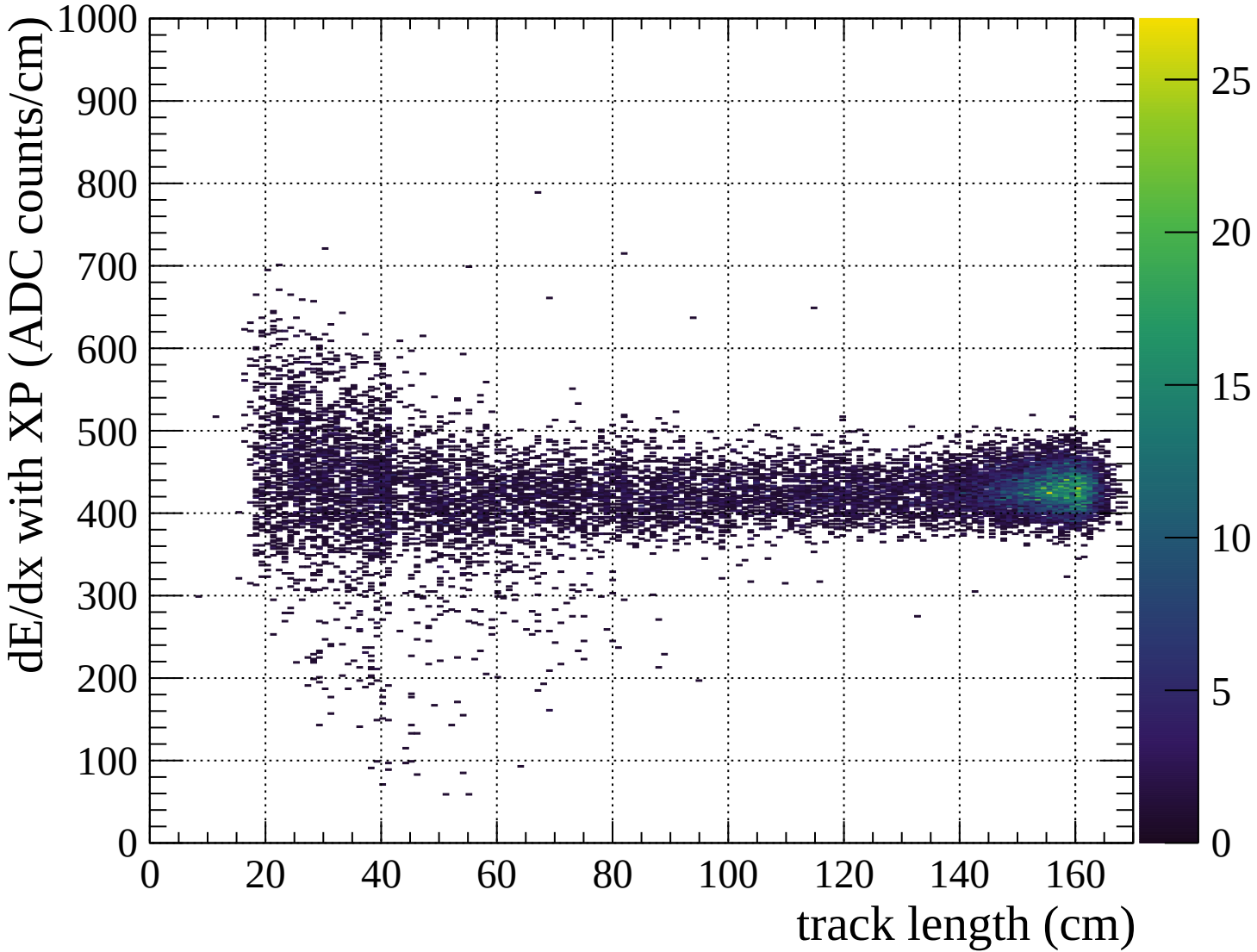


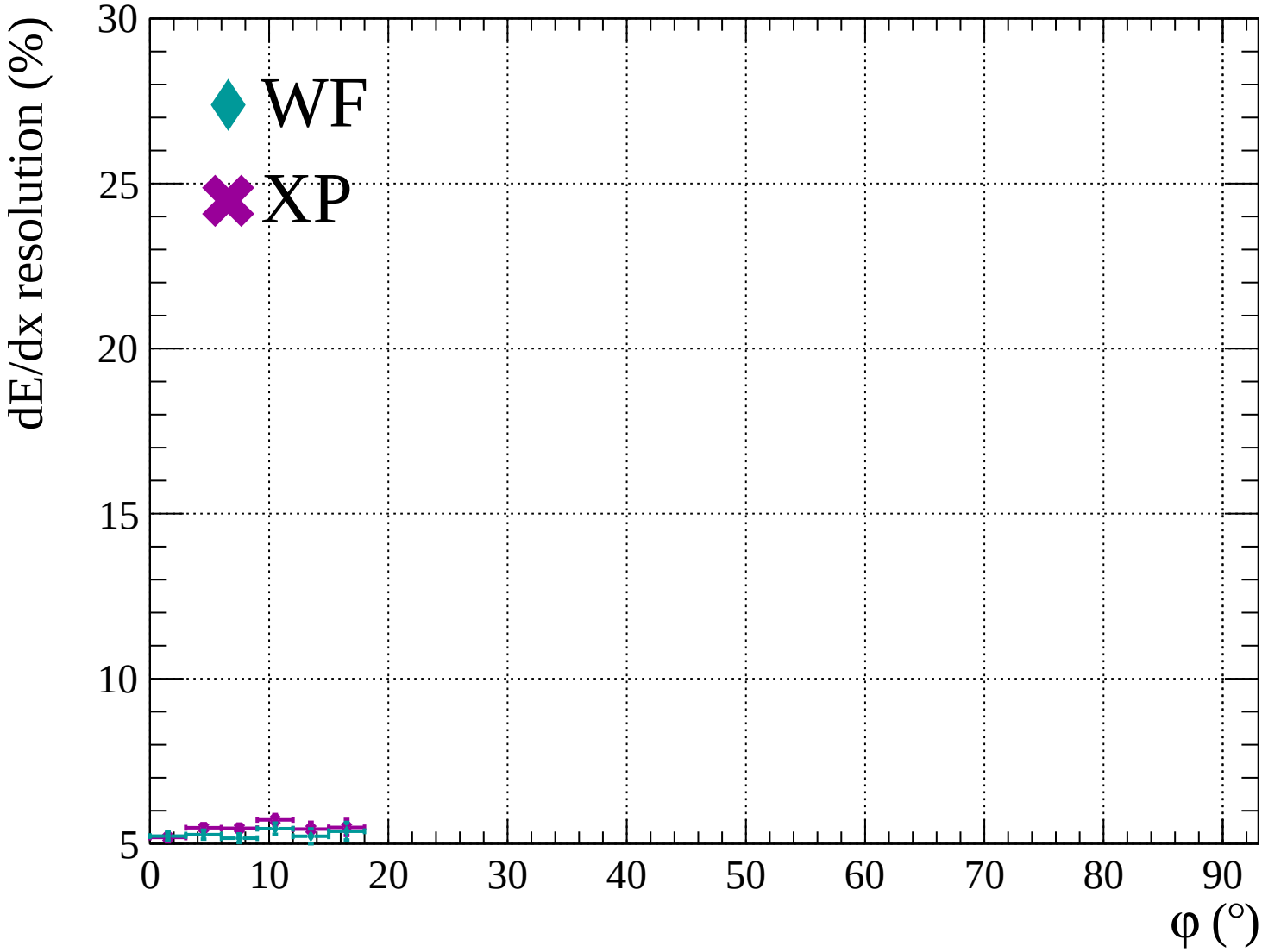


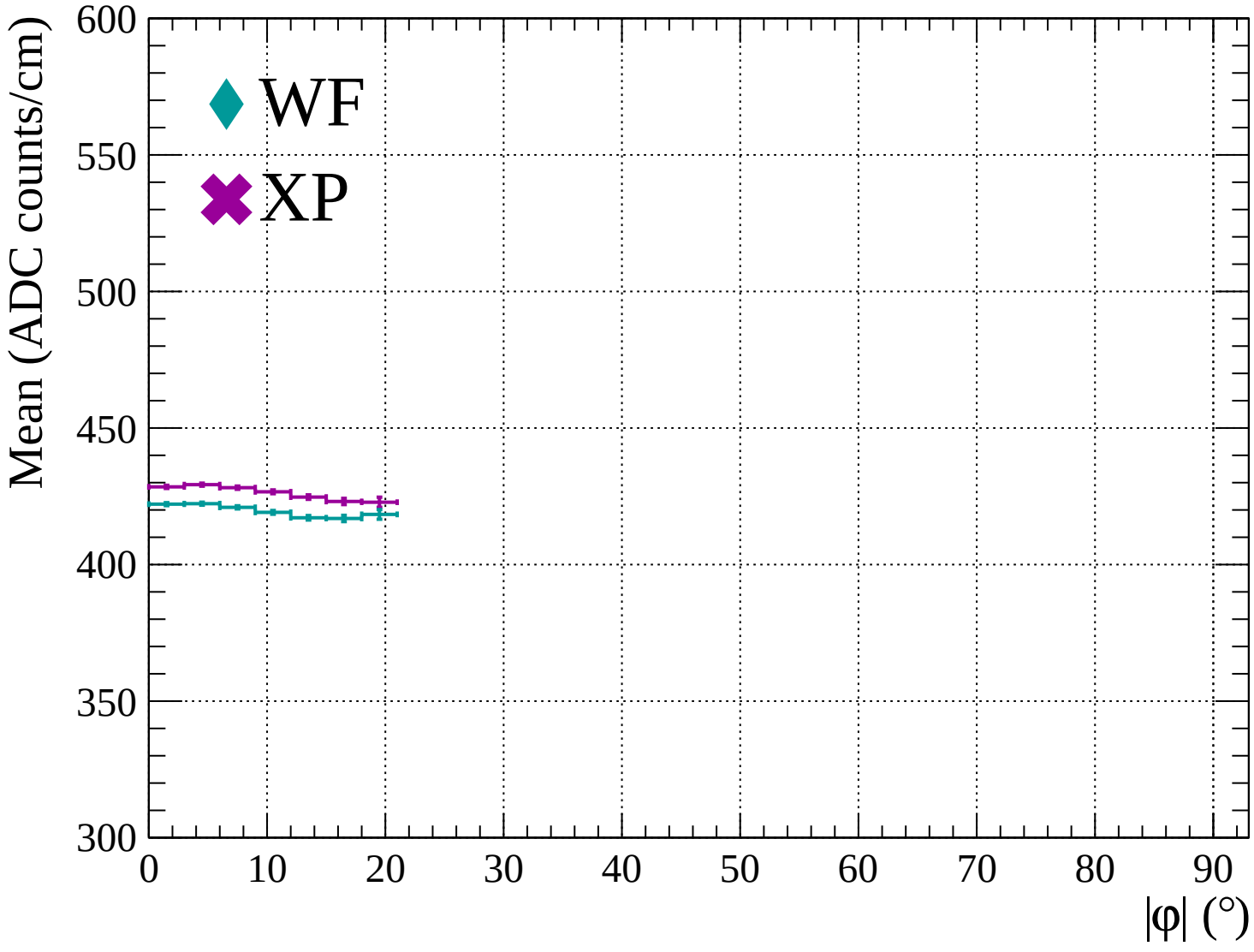


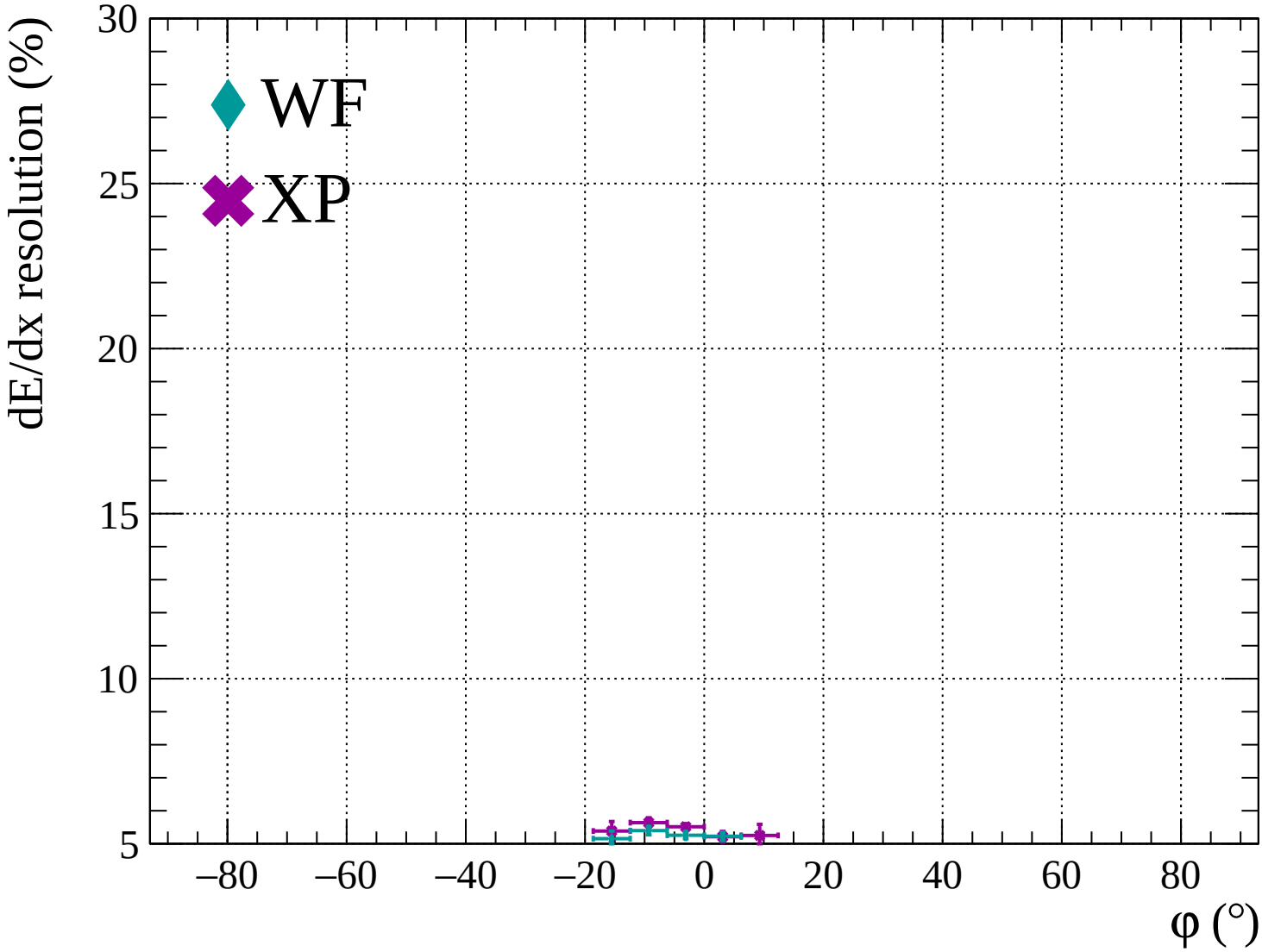


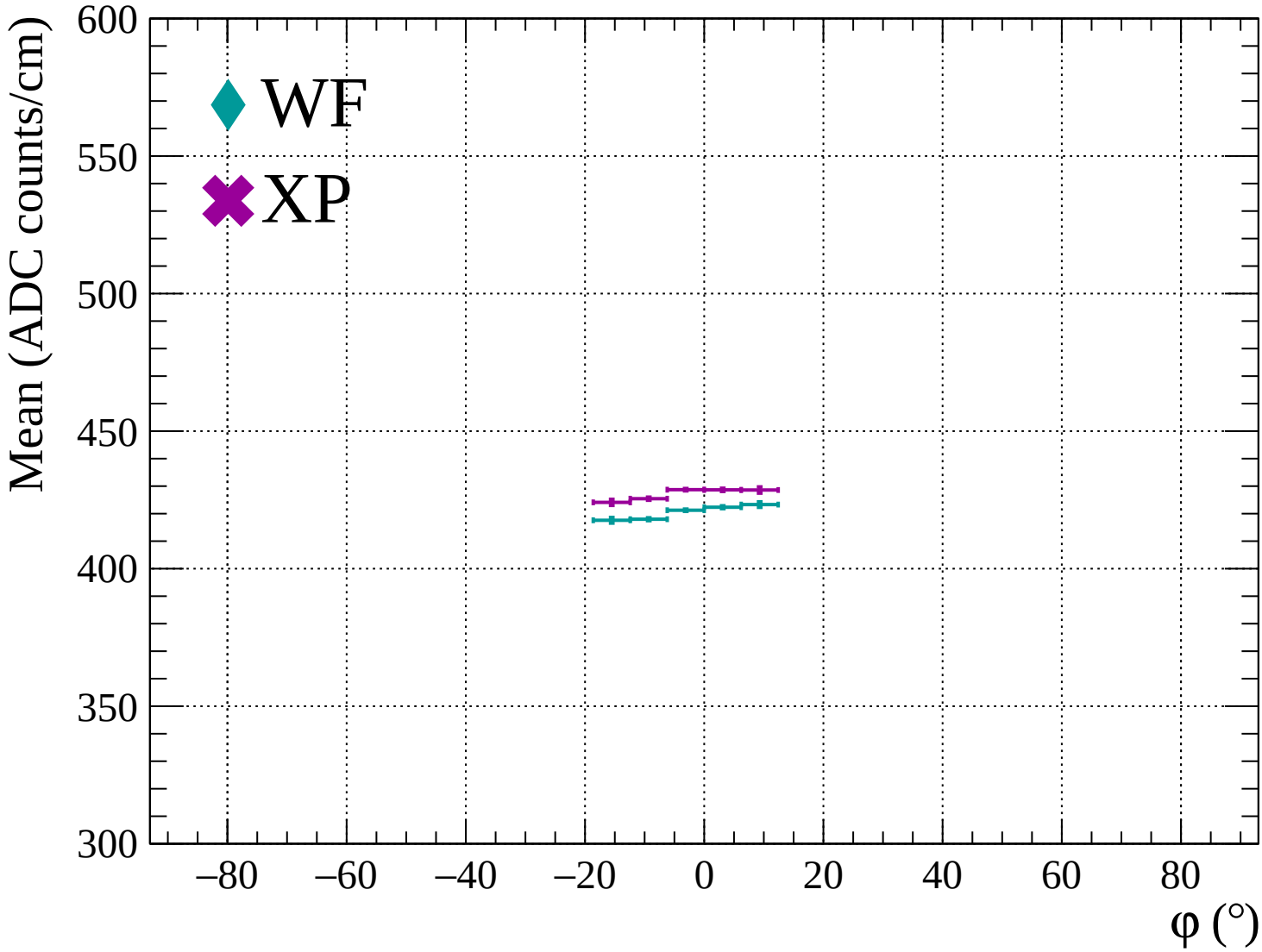


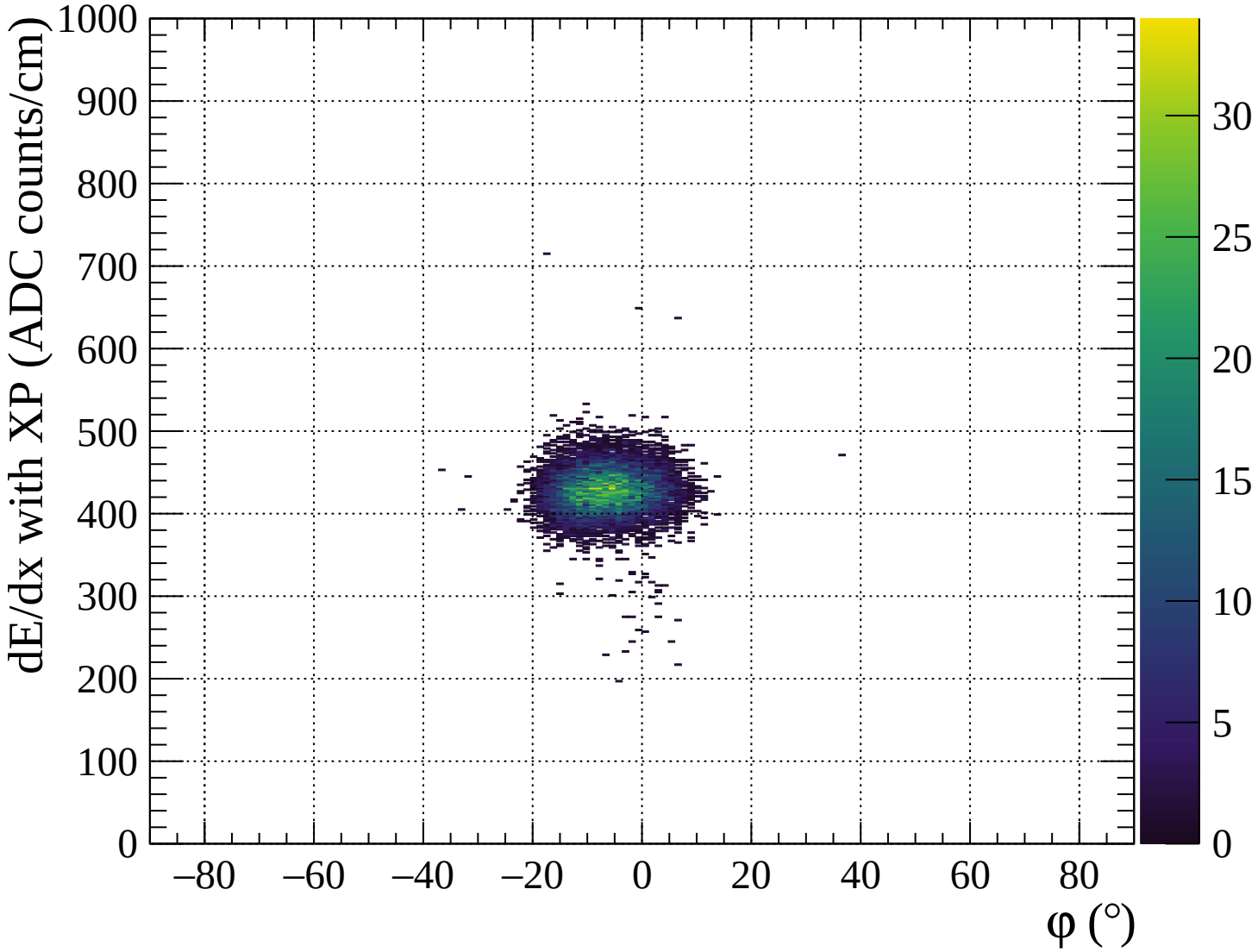


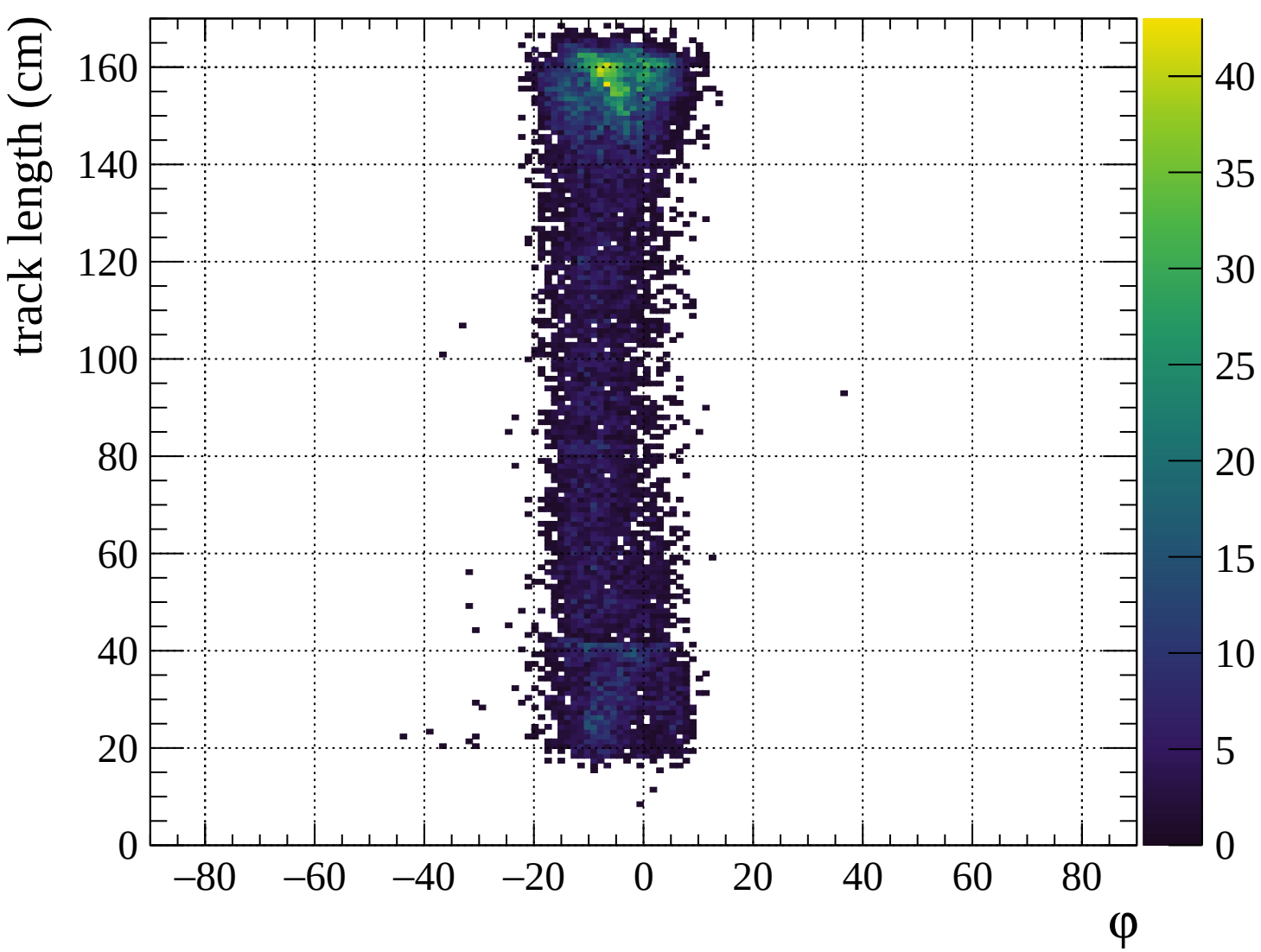


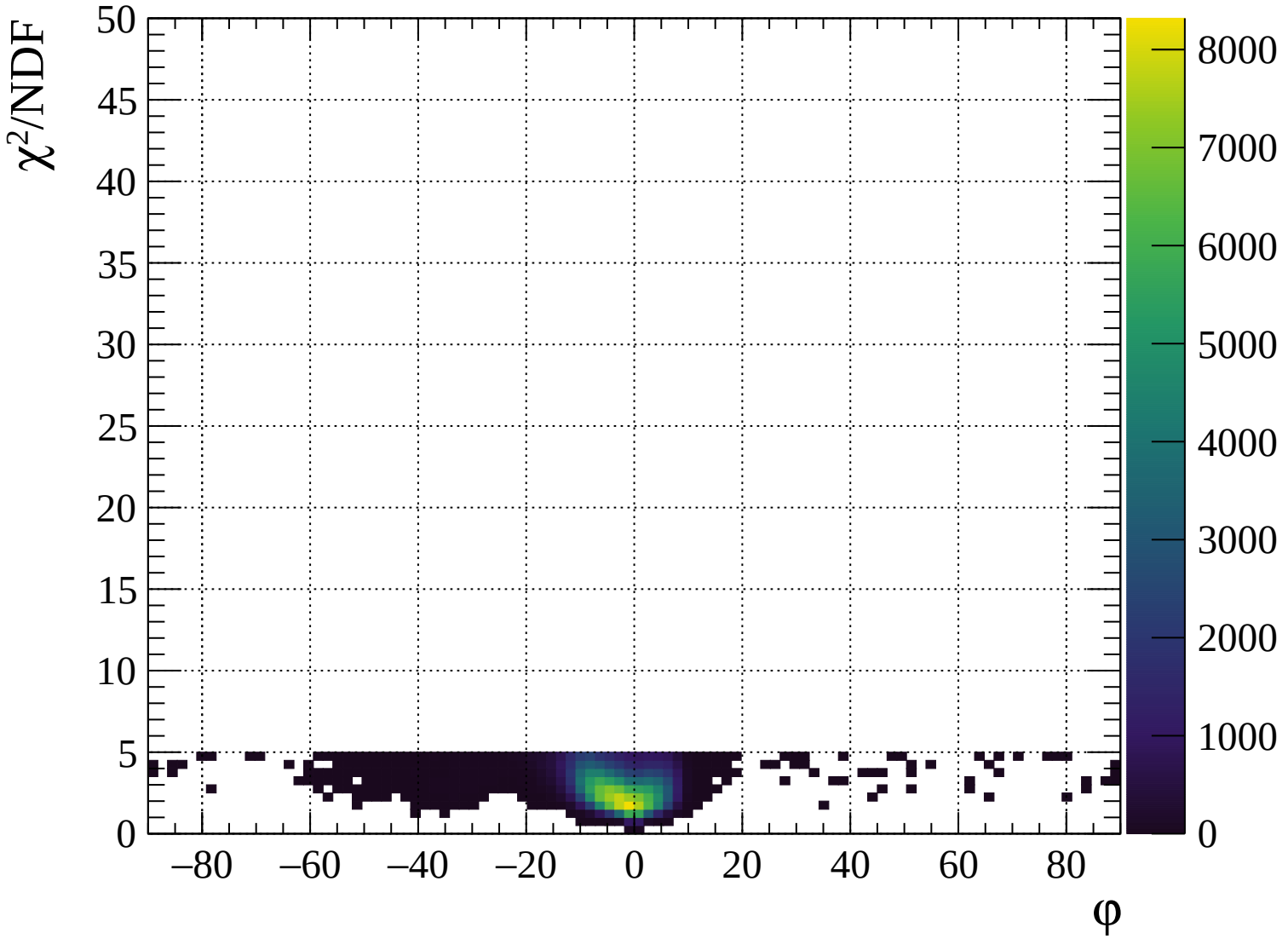


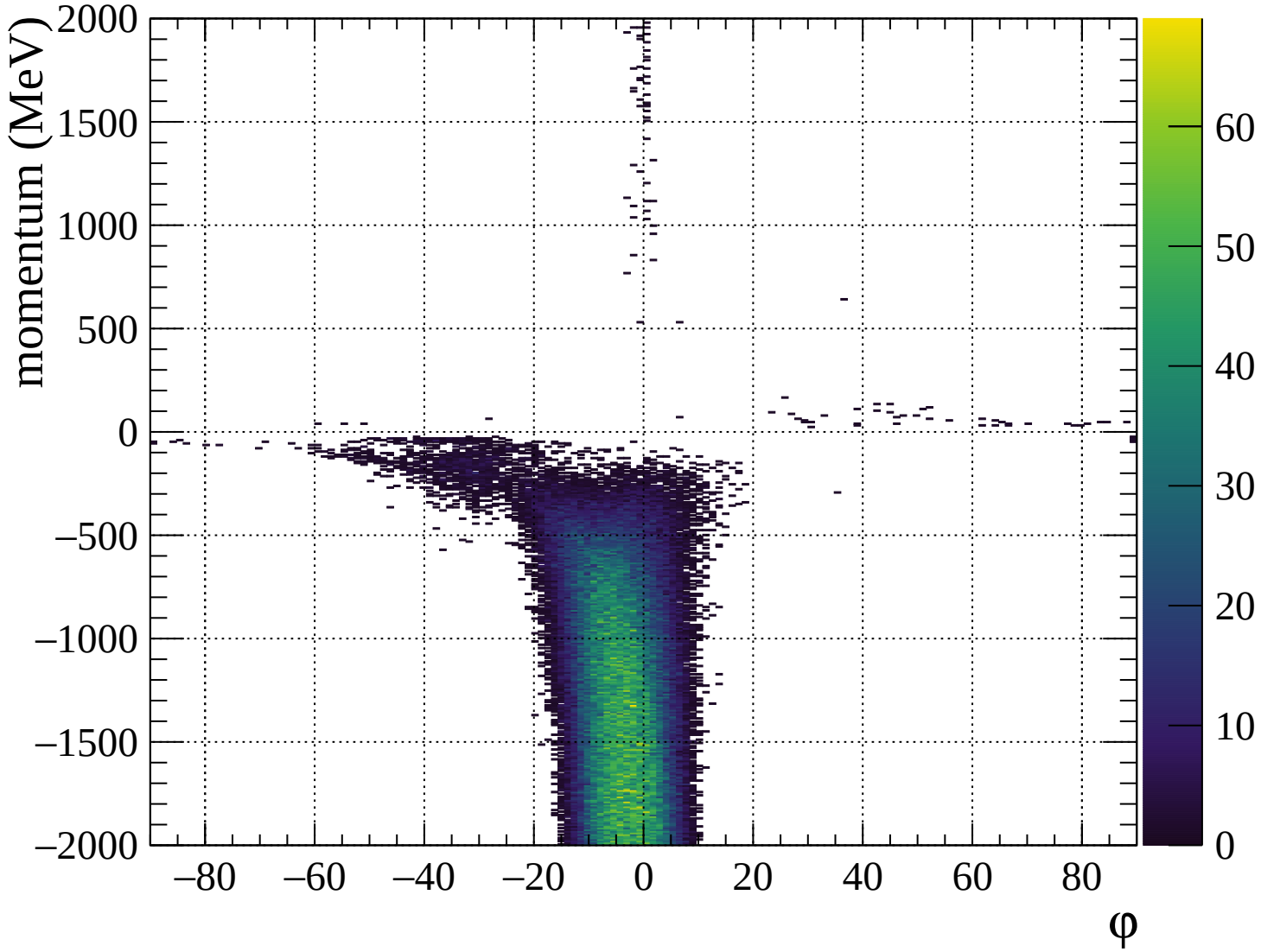


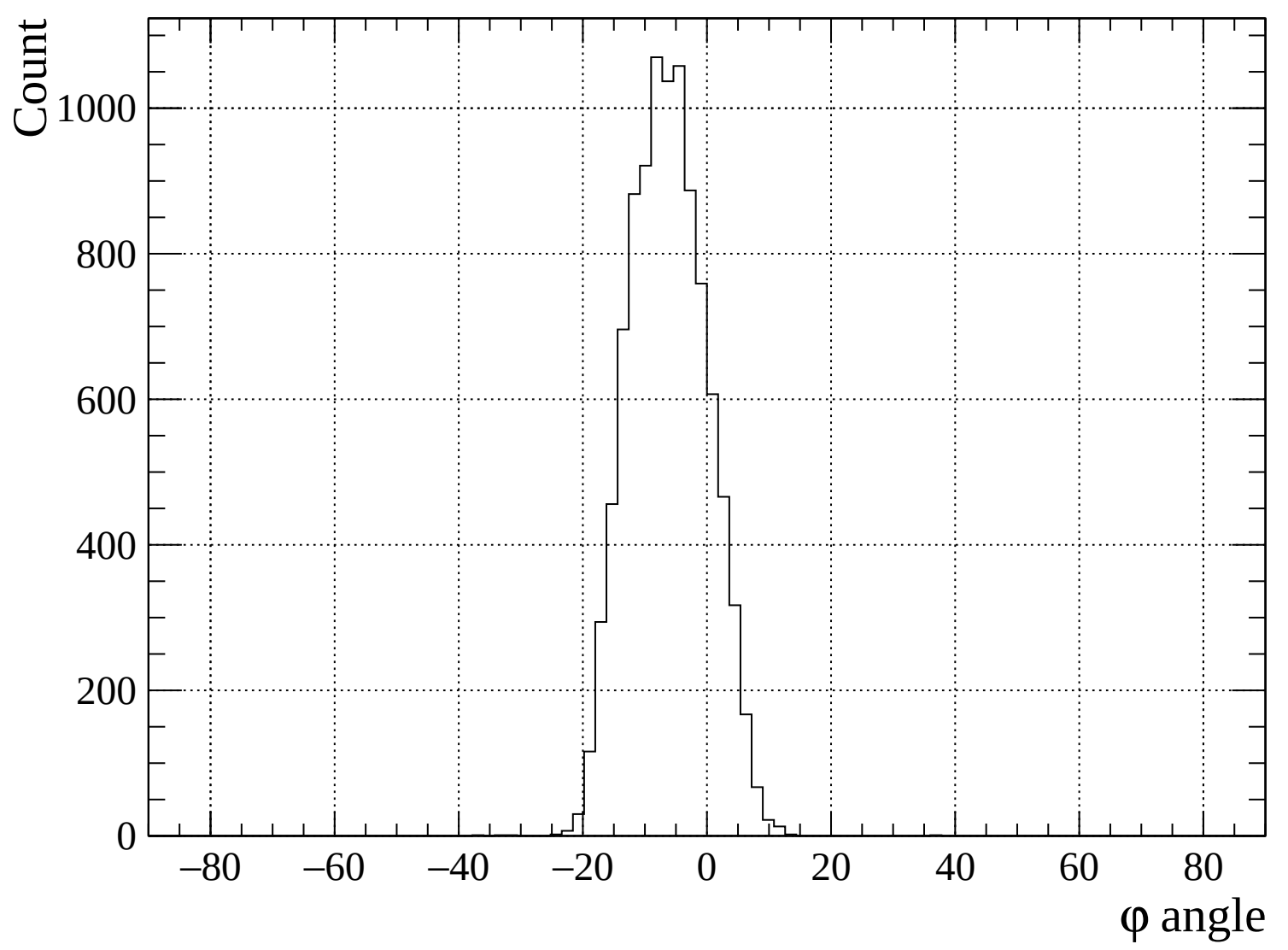


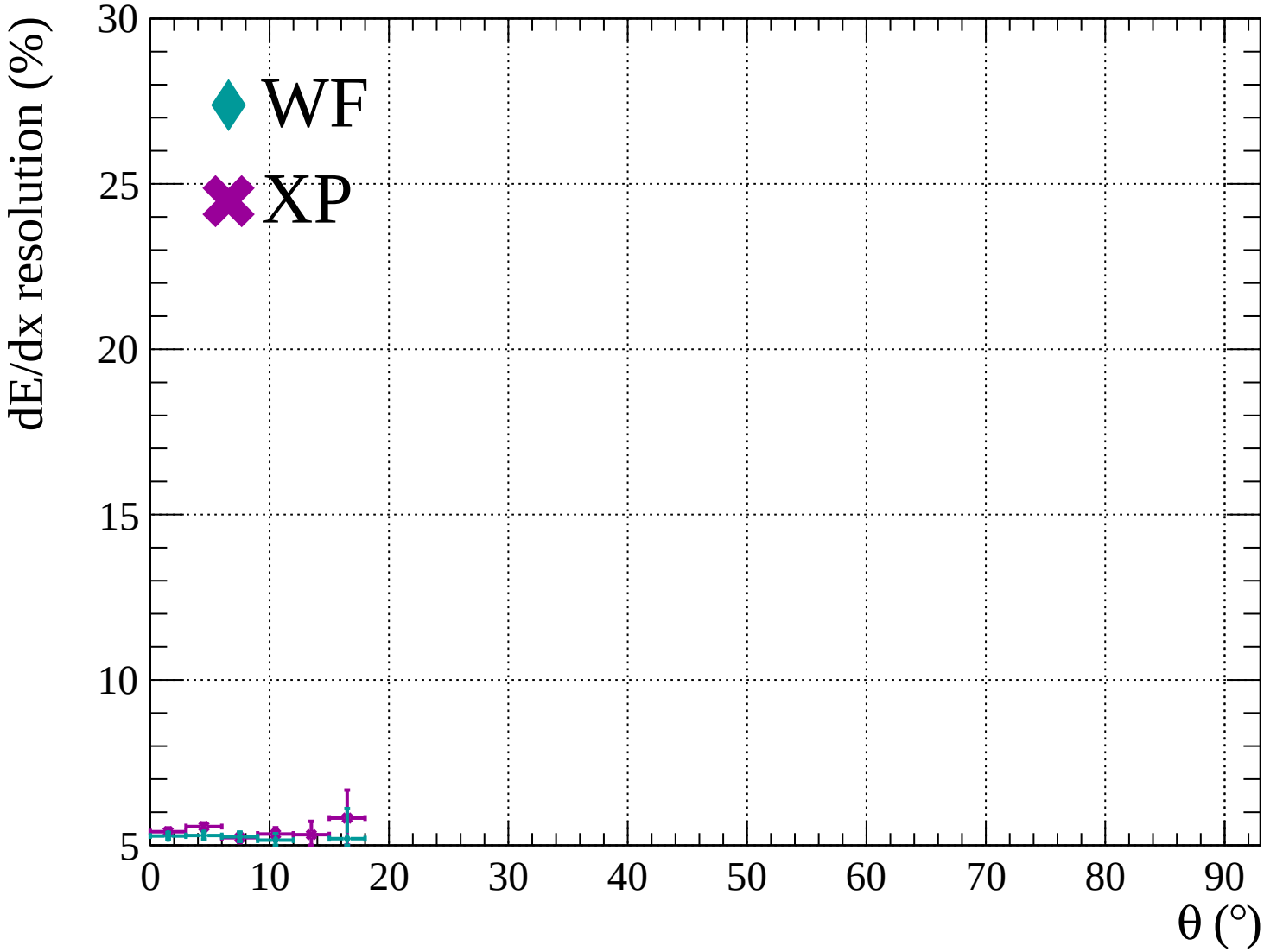


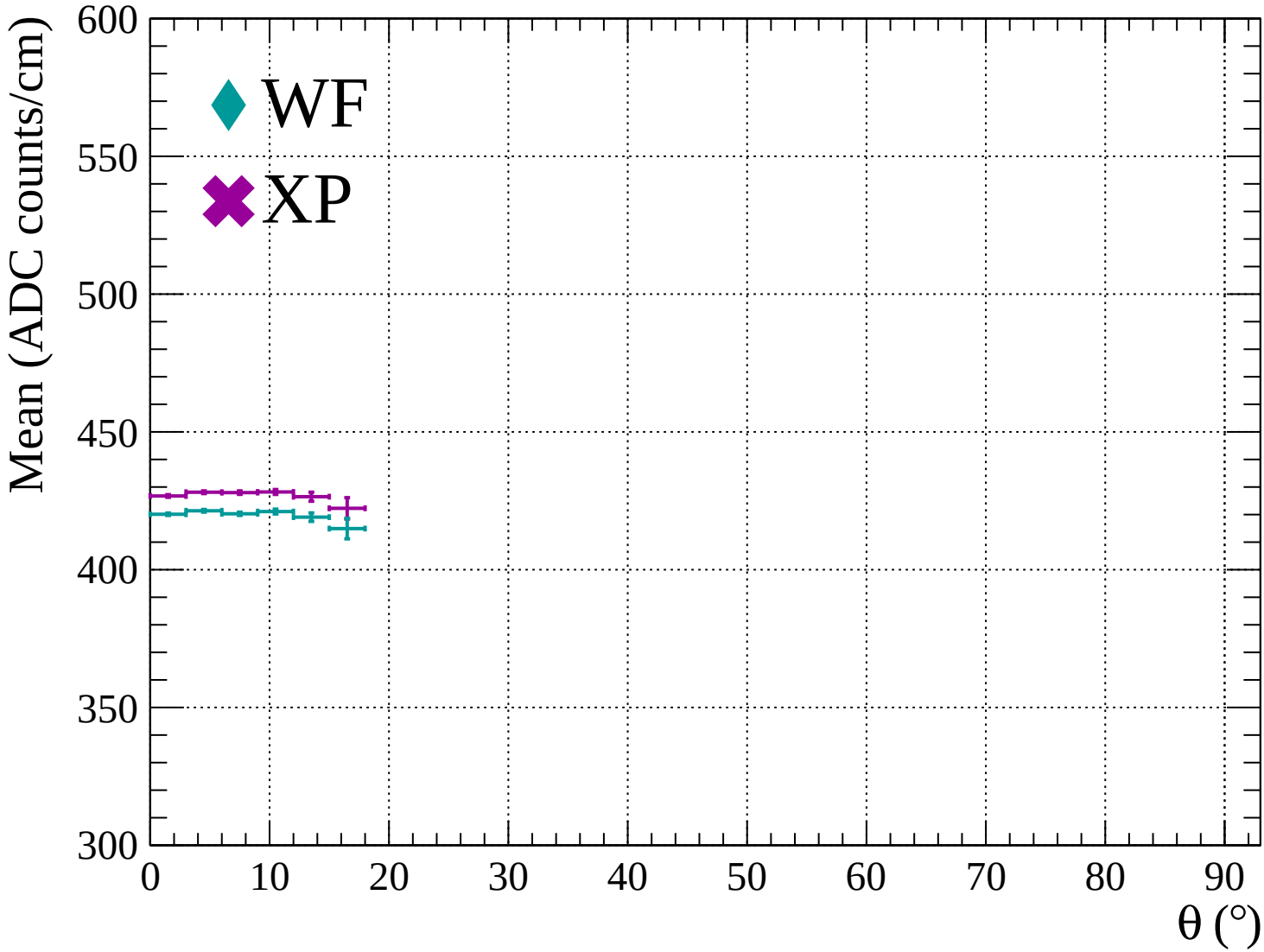


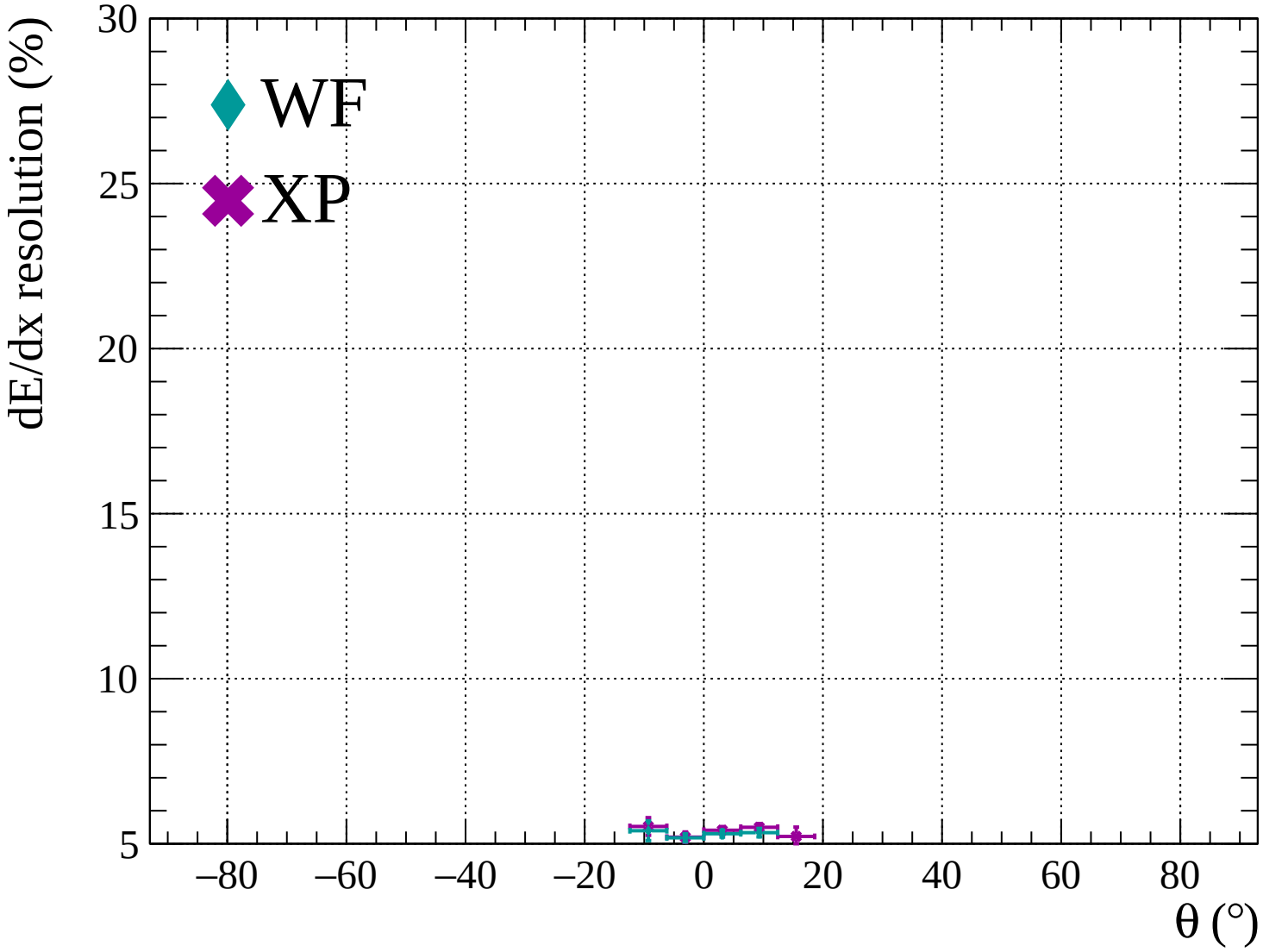


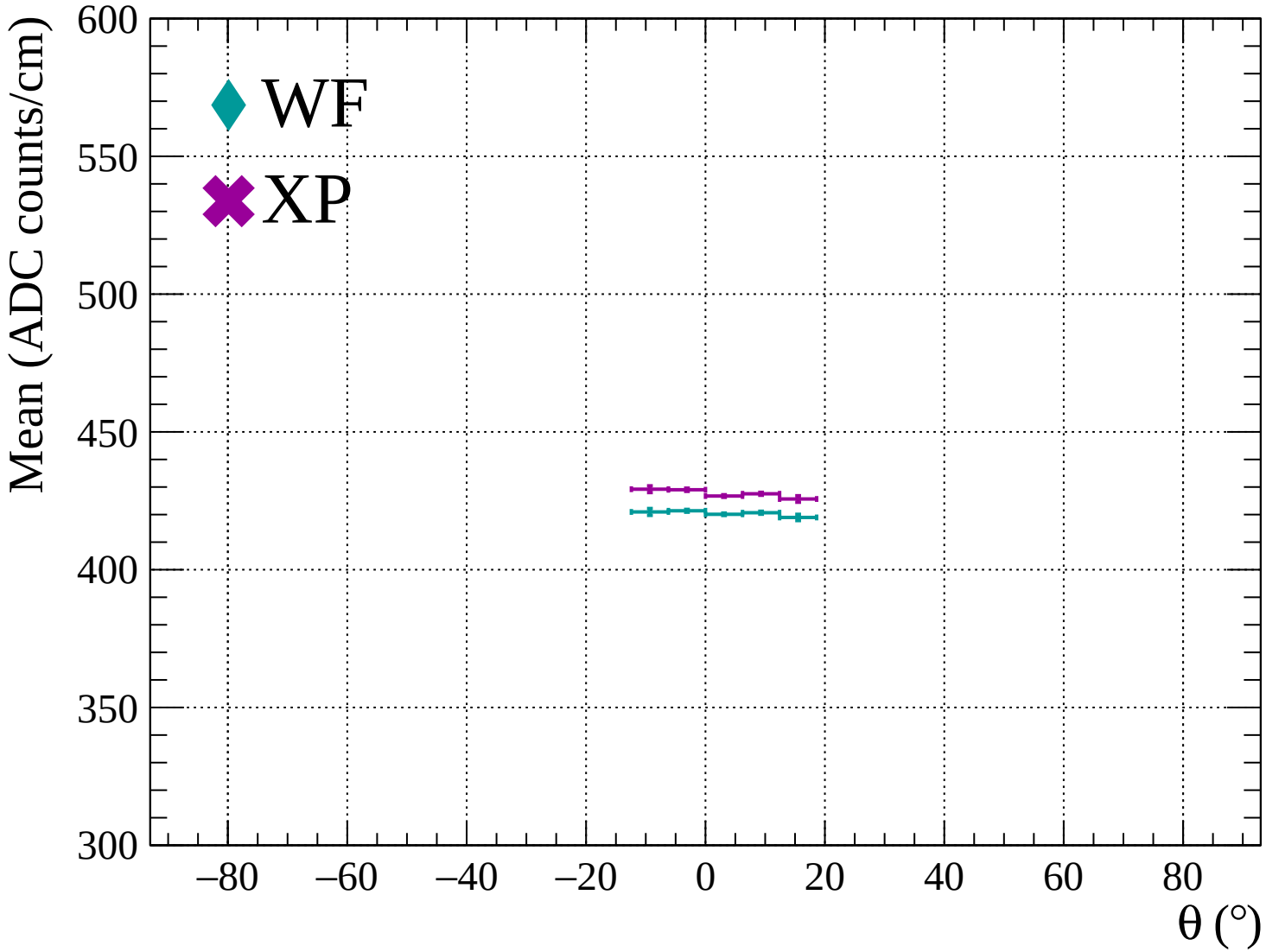


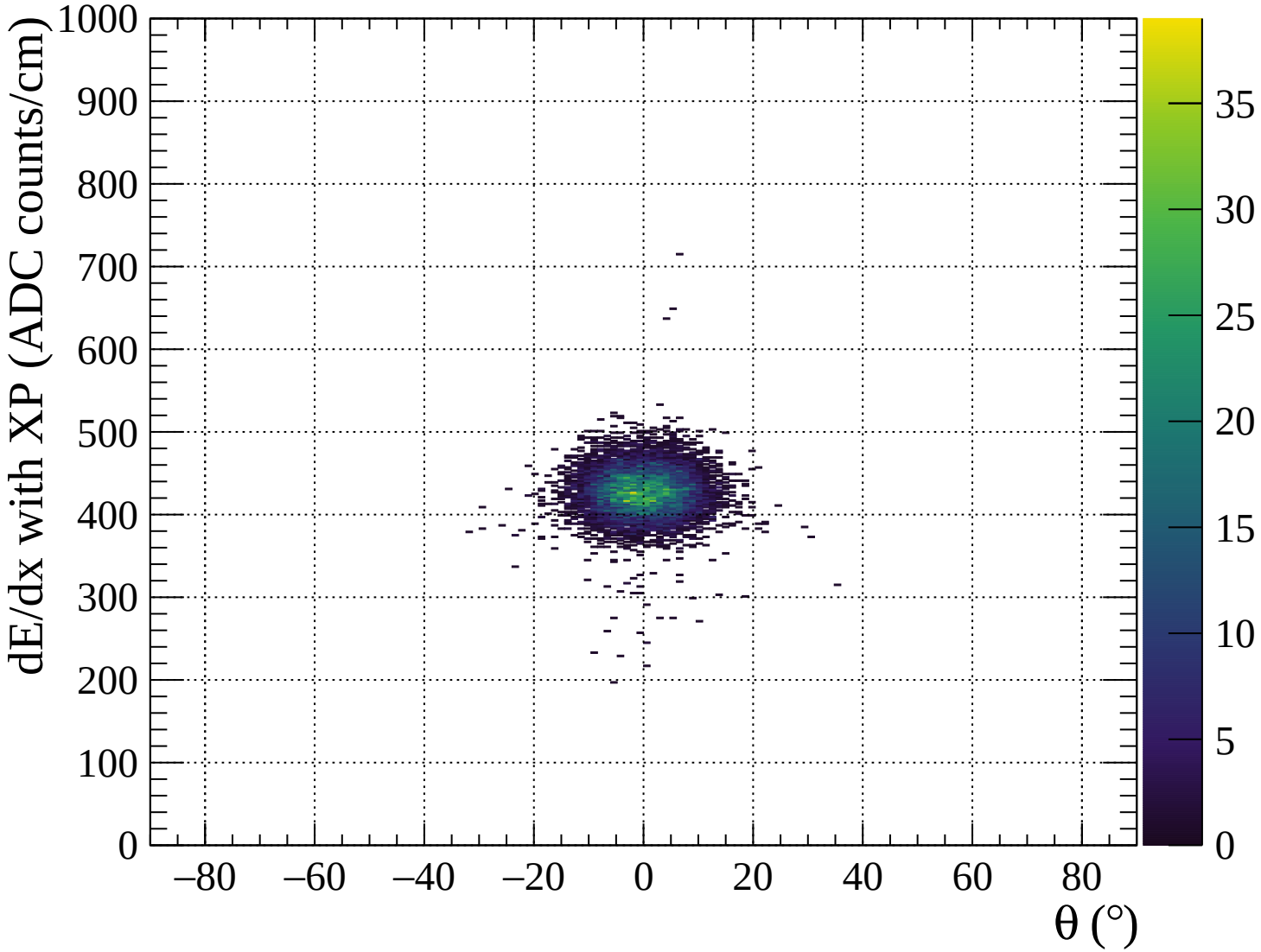


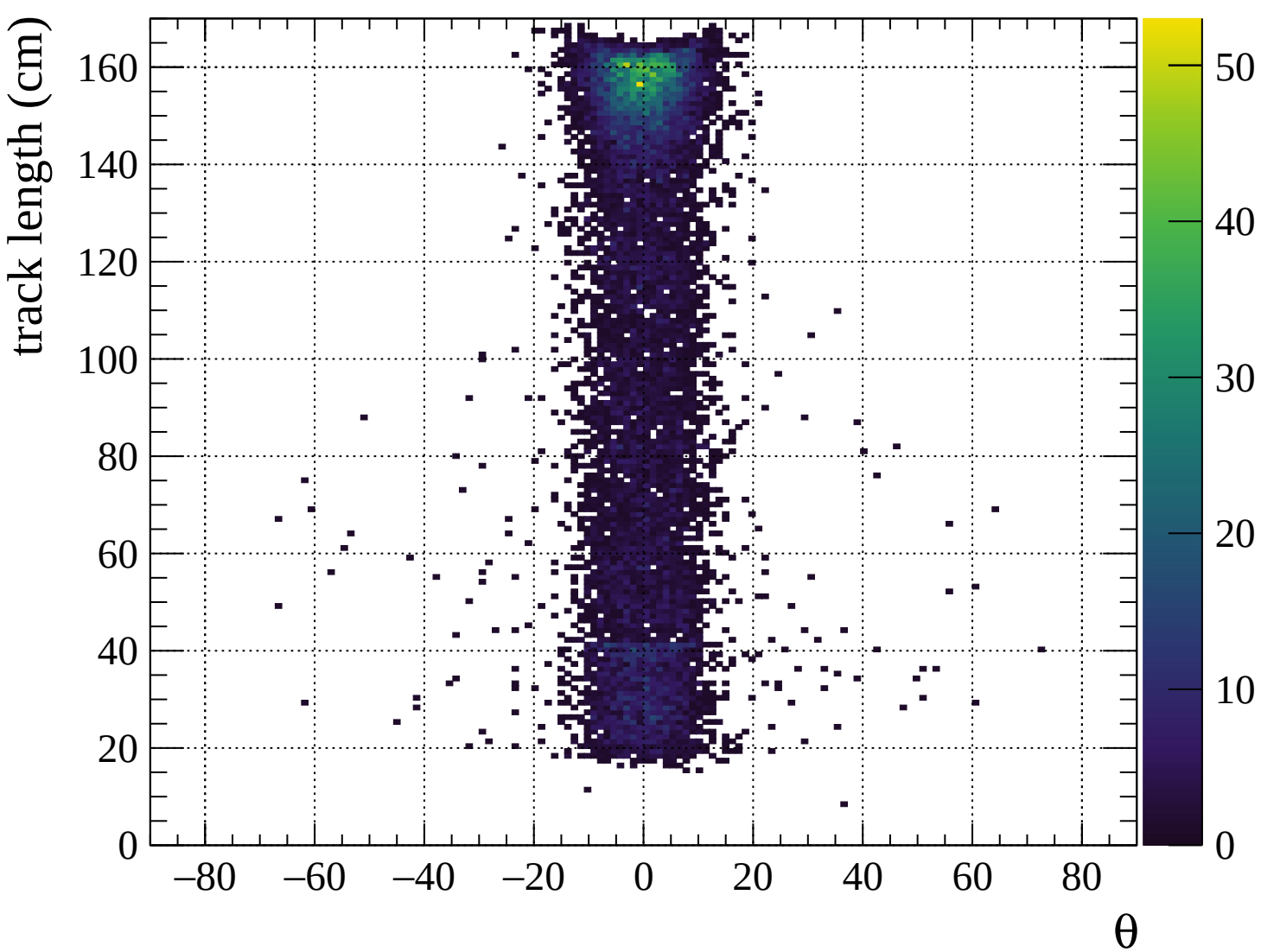


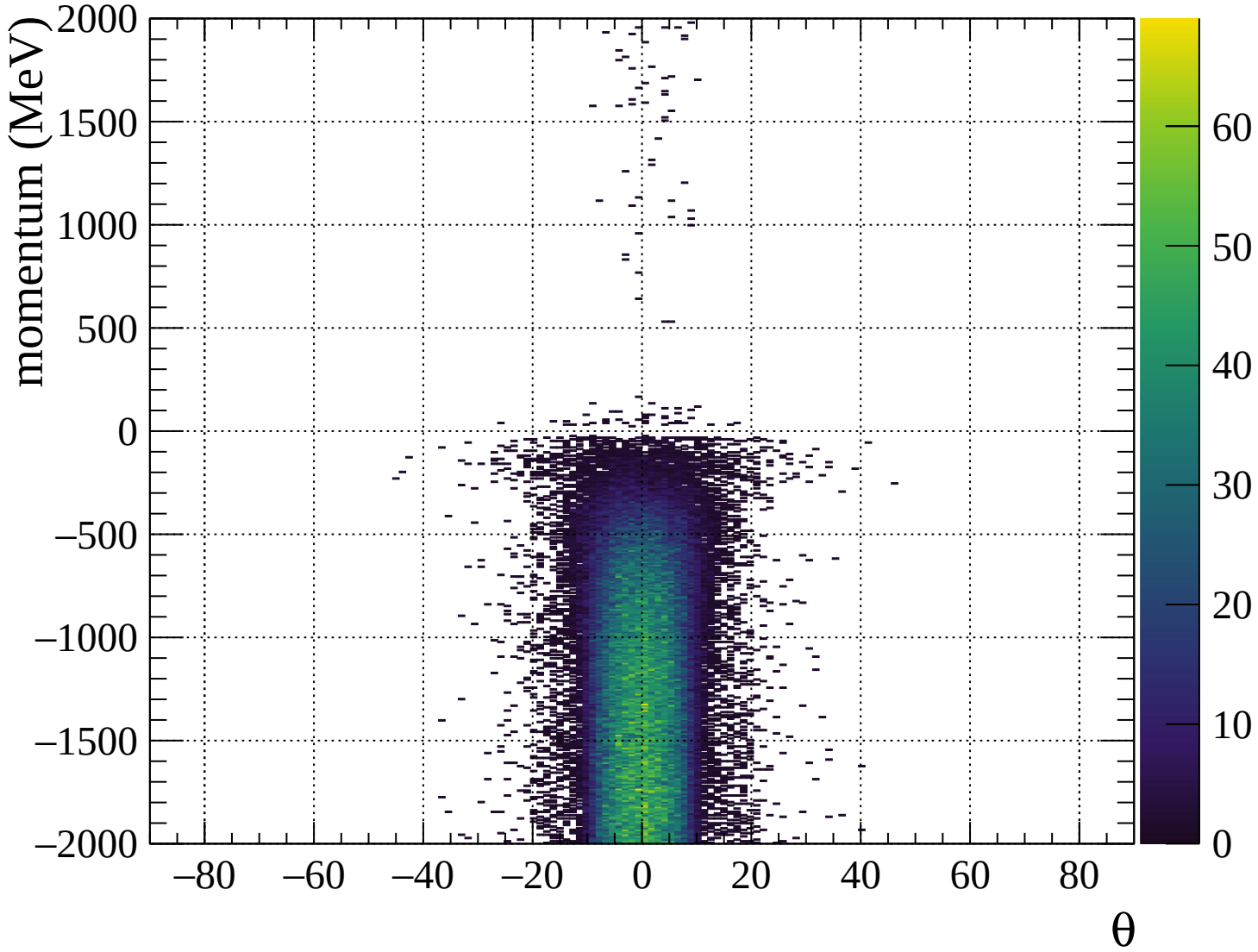


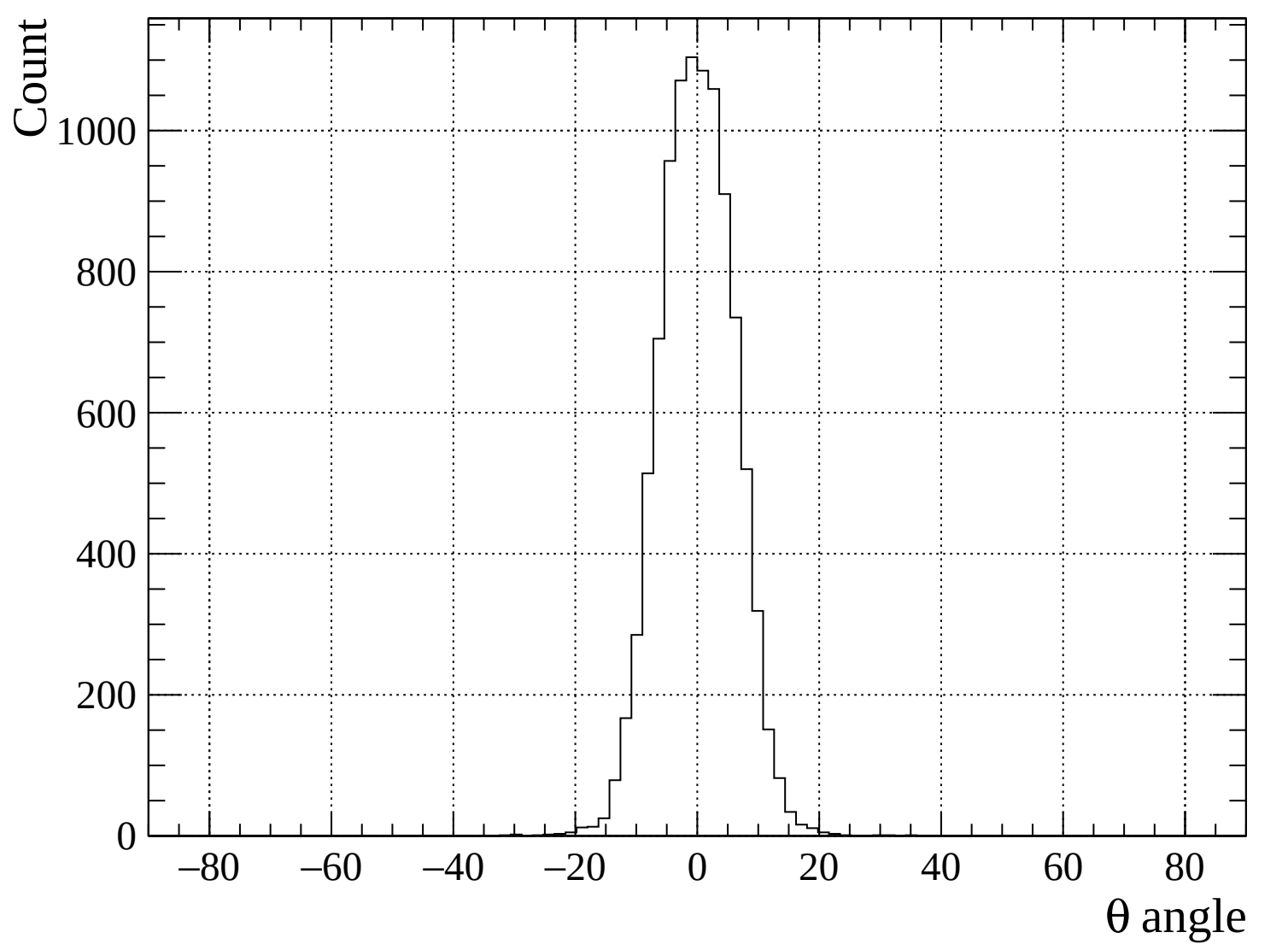


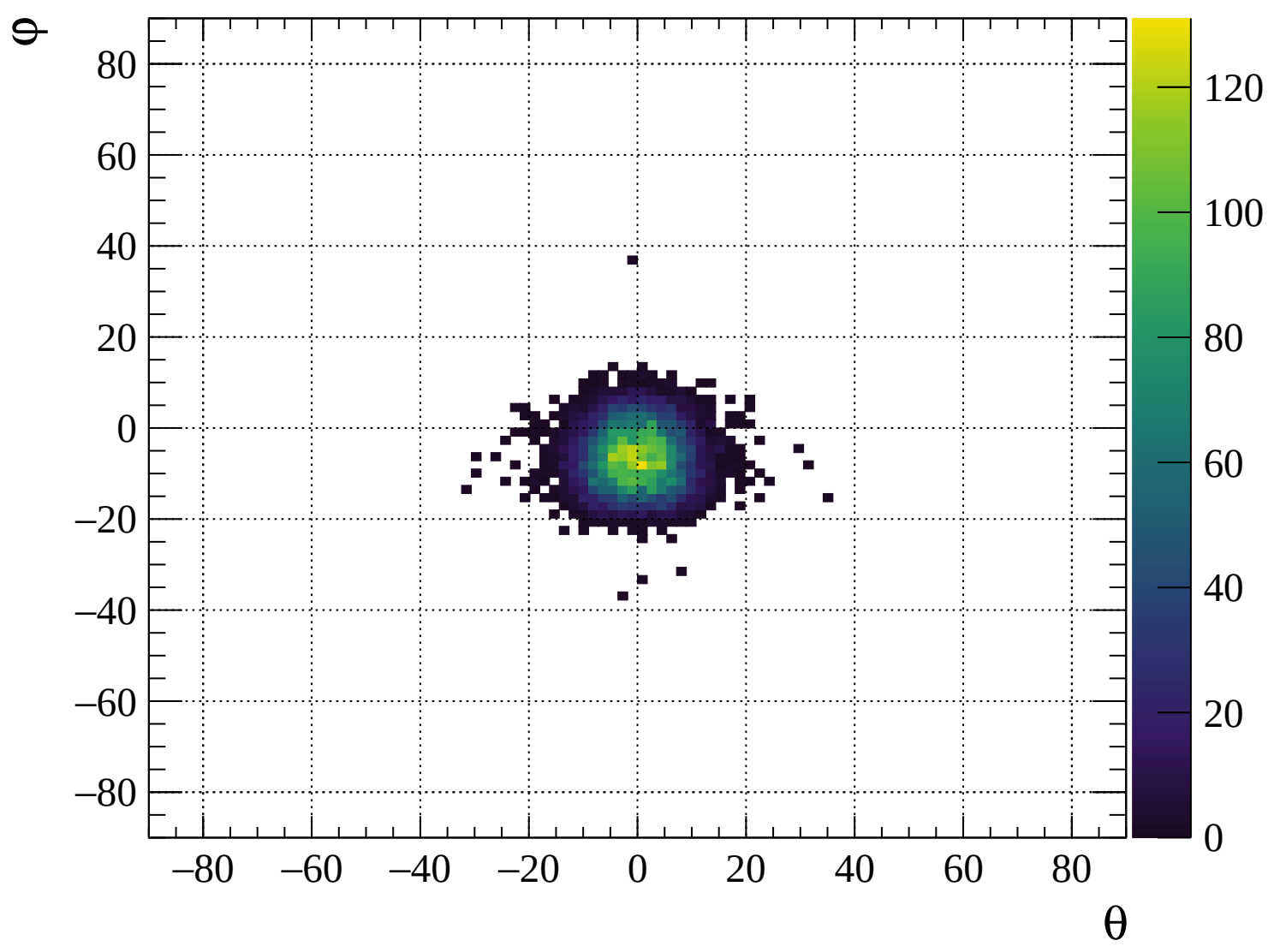


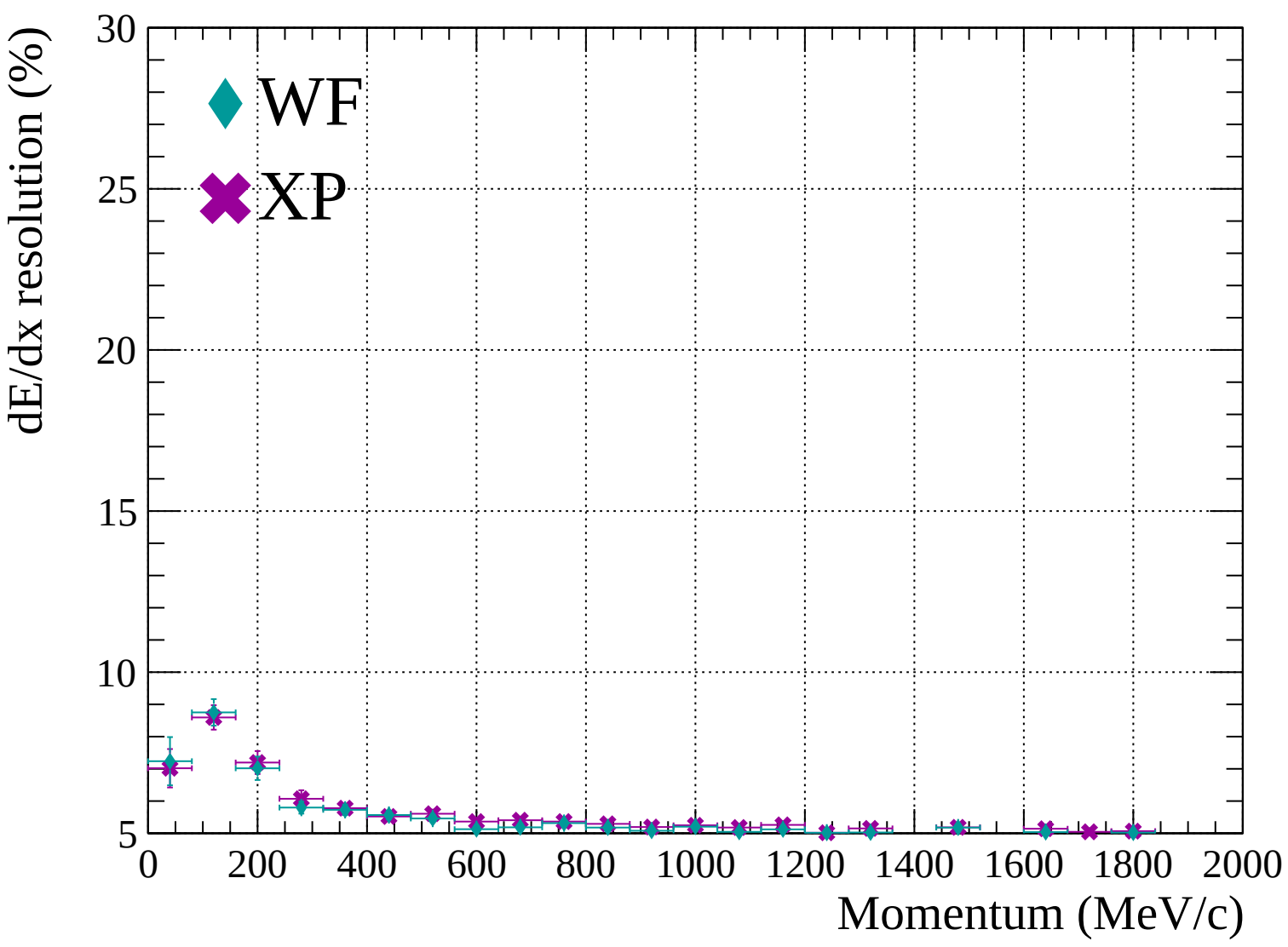


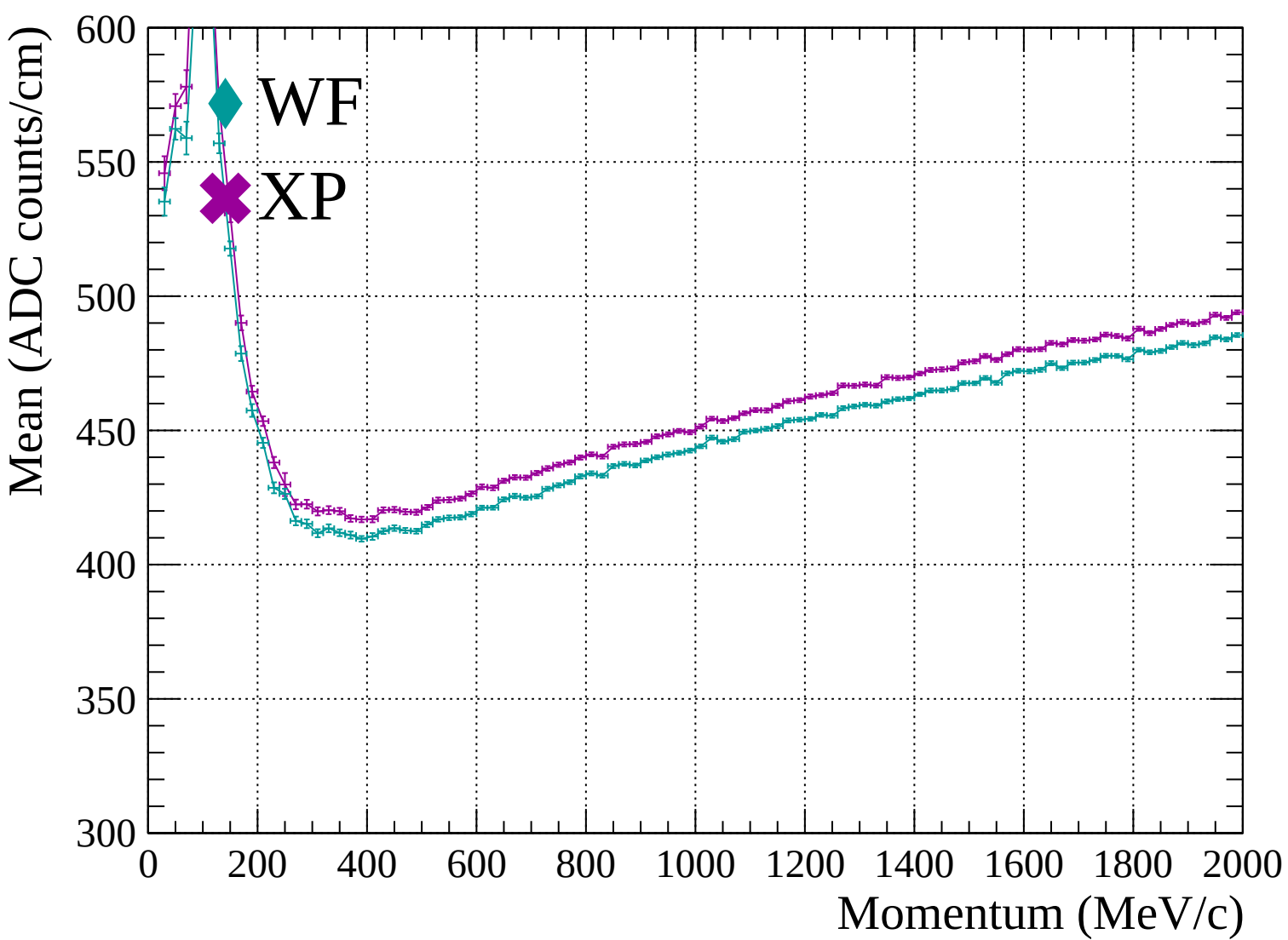


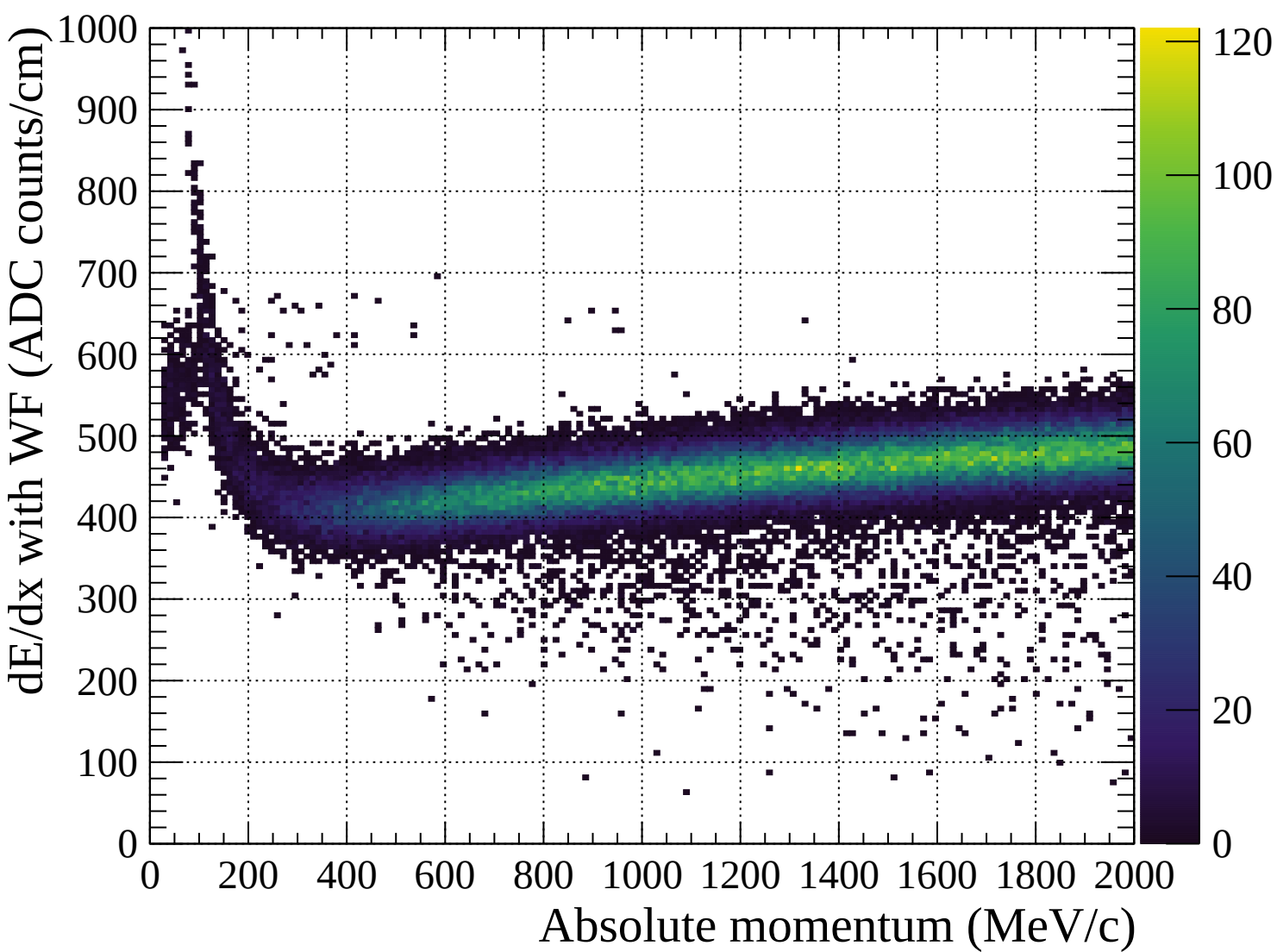


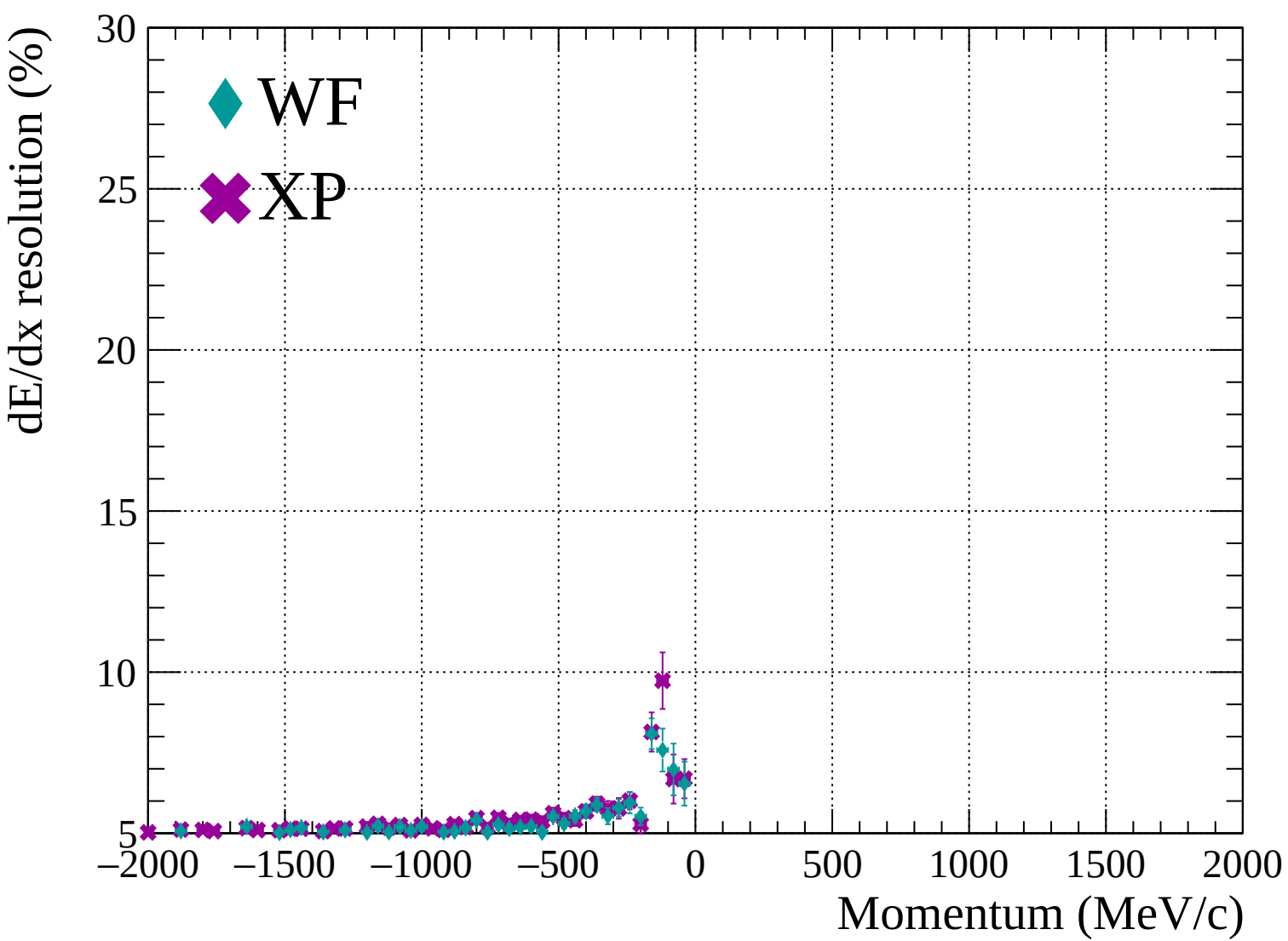


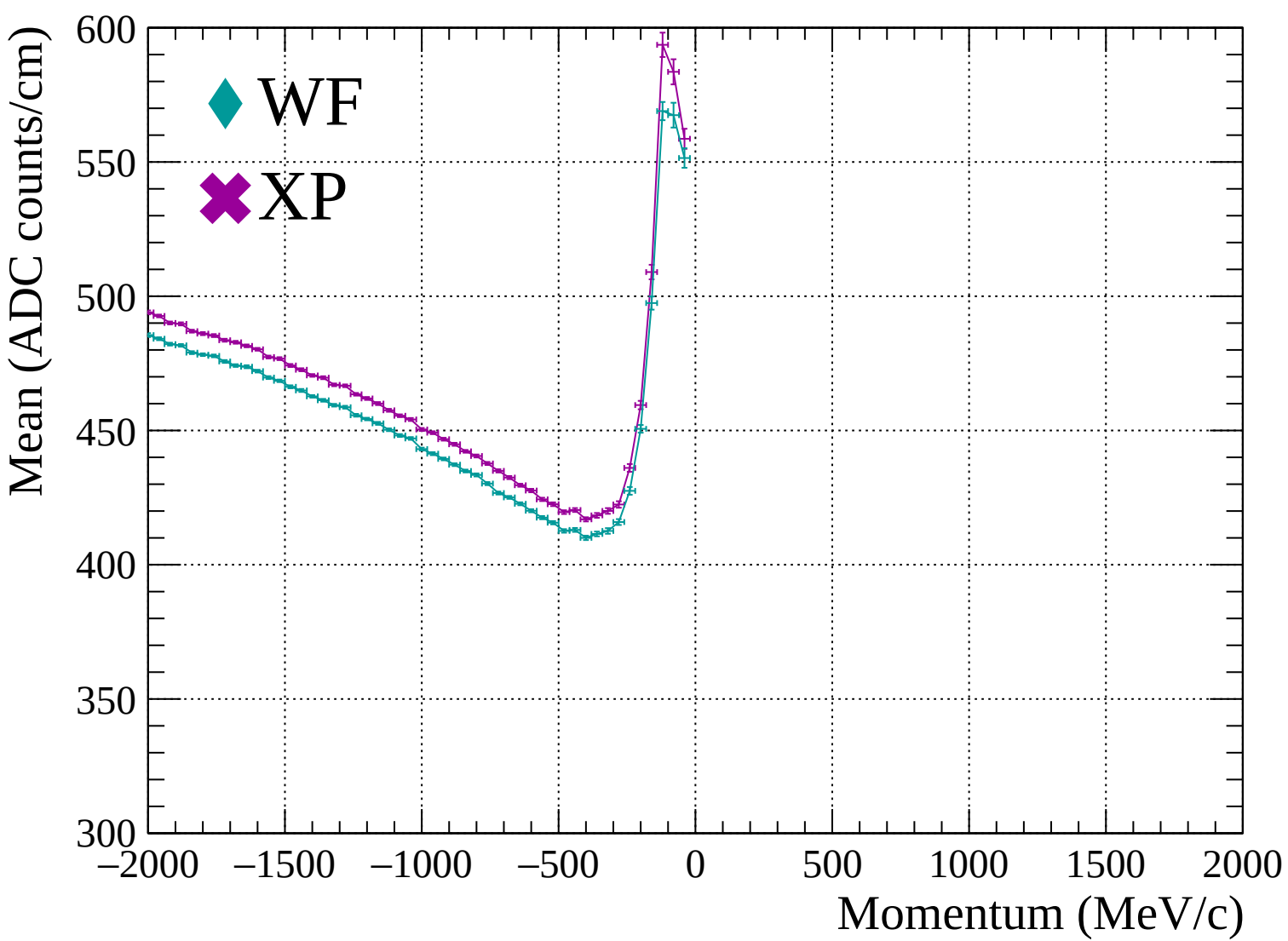


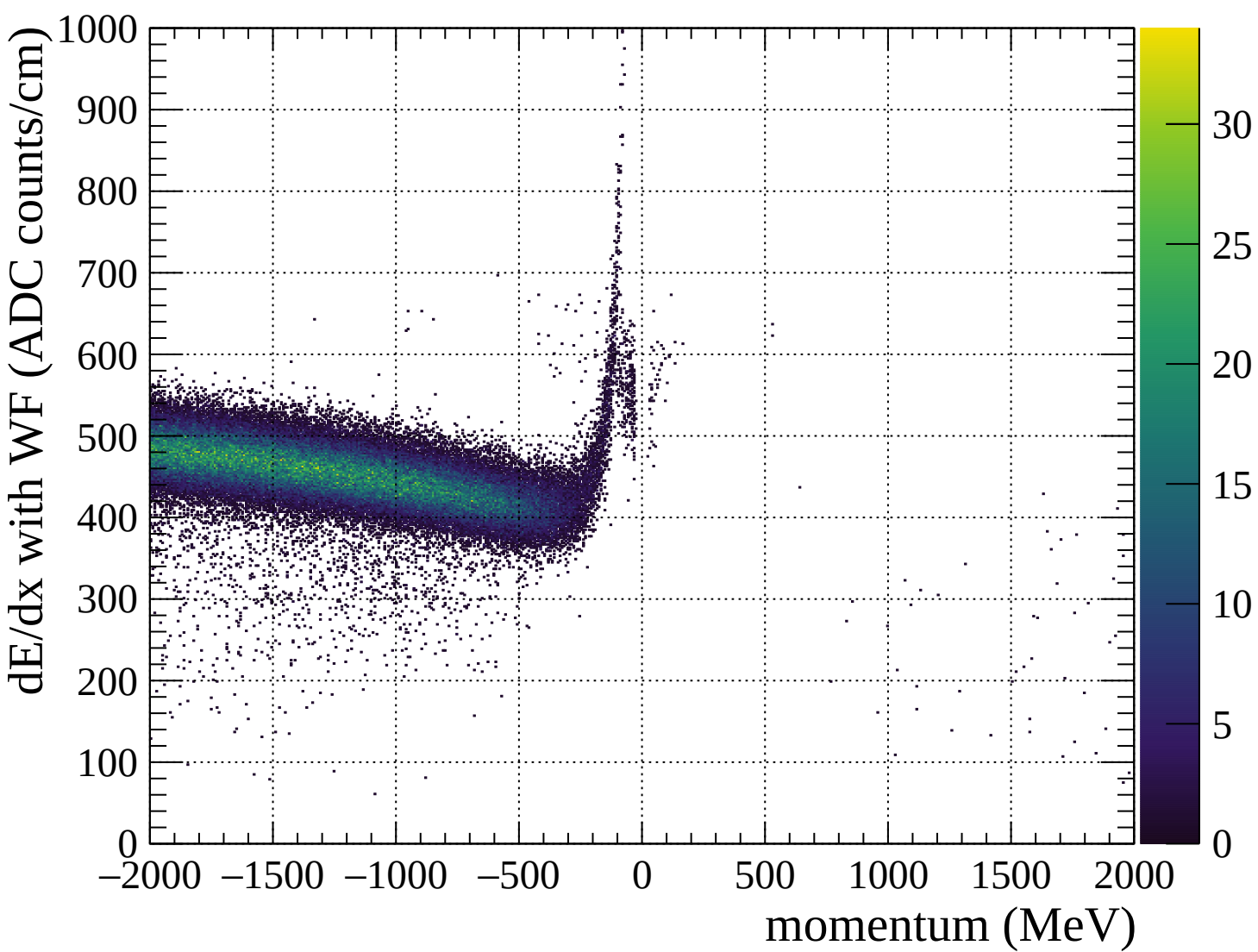


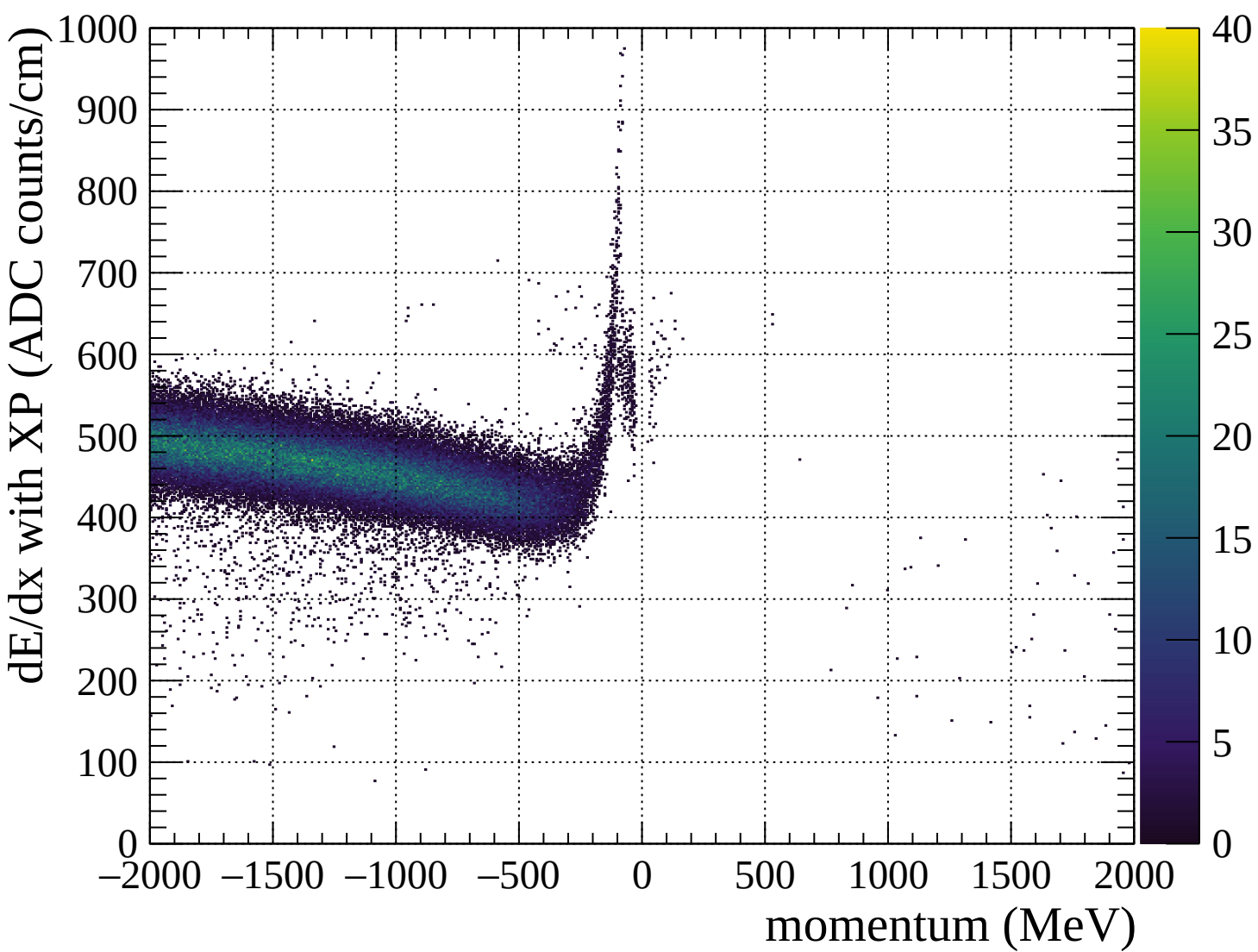


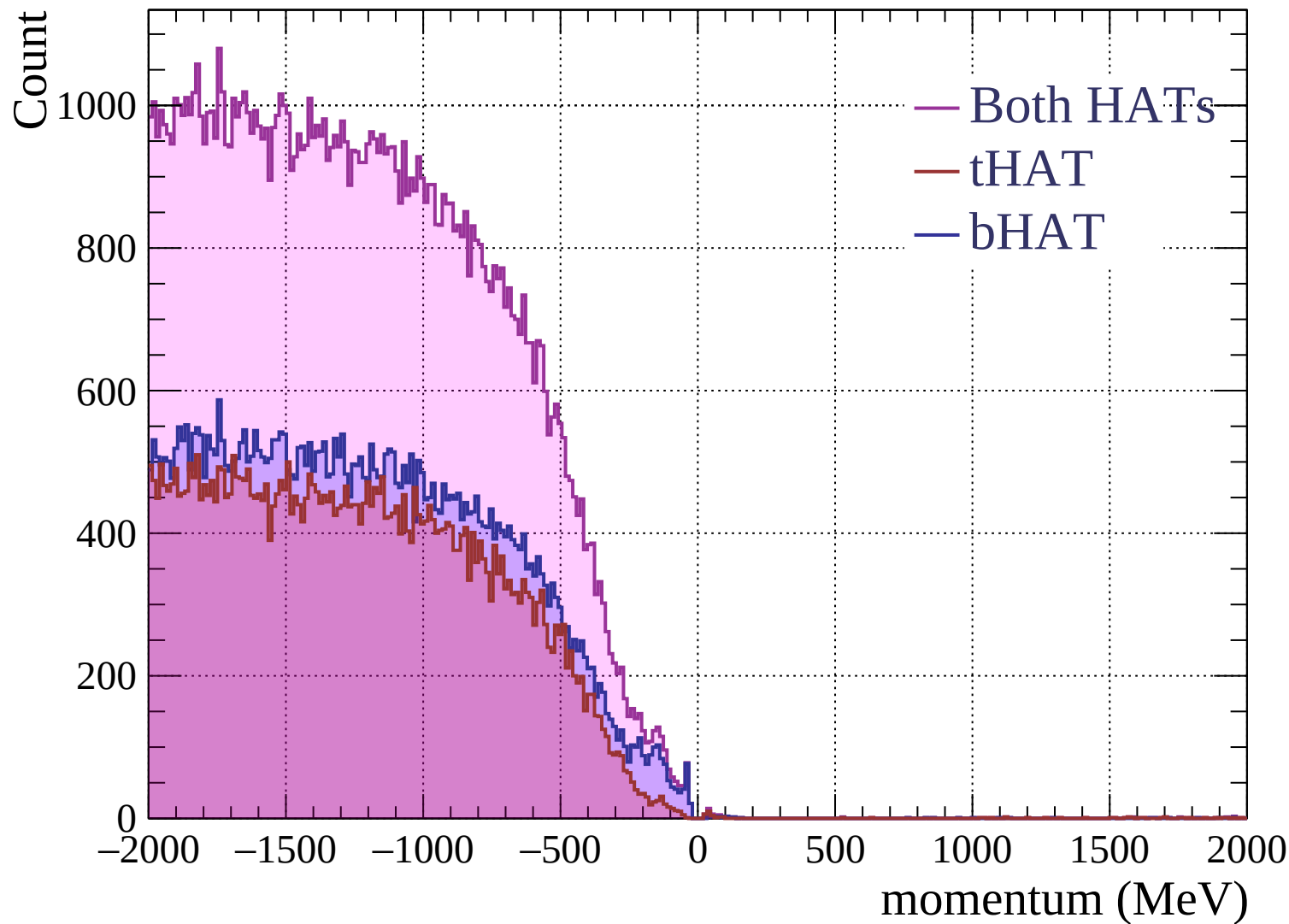


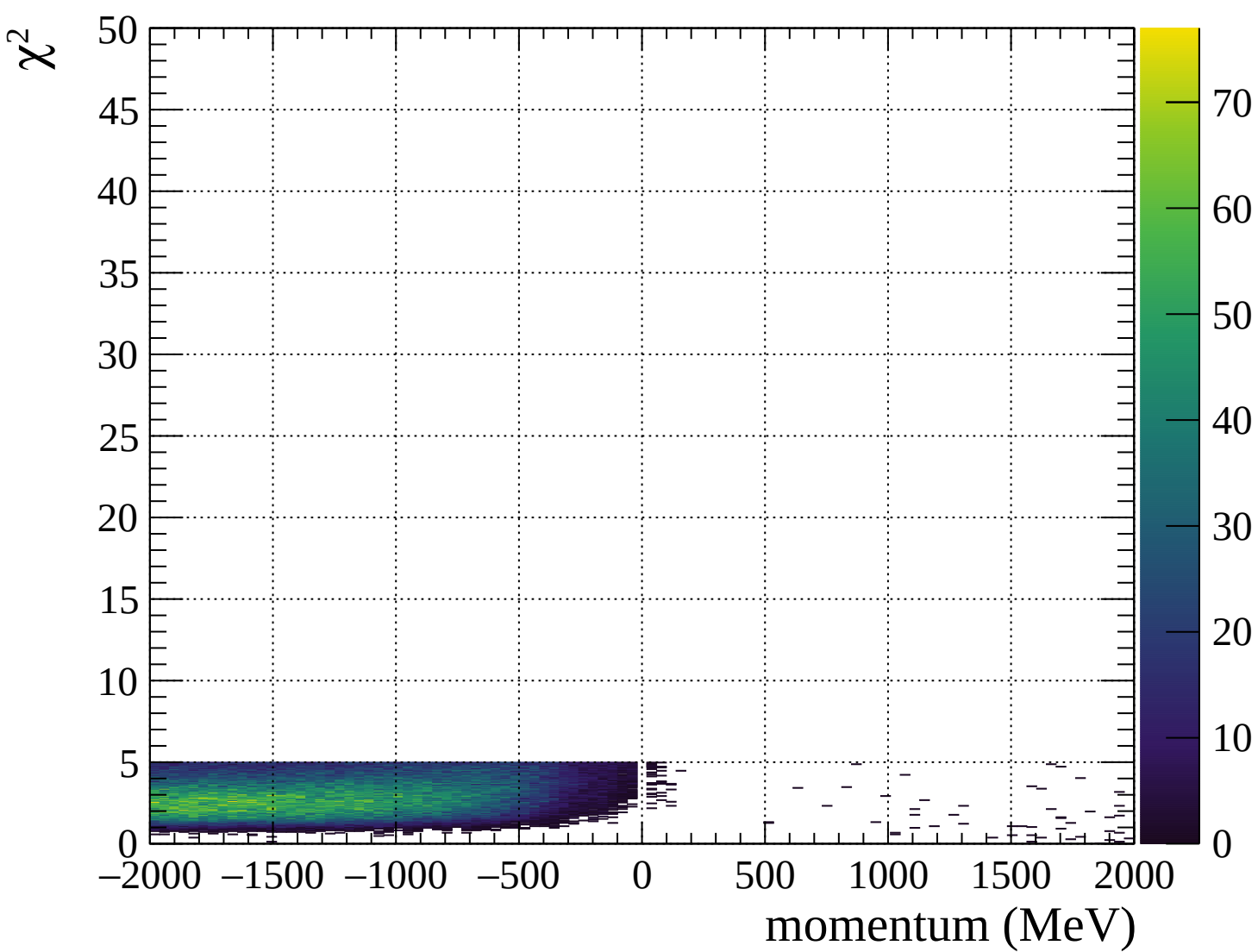


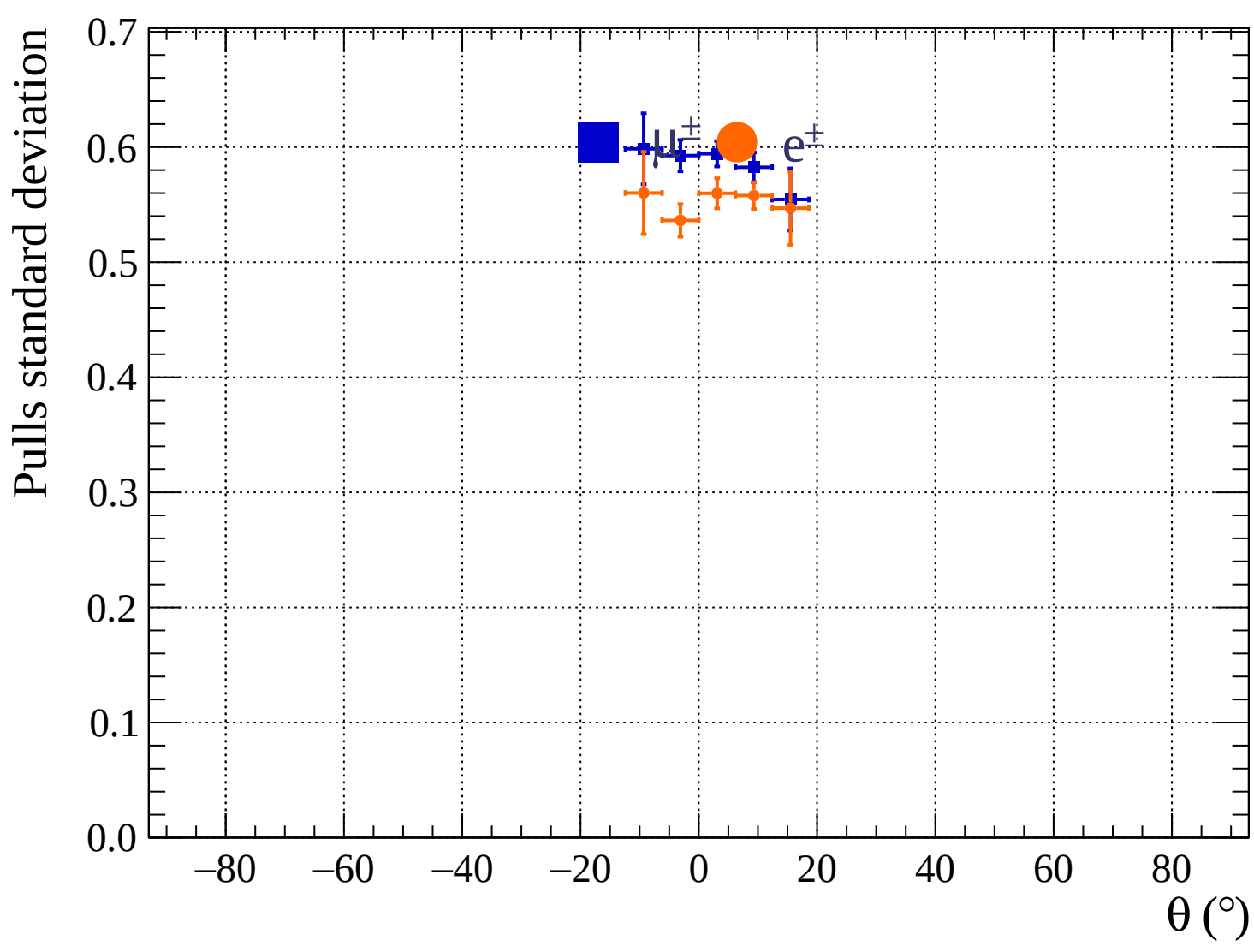


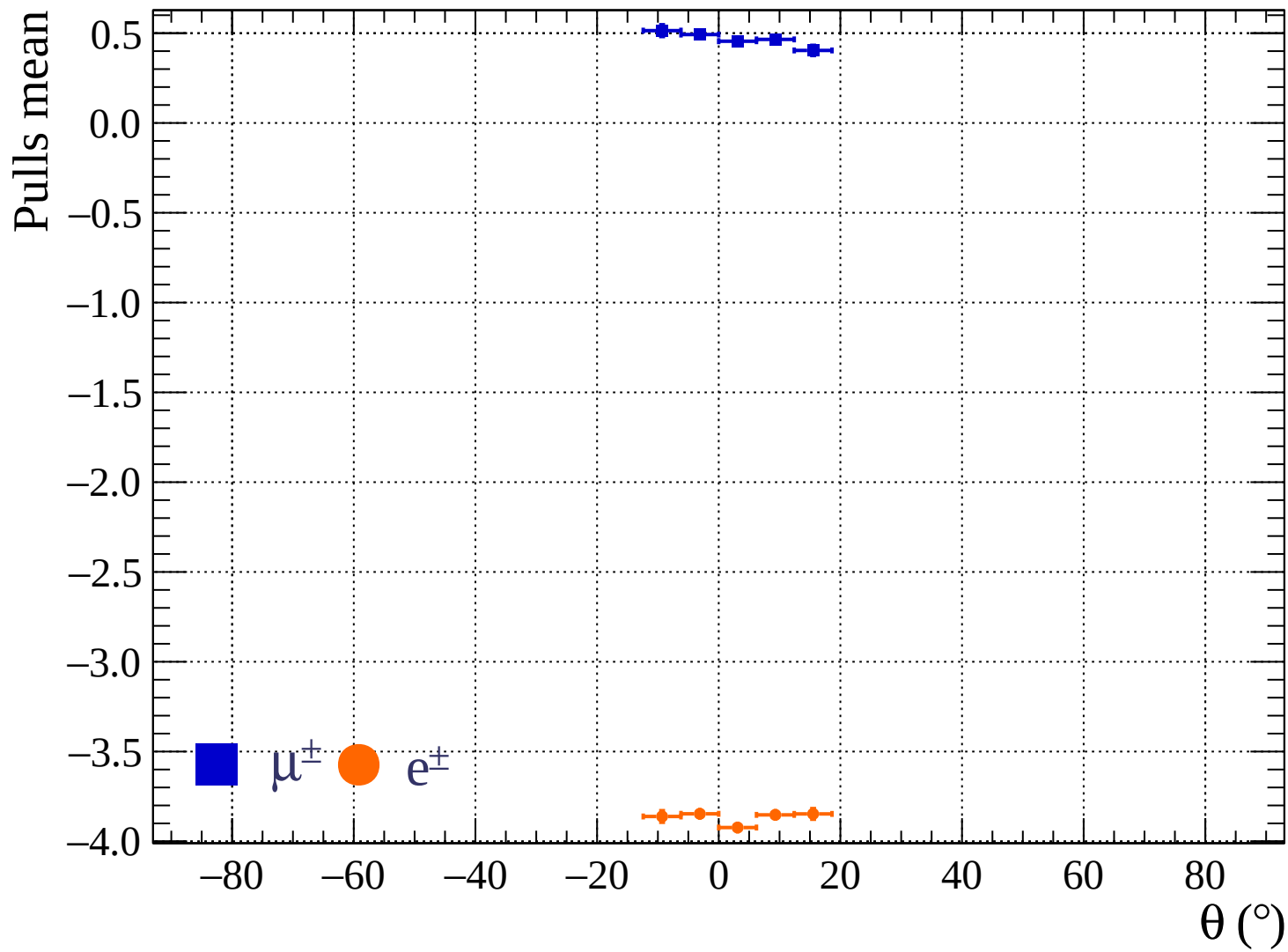




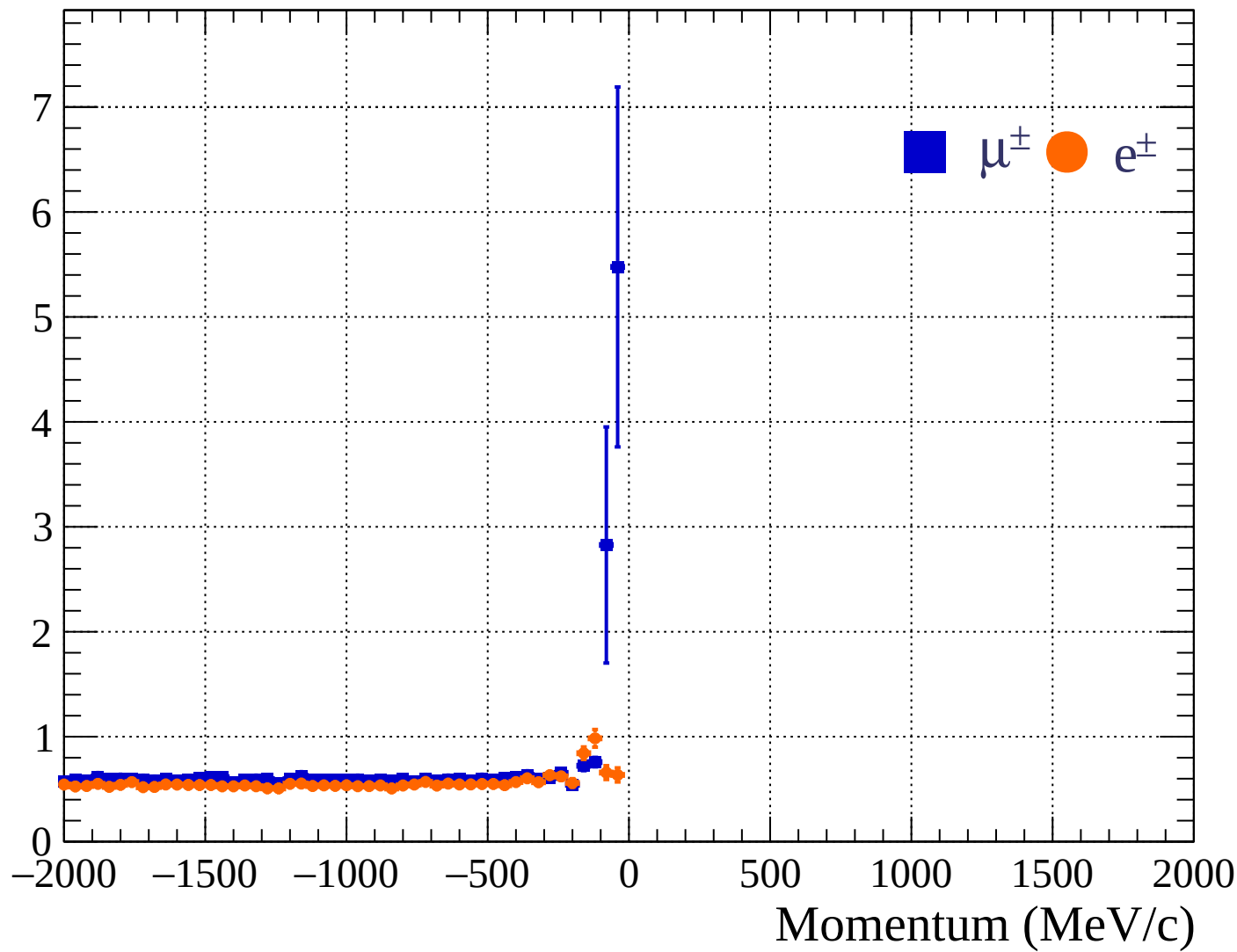




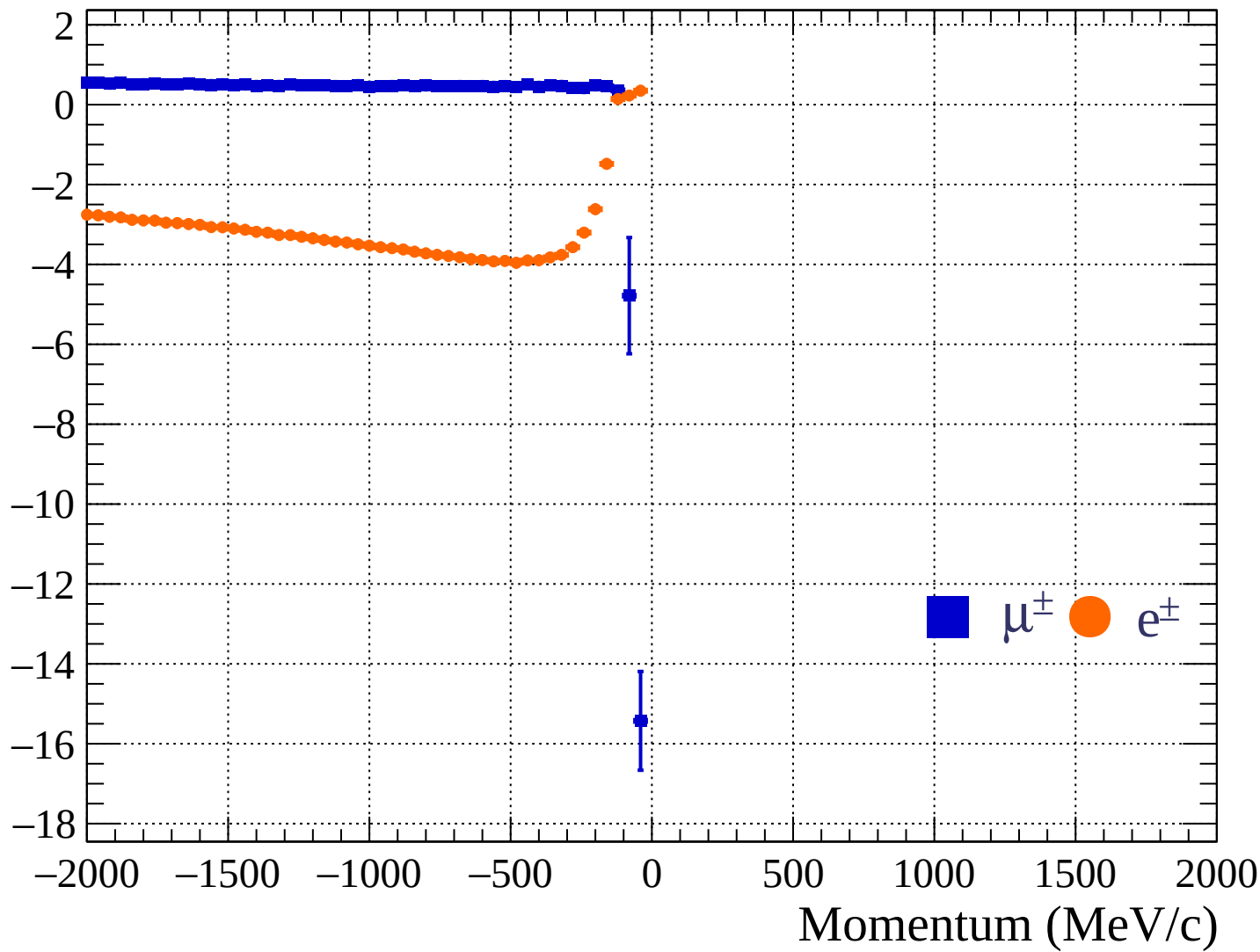


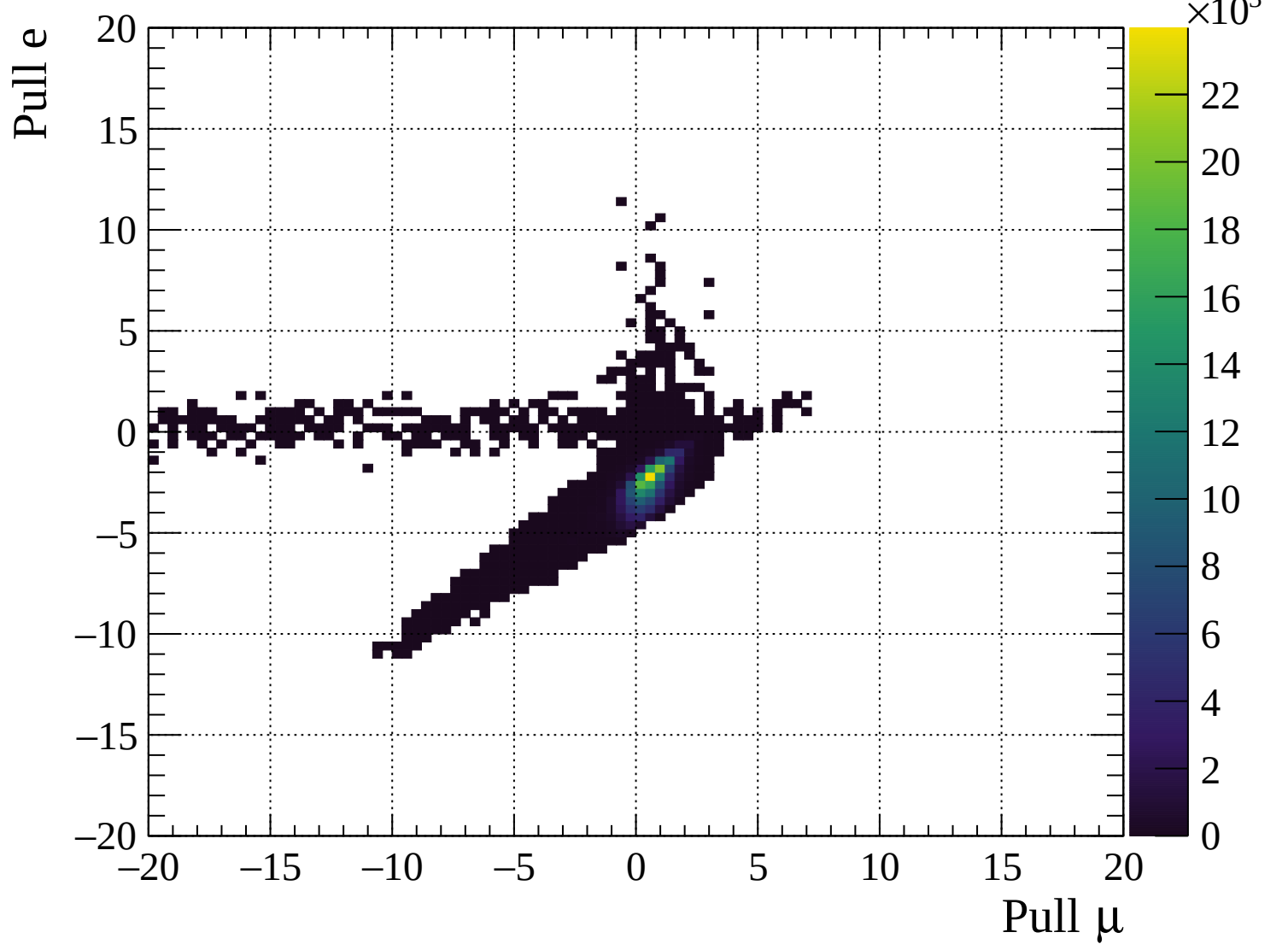


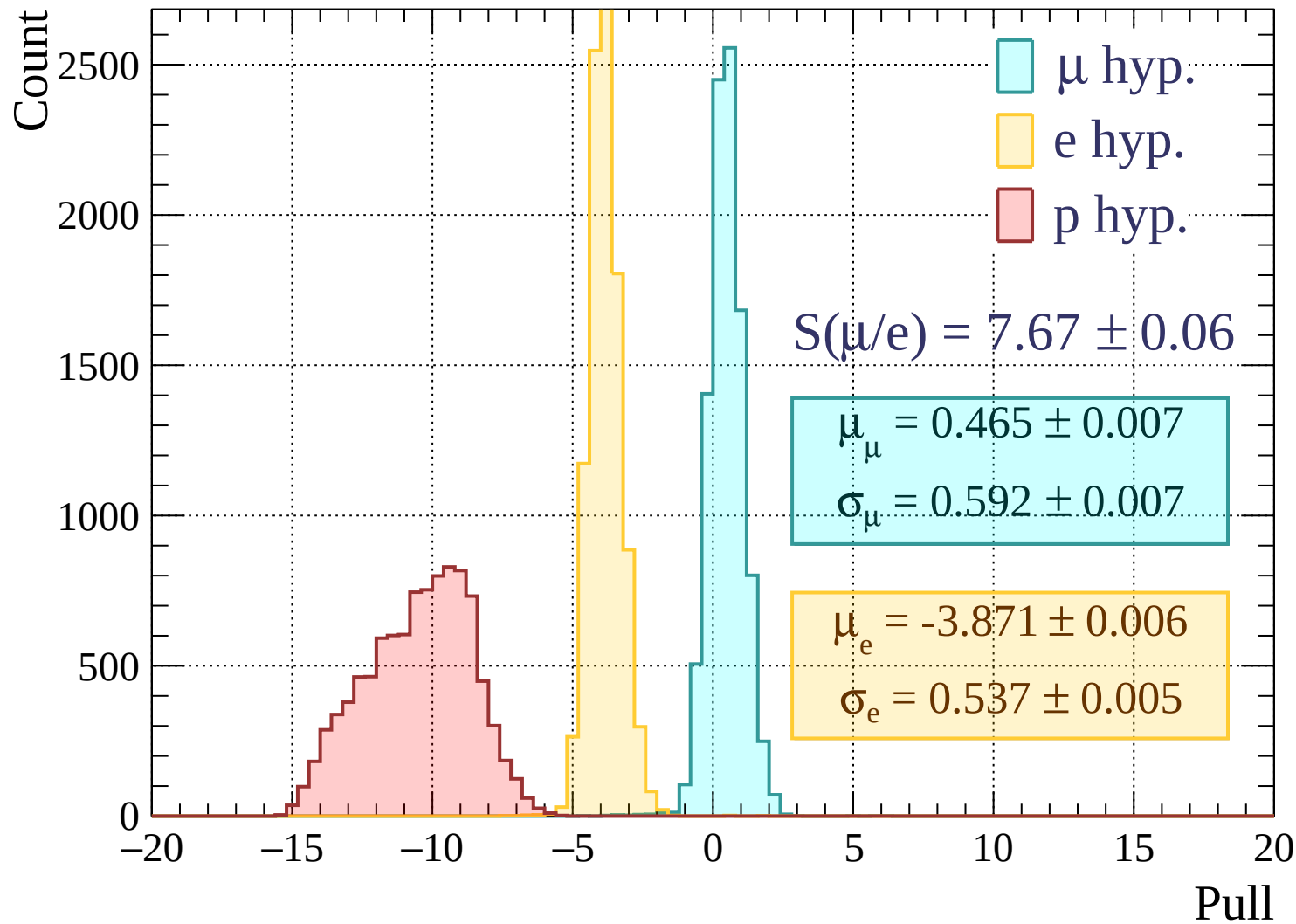
Pulls standard deviation

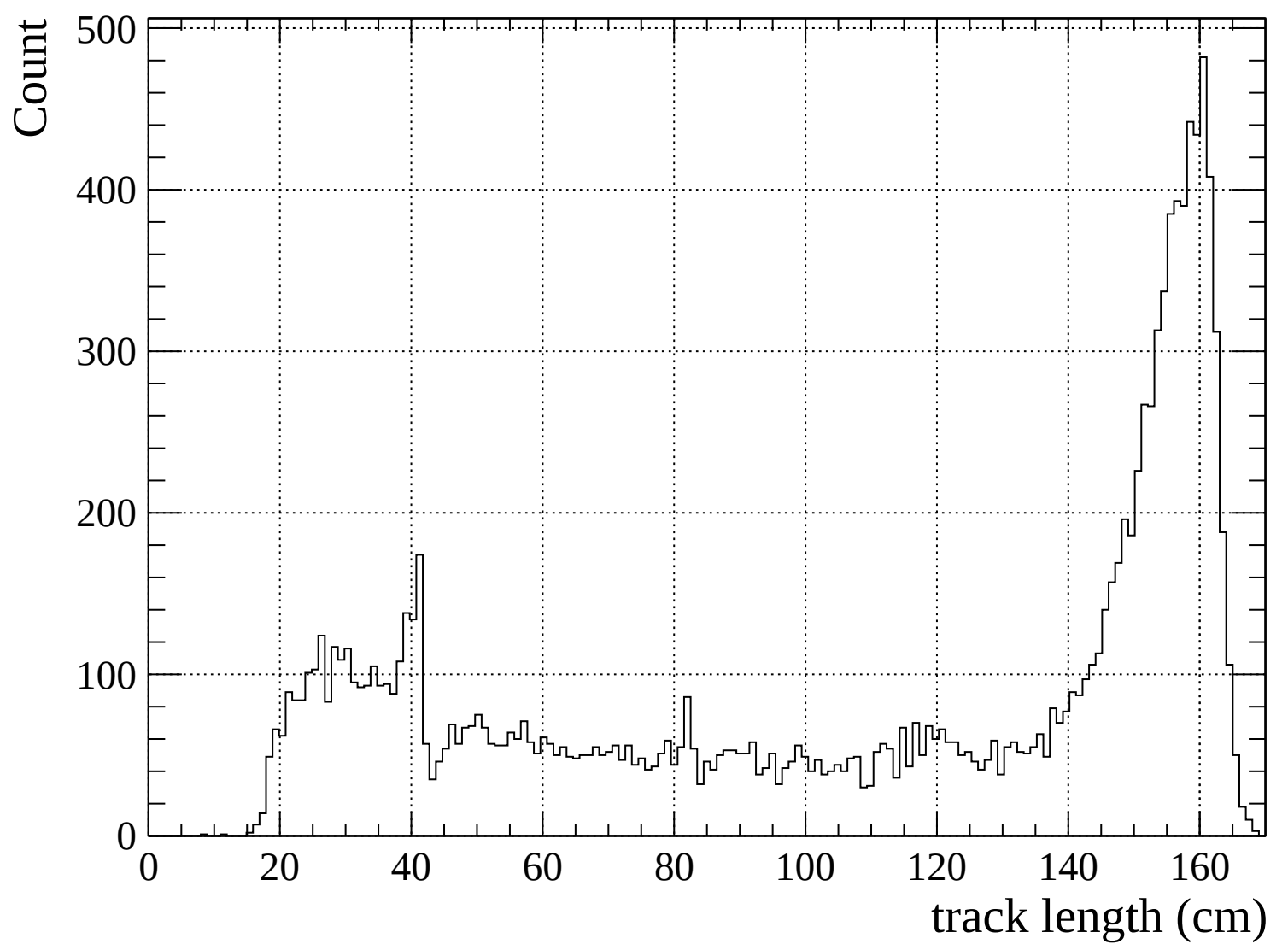


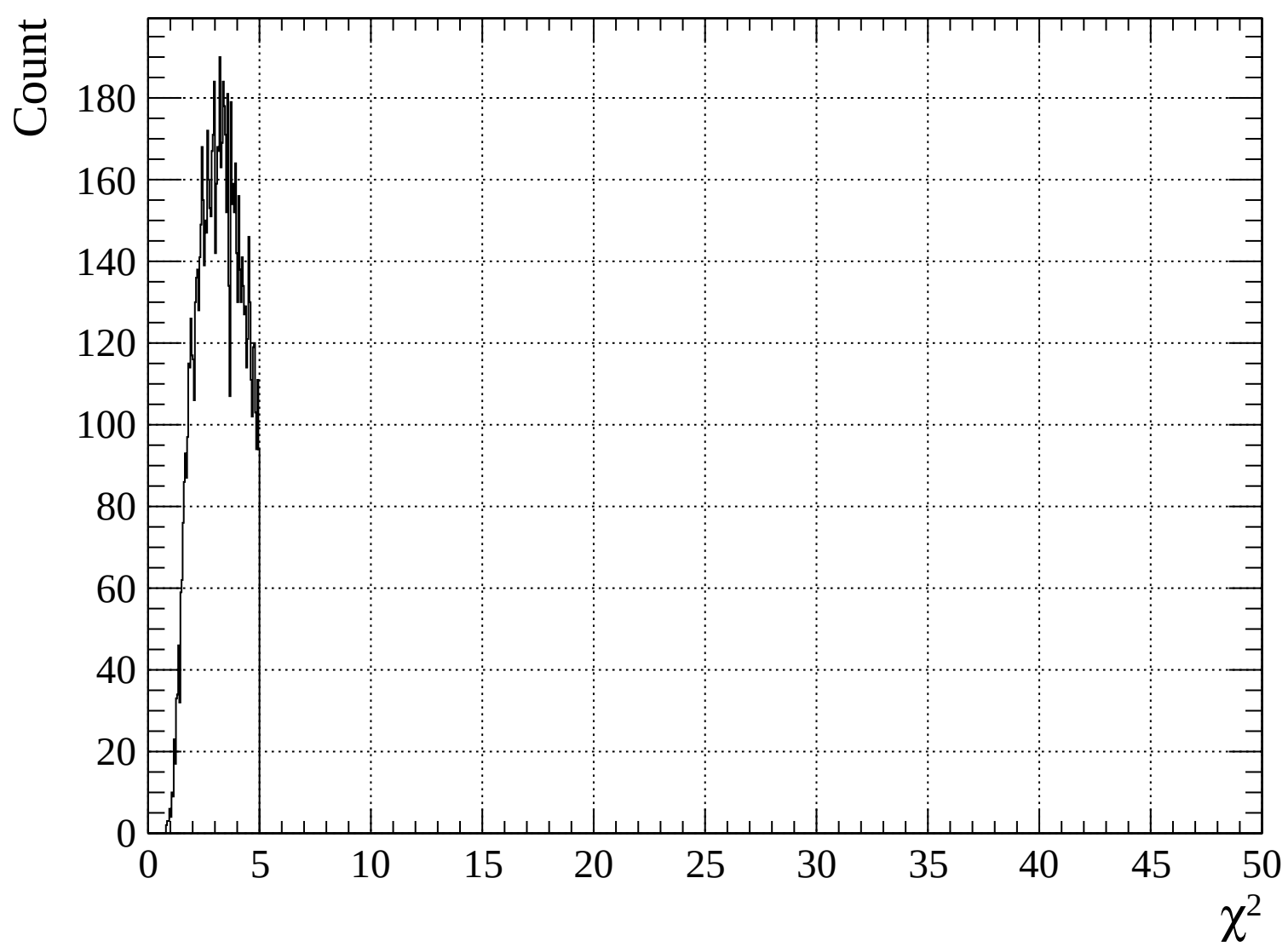
Pulls mean

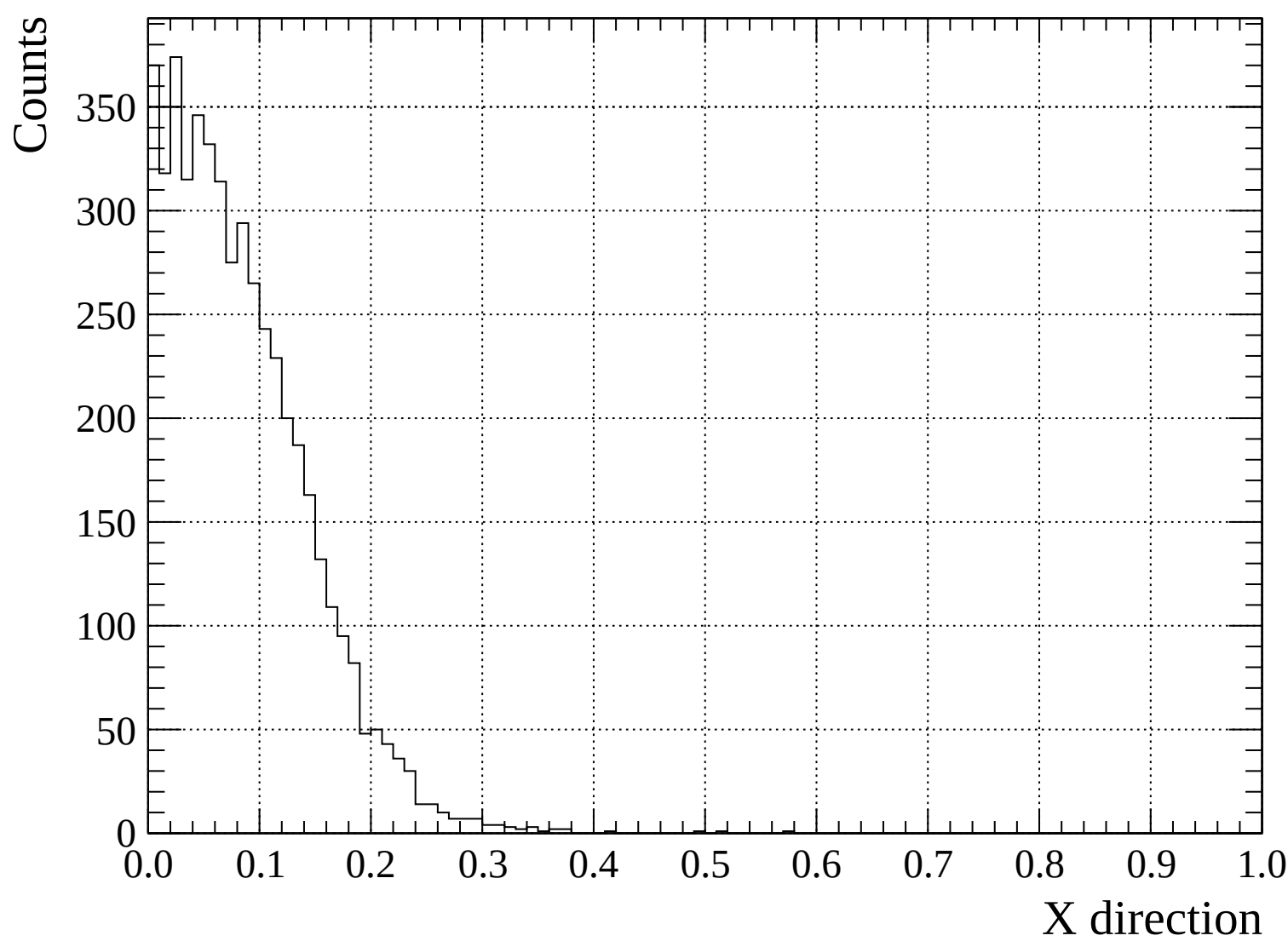


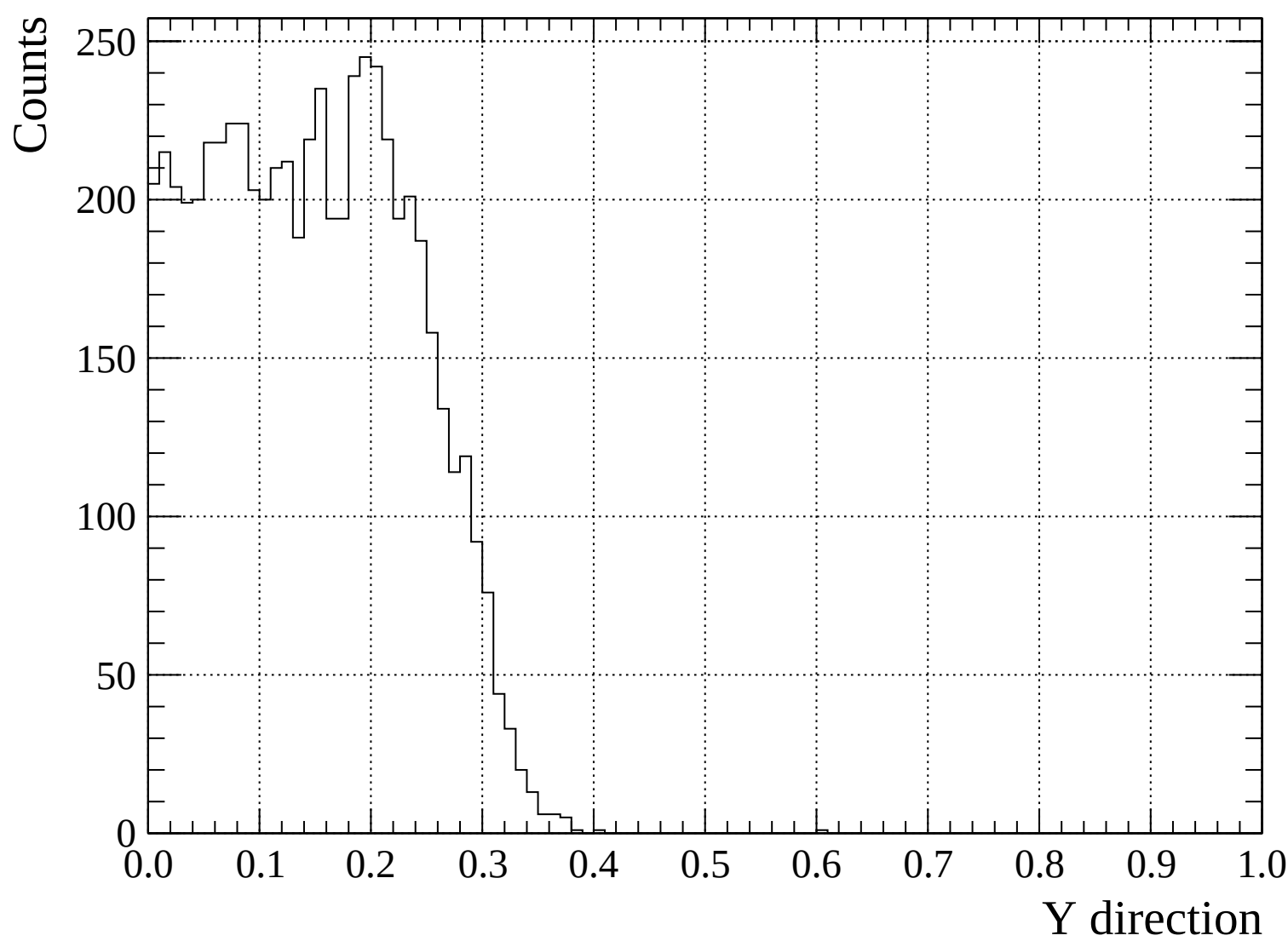


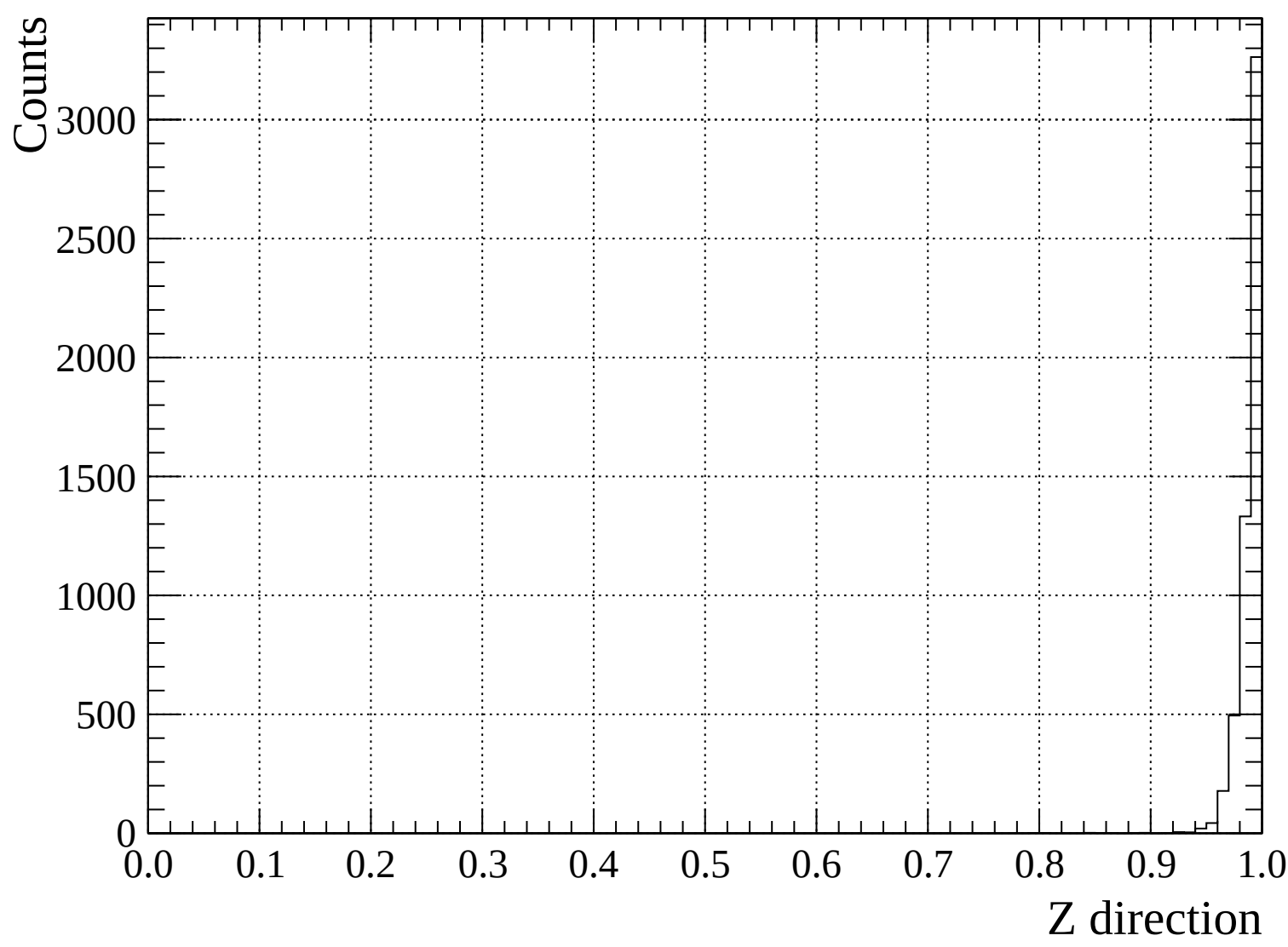


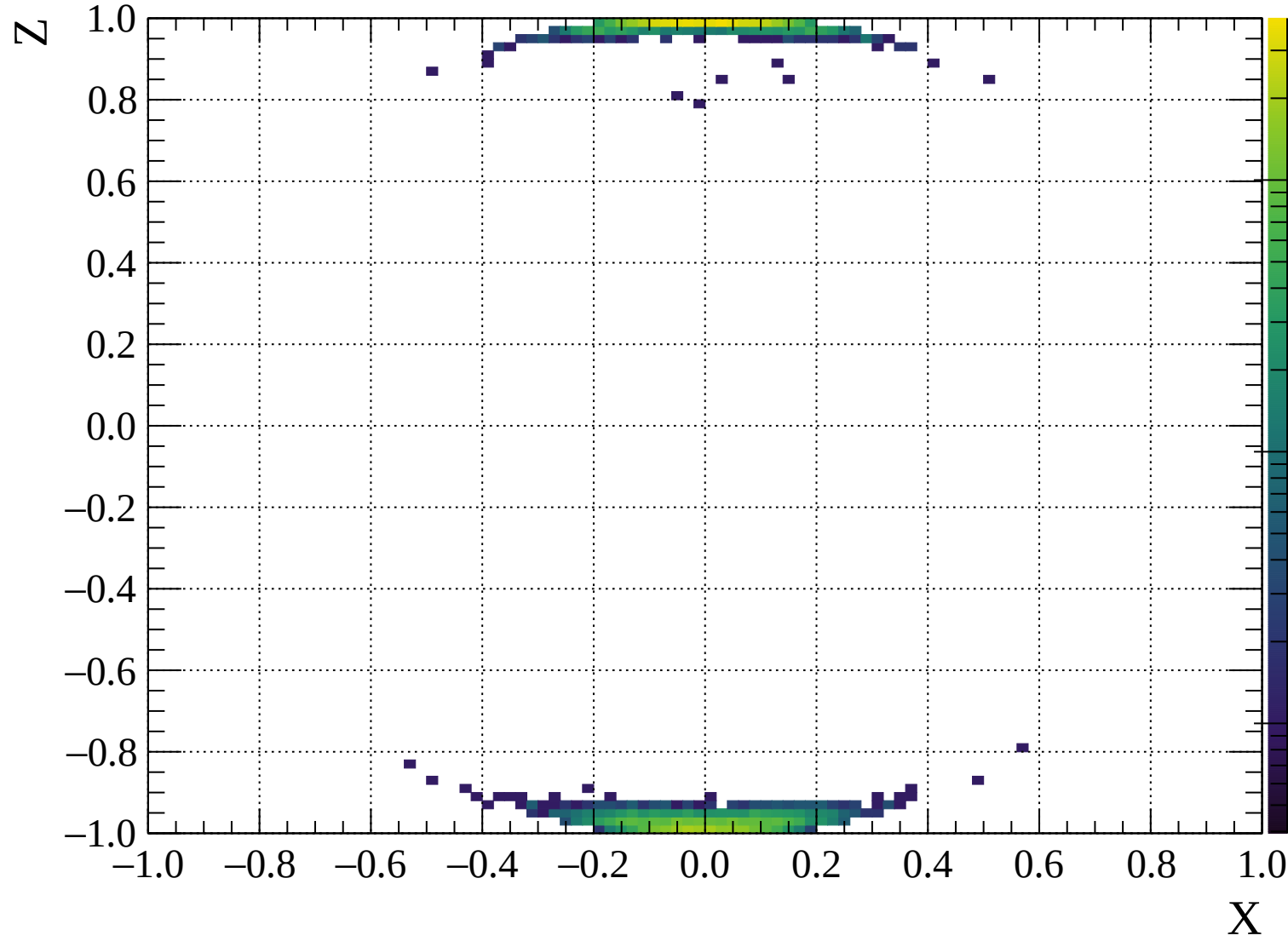


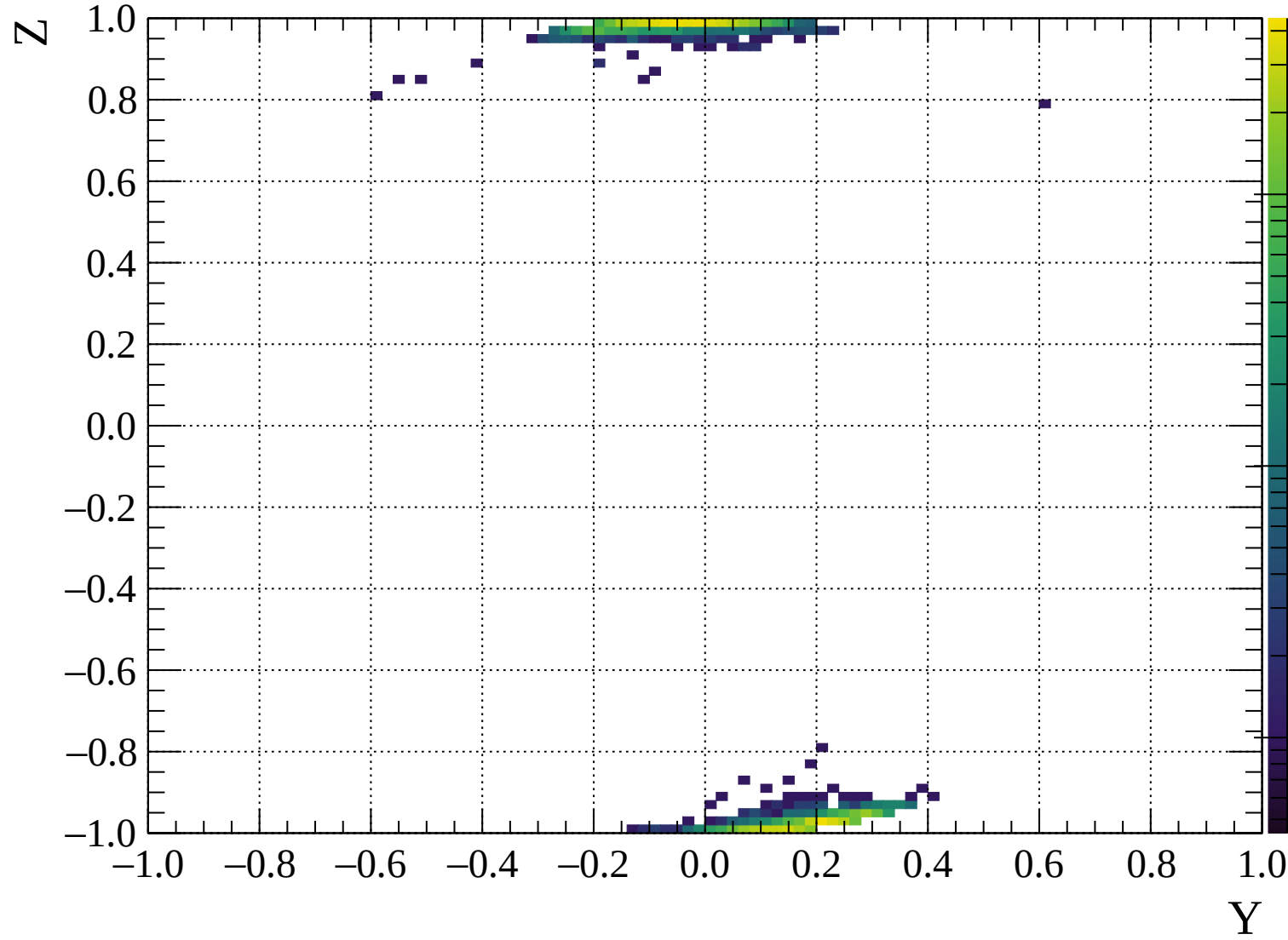


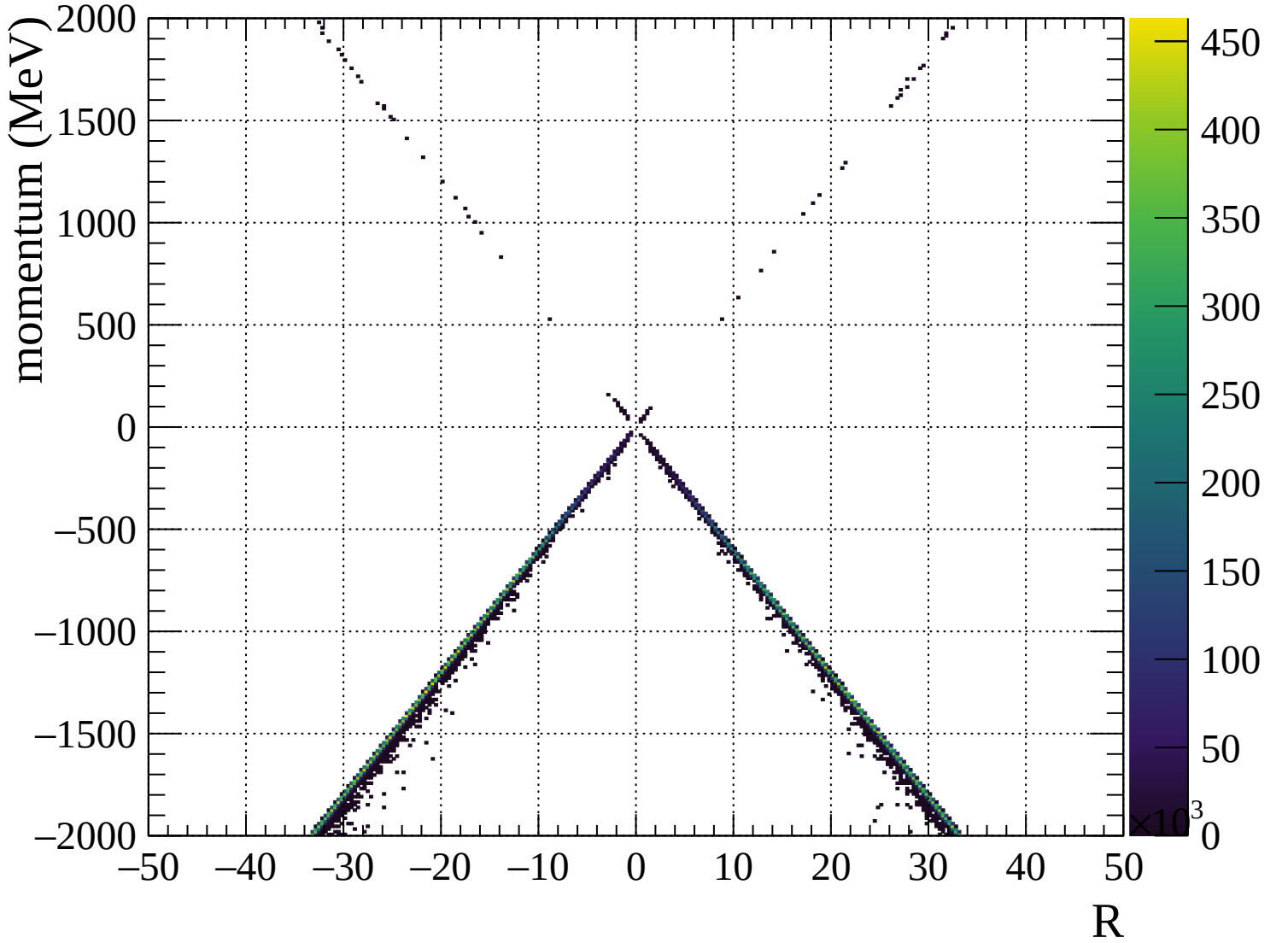


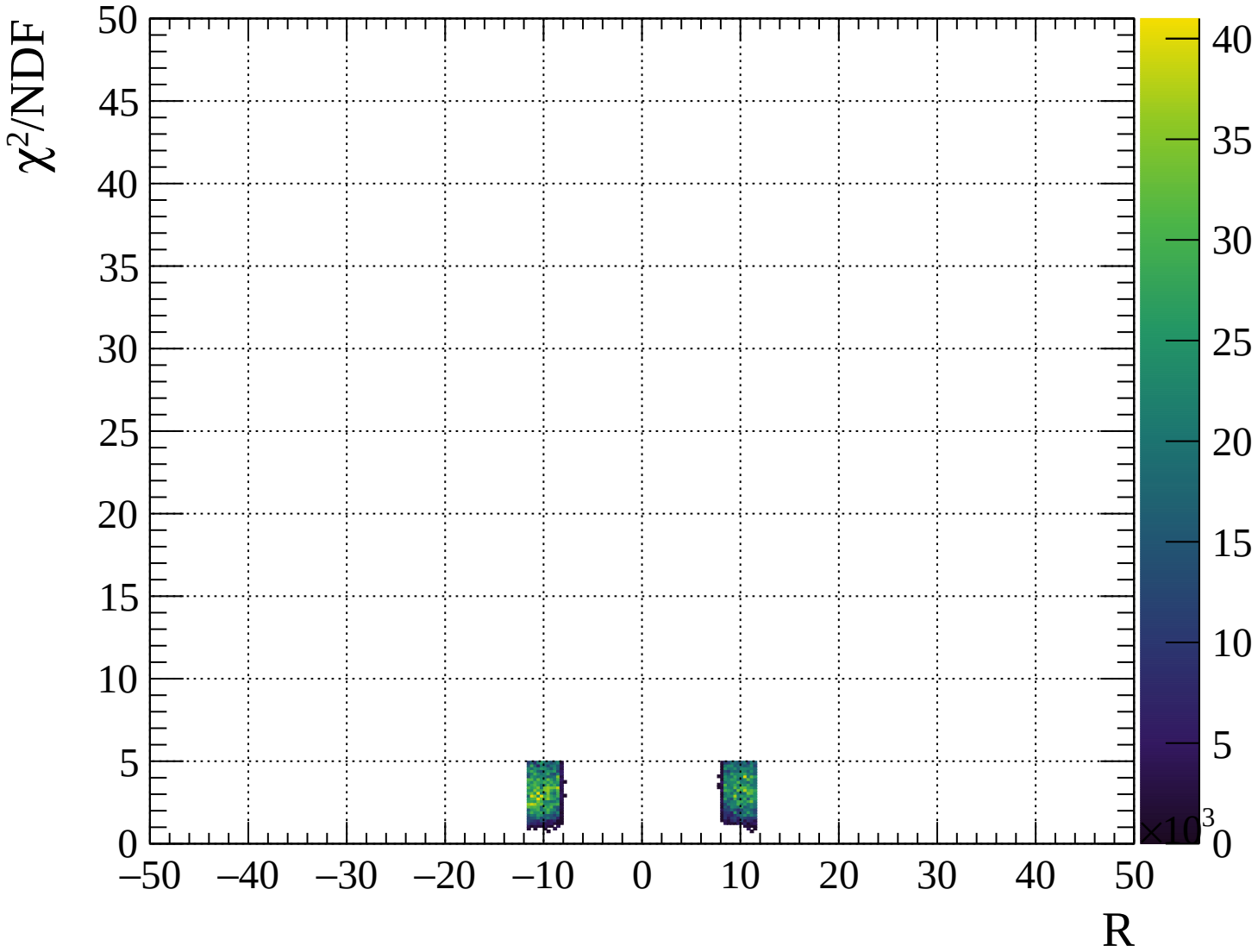






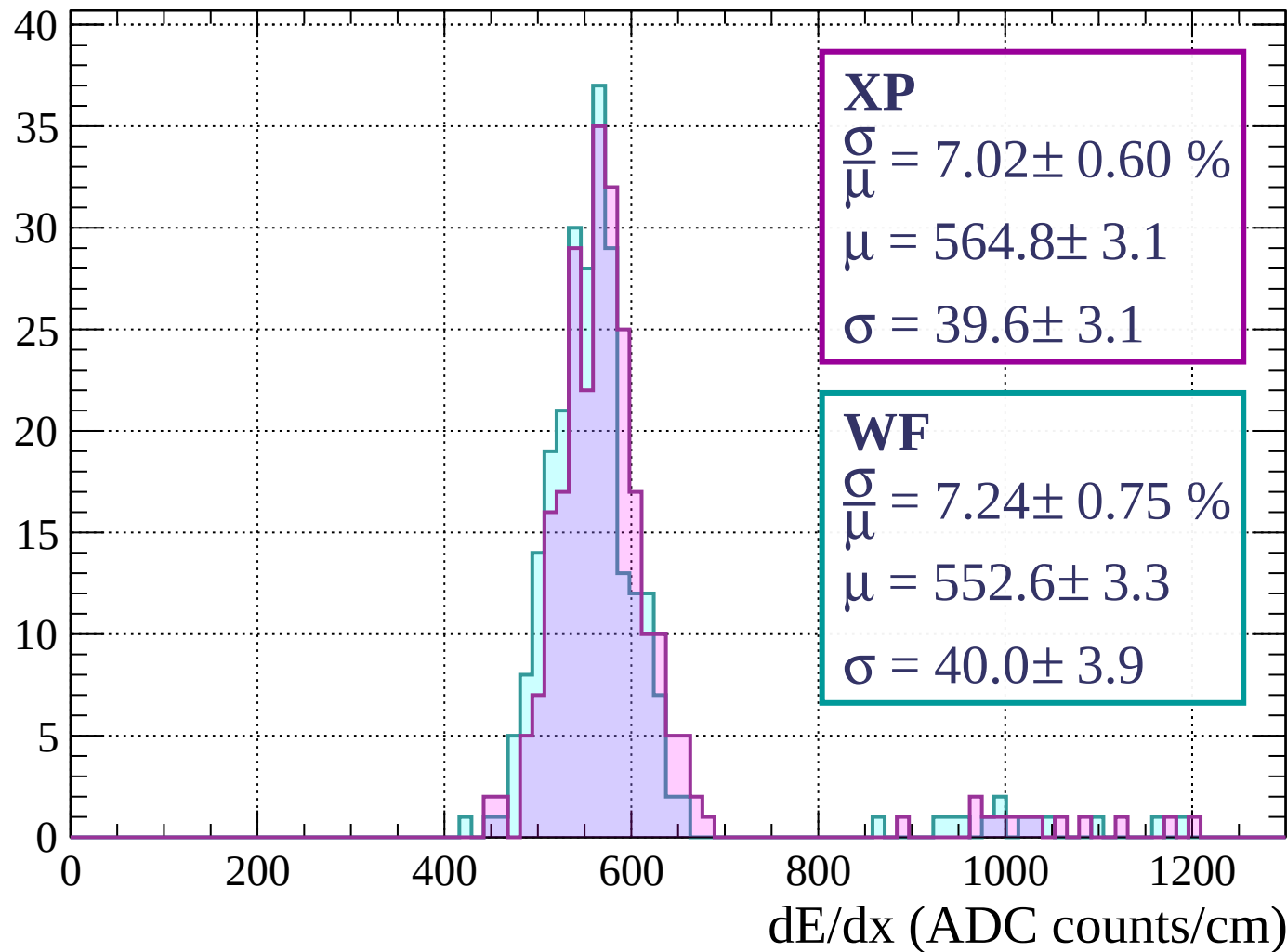






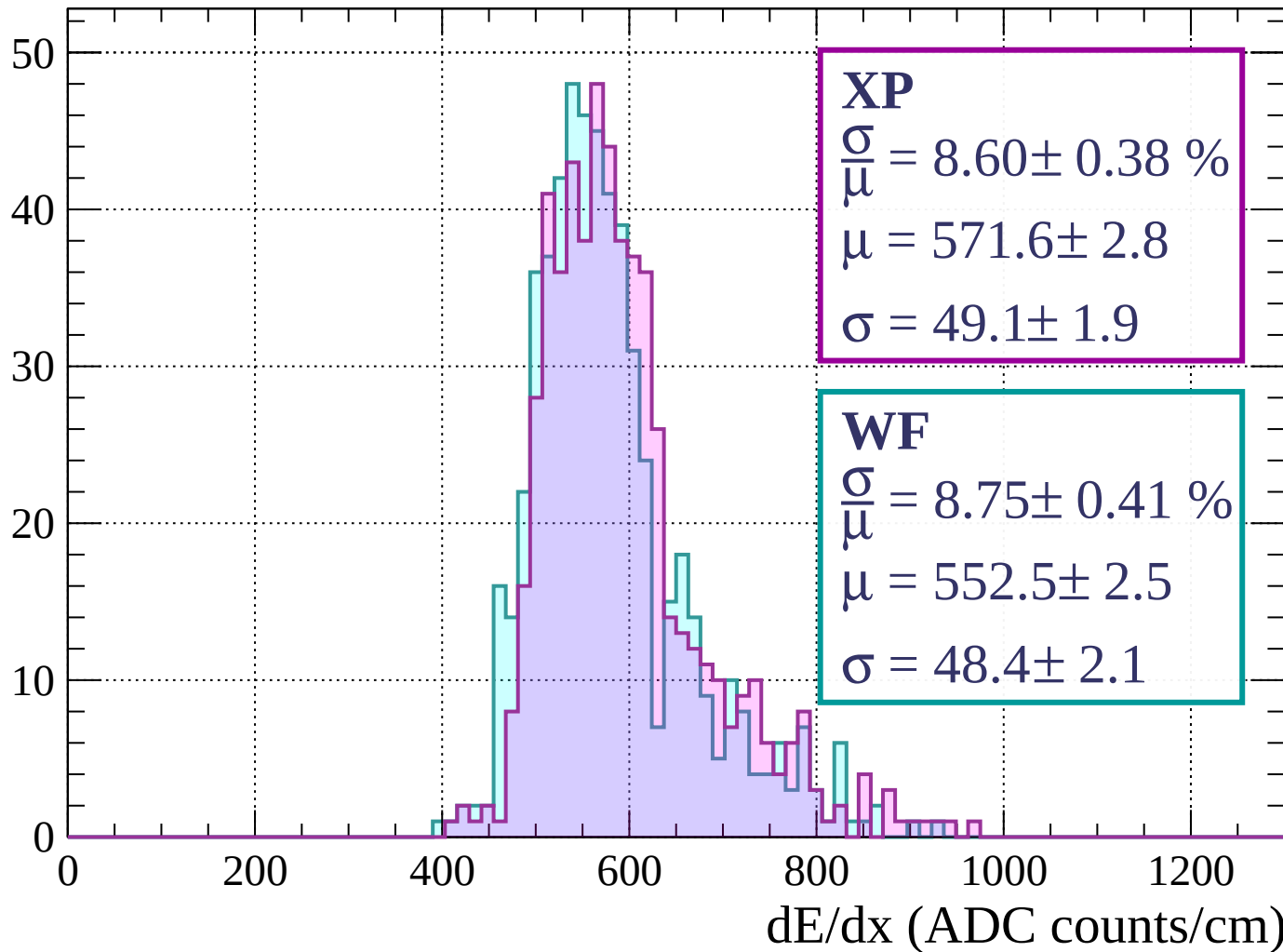
Energy loss | $0 < p < 80$

Count



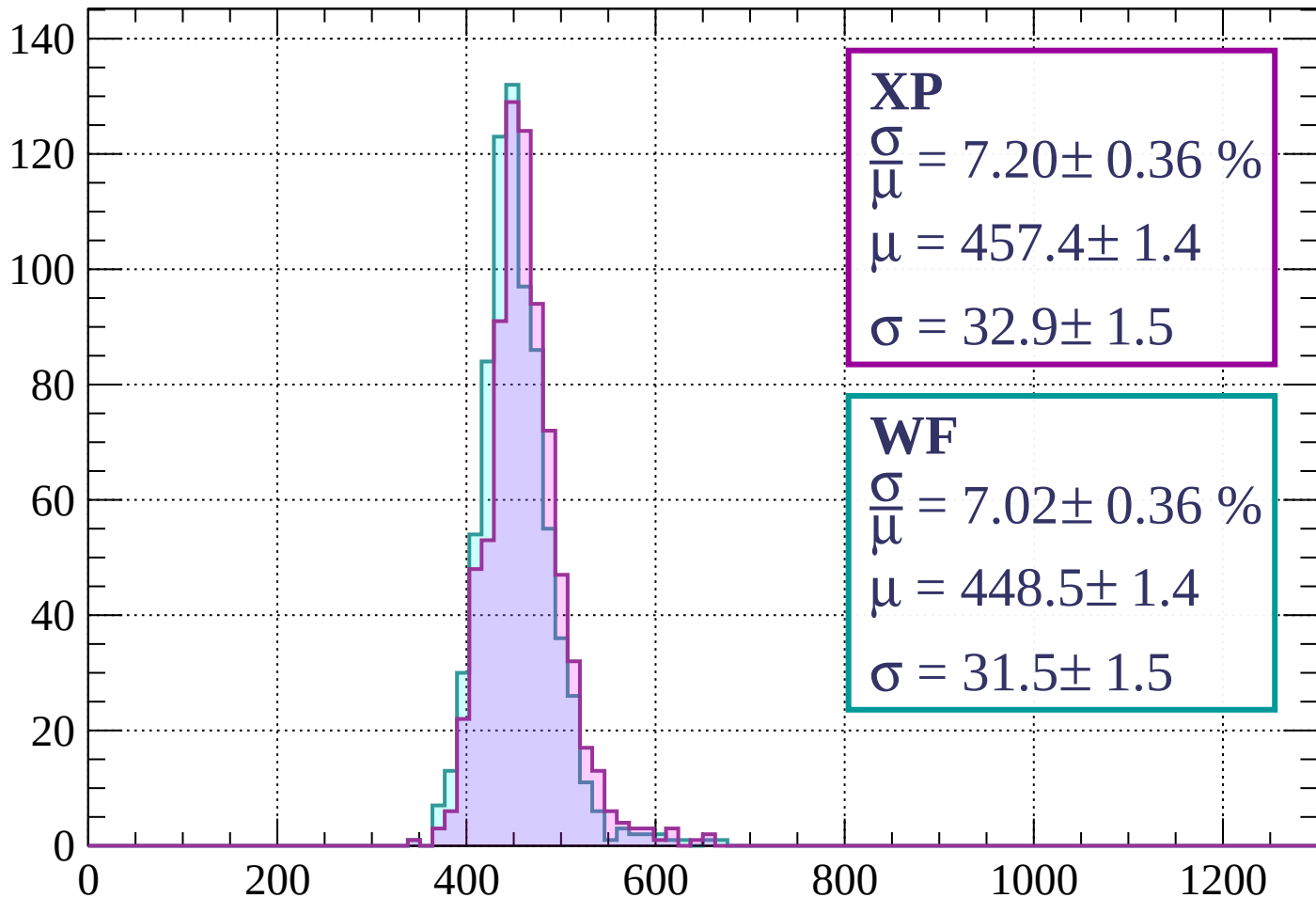
Energy loss | $80 < p < 160$

Count



Energy loss | $160 < p < 240$

Count



XP

$$\frac{\sigma}{\mu} = 7.20 \pm 0.36 \%$$

$$\mu = 457.4 \pm 1.4$$

$$\sigma = 32.9 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 7.02 \pm 0.36 \%$$

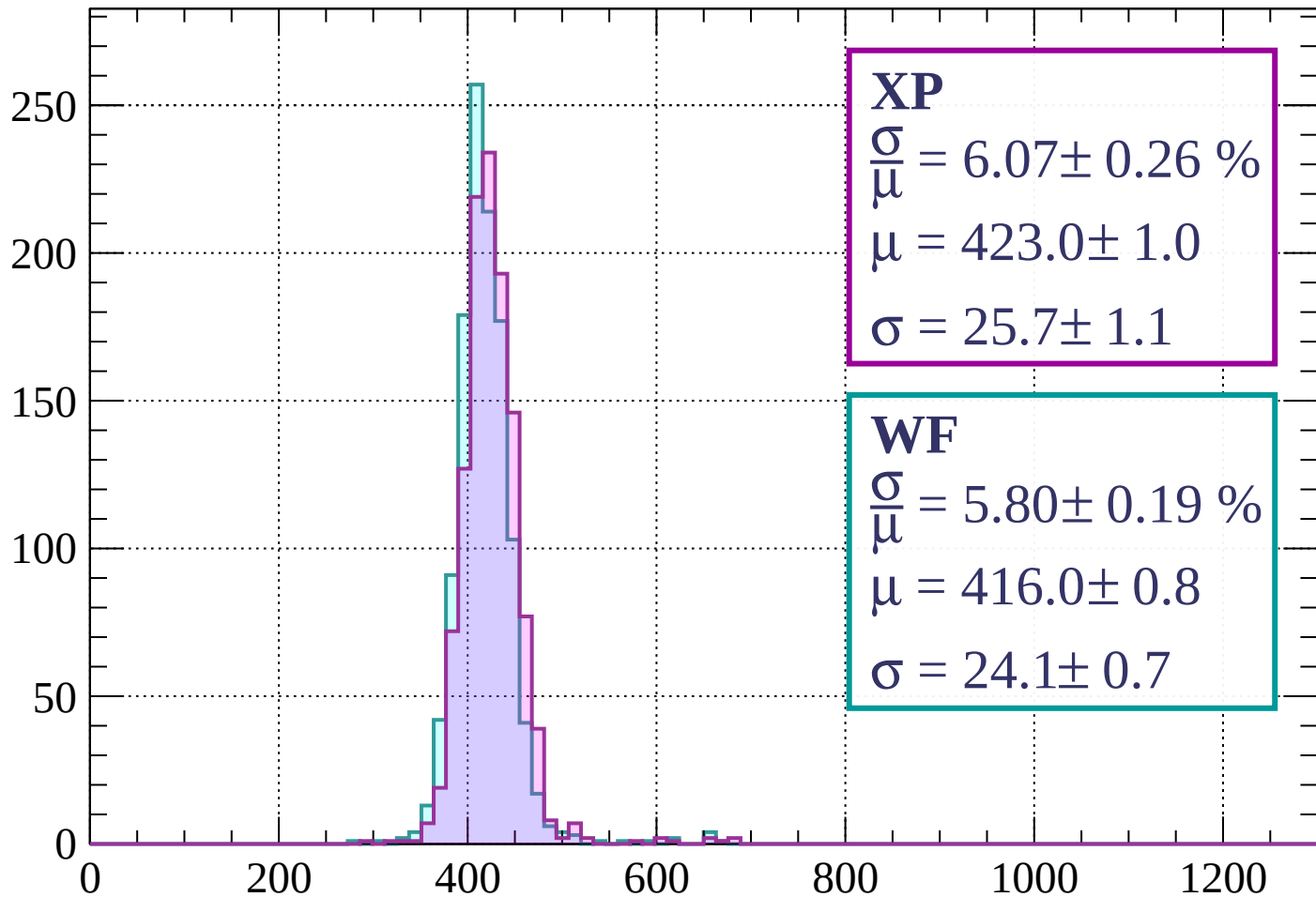
$$\mu = 448.5 \pm 1.4$$

$$\sigma = 31.5 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $240 < p < 320$

Count



XP

$$\frac{\sigma}{\mu} = 6.07 \pm 0.26 \%$$

$$\mu = 423.0 \pm 1.0$$

$$\sigma = 25.7 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 5.80 \pm 0.19 \%$$

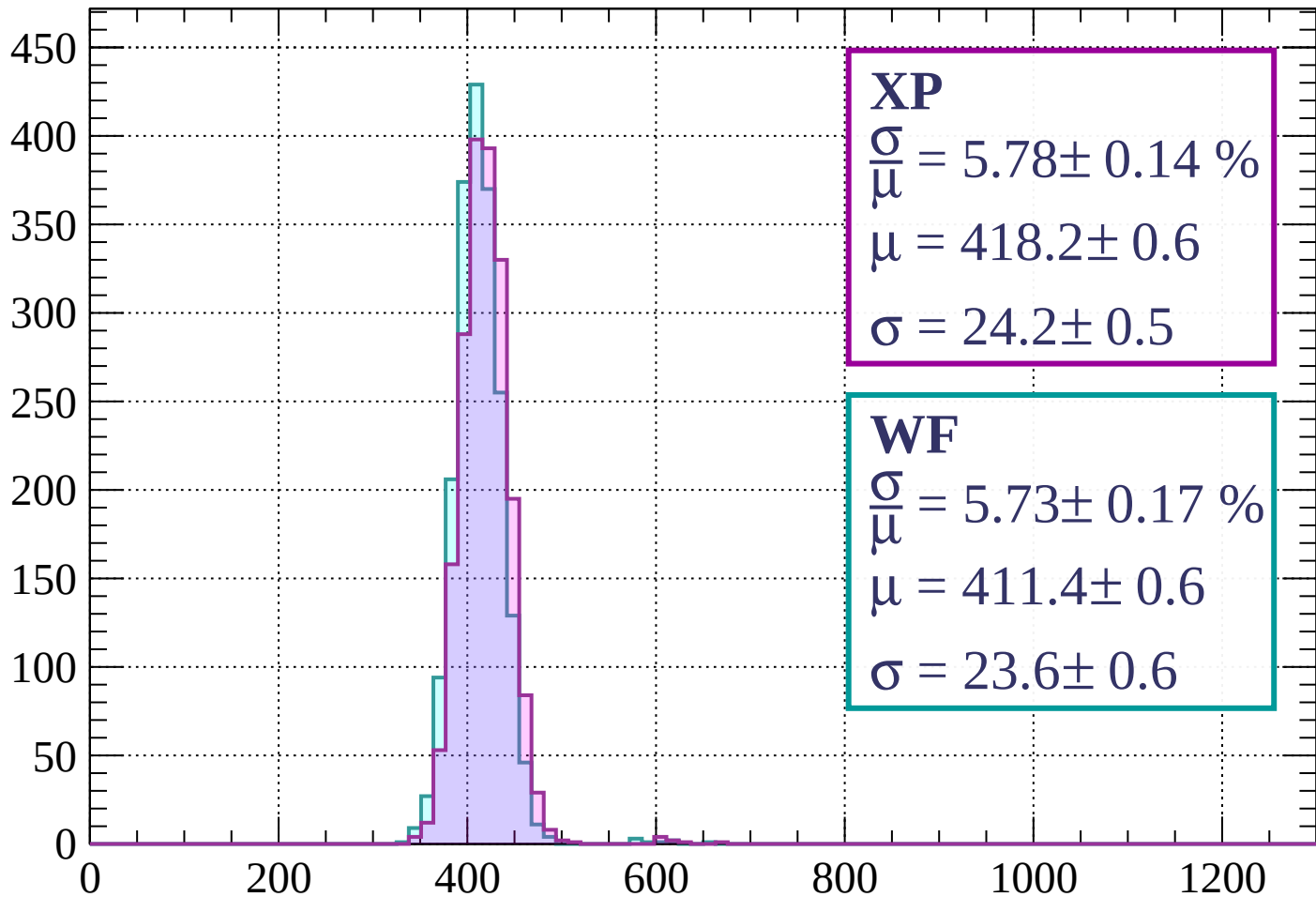
$$\mu = 416.0 \pm 0.8$$

$$\sigma = 24.1 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $320 < p < 400$

Count



XP

$$\frac{\sigma}{\mu} = 5.78 \pm 0.14 \%$$

$$\mu = 418.2 \pm 0.6$$

$$\sigma = 24.2 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 5.73 \pm 0.17 \%$$

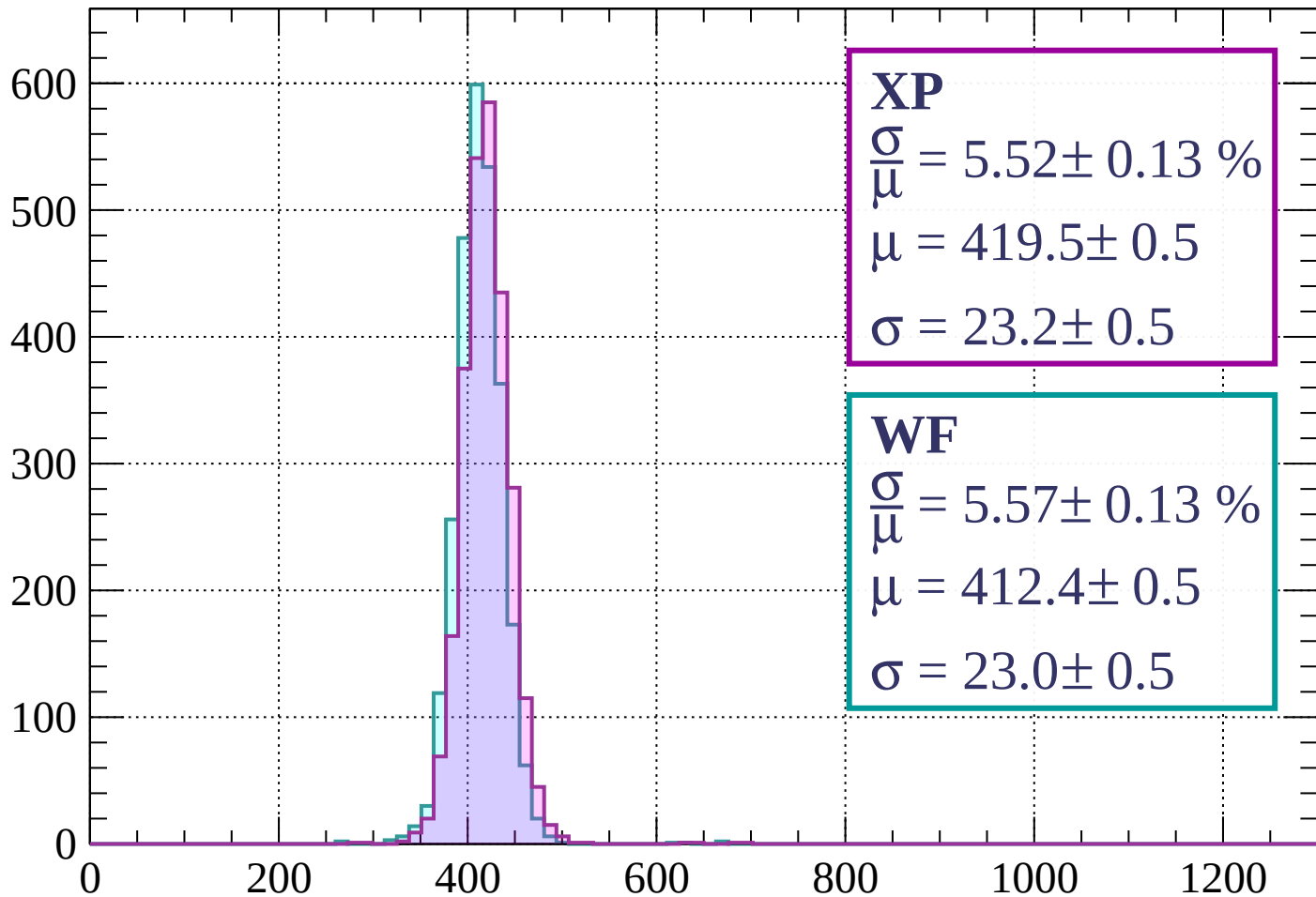
$$\mu = 411.4 \pm 0.6$$

$$\sigma = 23.6 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $400 < p < 480$

Count



XP

$$\frac{\sigma}{\mu} = 5.52 \pm 0.13 \%$$

$$\mu = 419.5 \pm 0.5$$

$$\sigma = 23.2 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 5.57 \pm 0.13 \%$$

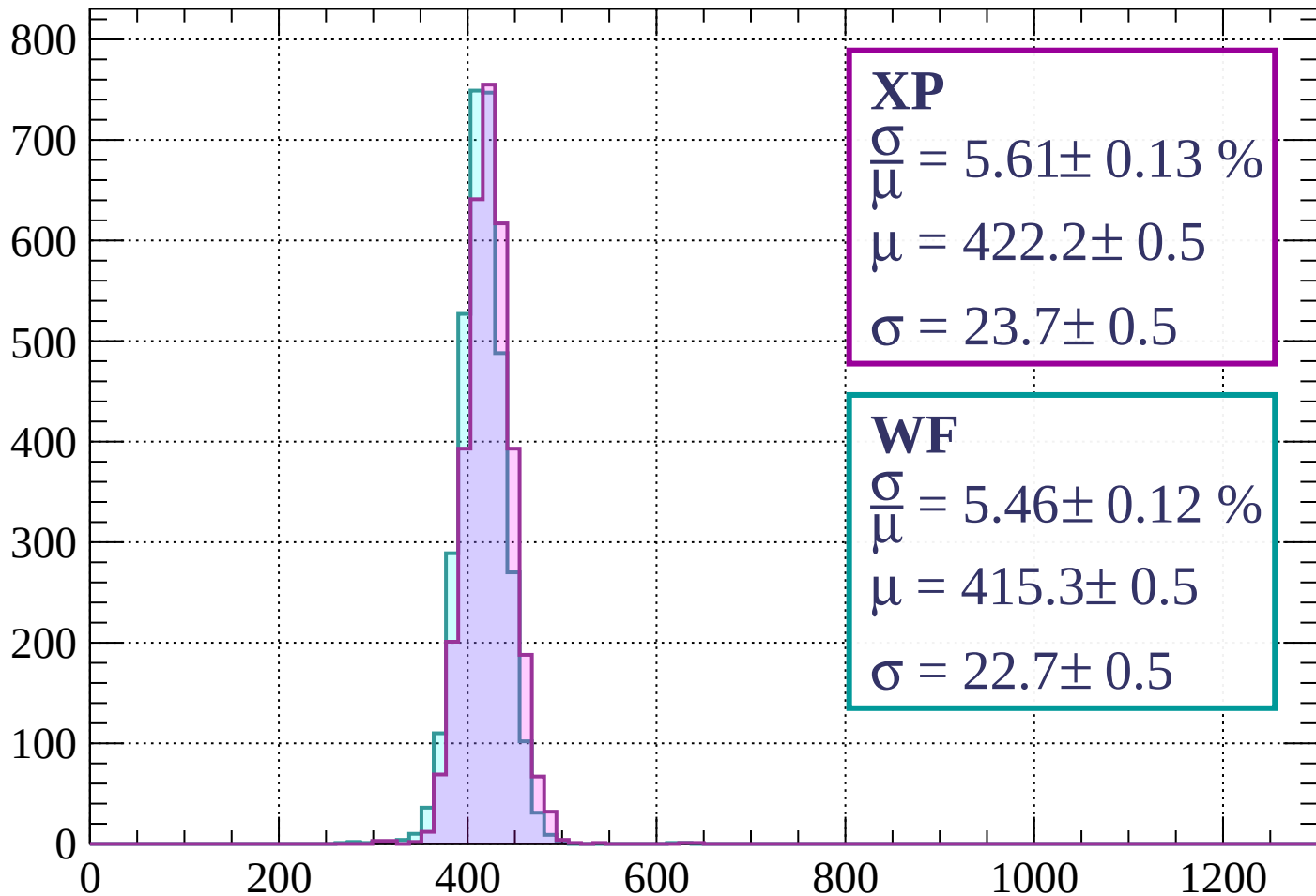
$$\mu = 412.4 \pm 0.5$$

$$\sigma = 23.0 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $480 < p < 560$

Count



XP

$$\frac{\sigma}{\mu} = 5.61 \pm 0.13 \%$$

$$\mu = 422.2 \pm 0.5$$

$$\sigma = 23.7 \pm 0.5$$

WF

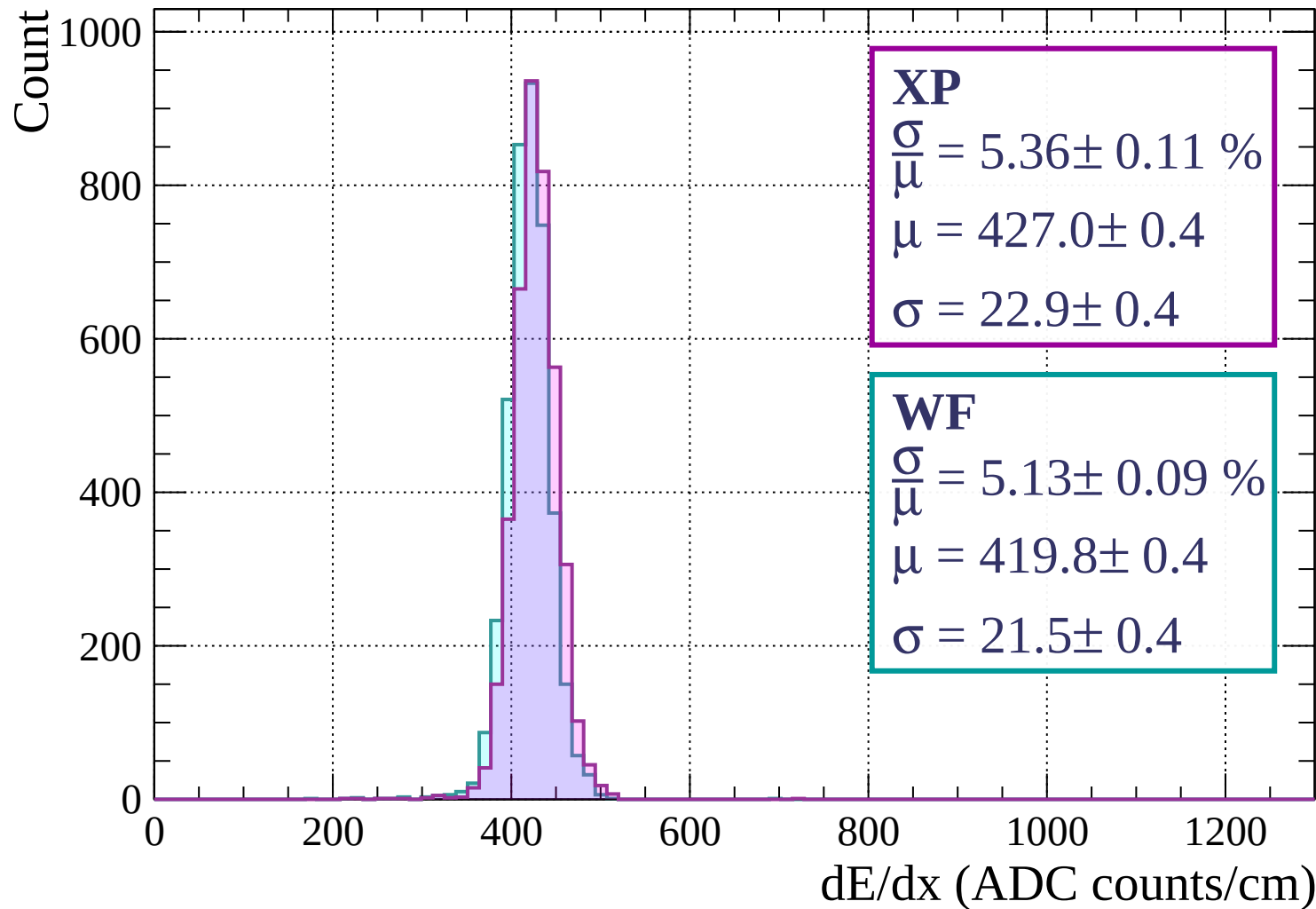
$$\frac{\sigma}{\mu} = 5.46 \pm 0.12 \%$$

$$\mu = 415.3 \pm 0.5$$

$$\sigma = 22.7 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $560 < p < 640$



Energy loss | $640 < p < 720$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 5.40 \pm 0.11 \%$$

$$\mu = 432.5 \pm 0.4$$

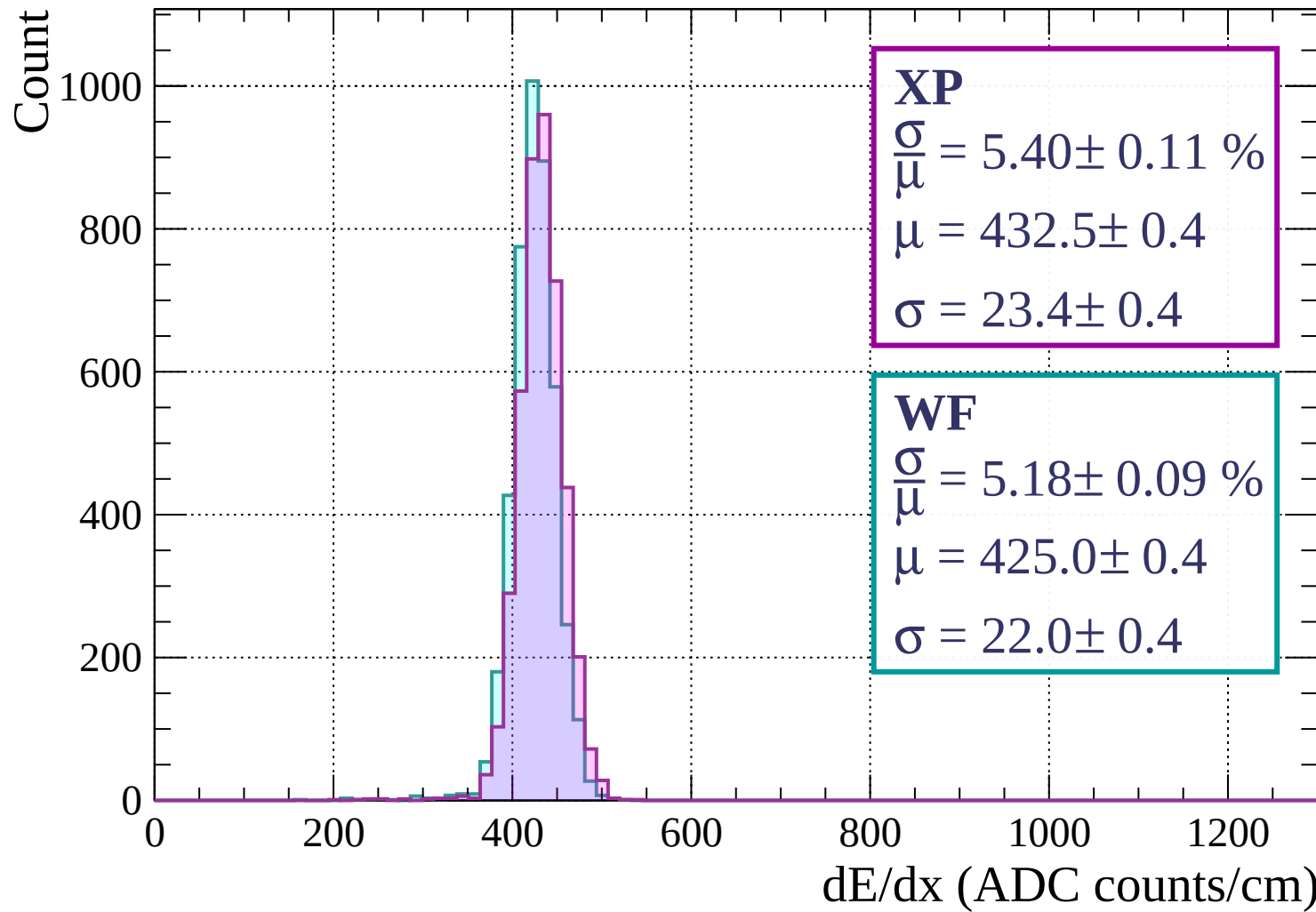
$$\sigma = 23.4 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 5.18 \pm 0.09 \%$$

$$\mu = 425.0 \pm 0.4$$

$$\sigma = 22.0 \pm 0.4$$



Energy loss | $720 < p < 800$

Count

1000

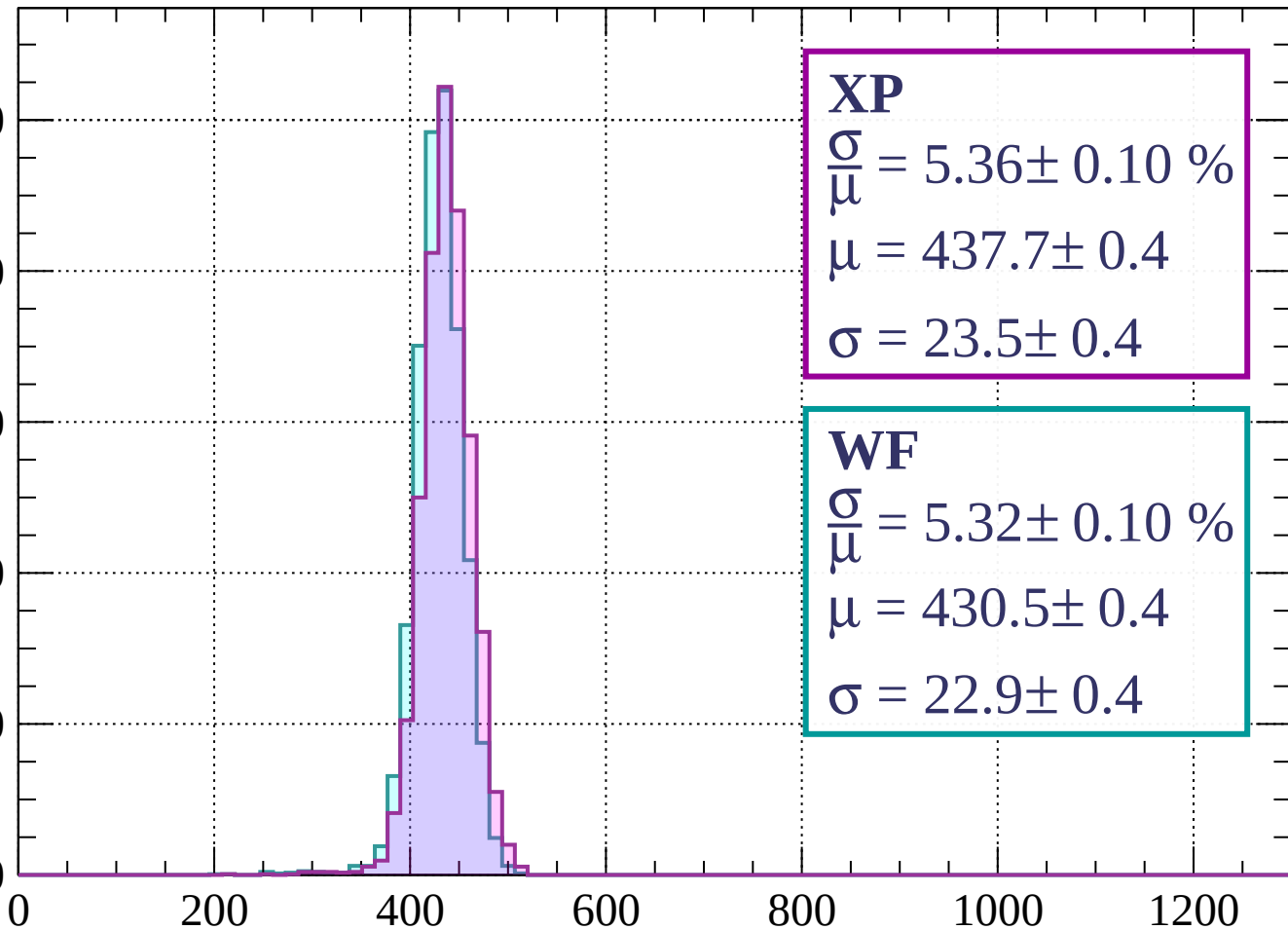
800

600

400

200

0



XP

$$\frac{\sigma}{\mu} = 5.36 \pm 0.10 \%$$

$$\mu = 437.7 \pm 0.4$$

$$\sigma = 23.5 \pm 0.4$$

WF

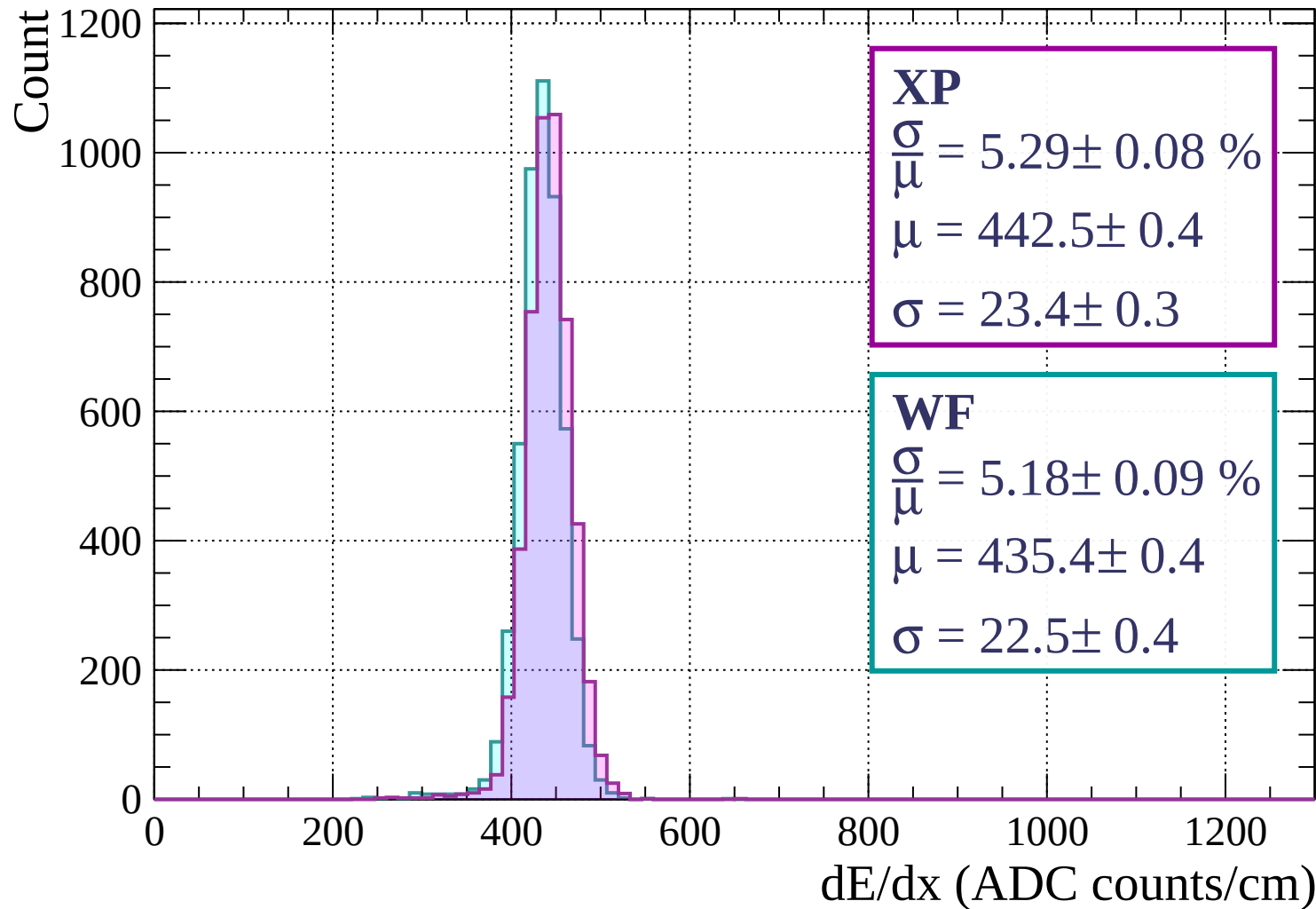
$$\frac{\sigma}{\mu} = 5.32 \pm 0.10 \%$$

$$\mu = 430.5 \pm 0.4$$

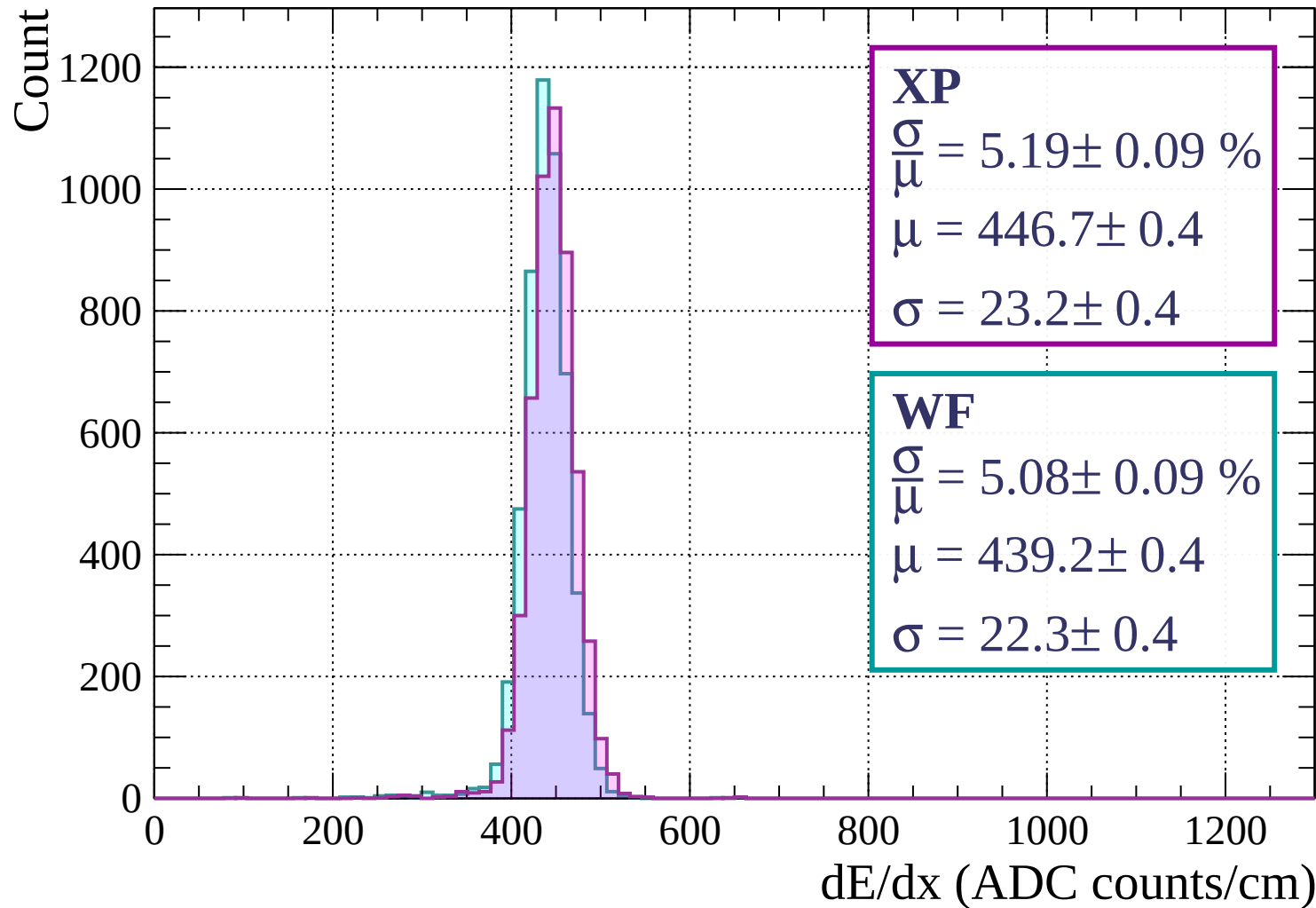
$$\sigma = 22.9 \pm 0.4$$

dE/dx (ADC counts/cm)

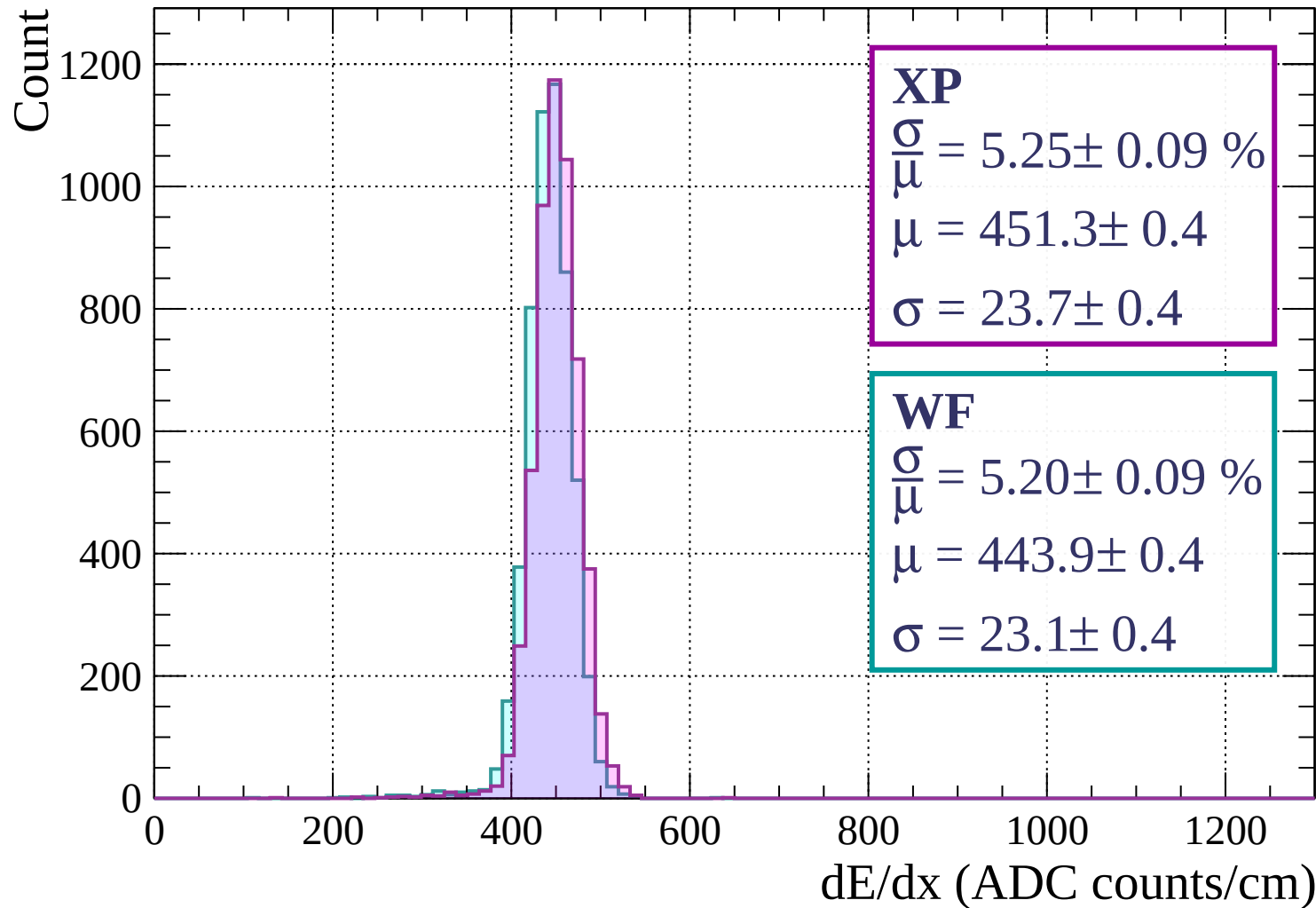
Energy loss | $800 < p < 880$



Energy loss | $880 < p < 960$



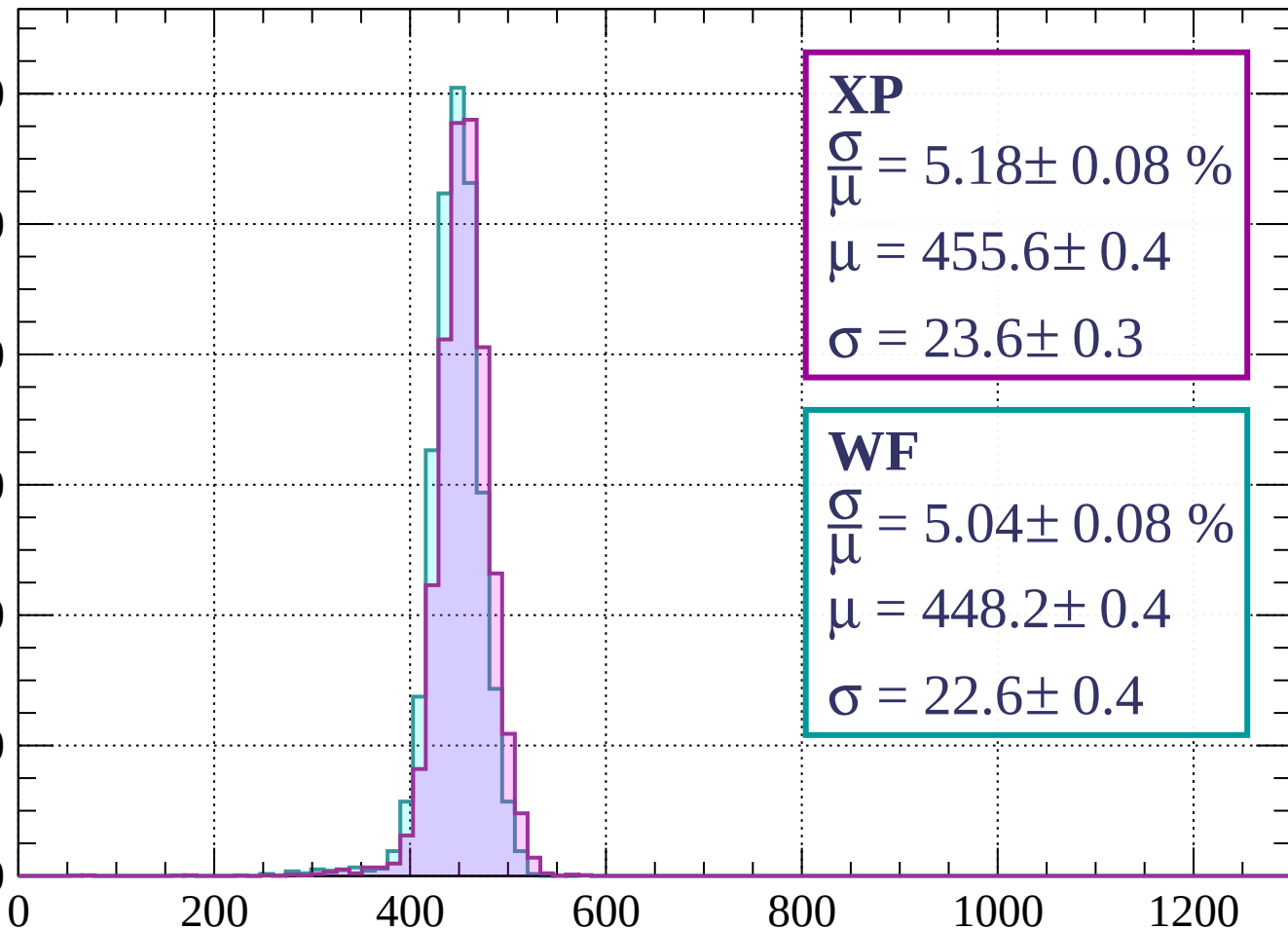
Energy loss | $960 < p < 1040$



Energy loss | $1040 < p < 1120$

Count

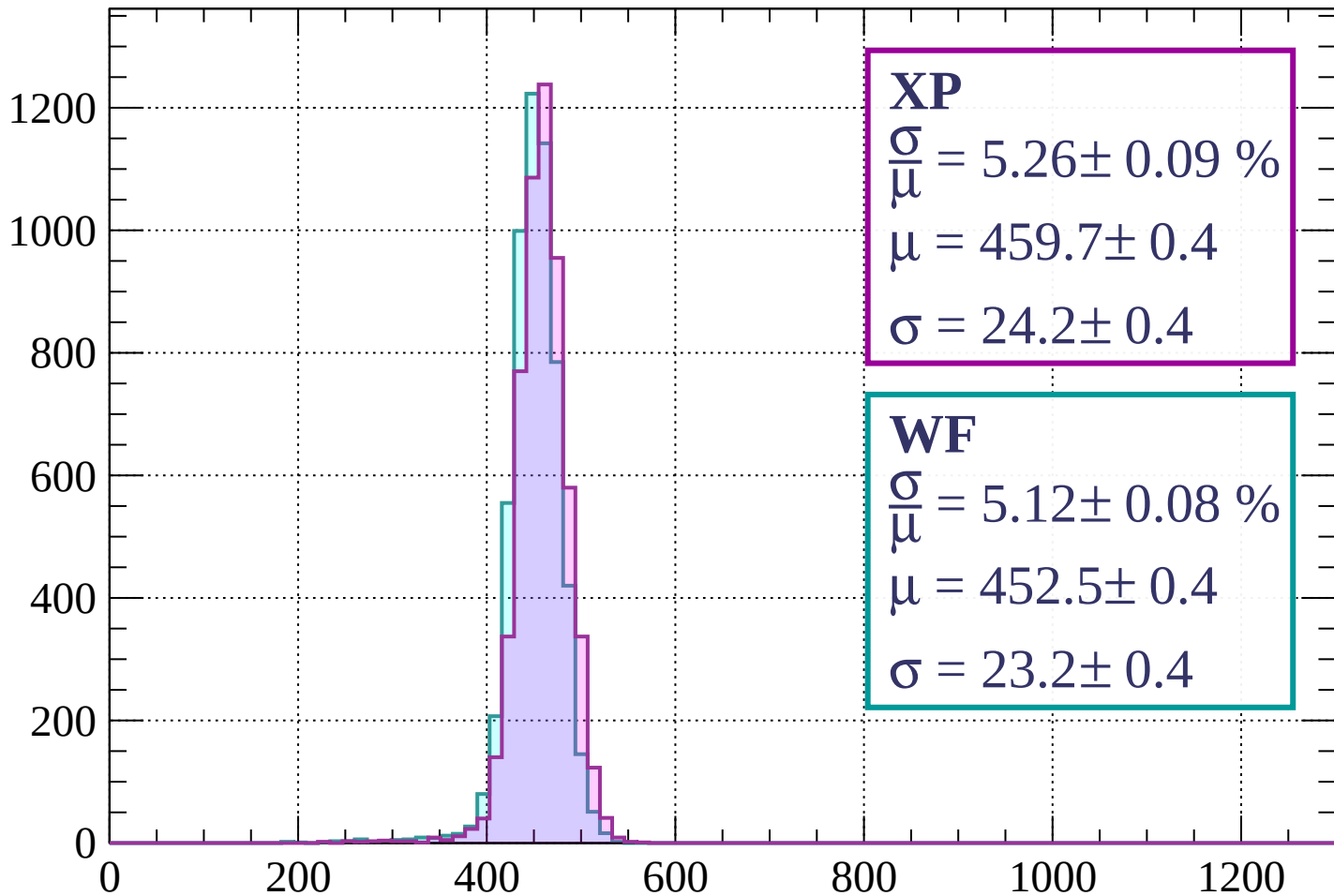
1200
1000
800
600
400
200
0



dE/dx (ADC counts/cm)

Energy loss | $1120 < p < 1200$

Count



XP

$$\frac{\sigma}{\mu} = 5.26 \pm 0.09 \%$$

$$\mu = 459.7 \pm 0.4$$

$$\sigma = 24.2 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 5.12 \pm 0.08 \%$$

$$\mu = 452.5 \pm 0.4$$

$$\sigma = 23.2 \pm 0.4$$

dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1280$

Count

1200

1000

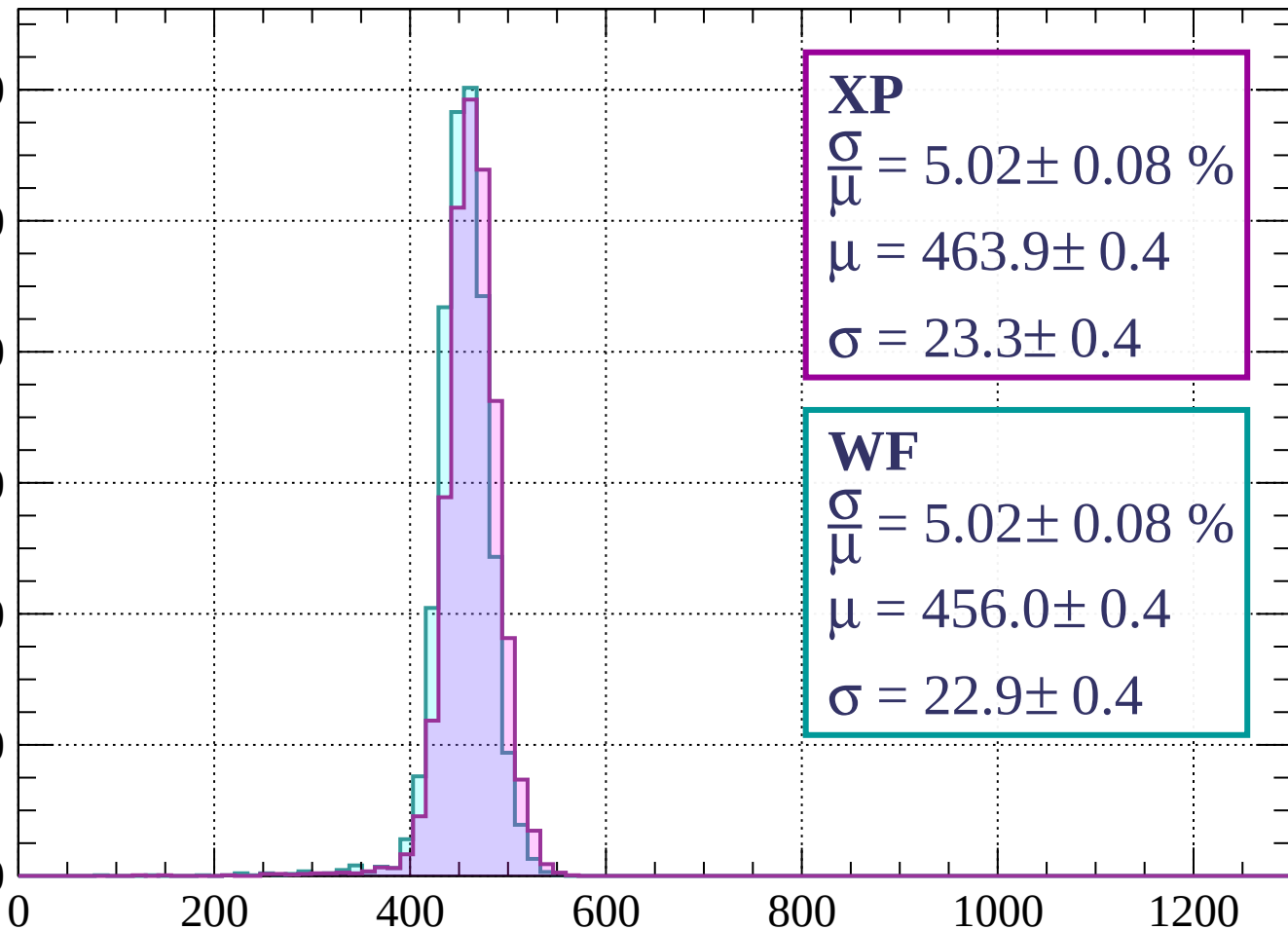
800

600

400

200

0



XP

$$\frac{\sigma}{\mu} = 5.02 \pm 0.08 \%$$

$$\mu = 463.9 \pm 0.4$$

$$\sigma = 23.3 \pm 0.4$$

WF

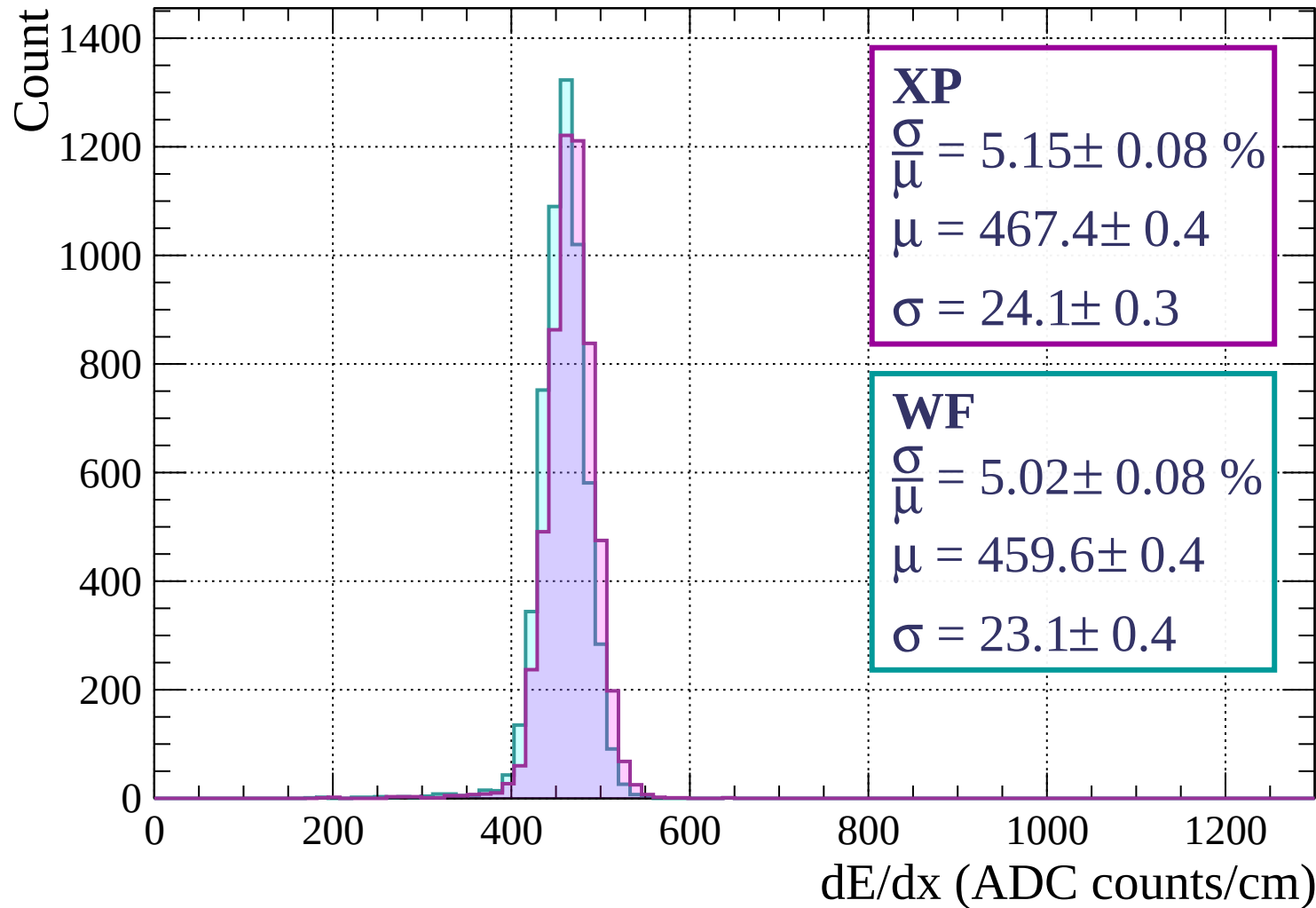
$$\frac{\sigma}{\mu} = 5.02 \pm 0.08 \%$$

$$\mu = 456.0 \pm 0.4$$

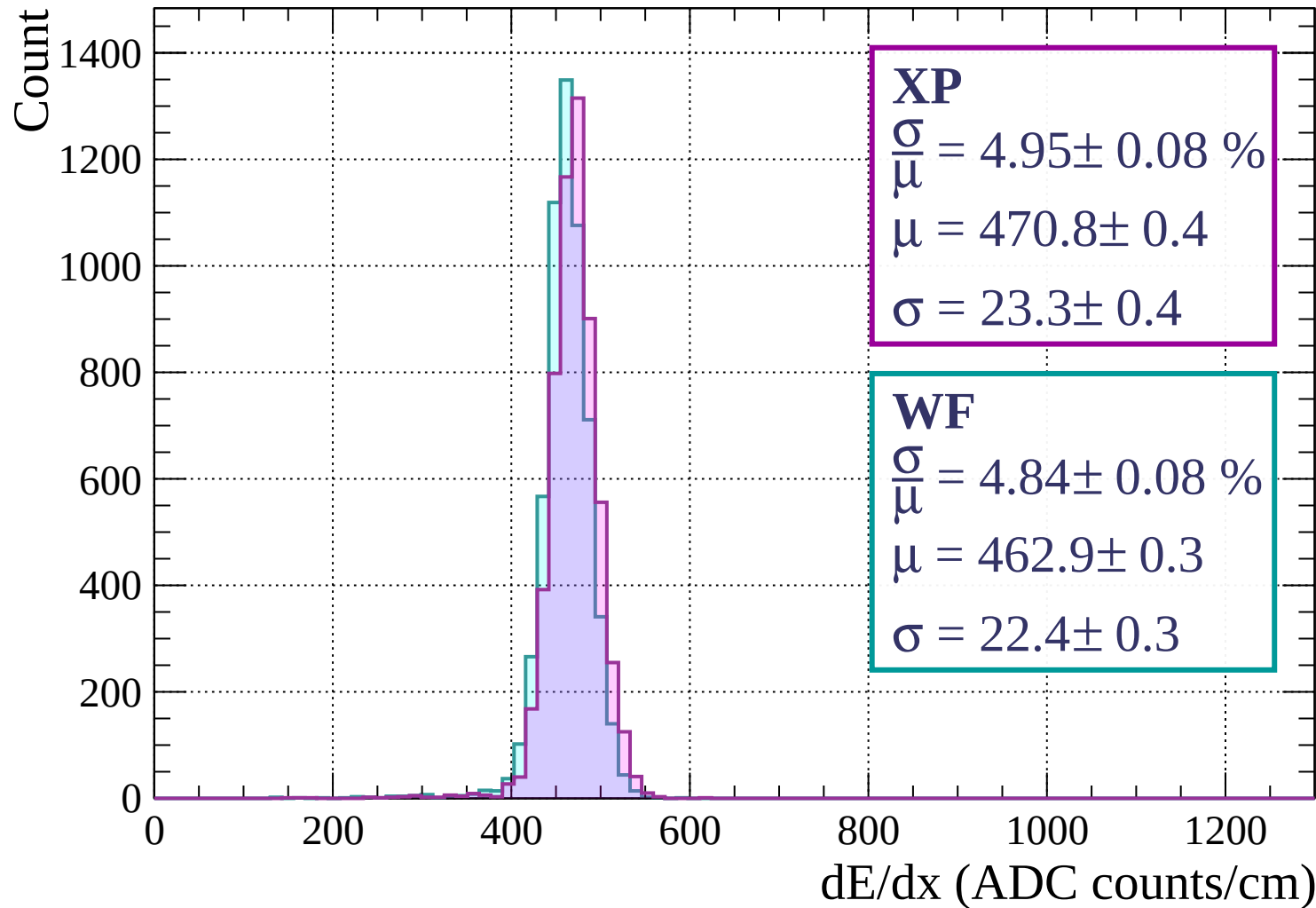
$$\sigma = 22.9 \pm 0.4$$

dE/dx (ADC counts/cm)

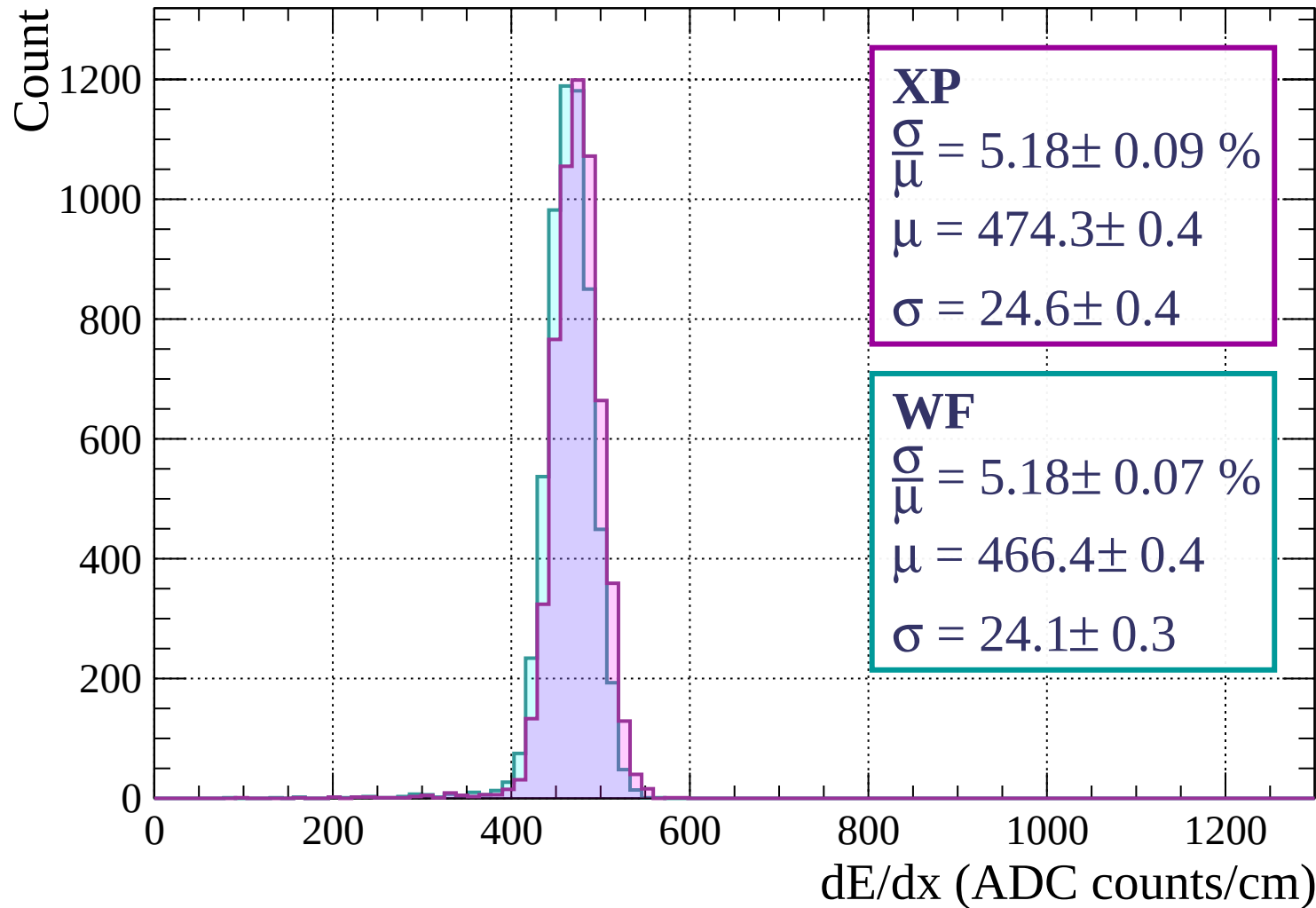
Energy loss | $1280 < p < 1360$



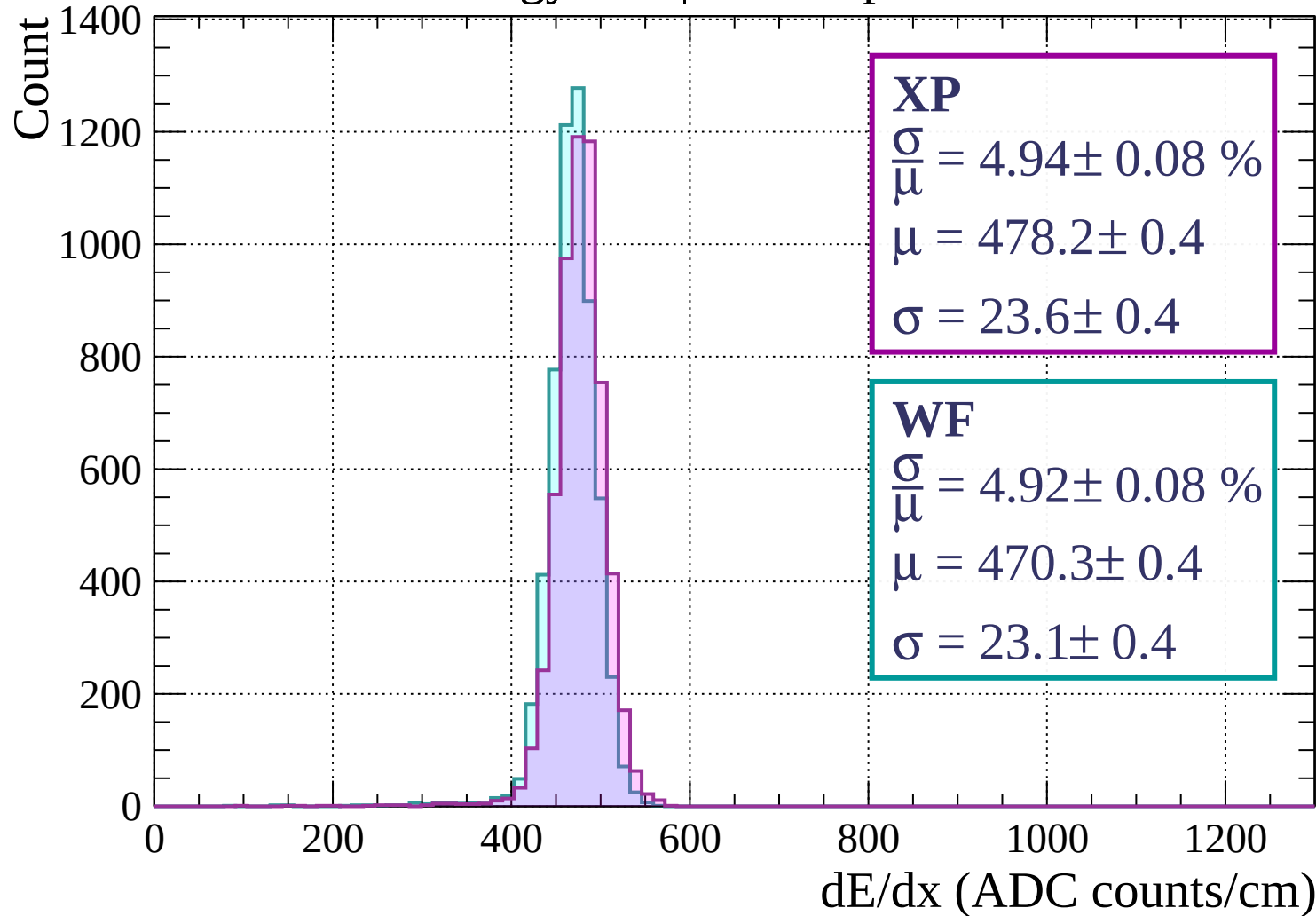
Energy loss | $1360 < p < 1440$



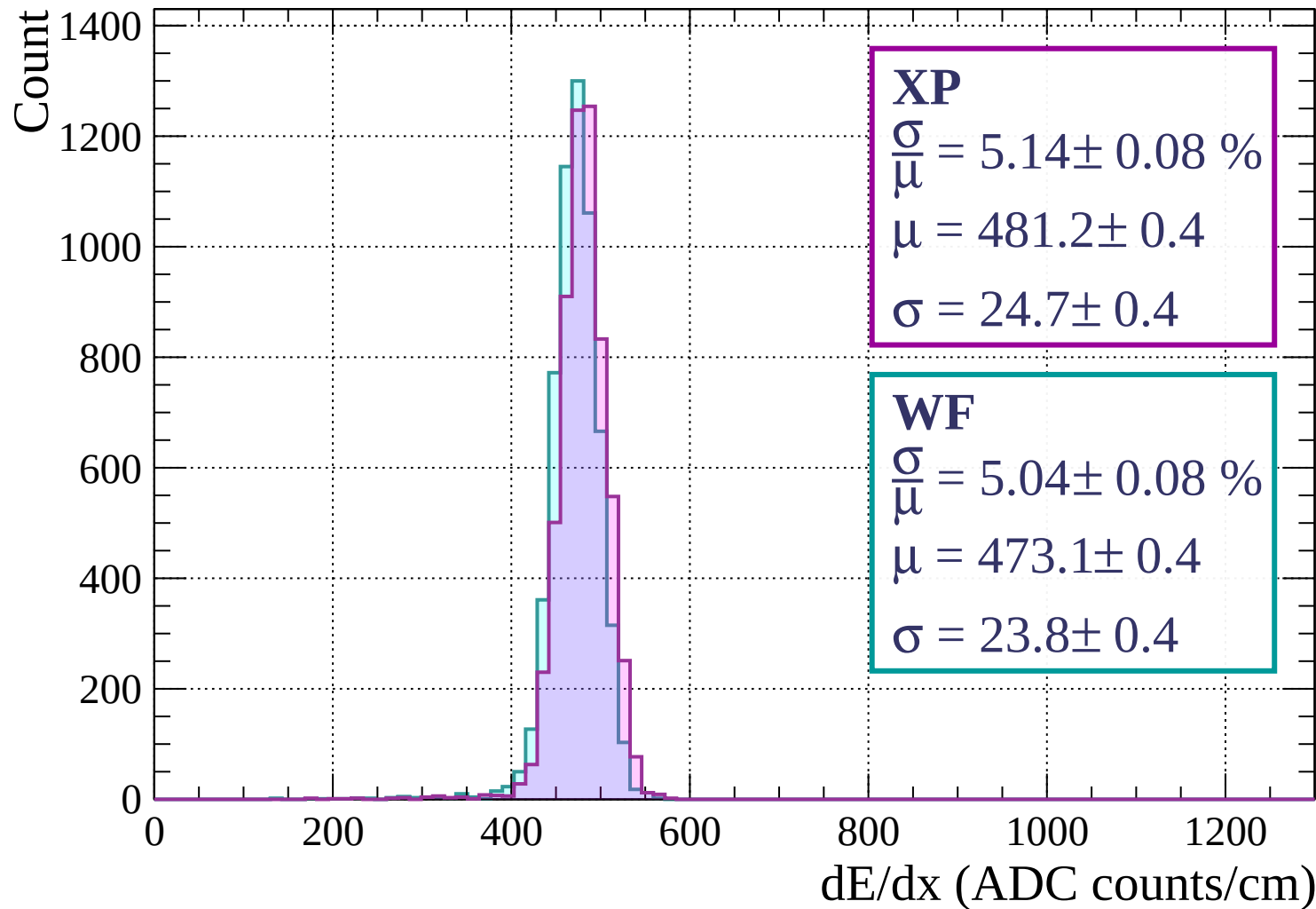
Energy loss | $1440 < p < 1520$



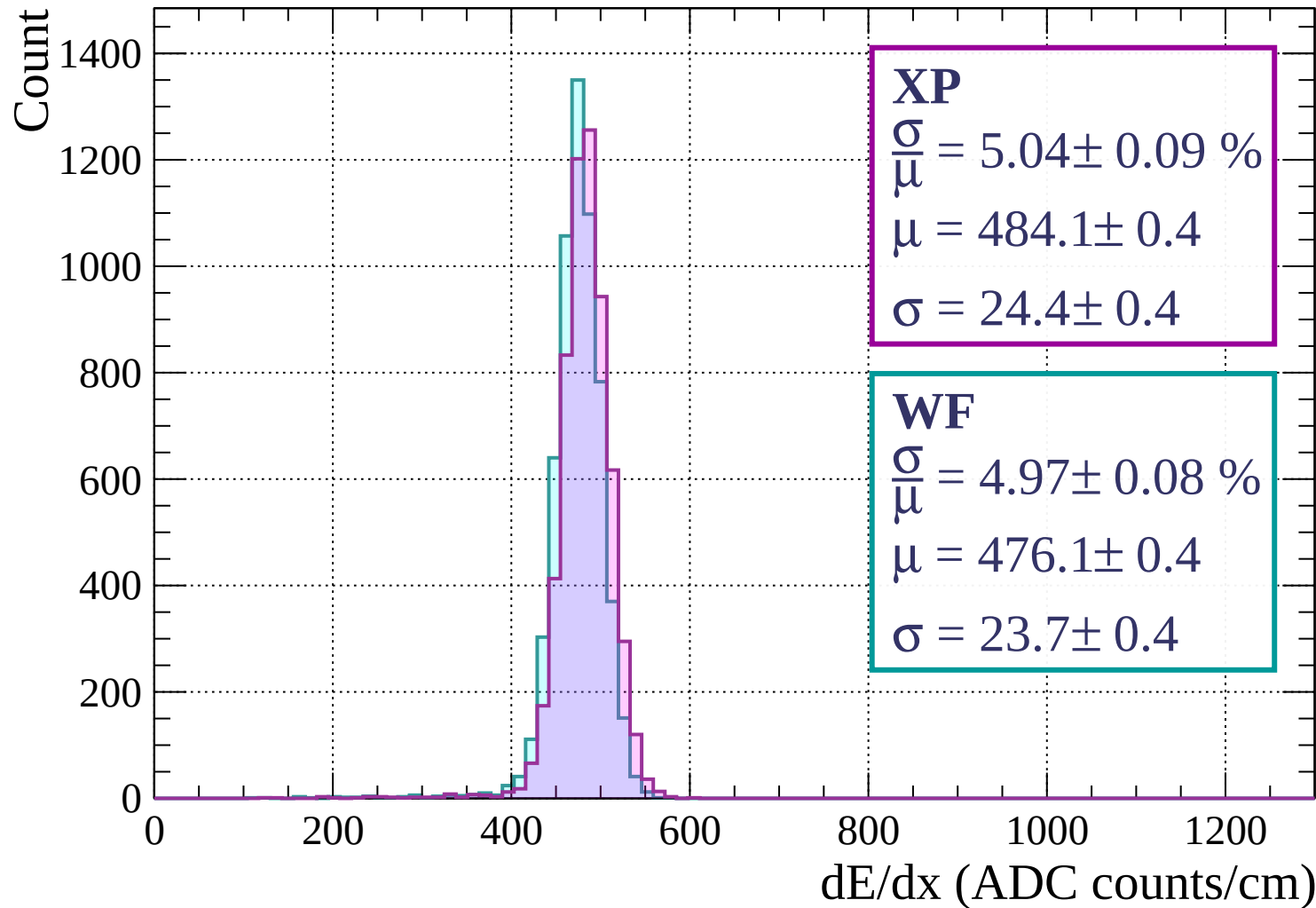
Energy loss | $1520 < p < 1600$



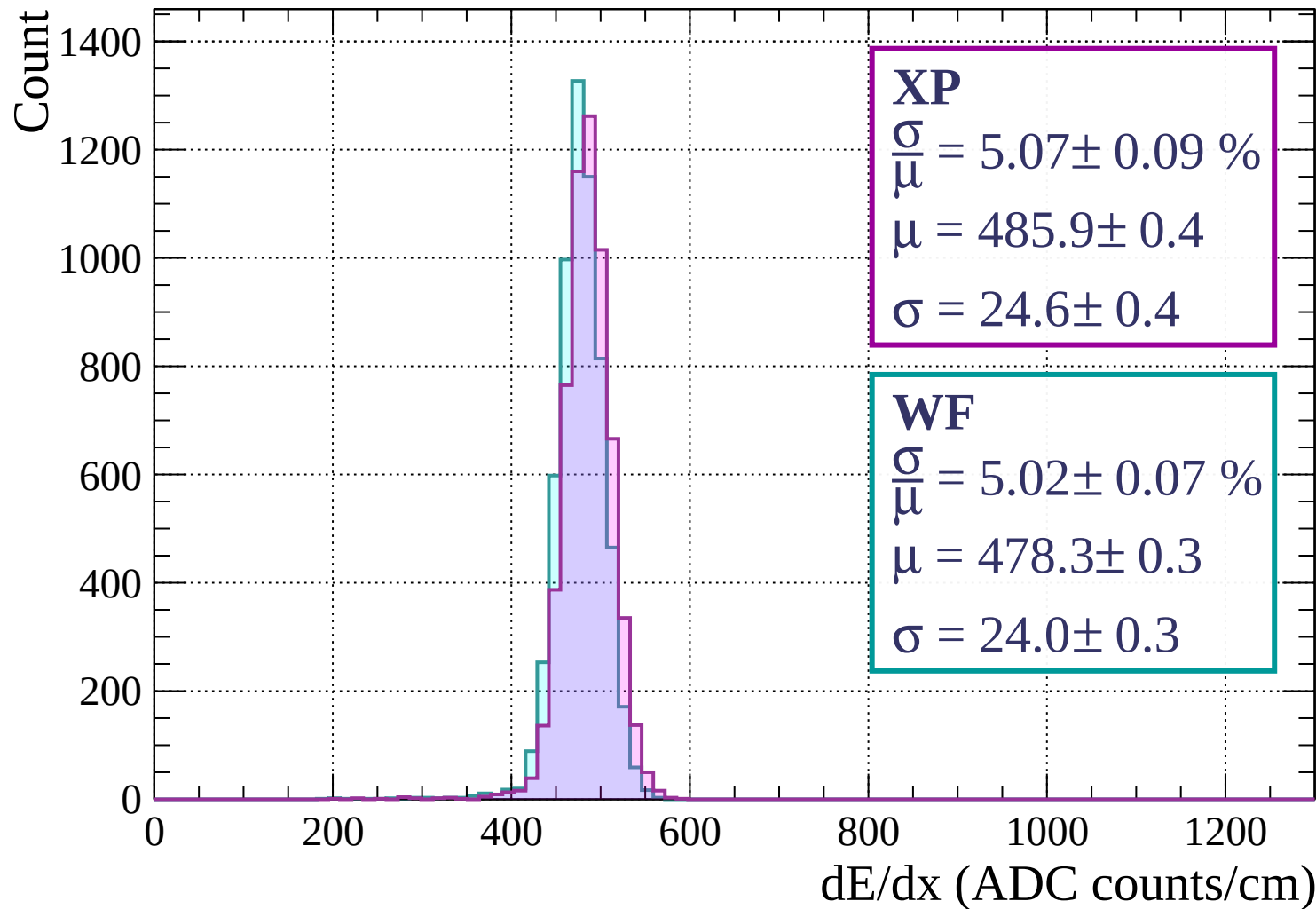
Energy loss | $1600 < p < 1680$



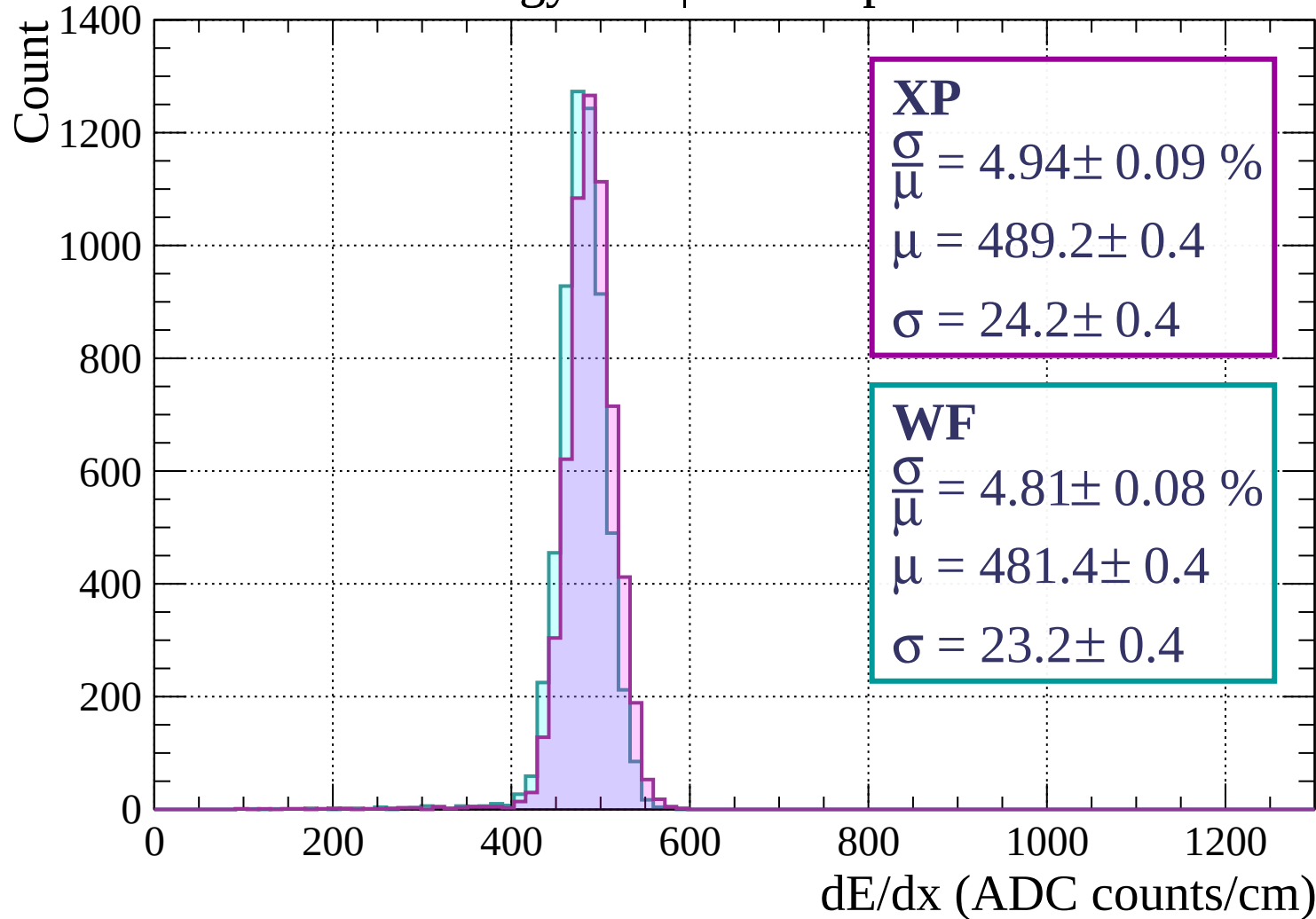
Energy loss | $1680 < p < 1760$



Energy loss | $1760 < p < 1840$

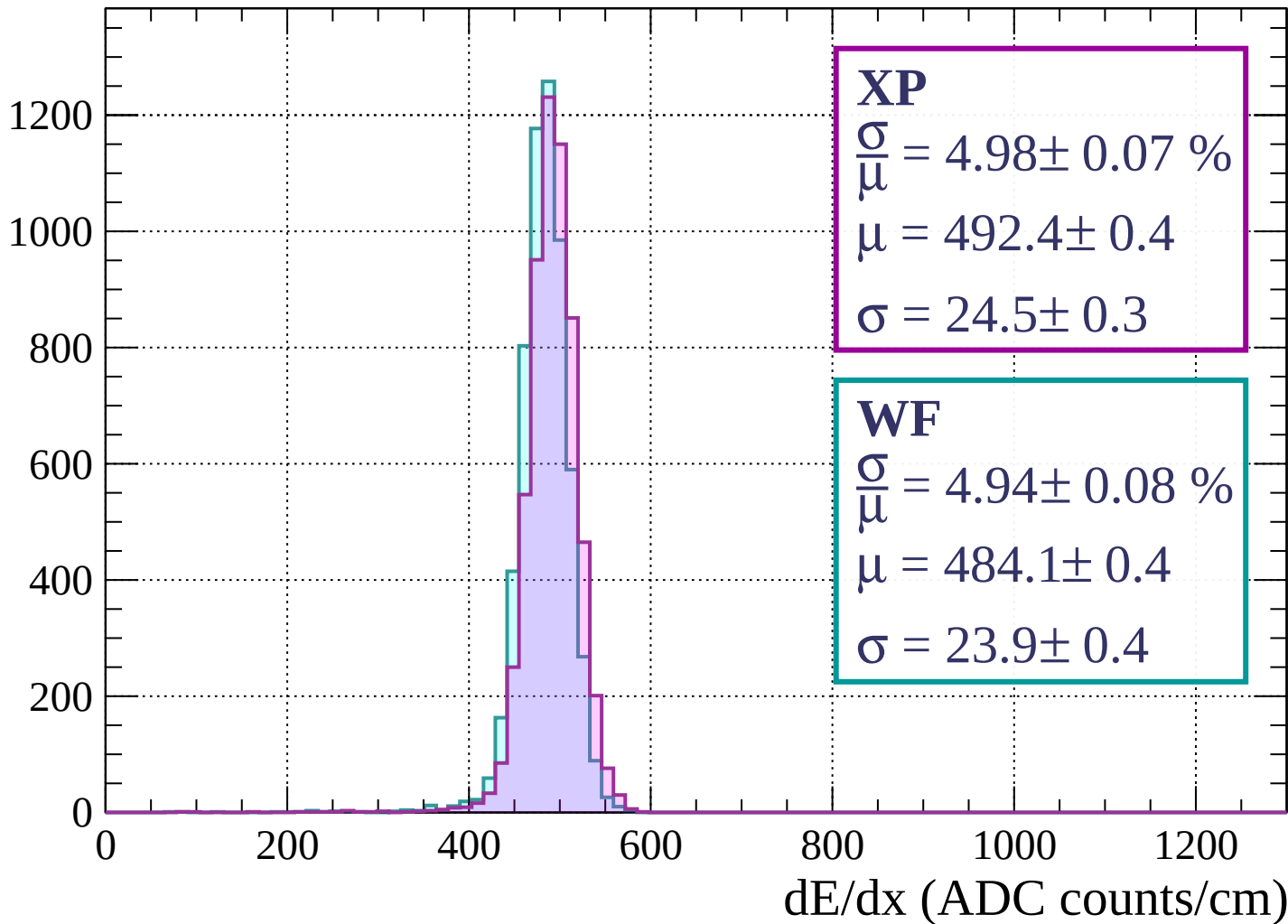


Energy loss | $1840 < p < 1920$



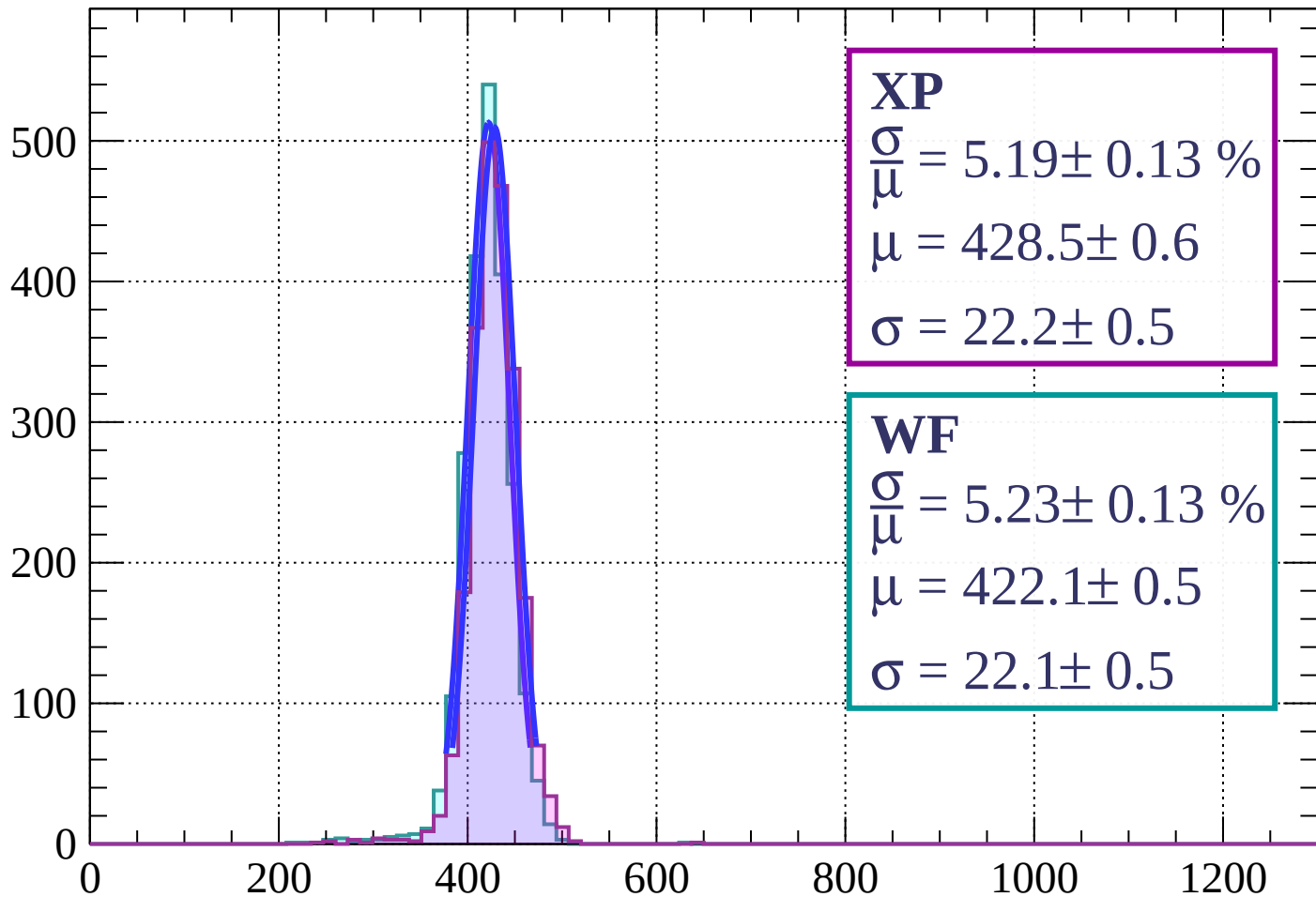
Energy loss | $1920 < p < 2000$

Count



Energy loss | $0 < \phi < 3$

Count



XP

$$\frac{\sigma}{\mu} = 5.19 \pm 0.13 \%$$

$$\mu = 428.5 \pm 0.6$$

$$\sigma = 22.2 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 5.23 \pm 0.13 \%$$

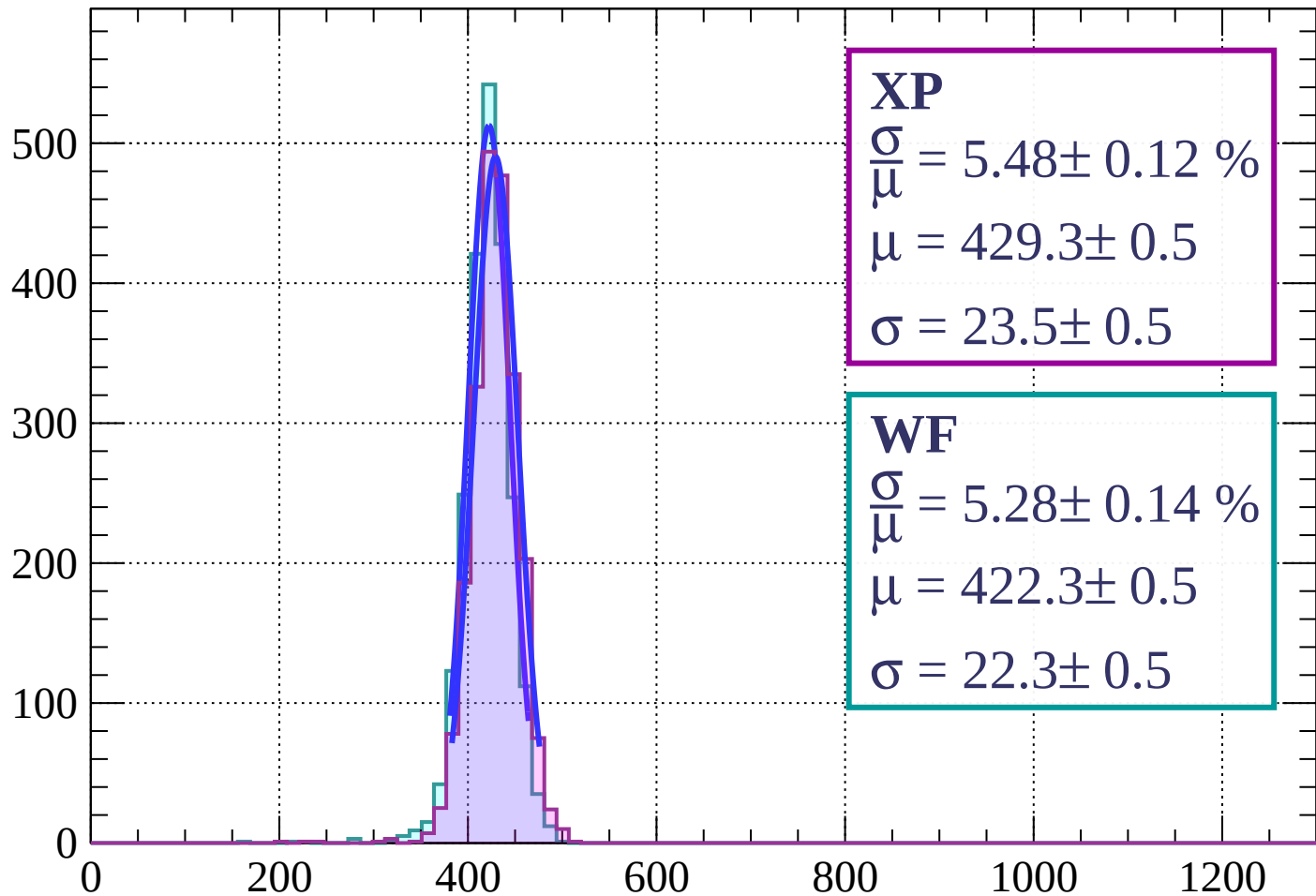
$$\mu = 422.1 \pm 0.5$$

$$\sigma = 22.1 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $3 < \phi < 6$

Count



XP

$$\frac{\sigma}{\mu} = 5.48 \pm 0.12 \%$$

$$\mu = 429.3 \pm 0.5$$

$$\sigma = 23.5 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 5.28 \pm 0.14 \%$$

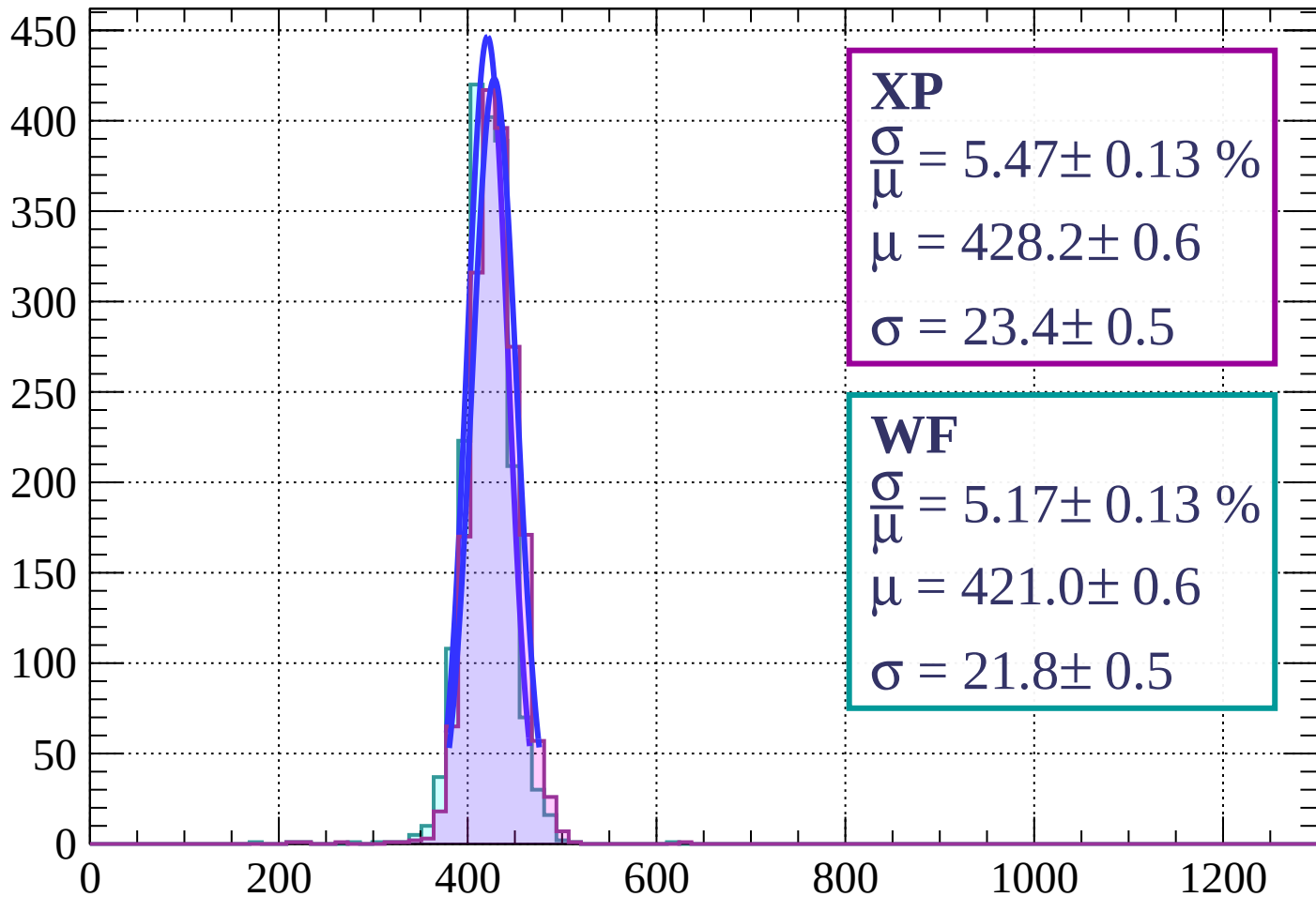
$$\mu = 422.3 \pm 0.5$$

$$\sigma = 22.3 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $6 < \phi < 9$

Count



XP

$$\frac{\sigma}{\mu} = 5.47 \pm 0.13 \%$$

$$\mu = 428.2 \pm 0.6$$

$$\sigma = 23.4 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 5.17 \pm 0.13 \%$$

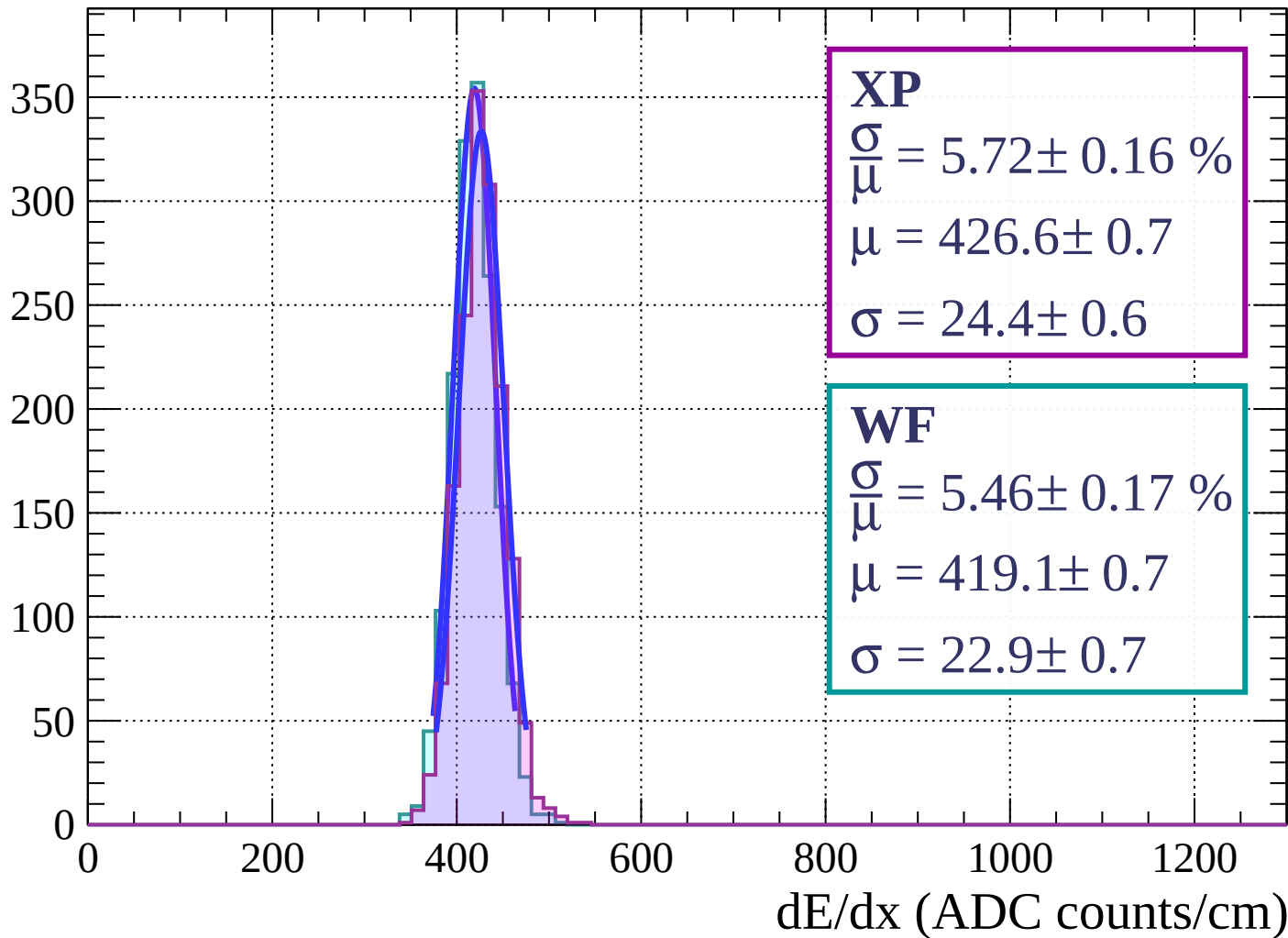
$$\mu = 421.0 \pm 0.6$$

$$\sigma = 21.8 \pm 0.5$$

dE/dx (ADC counts/cm)

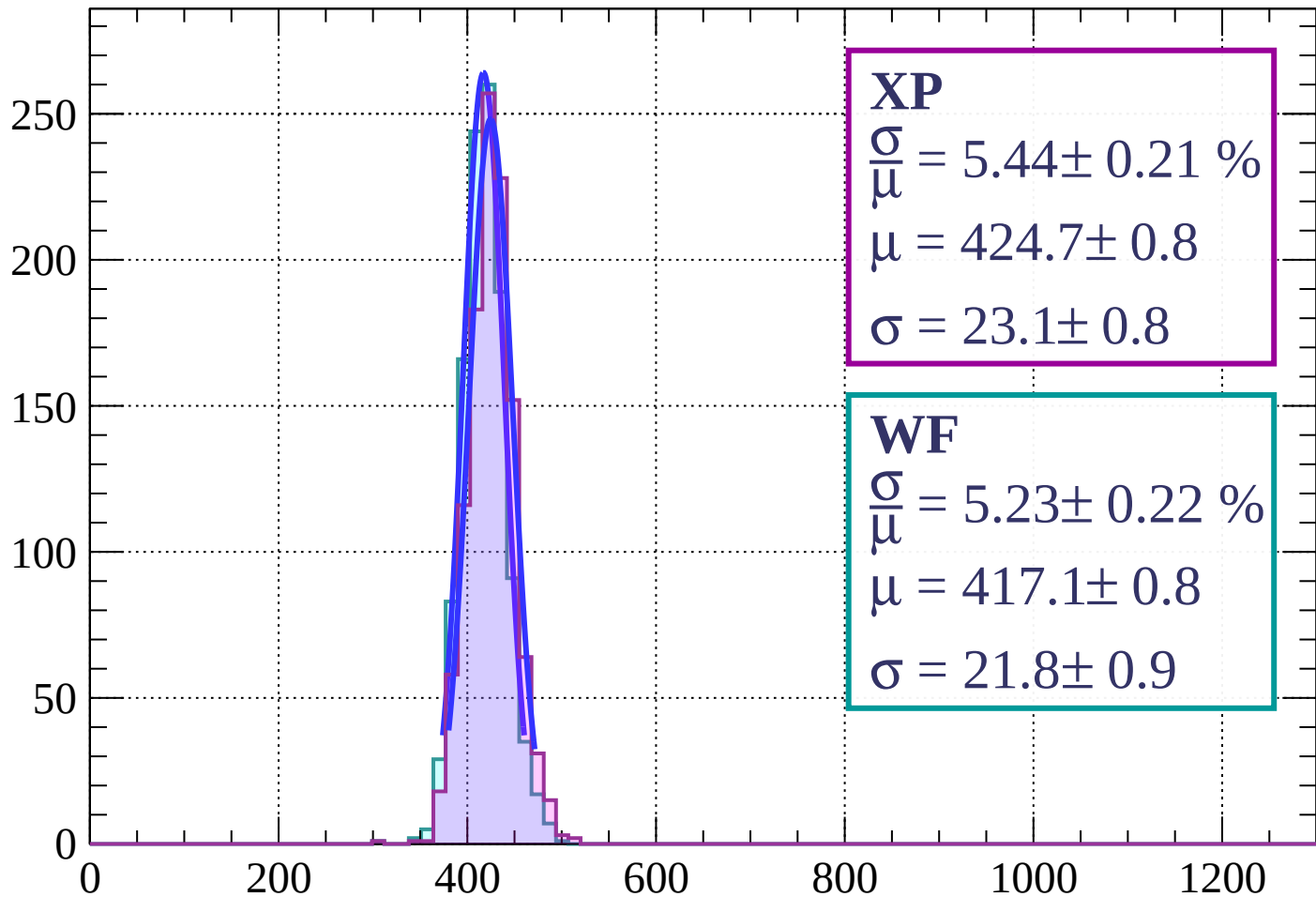
Energy loss | $9 < \phi < 12$

Count



Energy loss | $12 < \phi < 15$

Count



XP

$$\frac{\sigma}{\mu} = 5.44 \pm 0.21 \%$$

$$\mu = 424.7 \pm 0.8$$

$$\sigma = 23.1 \pm 0.8$$

WF

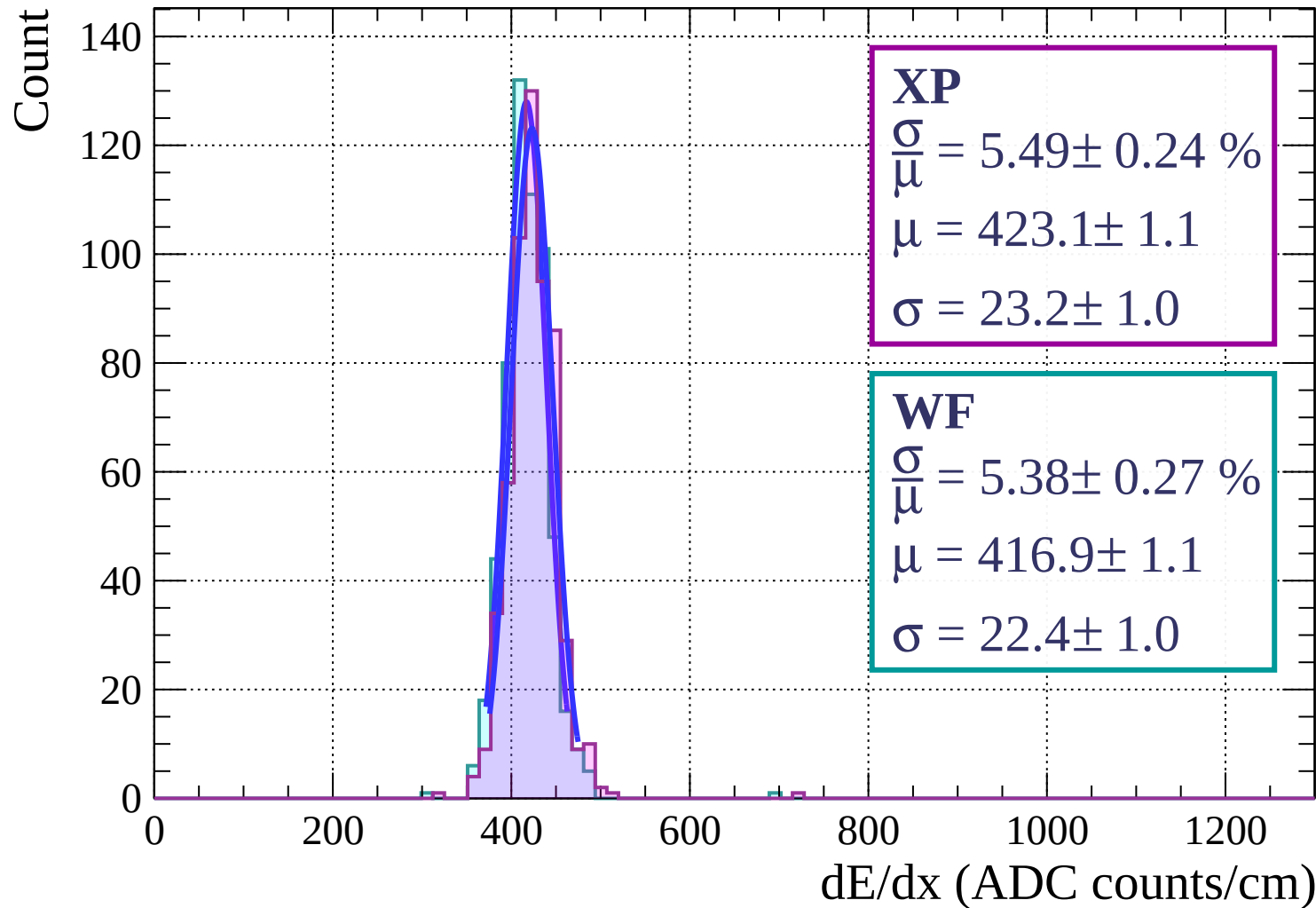
$$\frac{\sigma}{\mu} = 5.23 \pm 0.22 \%$$

$$\mu = 417.1 \pm 0.8$$

$$\sigma = 21.8 \pm 0.9$$

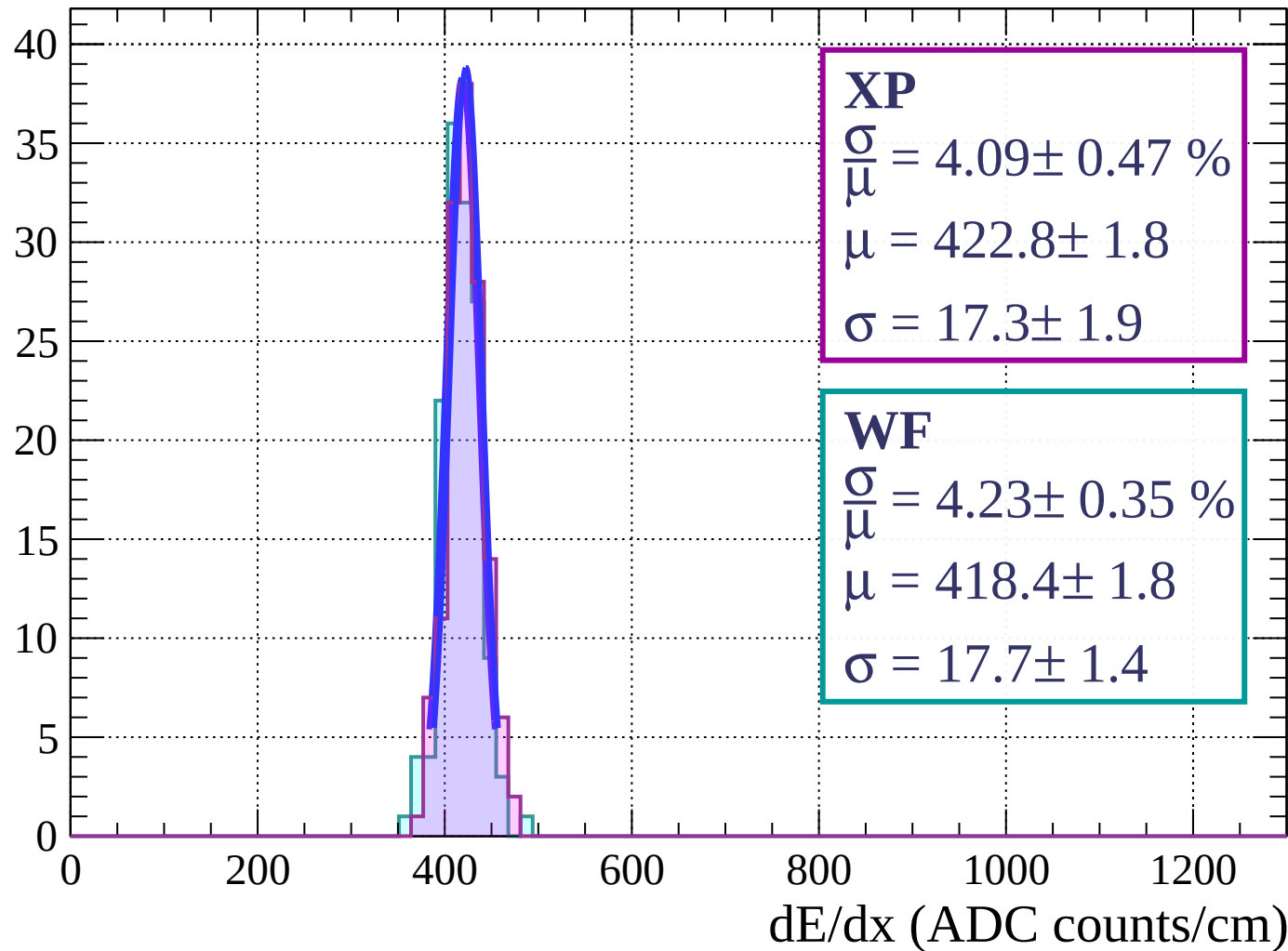
dE/dx (ADC counts/cm)

Energy loss | $15 < \phi < 18$



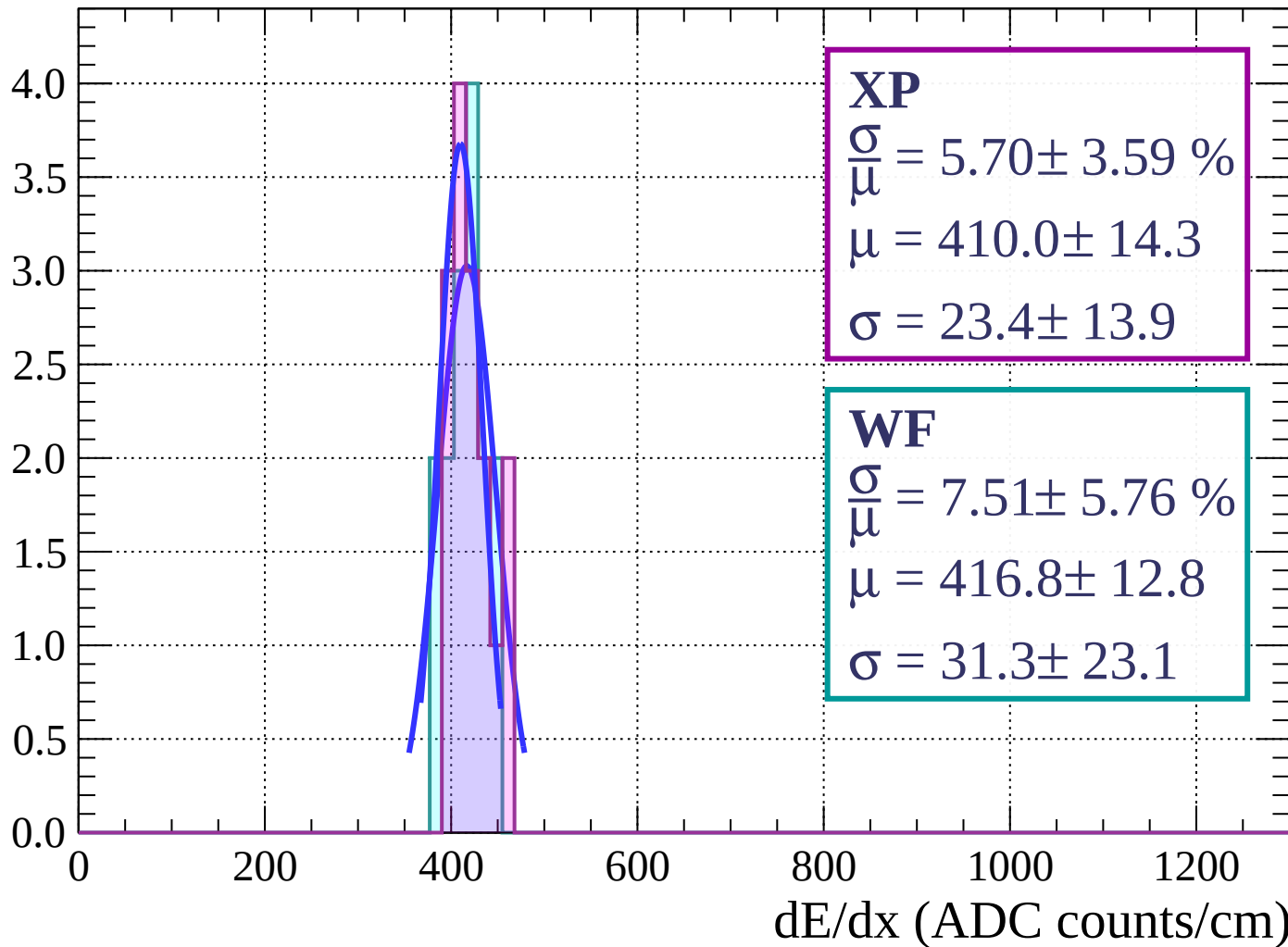
Energy loss | $18 < \phi < 21$

Count



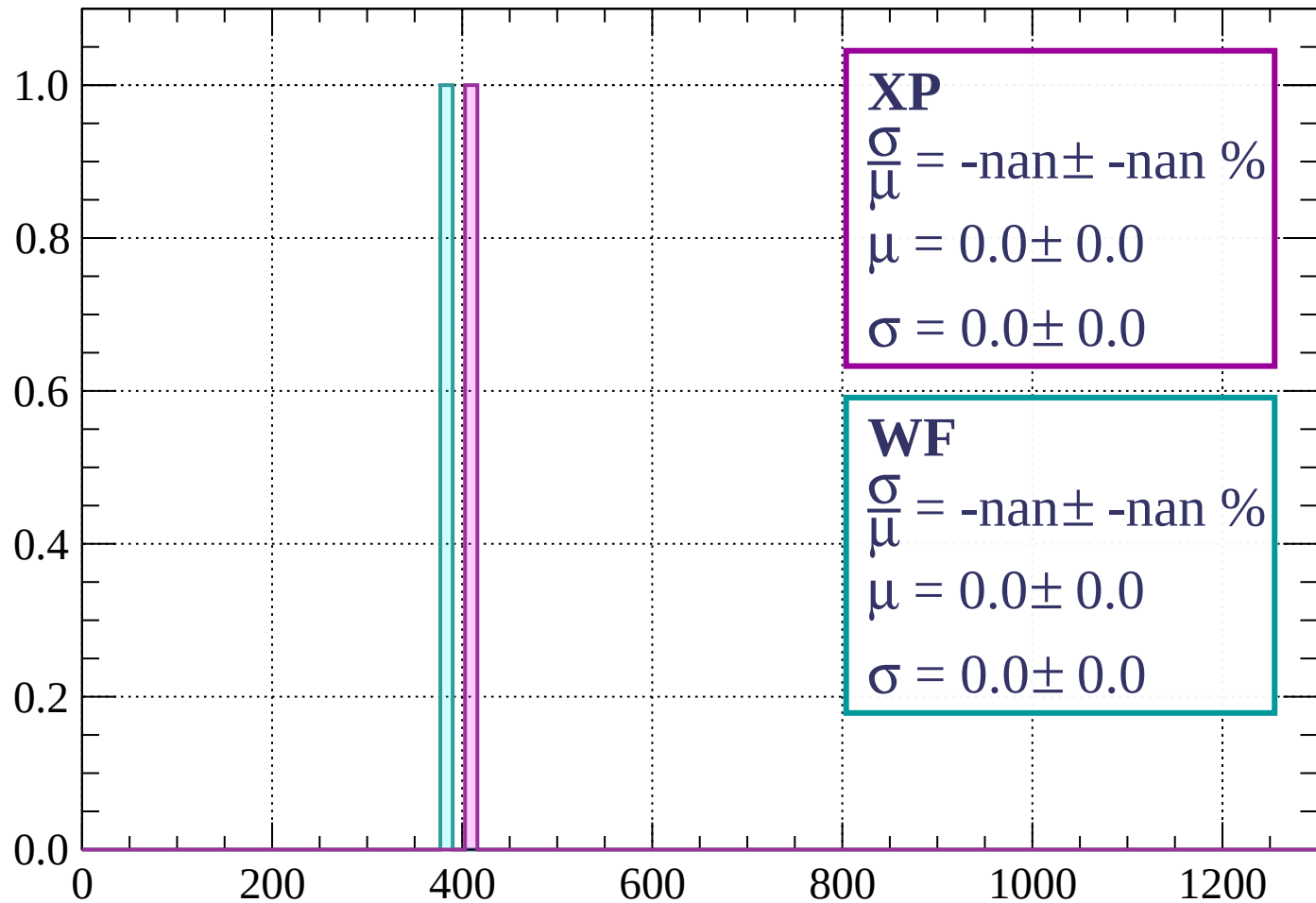
Energy loss | $21 < \phi < 24$

Count



Energy loss | $24 < \phi < 27$

Count



XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

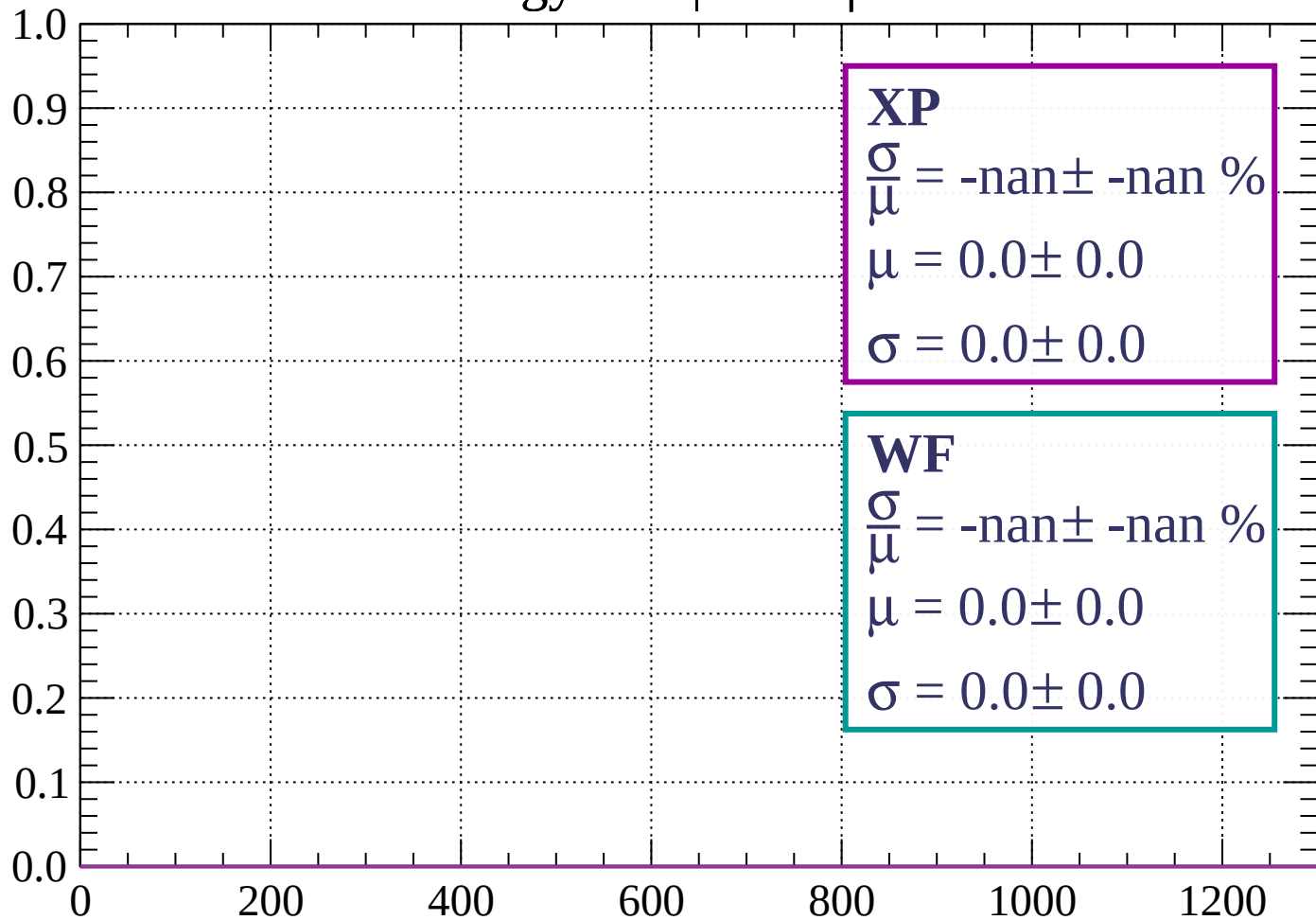
$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx (ADC counts/cm)

Energy loss | $27 < \phi < 30$

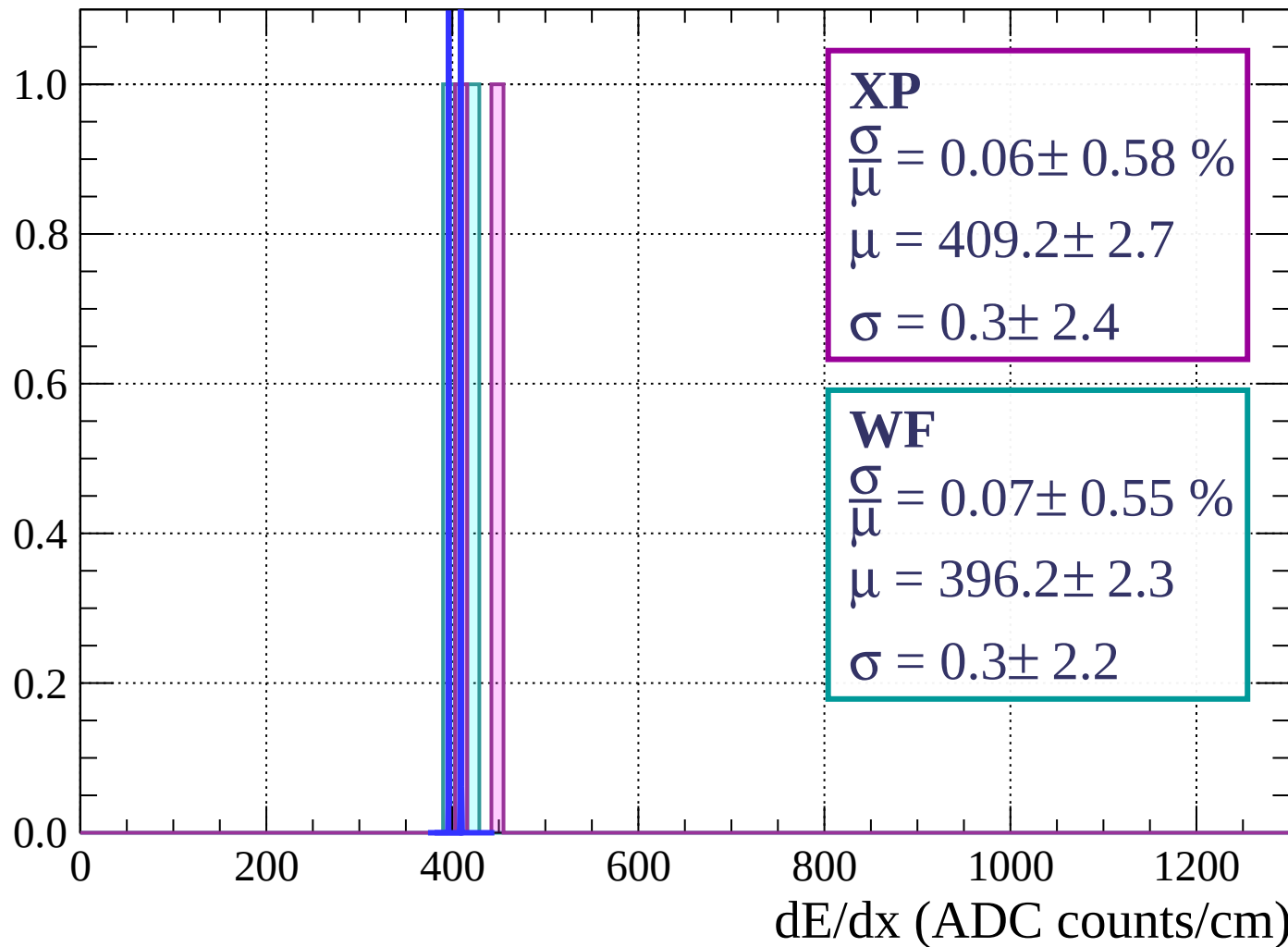
Count



dE/dx (ADC counts/cm)

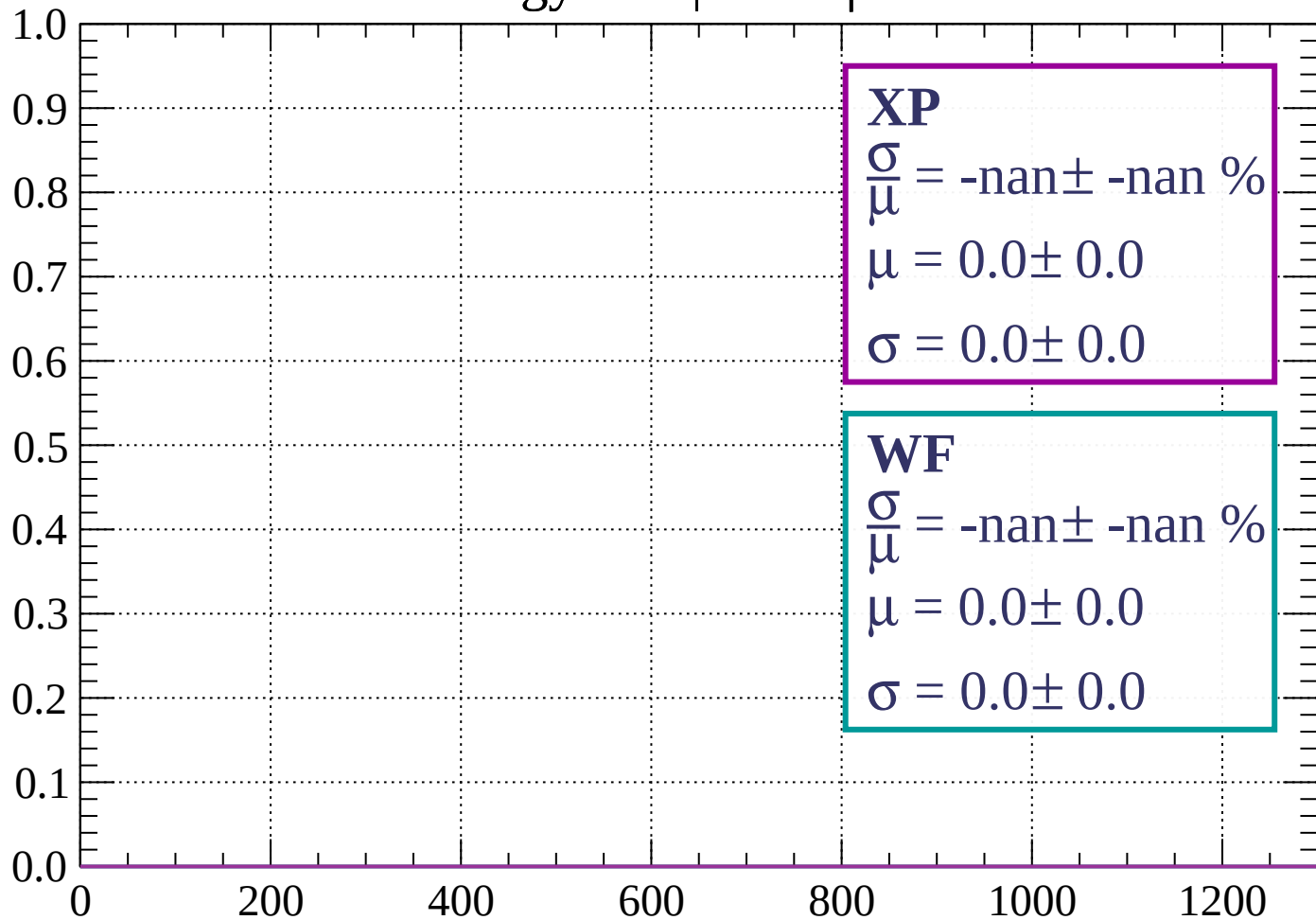
Energy loss | $30 < \phi < 33$

Count



Energy loss | $33 < \phi < 36$

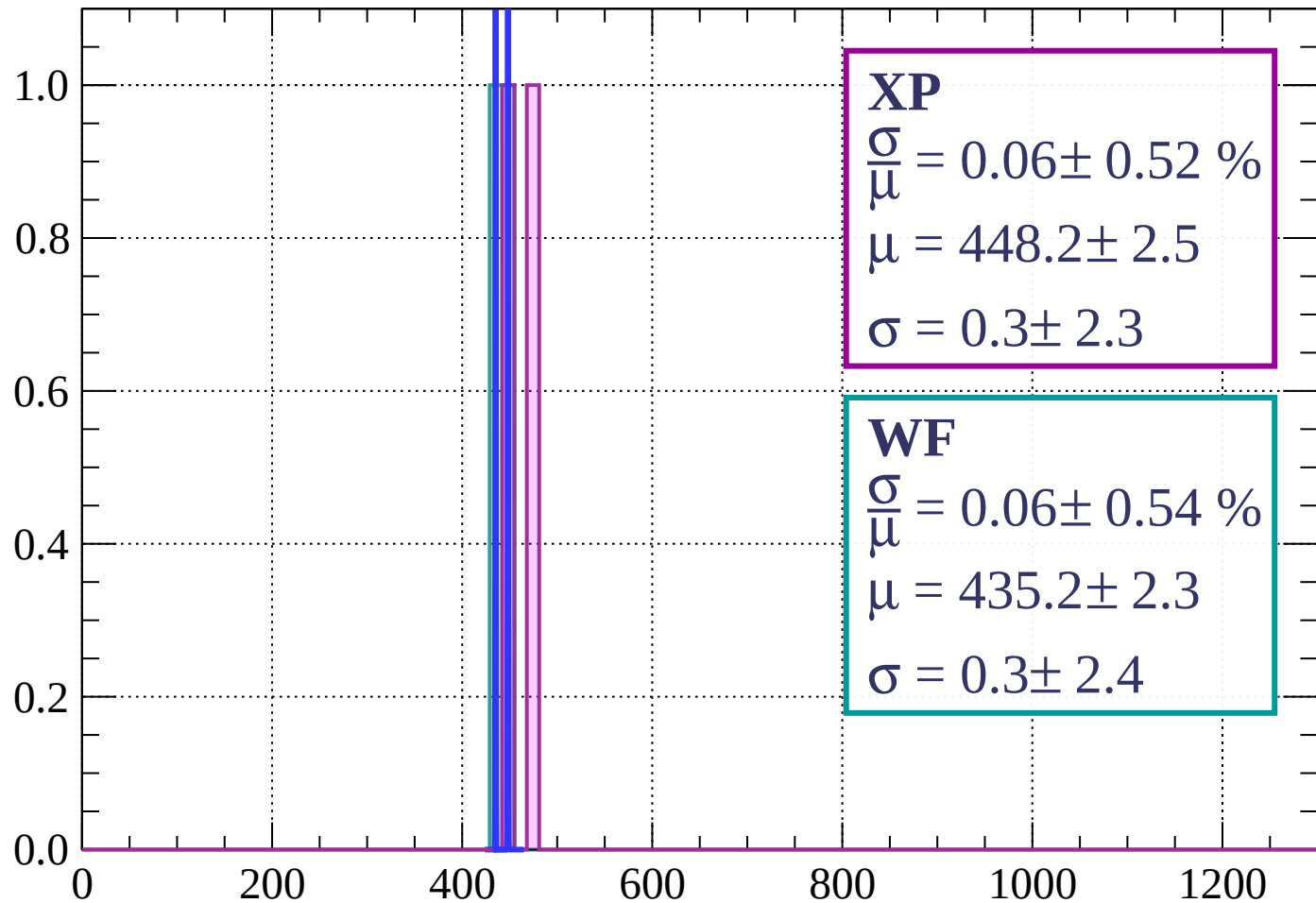
Count



dE/dx (ADC counts/cm)

Energy loss | $36 < \phi < 39$

Count



XP

$$\frac{\sigma}{\mu} = 0.06 \pm 0.52 \%$$

$$\mu = 448.2 \pm 2.5$$

$$\sigma = 0.3 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 0.06 \pm 0.54 \%$$

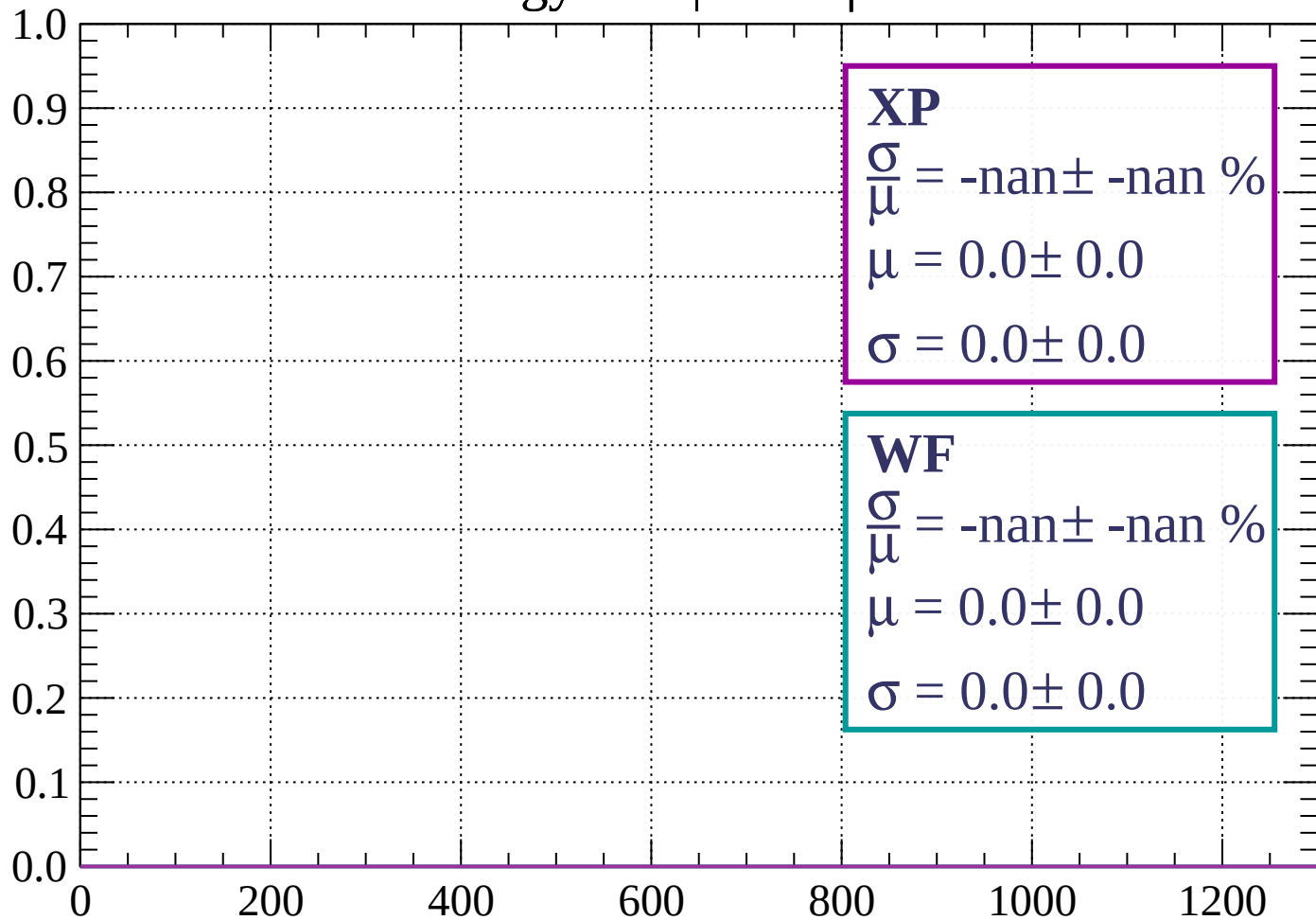
$$\mu = 435.2 \pm 2.3$$

$$\sigma = 0.3 \pm 2.4$$

dE/dx (ADC counts/cm)

Energy loss | $39 < \phi < 42$

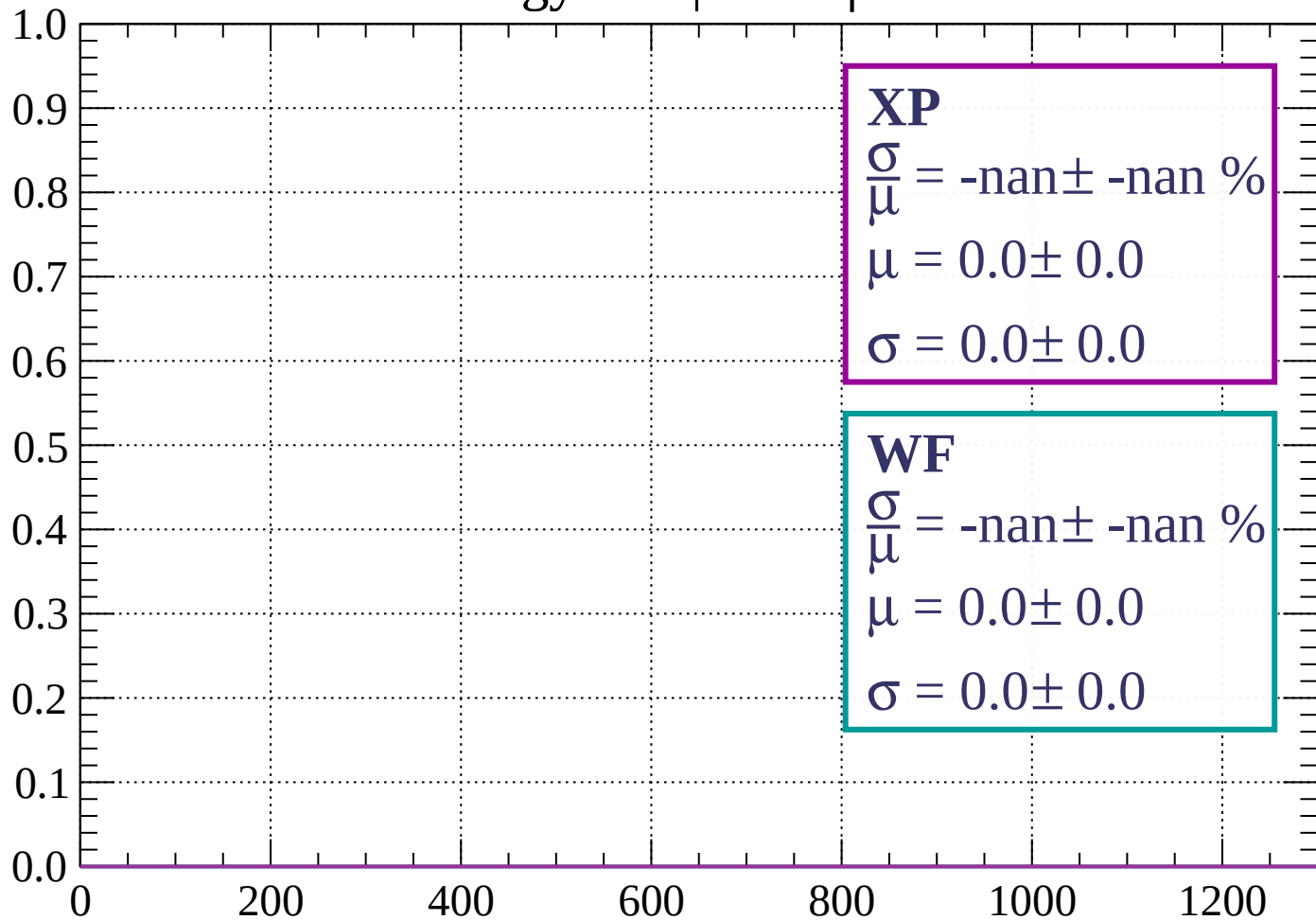
Count



dE/dx (ADC counts/cm)

Energy loss | $42 < \phi < 45$

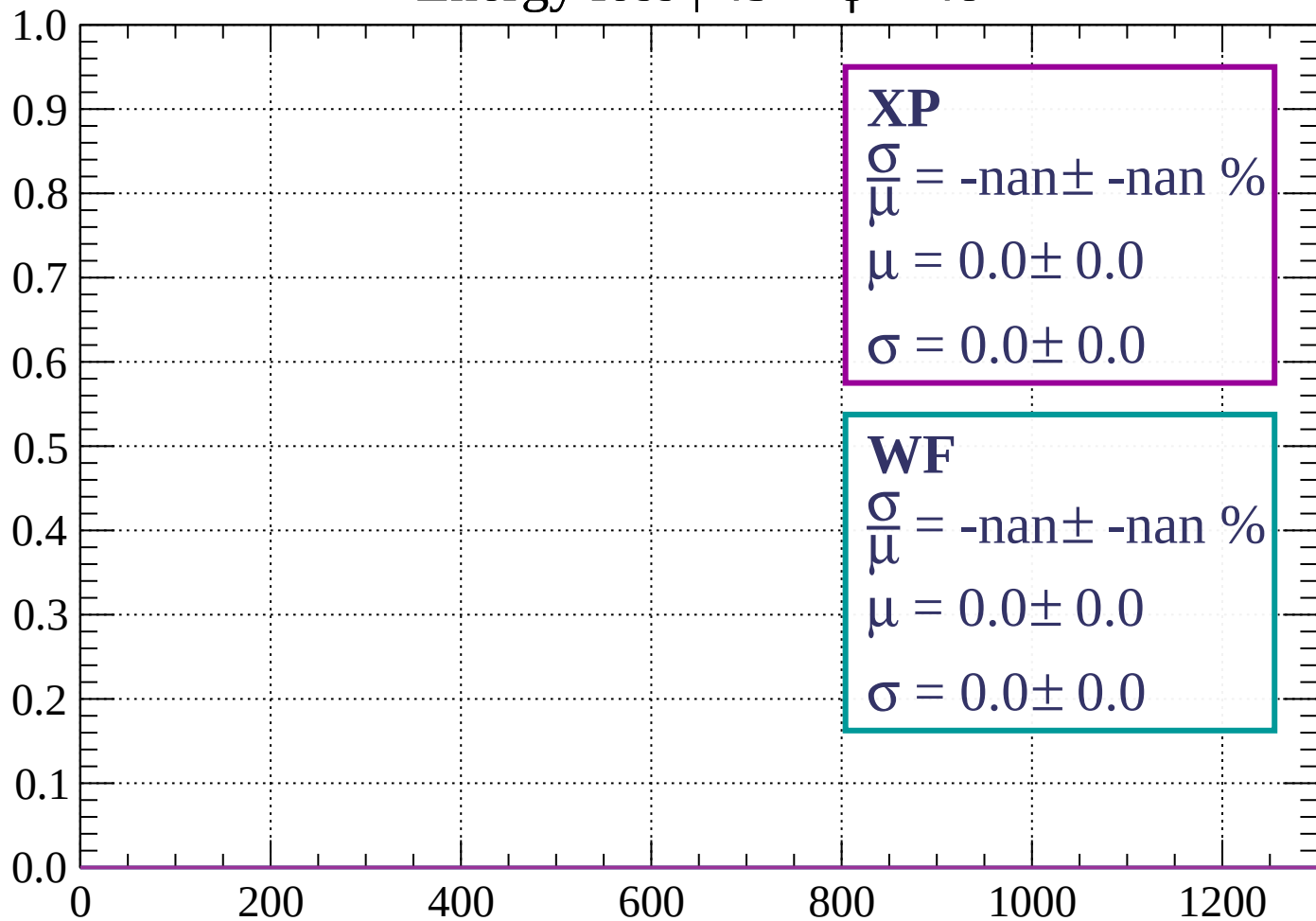
Count



dE/dx (ADC counts/cm)

Energy loss | $45 < \phi < 48$

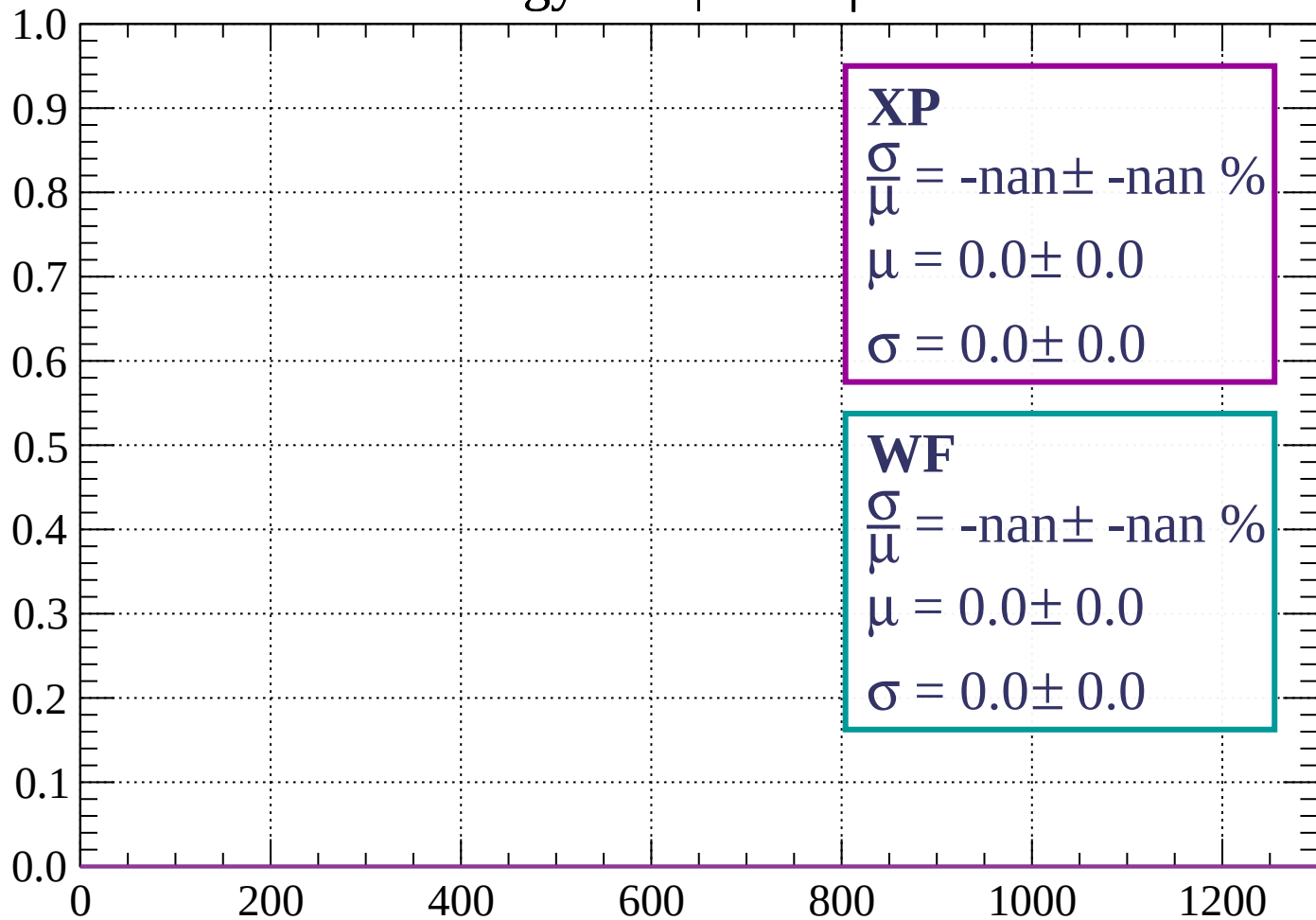
Count



dE/dx (ADC counts/cm)

Energy loss | $48 < \phi < 51$

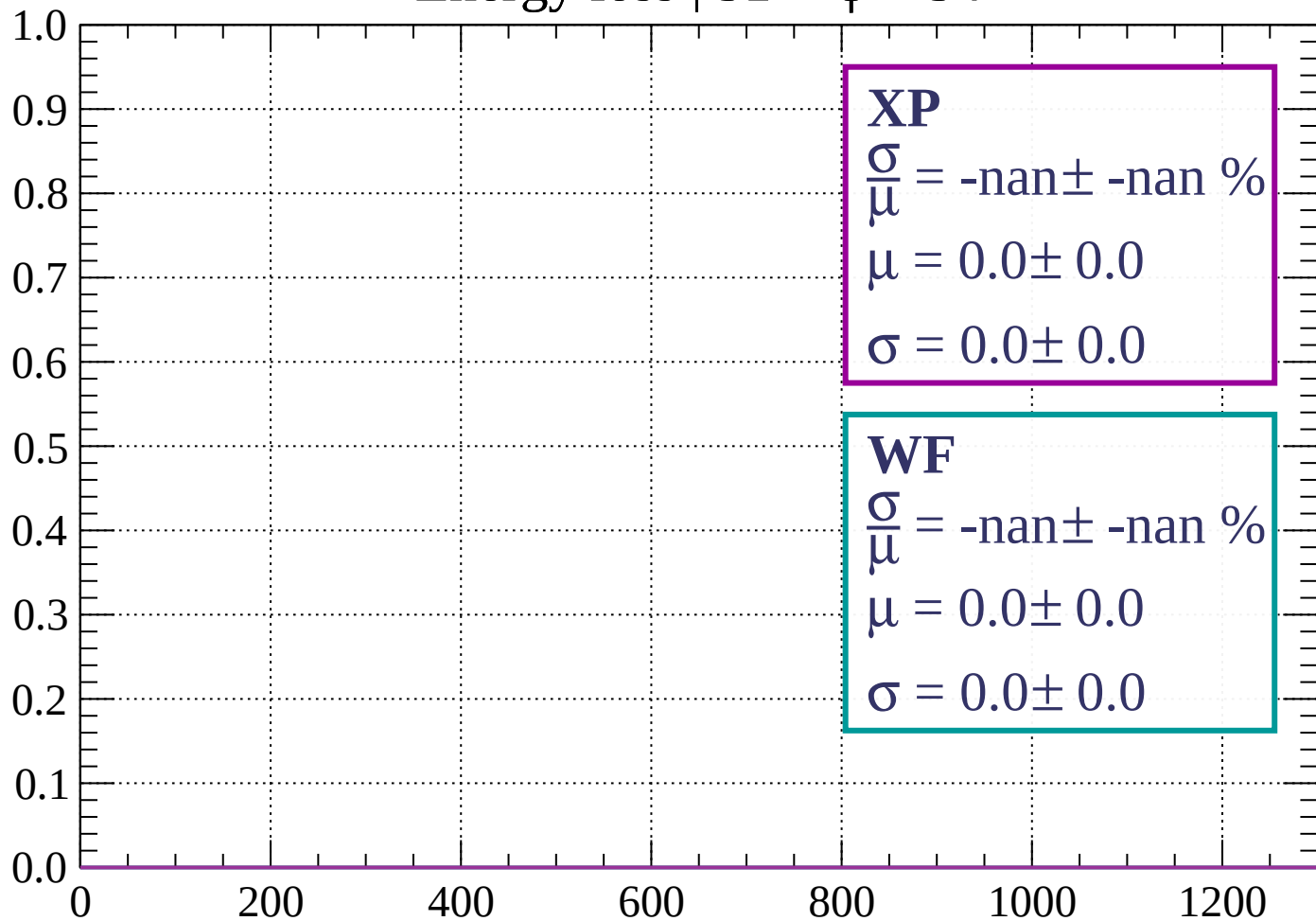
Count



dE/dx (ADC counts/cm)

Energy loss | $51 < \phi < 54$

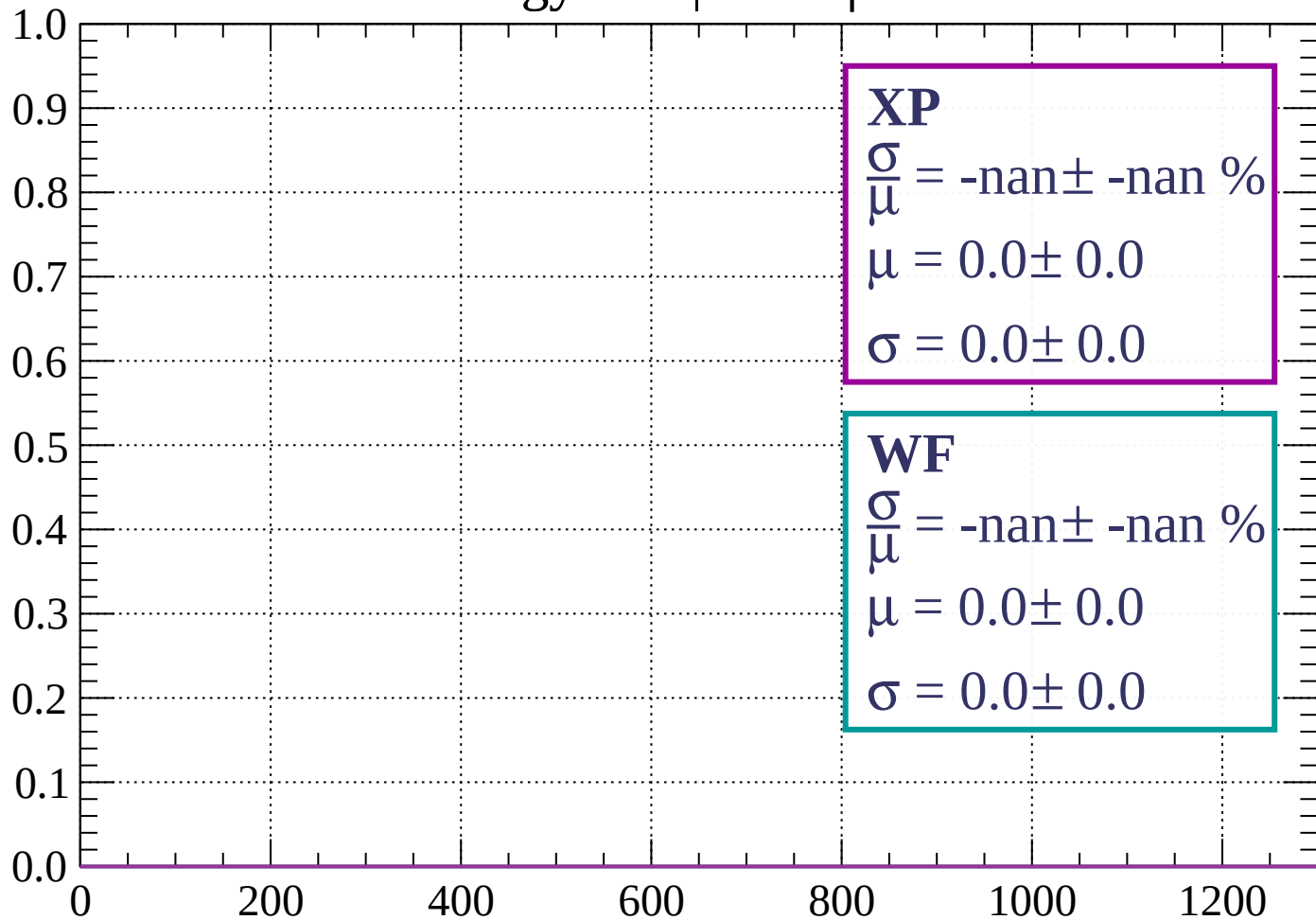
Count



dE/dx (ADC counts/cm)

Energy loss | $54 < \phi < 57$

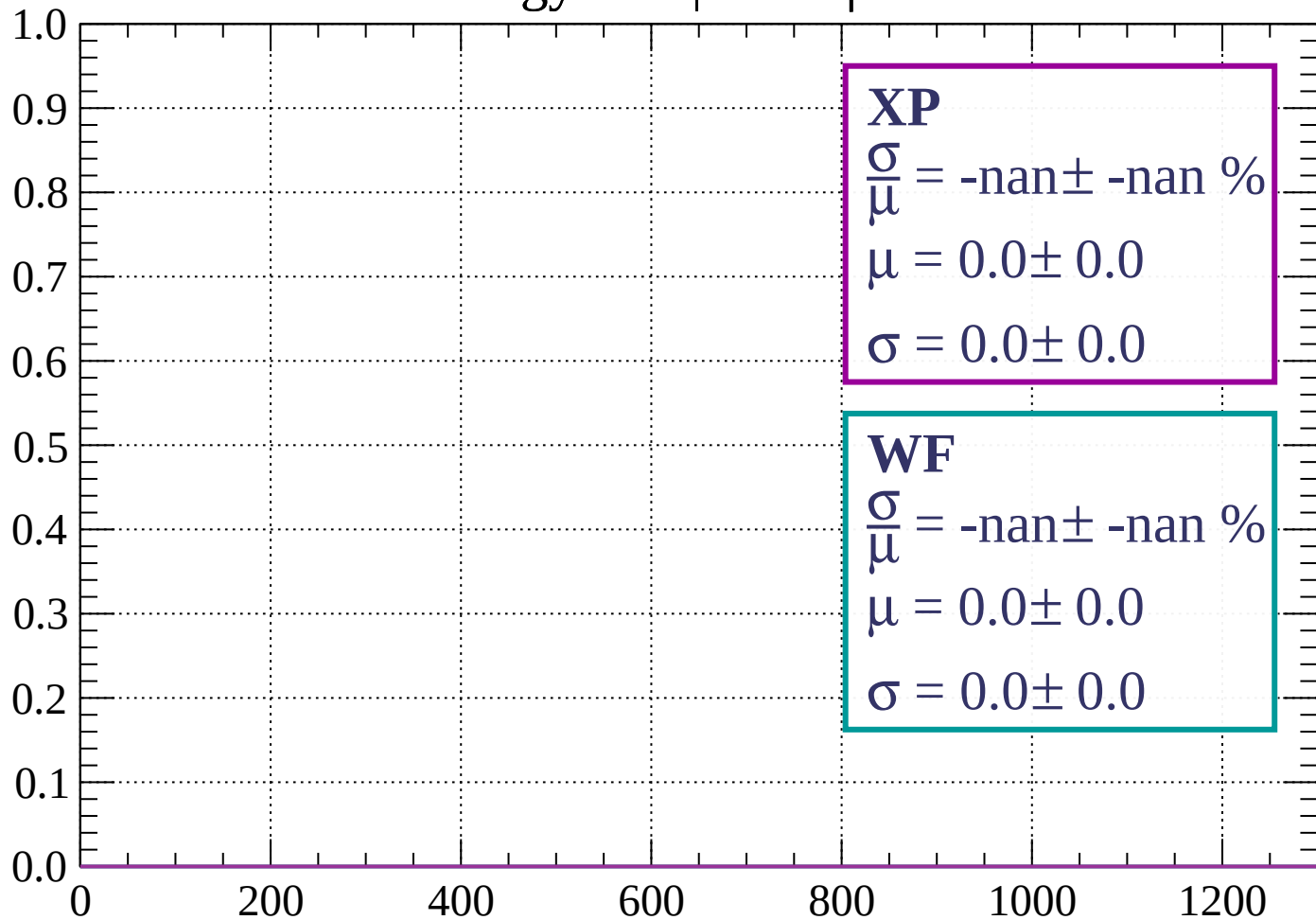
Count



dE/dx (ADC counts/cm)

Energy loss | $57 < \phi < 60$

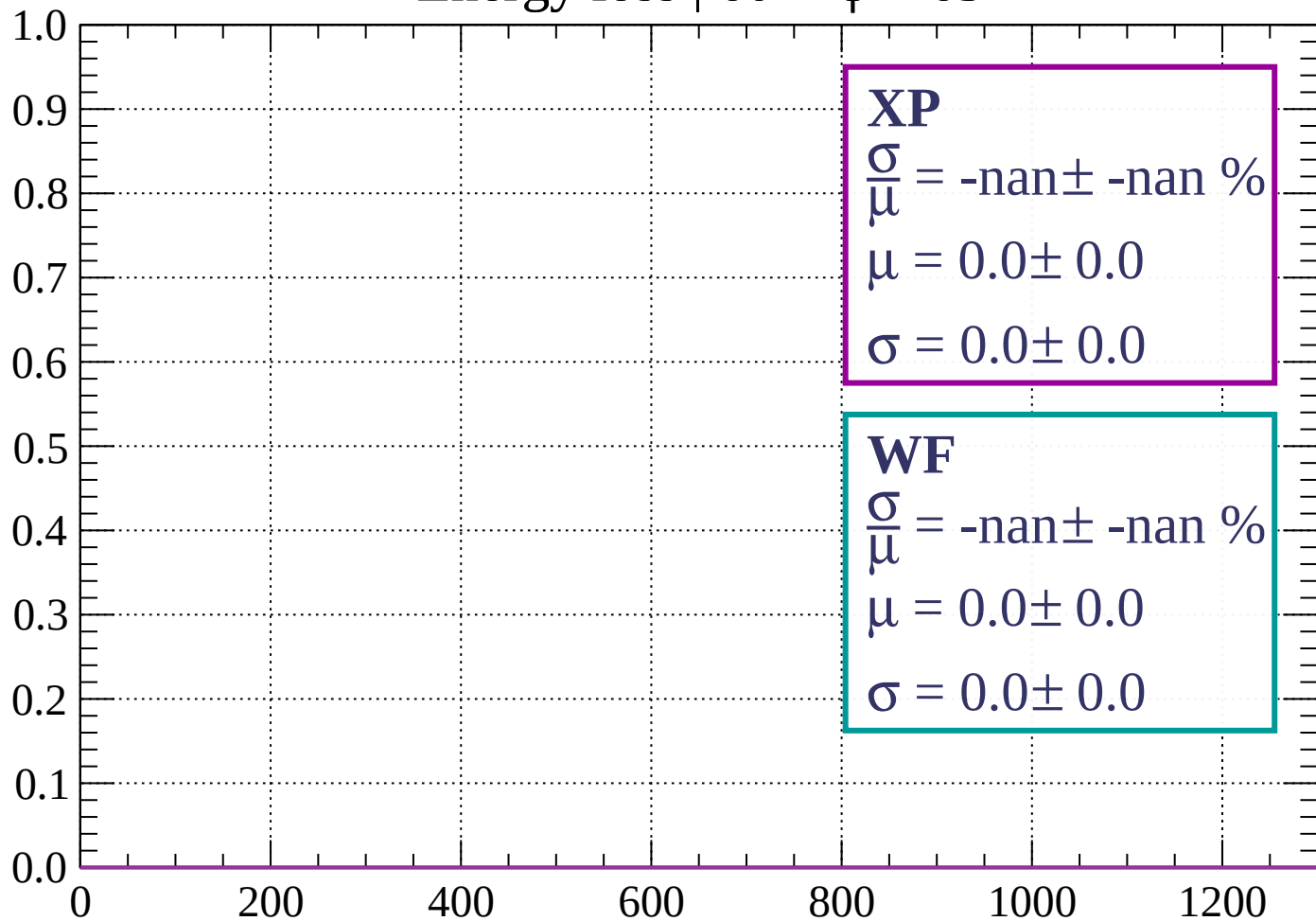
Count



dE/dx (ADC counts/cm)

Energy loss | $60 < \phi < 63$

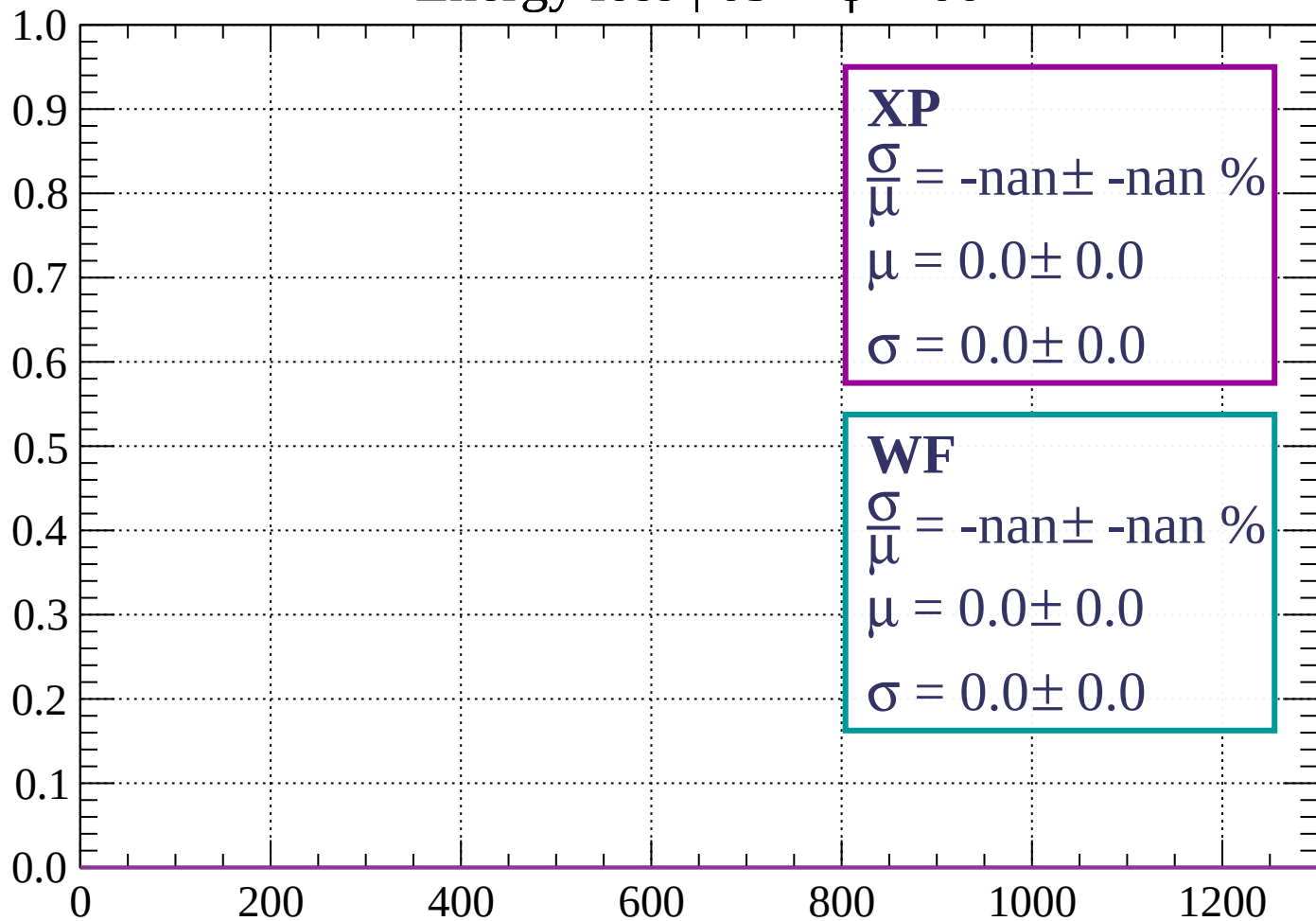
Count



dE/dx (ADC counts/cm)

Energy loss | $63 < \phi < 66$

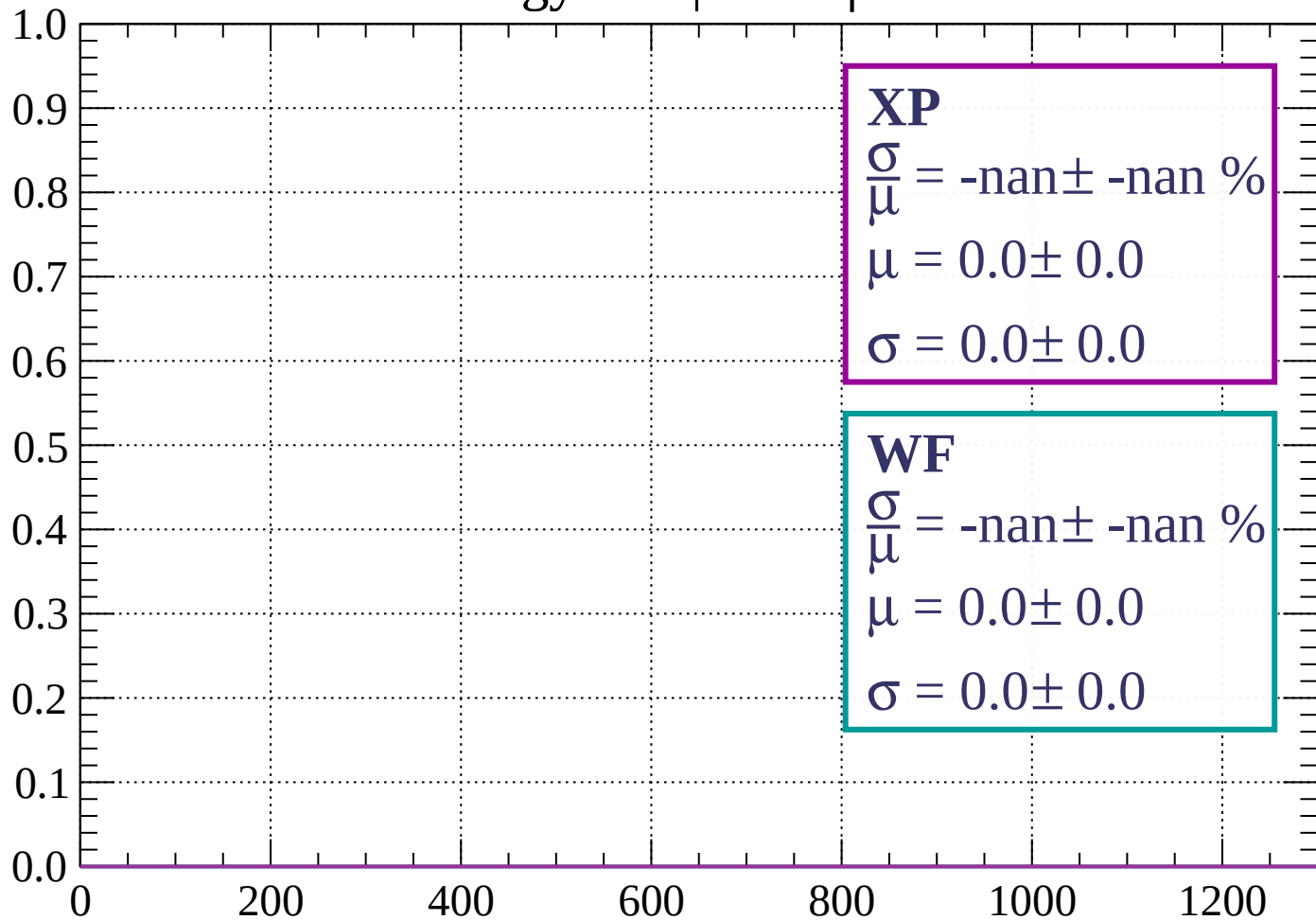
Count



dE/dx (ADC counts/cm)

Energy loss | $66 < \phi < 69$

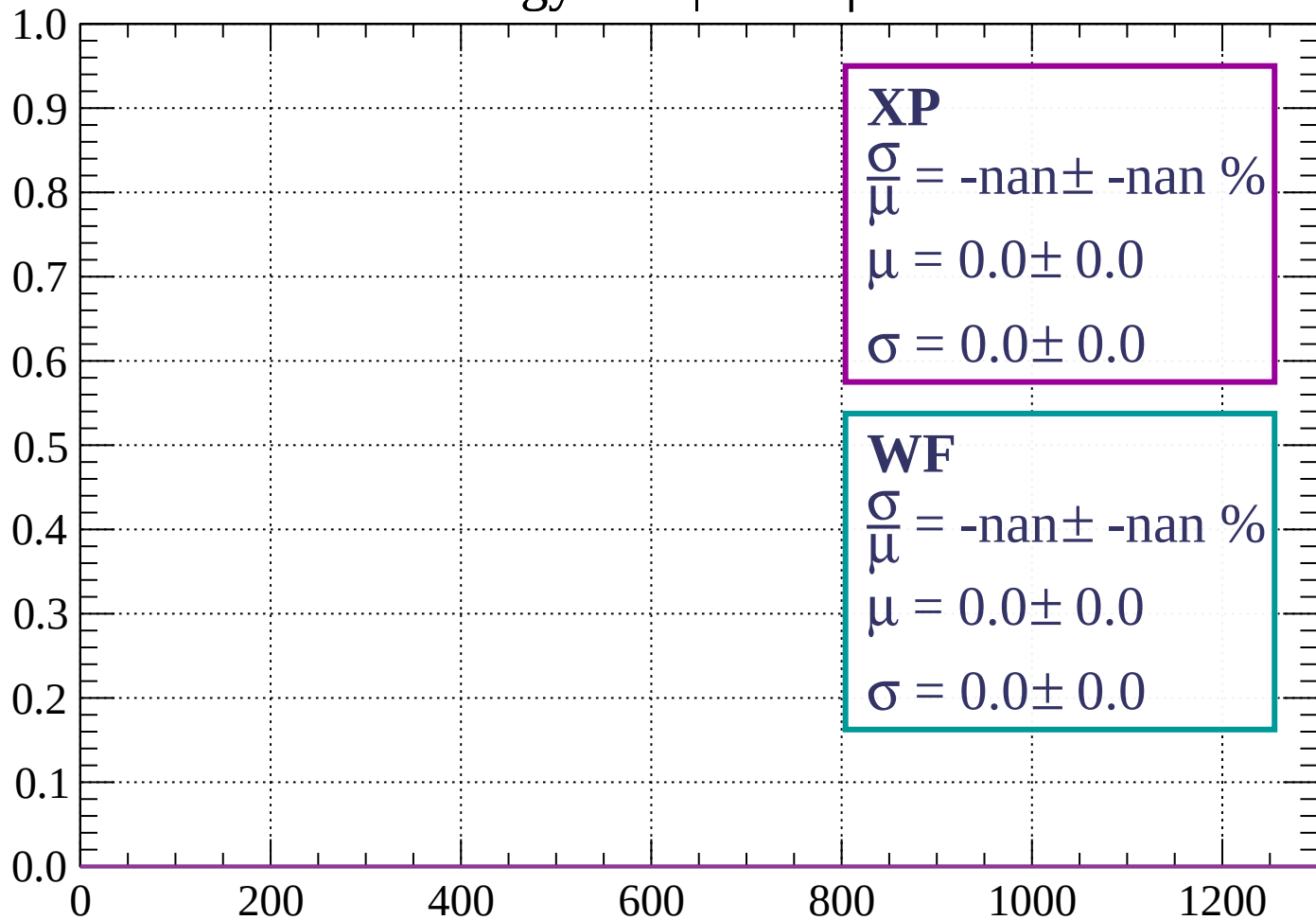
Count



dE/dx (ADC counts/cm)

Energy loss | $69 < \phi < 72$

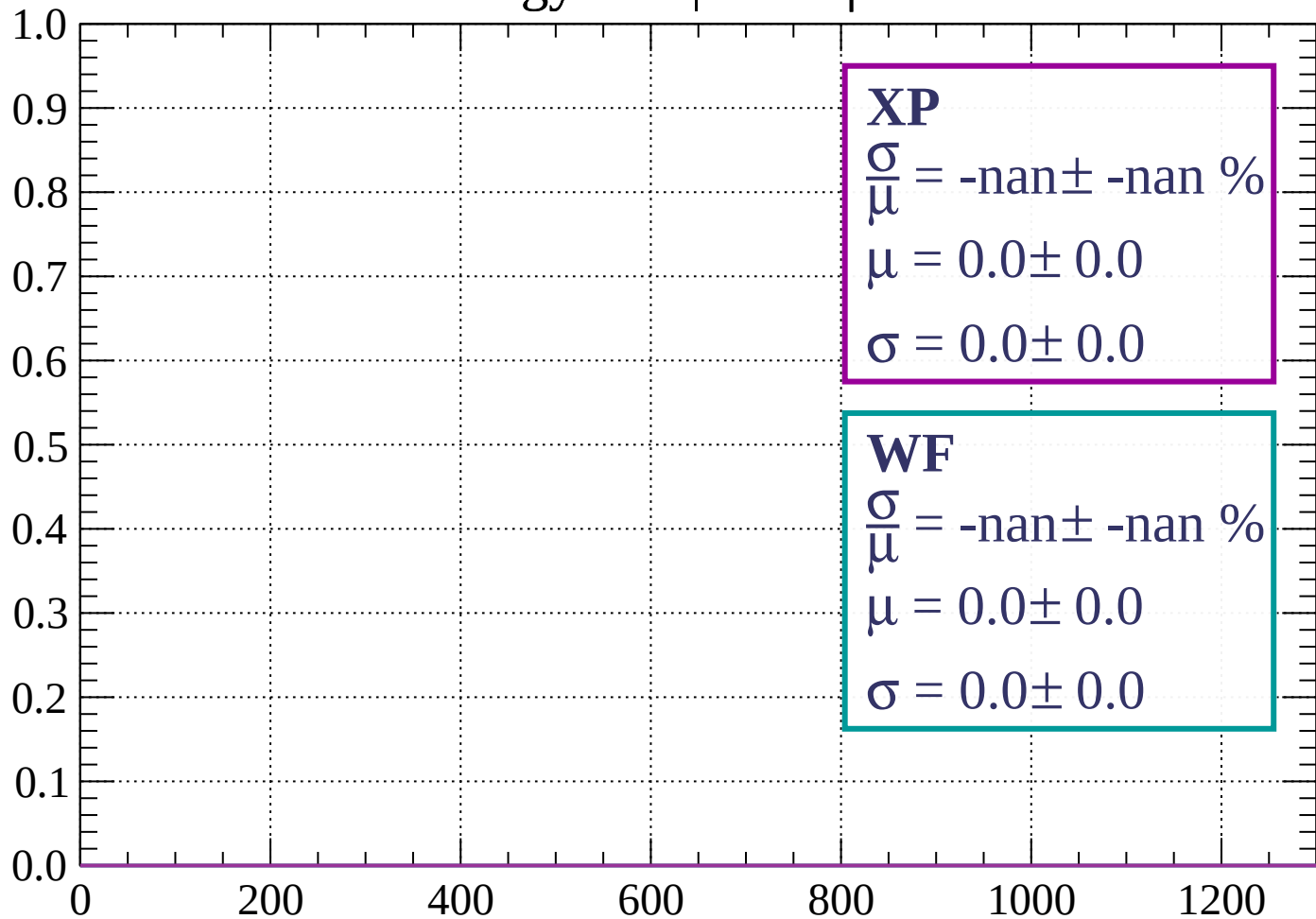
Count



dE/dx (ADC counts/cm)

Energy loss | $72 < \phi < 75$

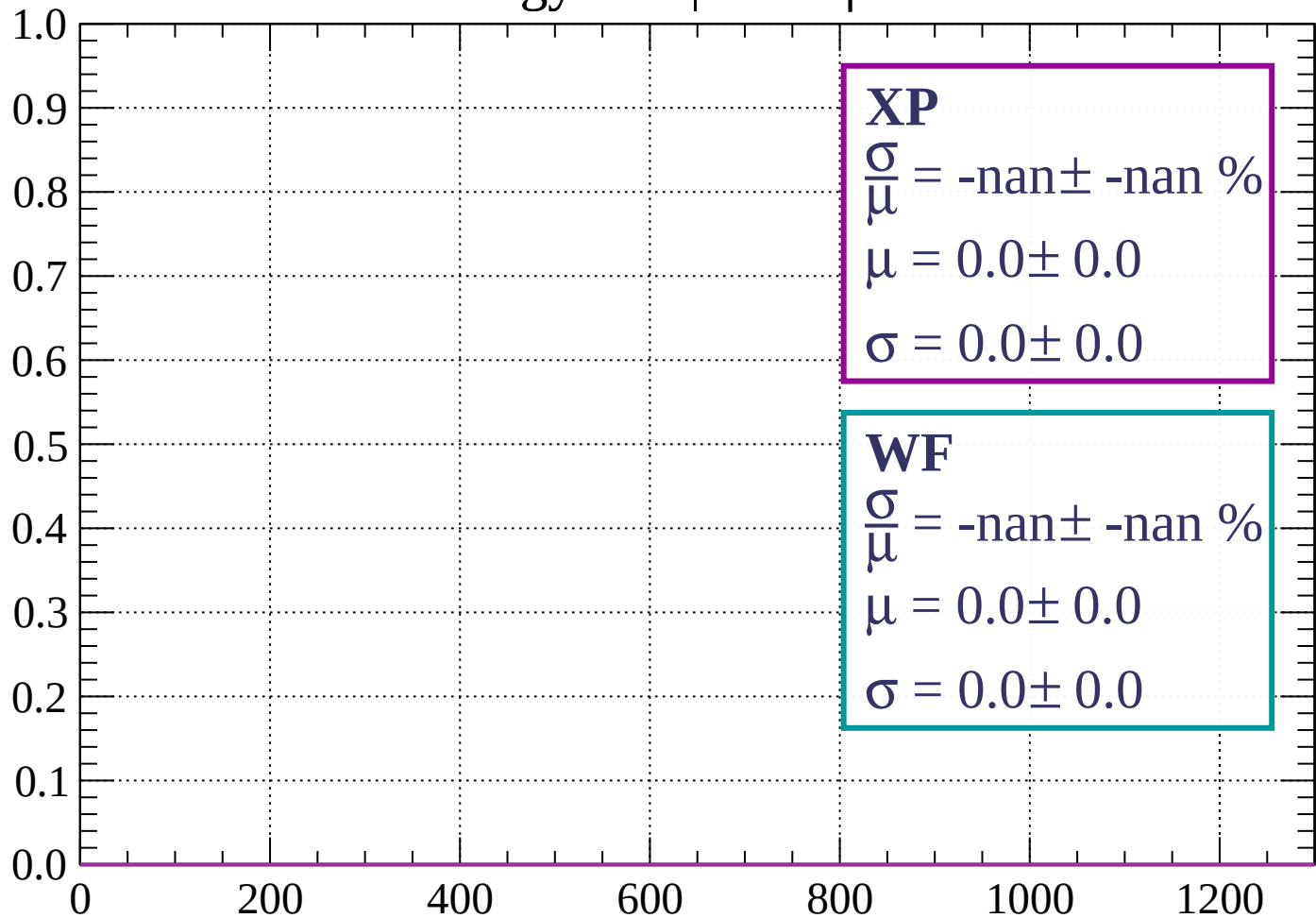
Count



dE/dx (ADC counts/cm)

Energy loss | $75 < \phi < 78$

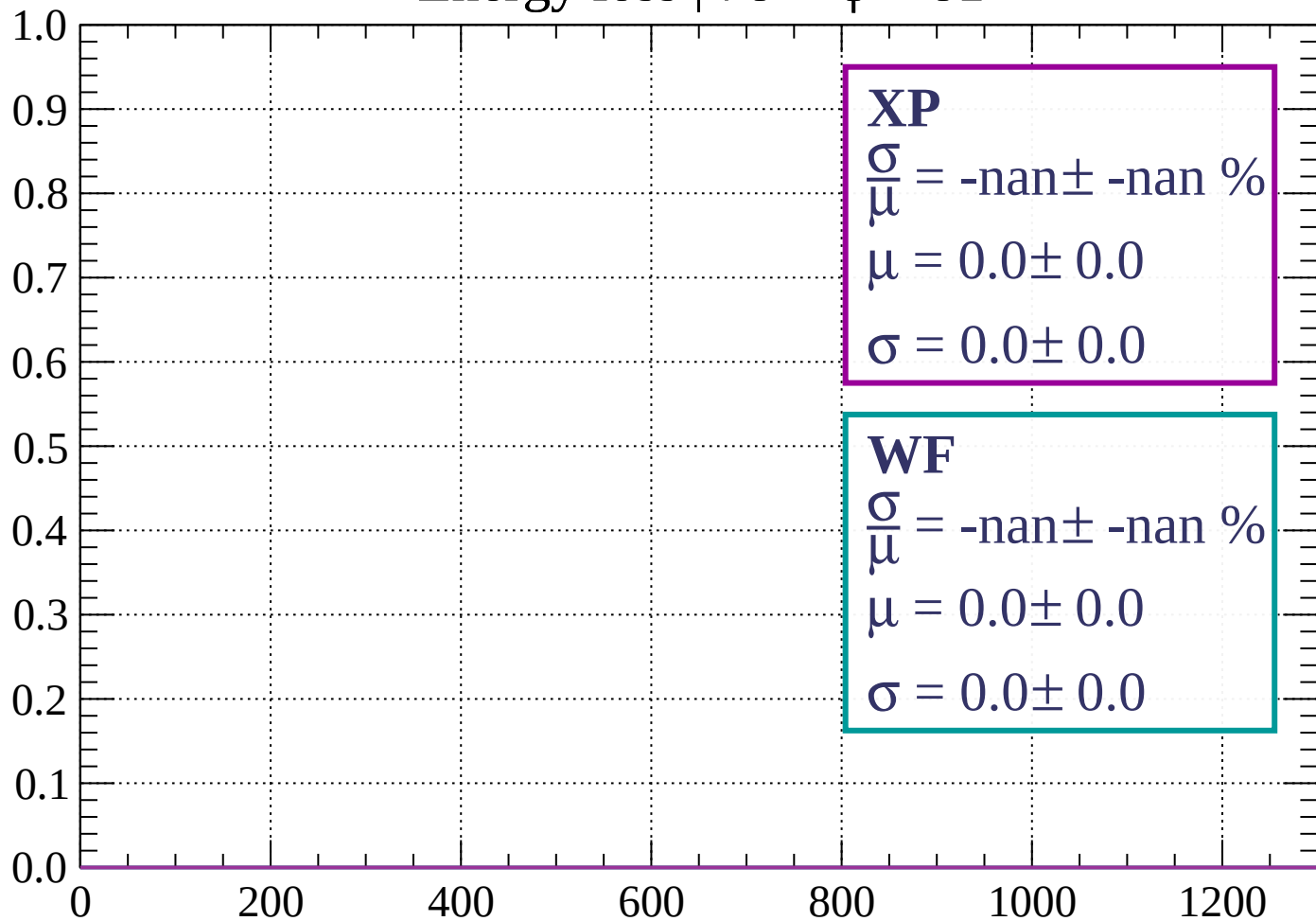
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \phi < 81$

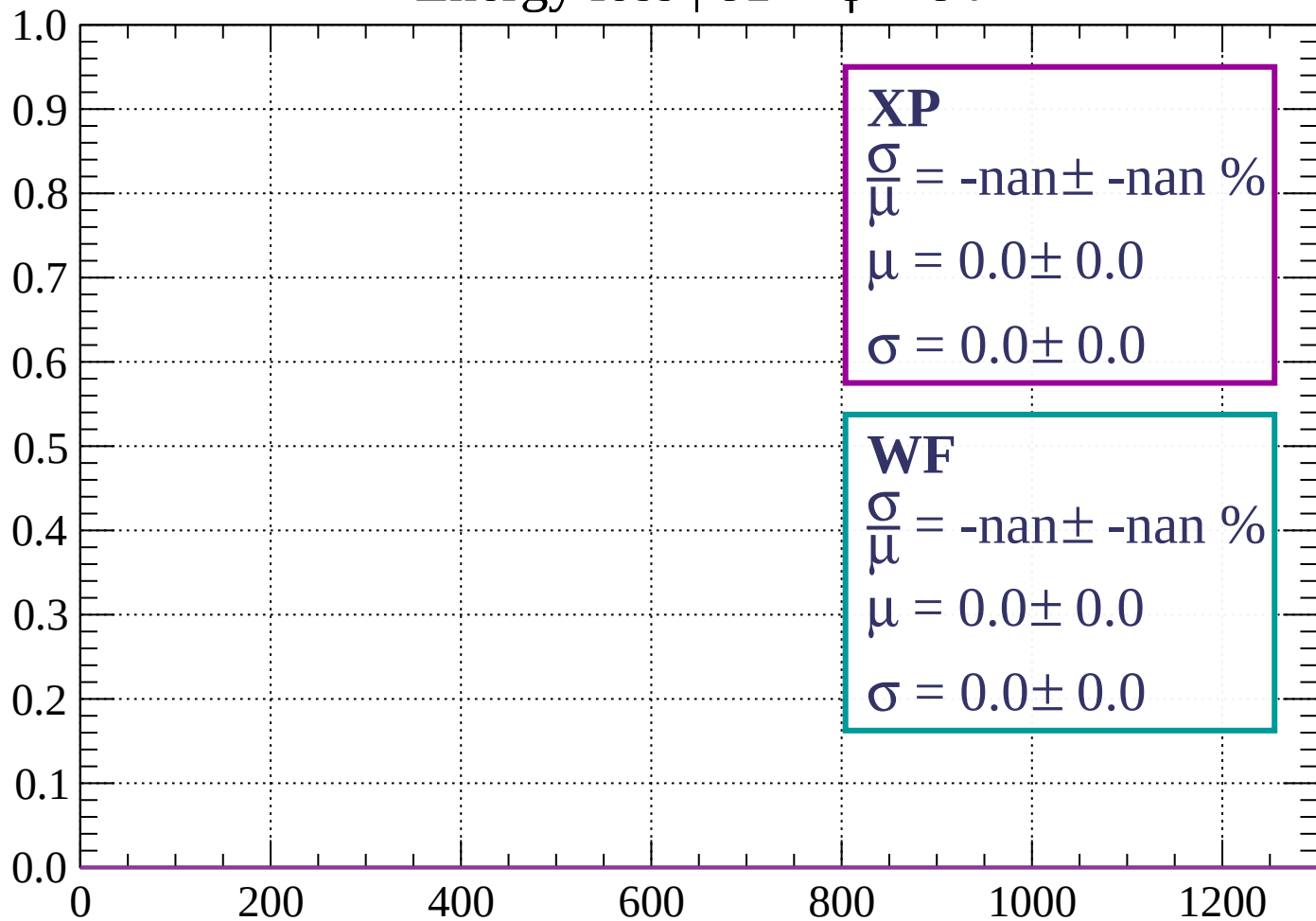
Count



dE/dx (ADC counts/cm)

Energy loss | $81 < \phi < 84$

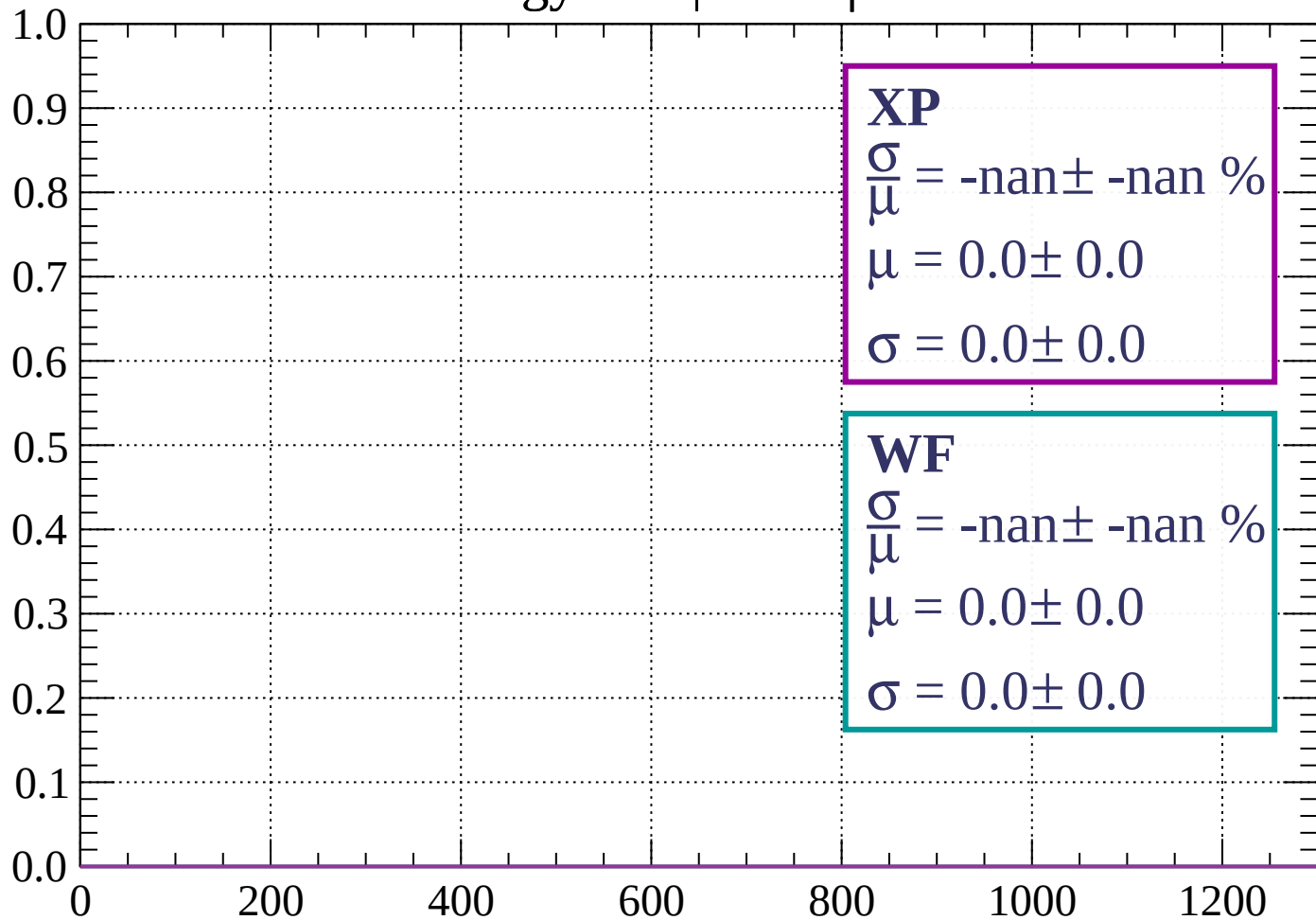
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \phi < 87$

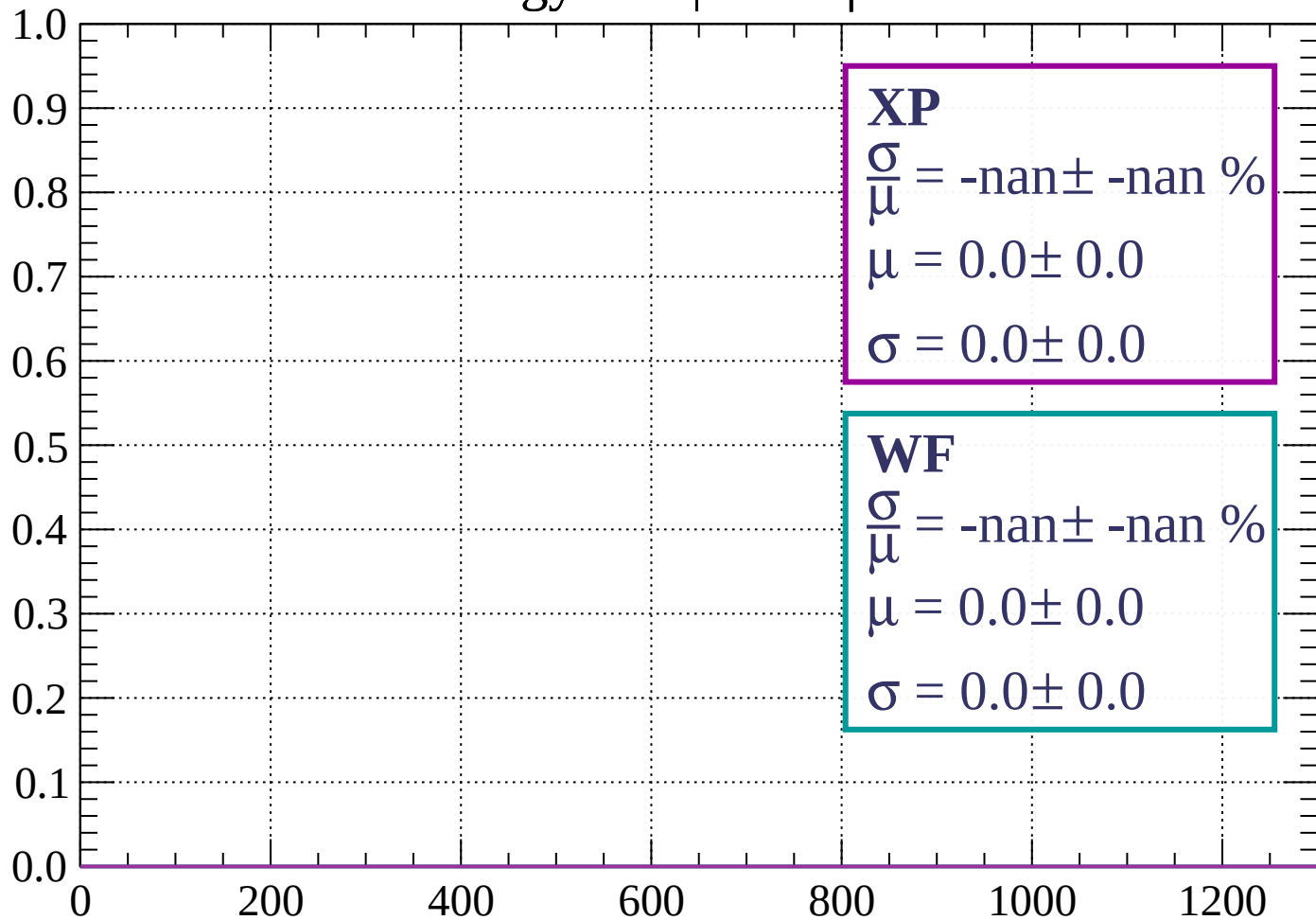
Count



dE/dx (ADC counts/cm)

Energy loss | $87 < \phi < 90$

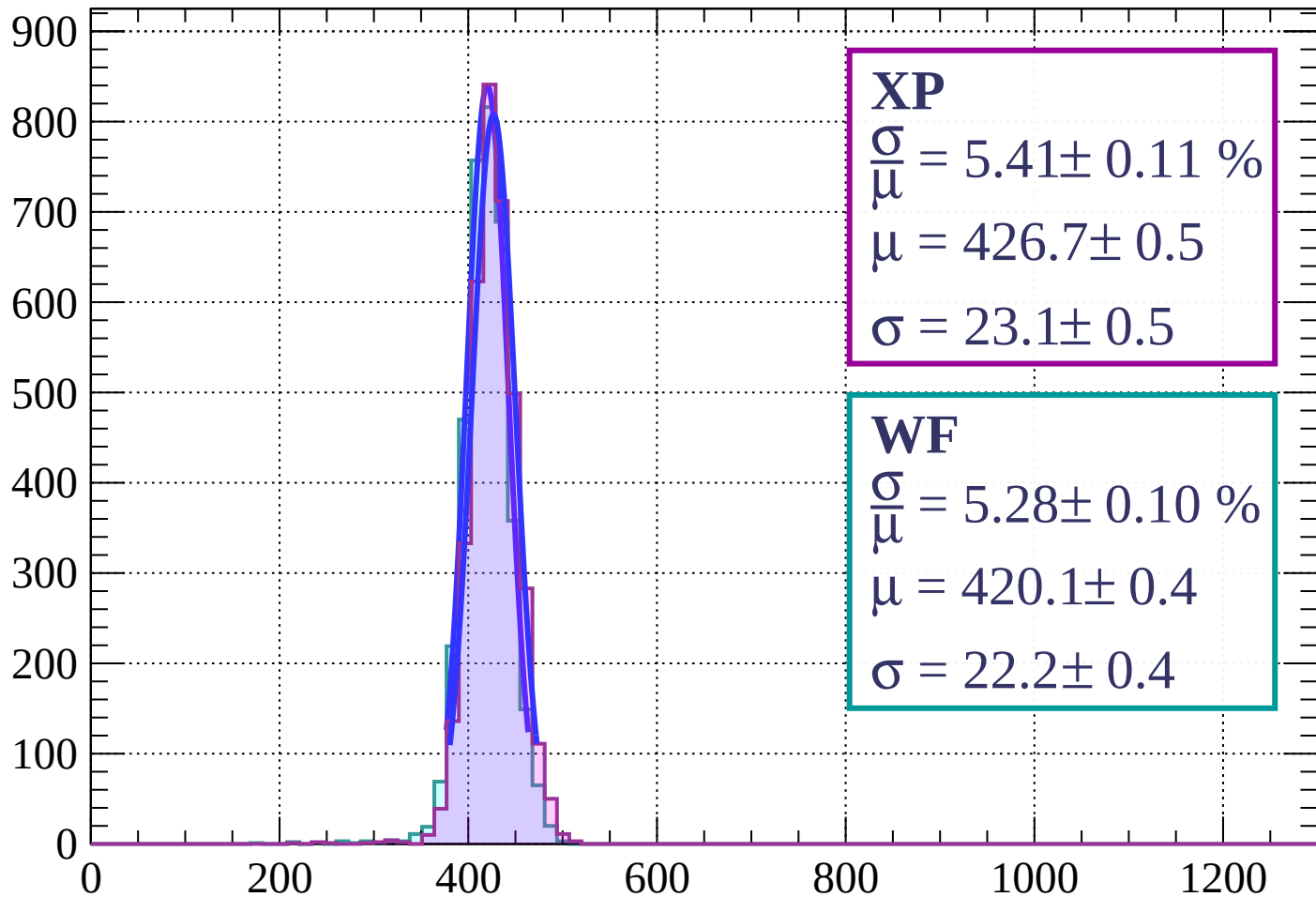
Count



dE/dx (ADC counts/cm)

Energy loss | $0 < \theta < 3$

Count



XP

$$\frac{\sigma}{\mu} = 5.41 \pm 0.11 \%$$

$$\mu = 426.7 \pm 0.5$$

$$\sigma = 23.1 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 5.28 \pm 0.10 \%$$

$$\mu = 420.1 \pm 0.4$$

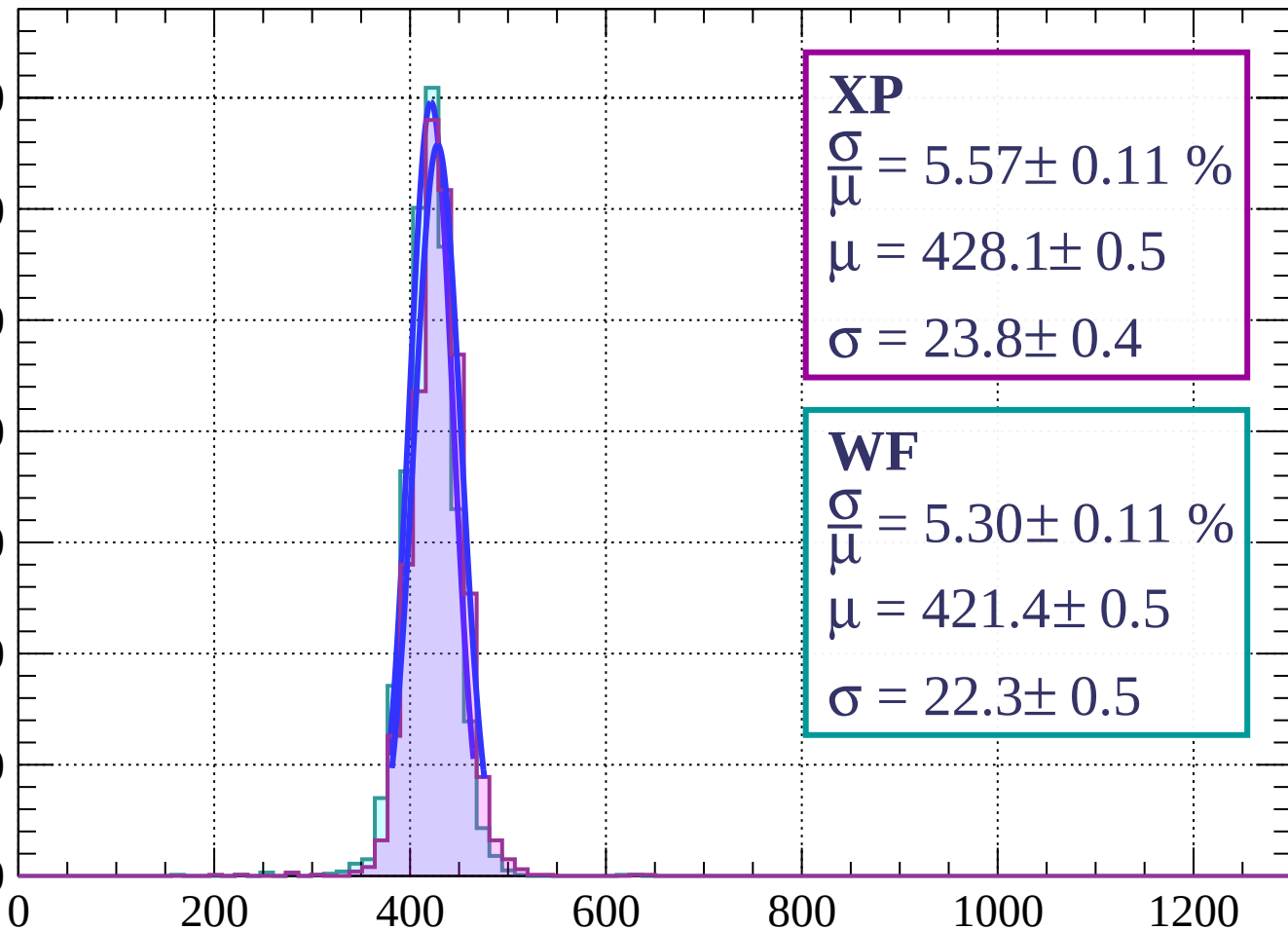
$$\sigma = 22.2 \pm 0.4$$

dE/dx (ADC counts/cm)

Energy loss | $3 < \theta < 6$

Count

700
600
500
400
300
200
100
0



XP

$$\frac{\sigma}{\mu} = 5.57 \pm 0.11 \%$$

$$\mu = 428.1 \pm 0.5$$

$$\sigma = 23.8 \pm 0.4$$

WF

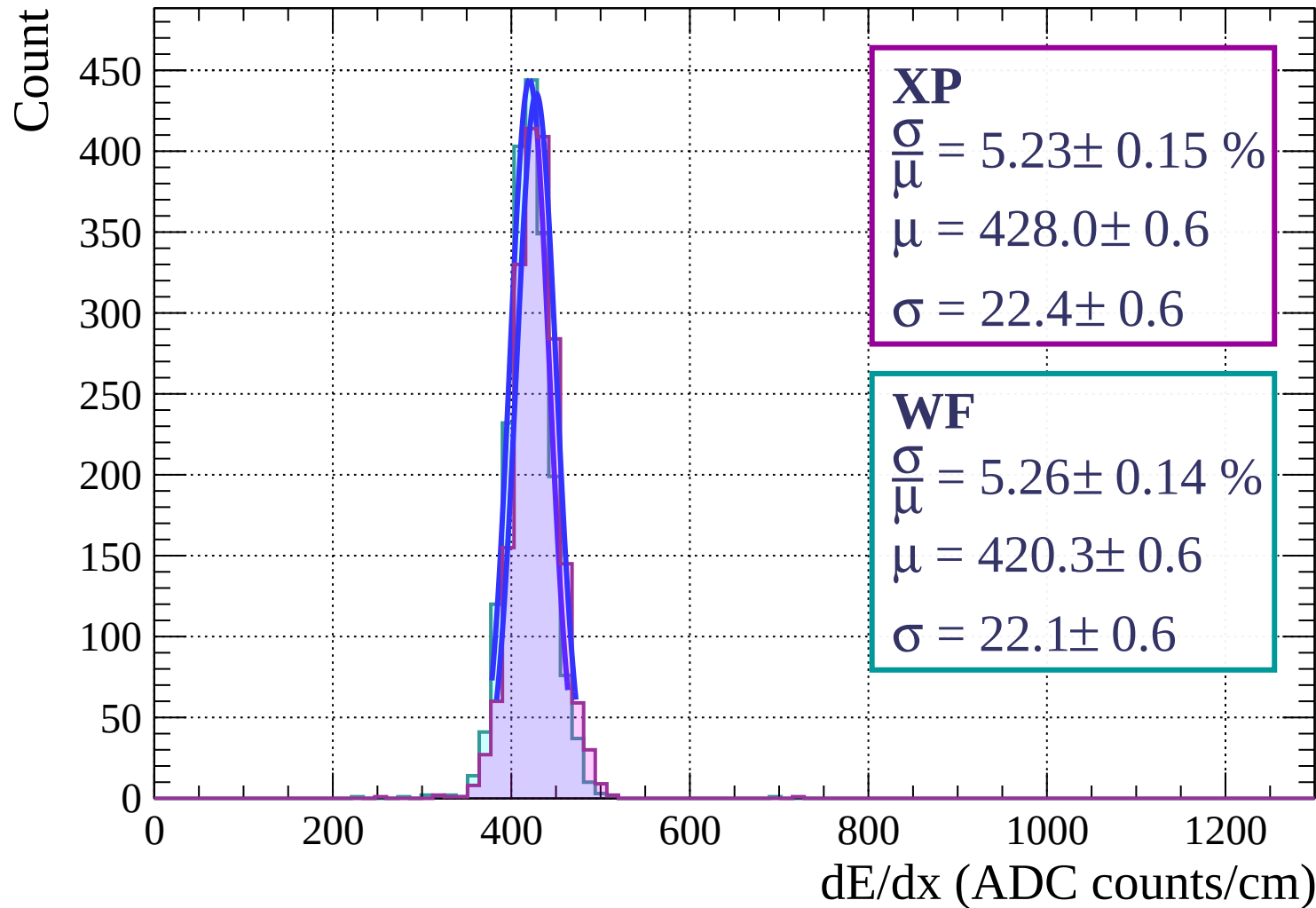
$$\frac{\sigma}{\mu} = 5.30 \pm 0.11 \%$$

$$\mu = 421.4 \pm 0.5$$

$$\sigma = 22.3 \pm 0.5$$

dE/dx (ADC counts/cm)

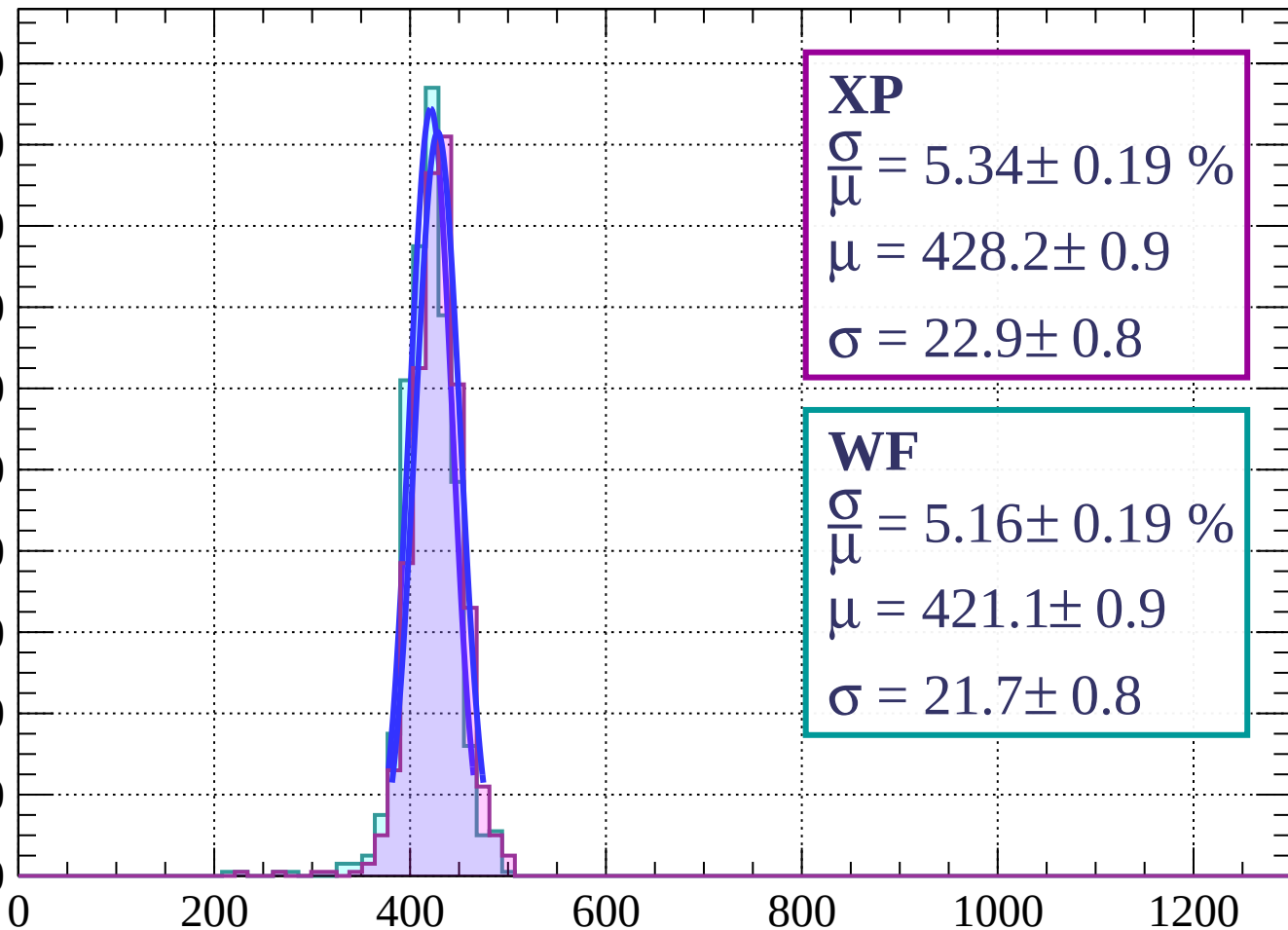
Energy loss | $6 < \theta < 9$



Energy loss | $9 < \theta < 12$

Count

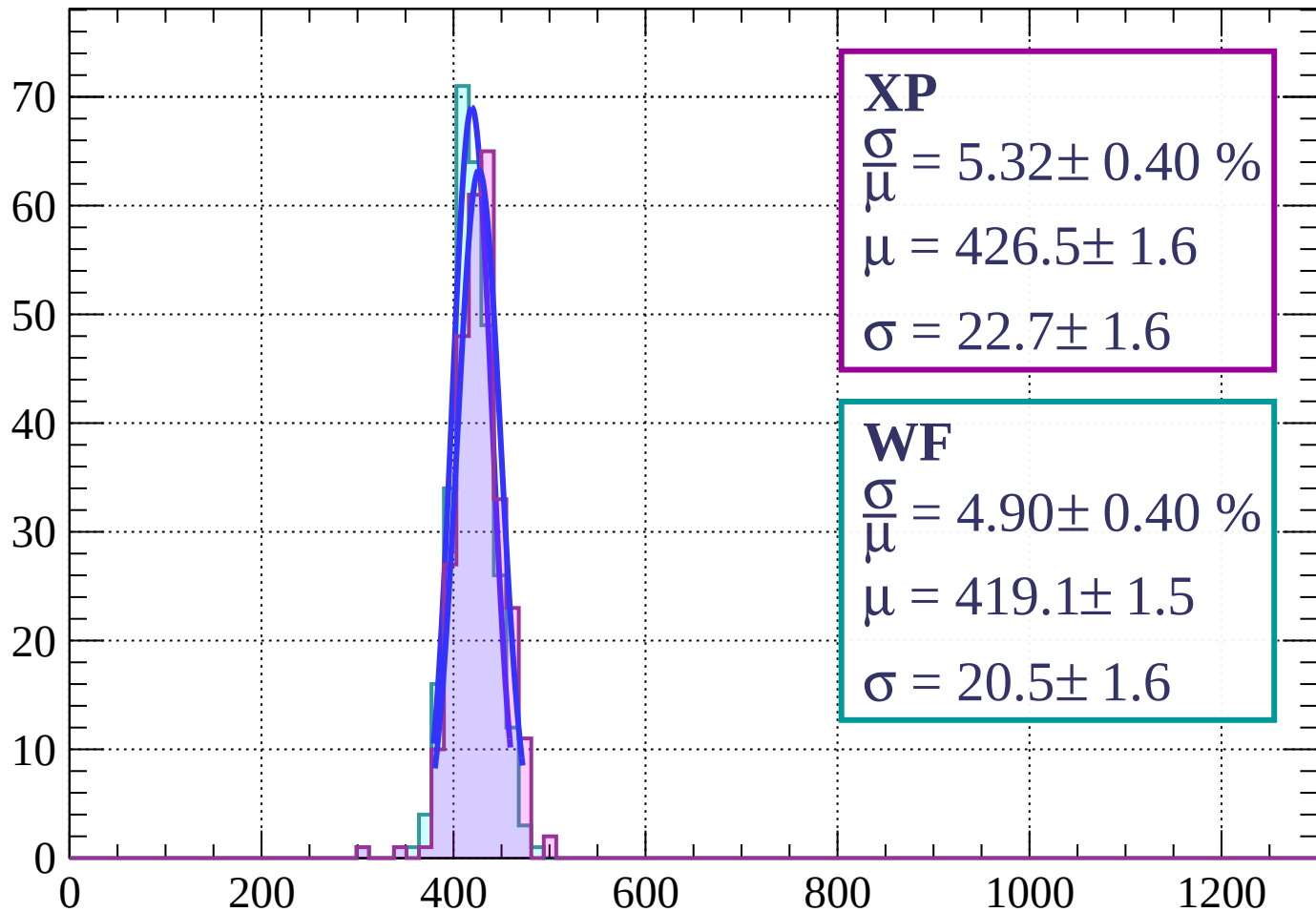
200
180
160
140
120
100
80
60
40
20
0



dE/dx (ADC counts/cm)

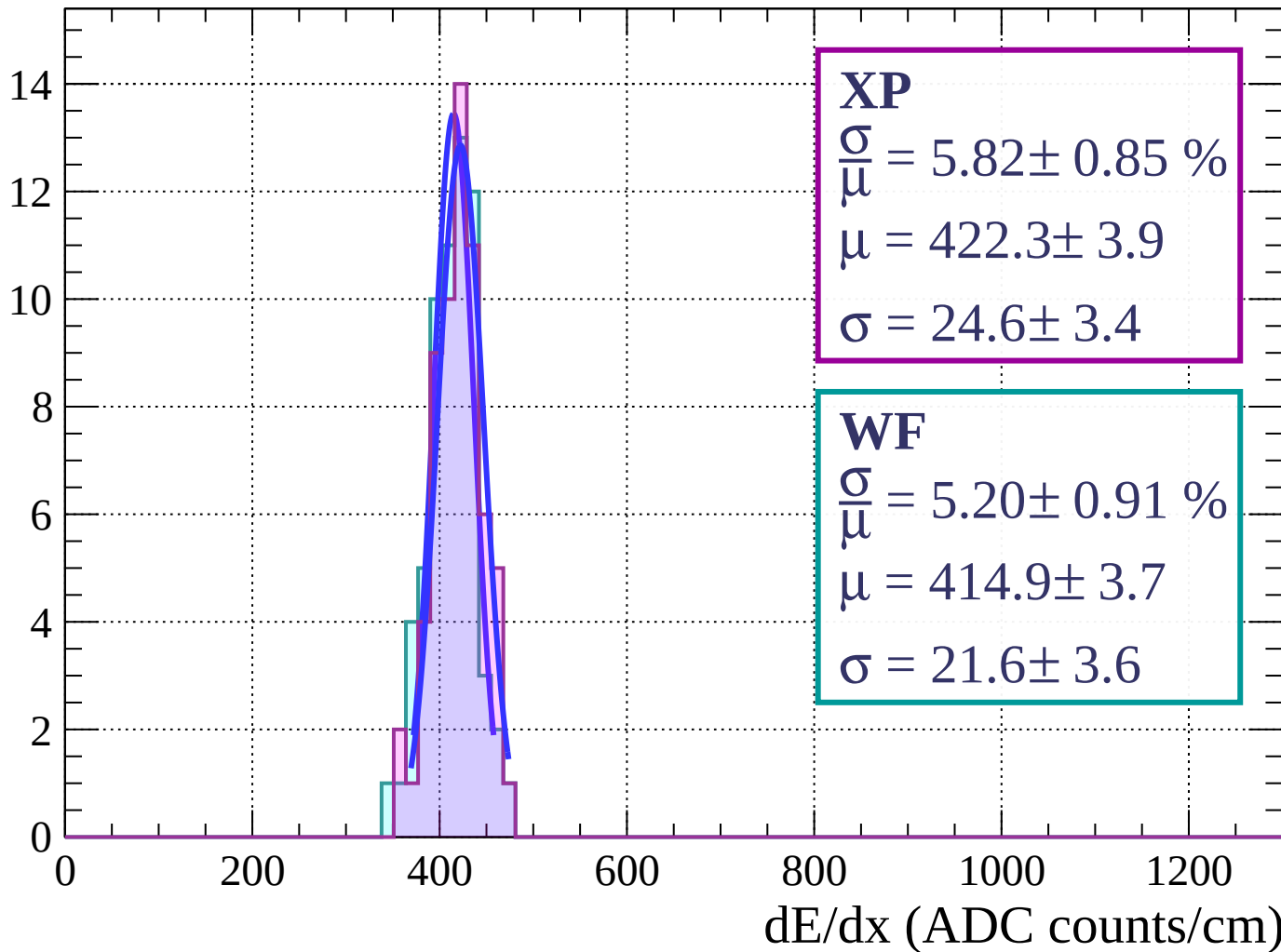
Energy loss | $12 < \theta < 15$

Count



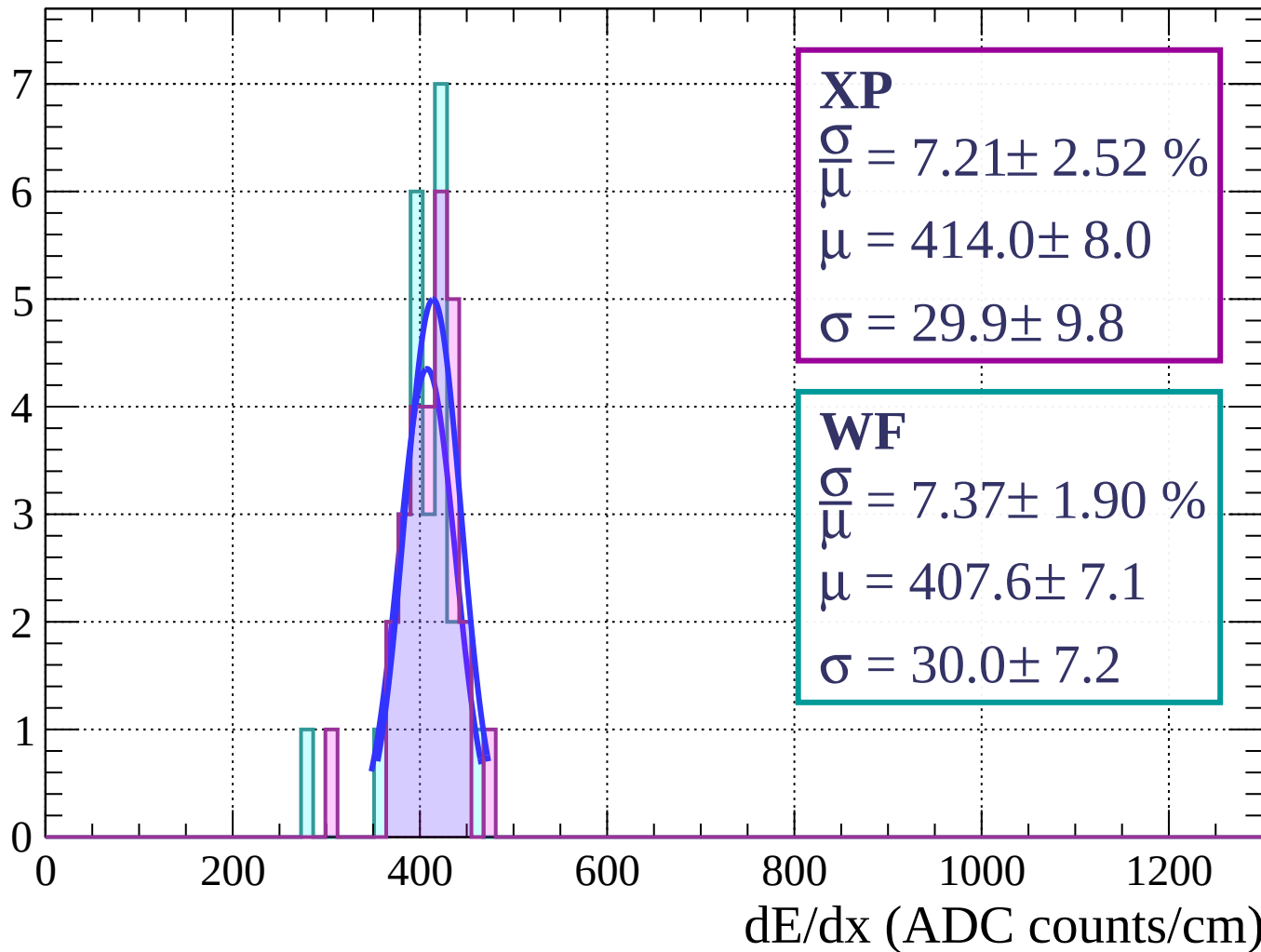
Energy loss | $15 < \theta < 18$

Count



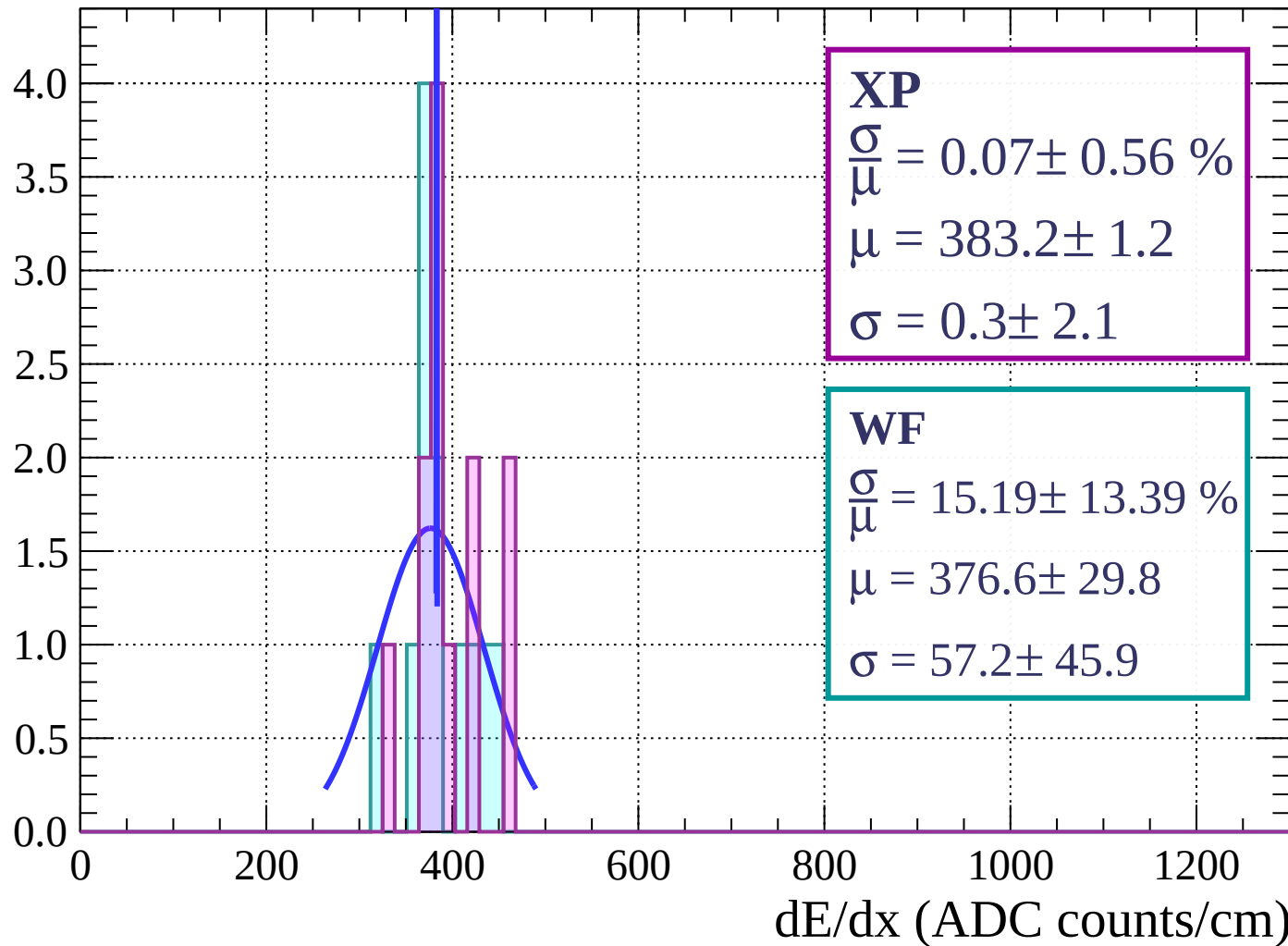
Energy loss | $18 < \theta < 21$

Count



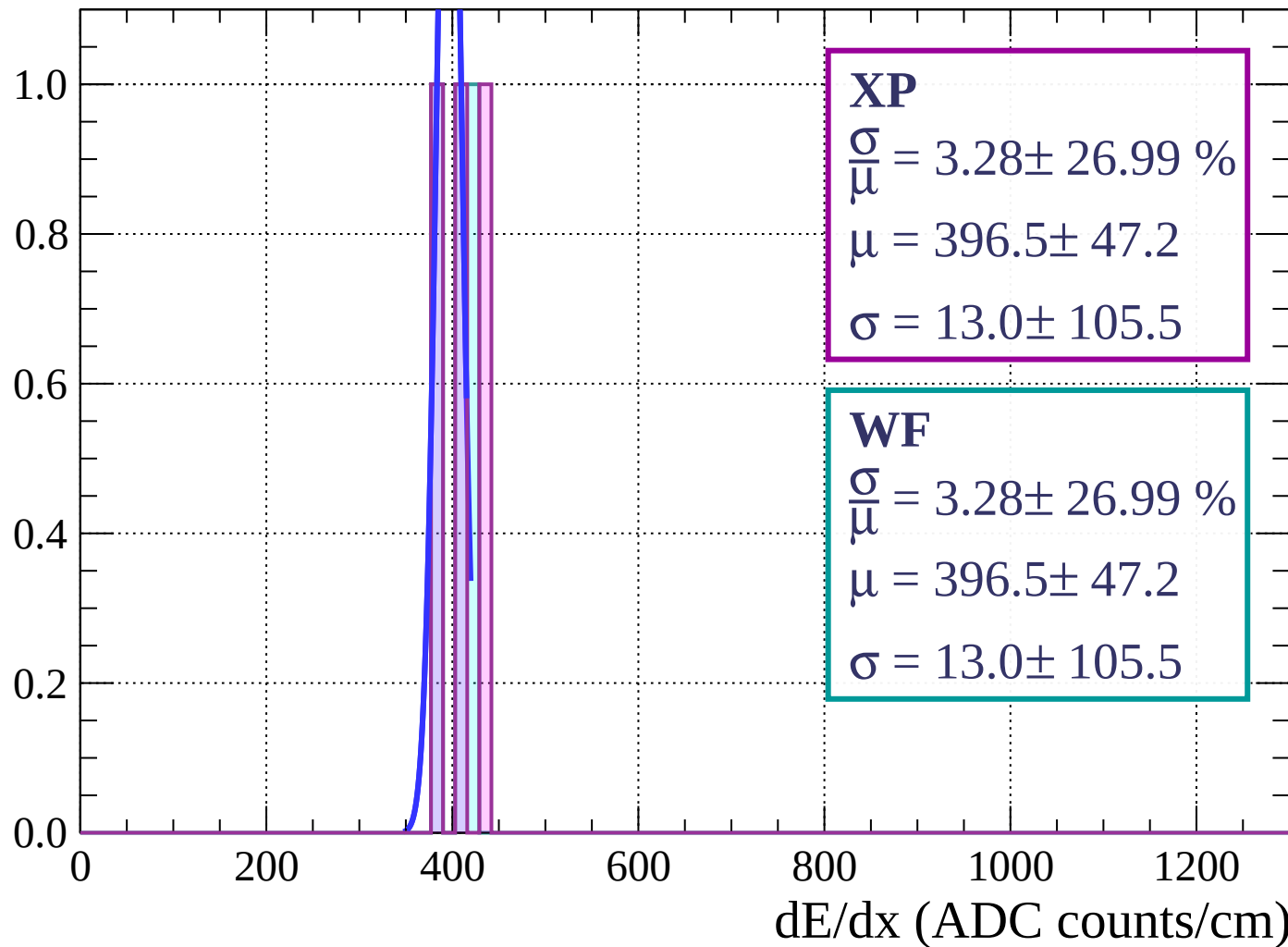
Energy loss | $21 < \theta < 24$

Count



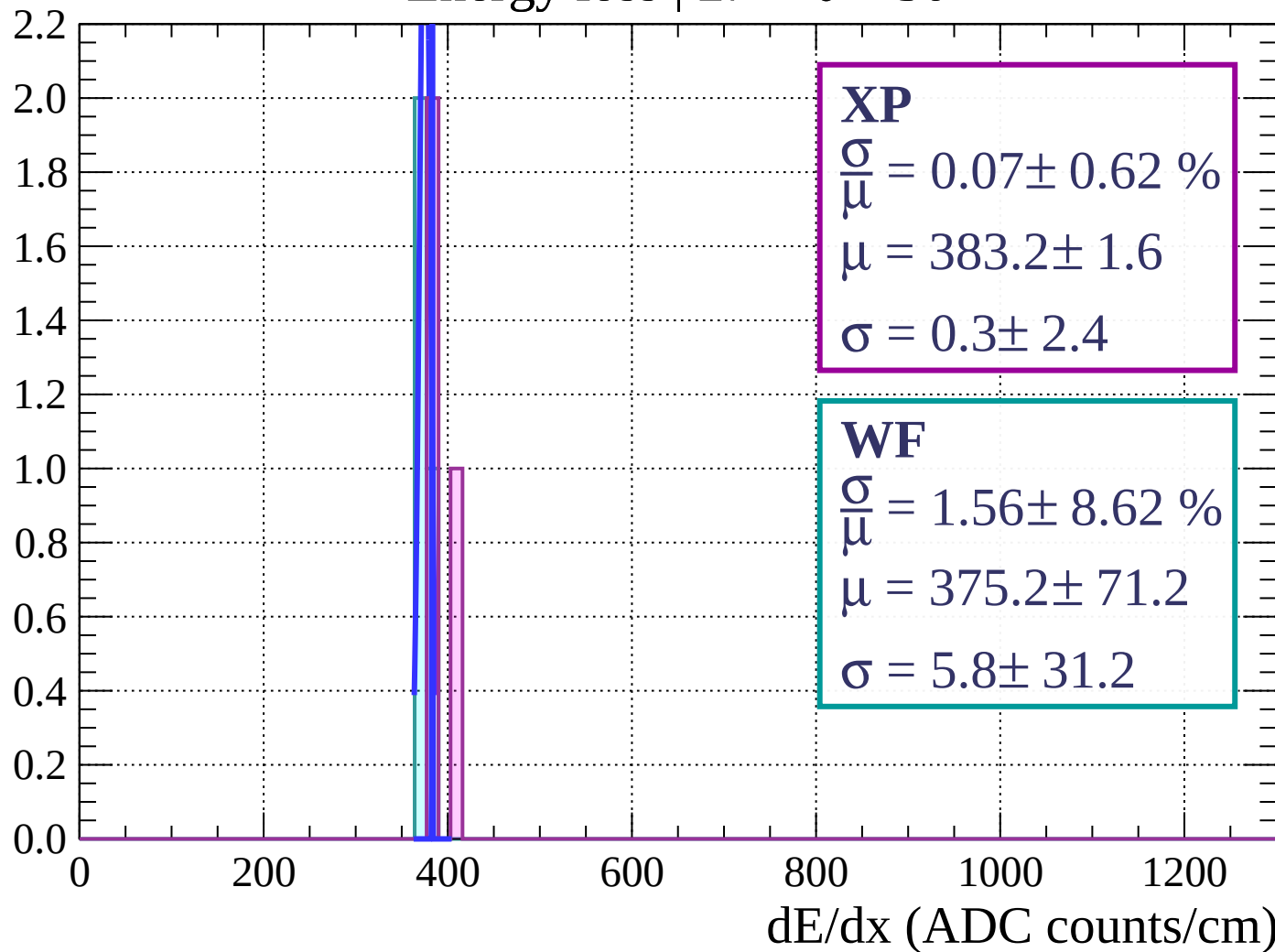
Energy loss | $24 < \theta < 27$

Count



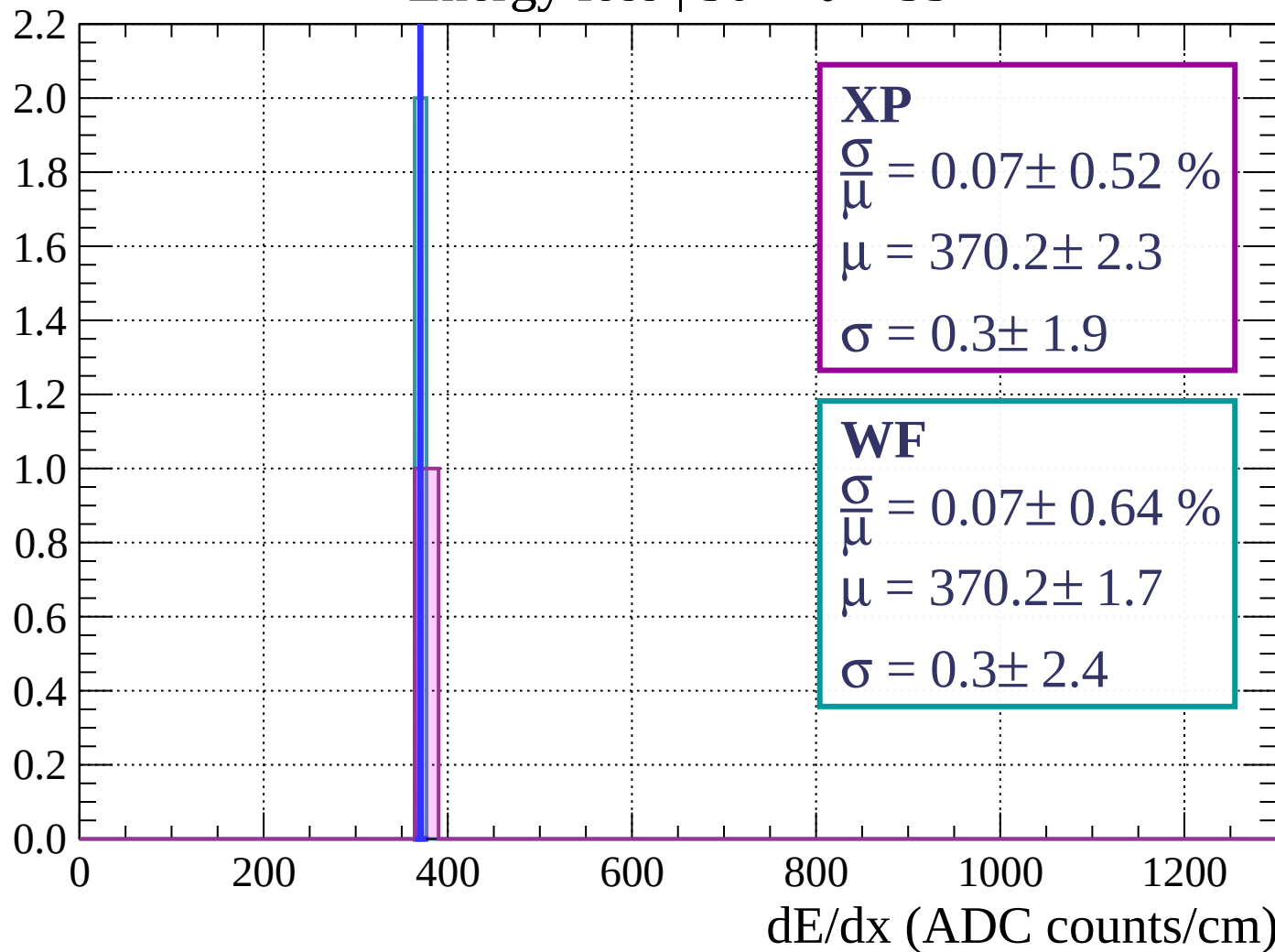
Energy loss | $27 < \theta < 30$

Count



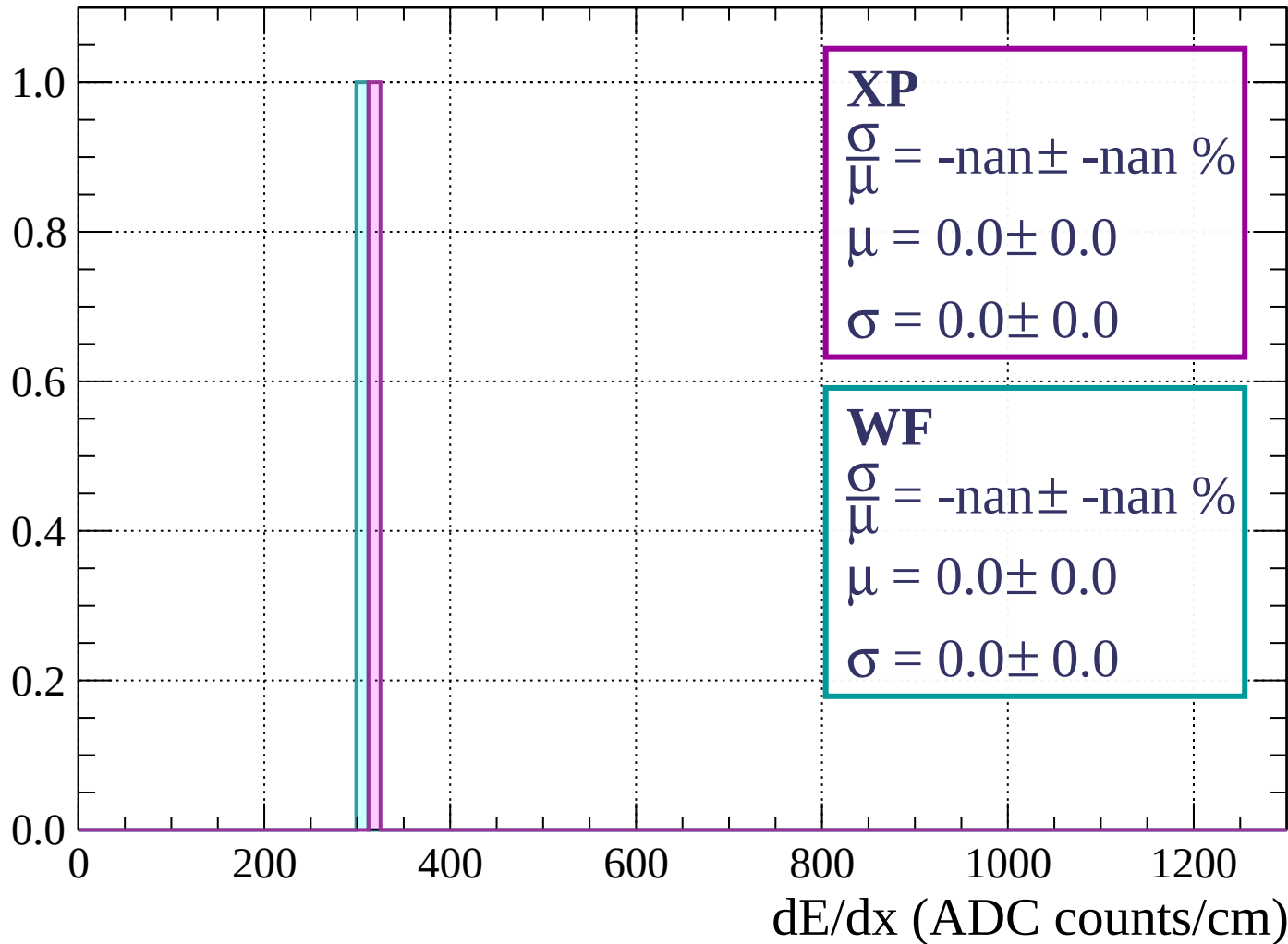
Energy loss | $30 < \theta < 33$

Count



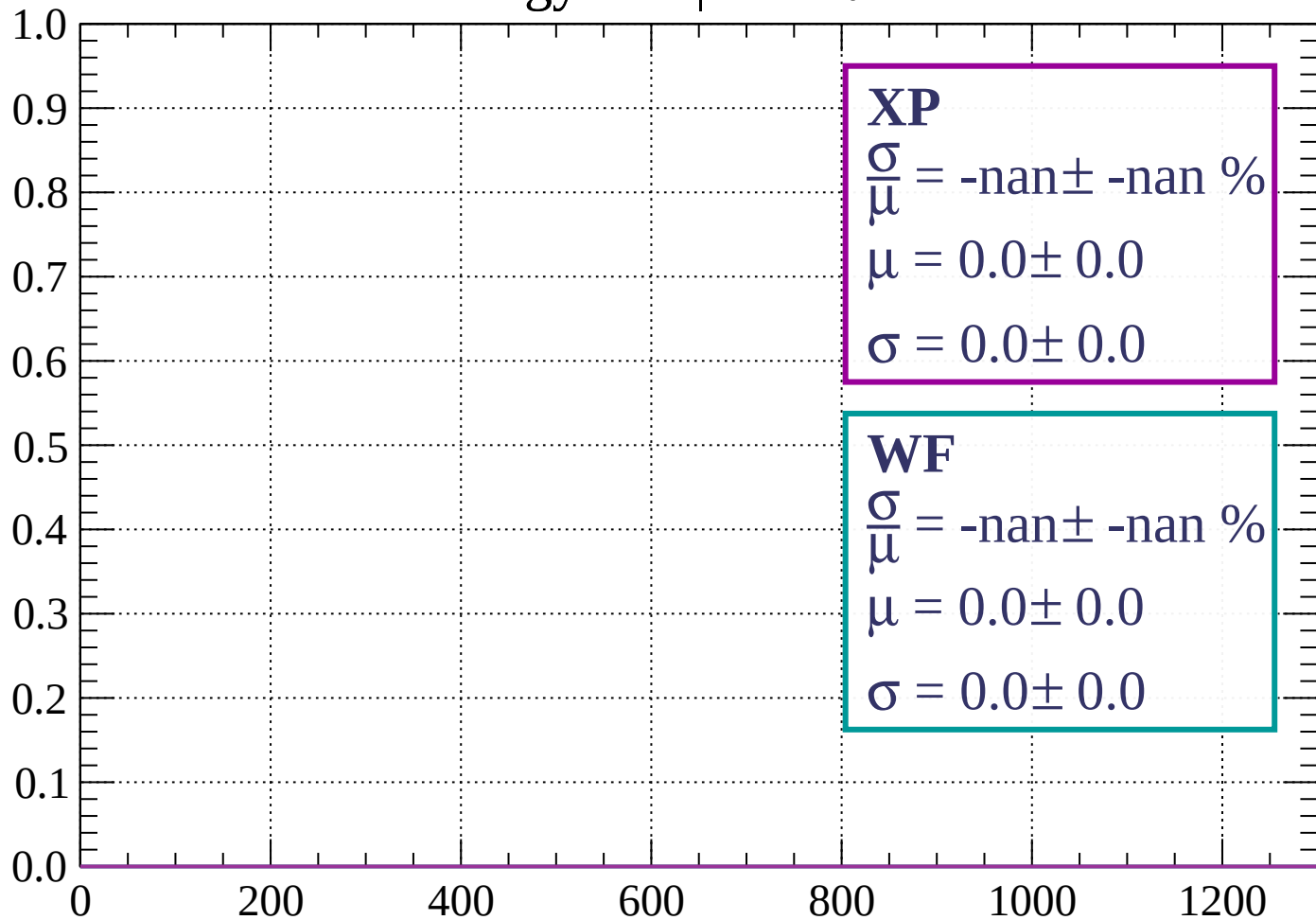
Energy loss | $33 < \theta < 36$

Count



Energy loss | $36 < \theta < 39$

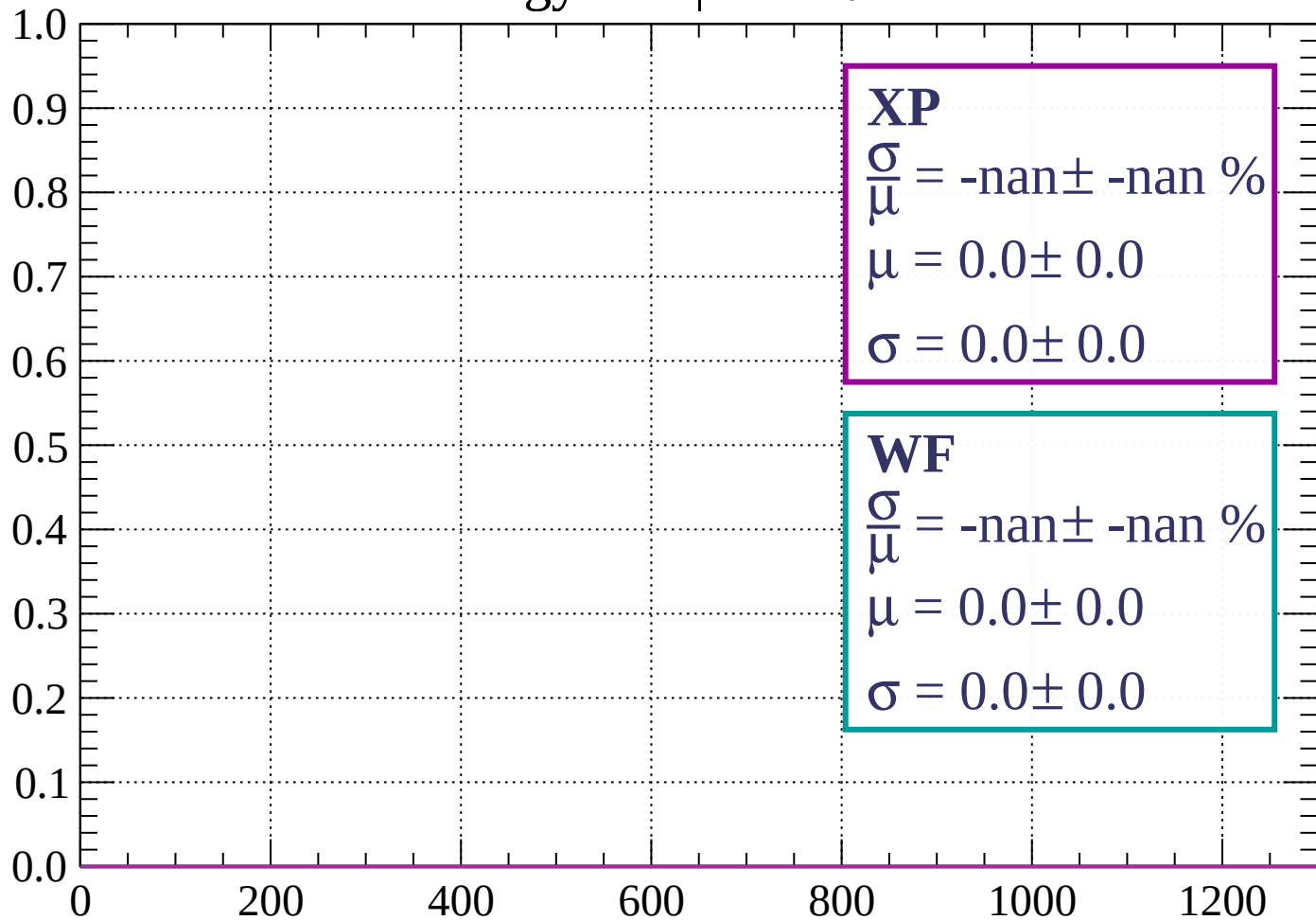
Count



dE/dx (ADC counts/cm)

Energy loss | $39 < \theta < 42$

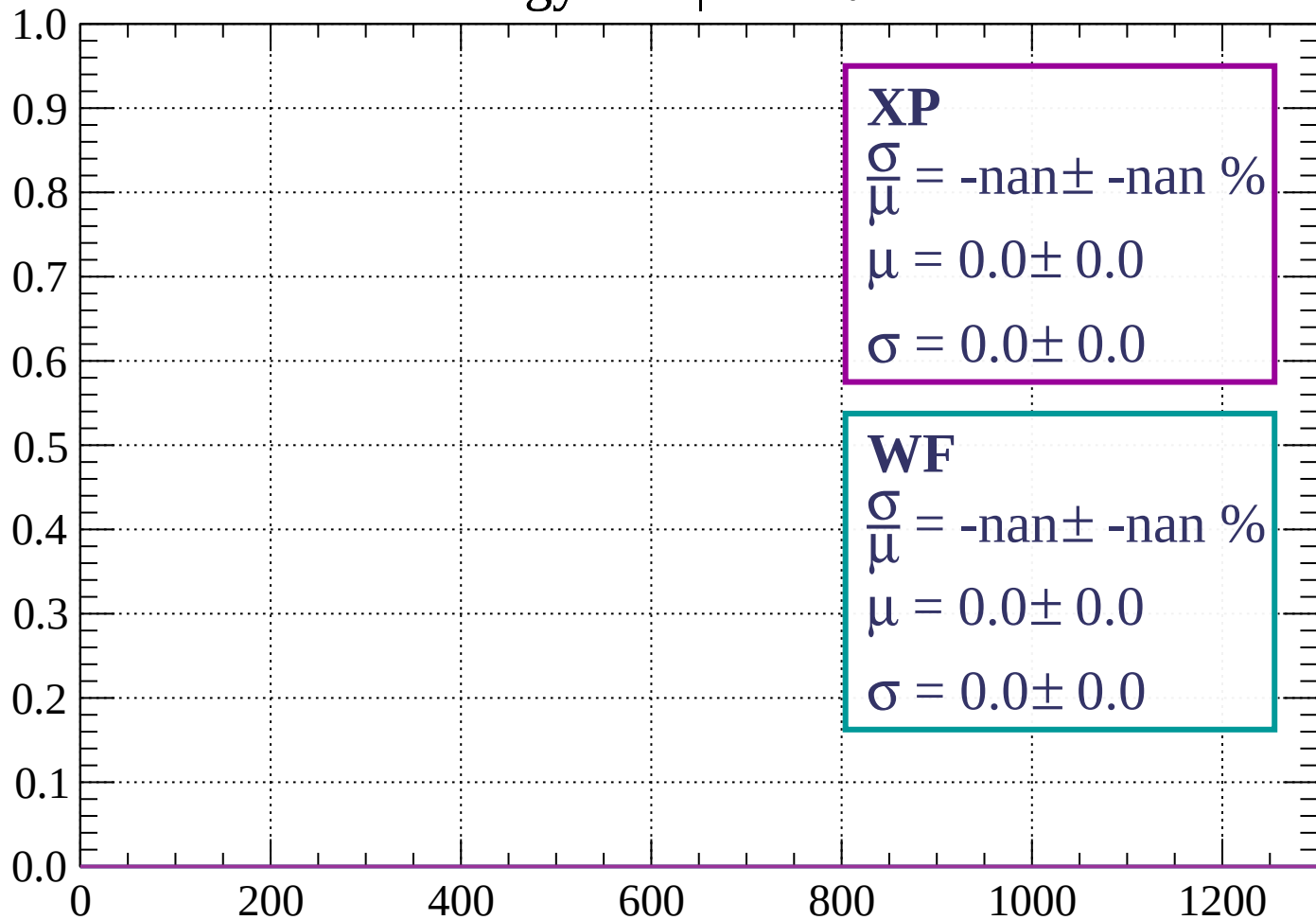
Count



dE/dx (ADC counts/cm)

Energy loss | $42 < \theta < 45$

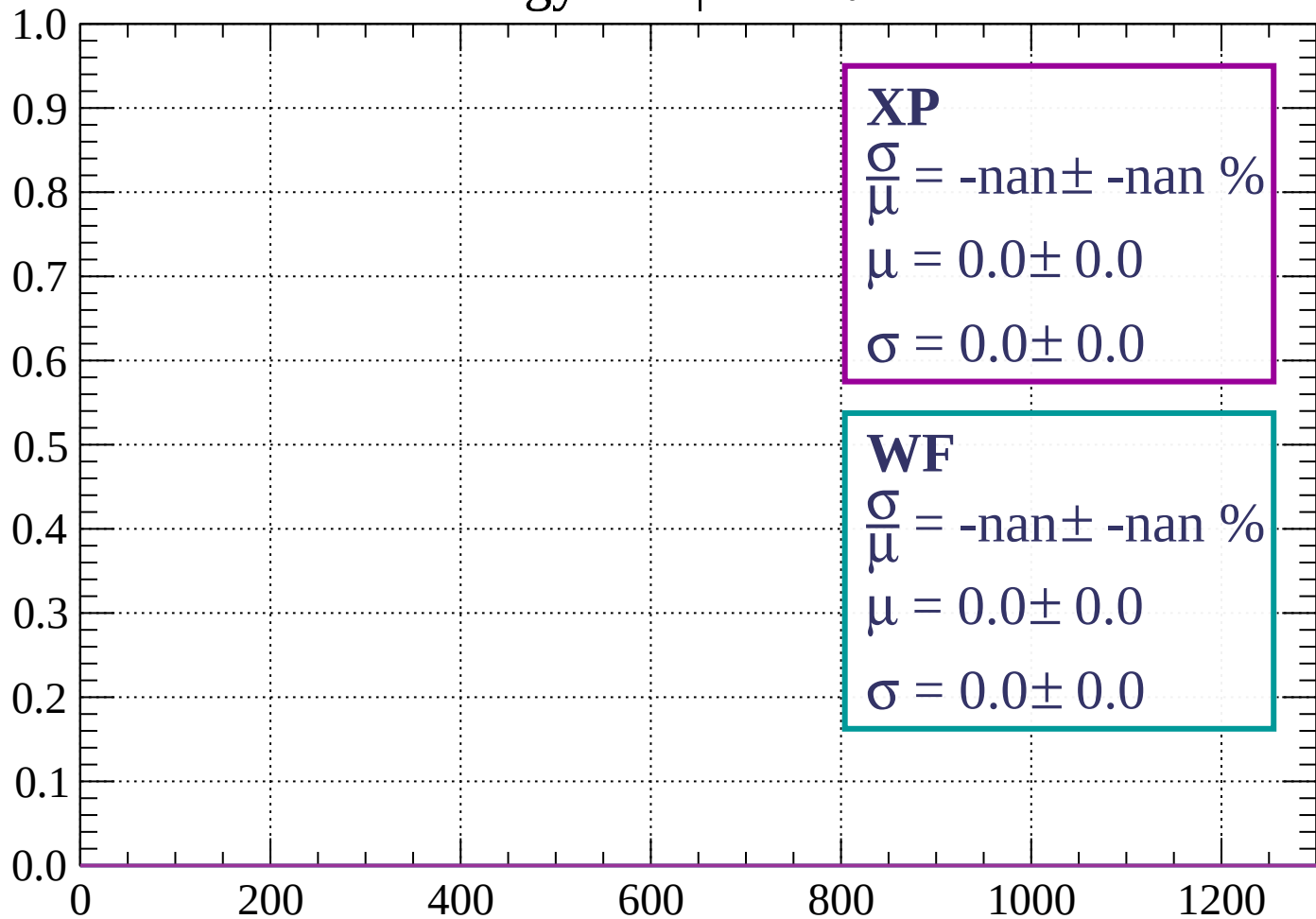
Count



dE/dx (ADC counts/cm)

Energy loss | $45 < \theta < 48$

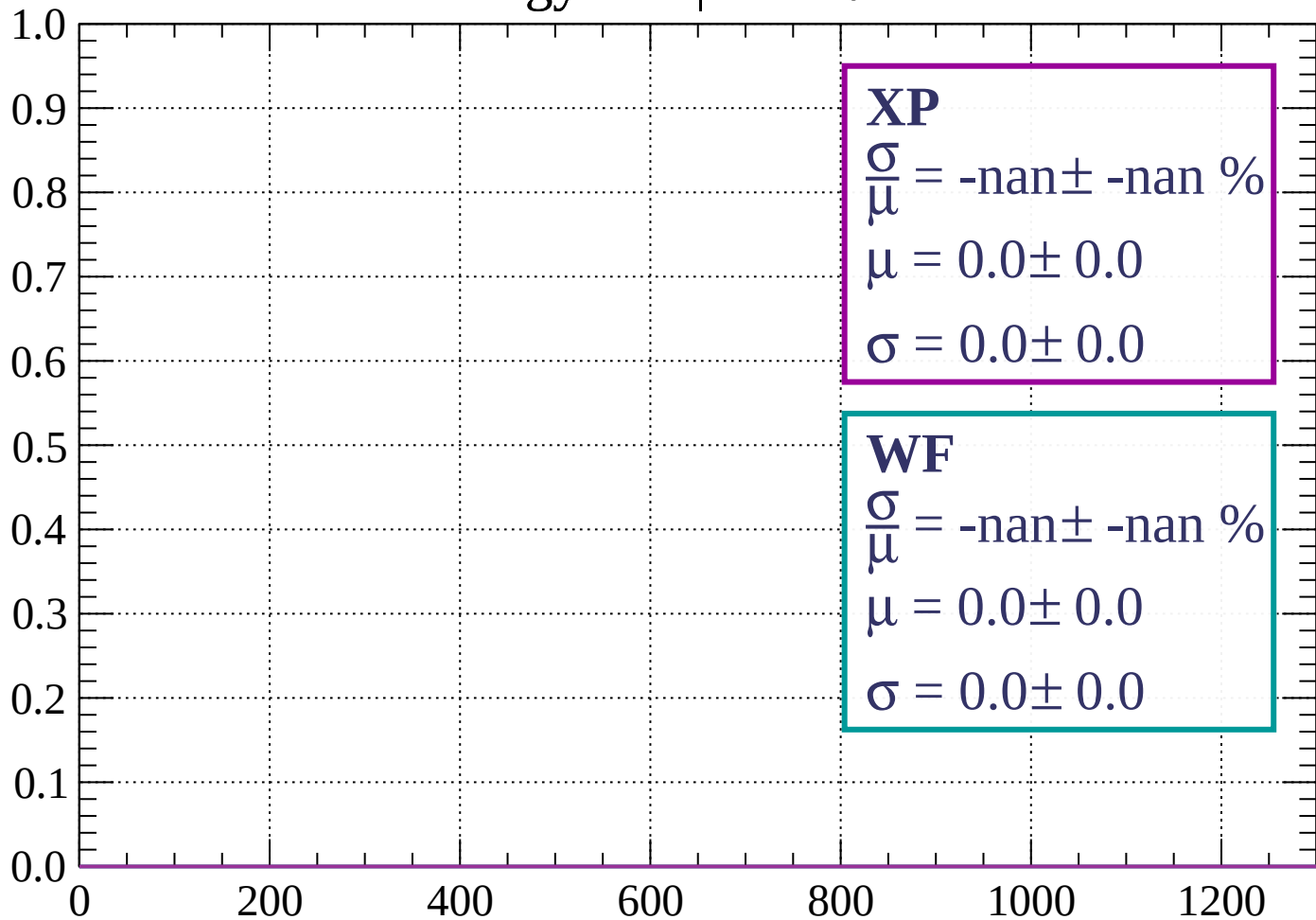
Count



dE/dx (ADC counts/cm)

Energy loss | $48 < \theta < 51$

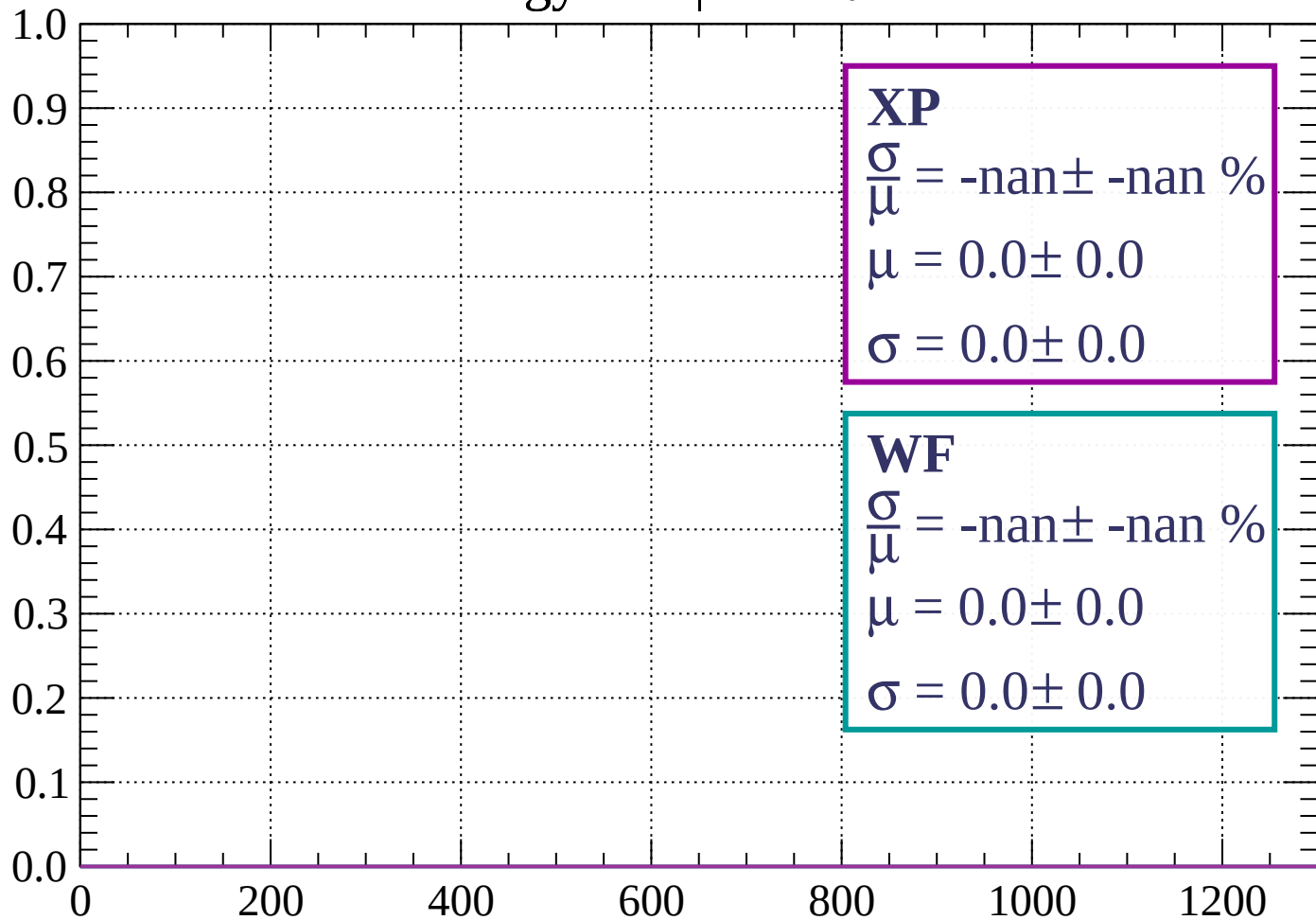
Count



dE/dx (ADC counts/cm)

Energy loss | $51 < \theta < 54$

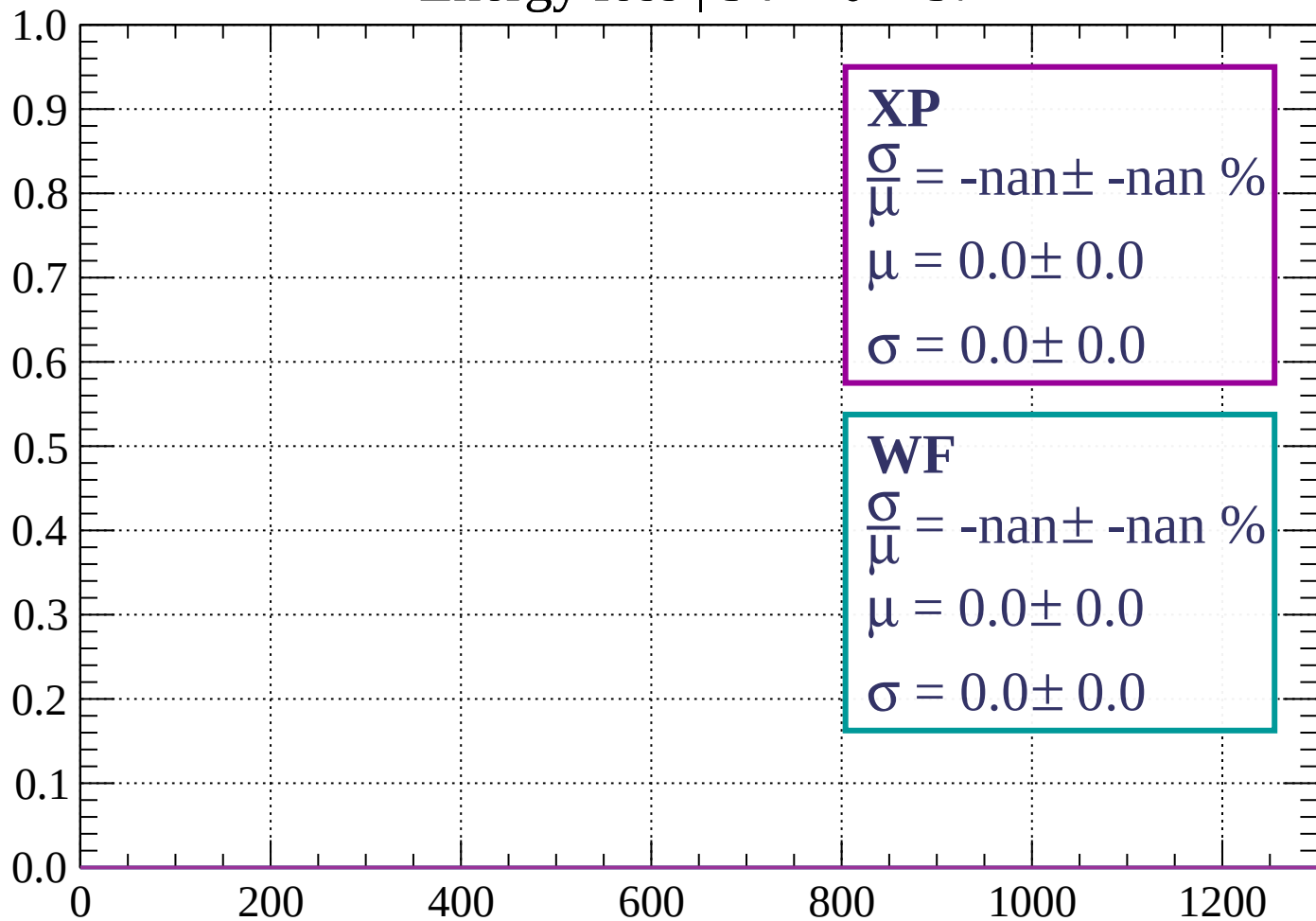
Count



dE/dx (ADC counts/cm)

Energy loss | $54 < \theta < 57$

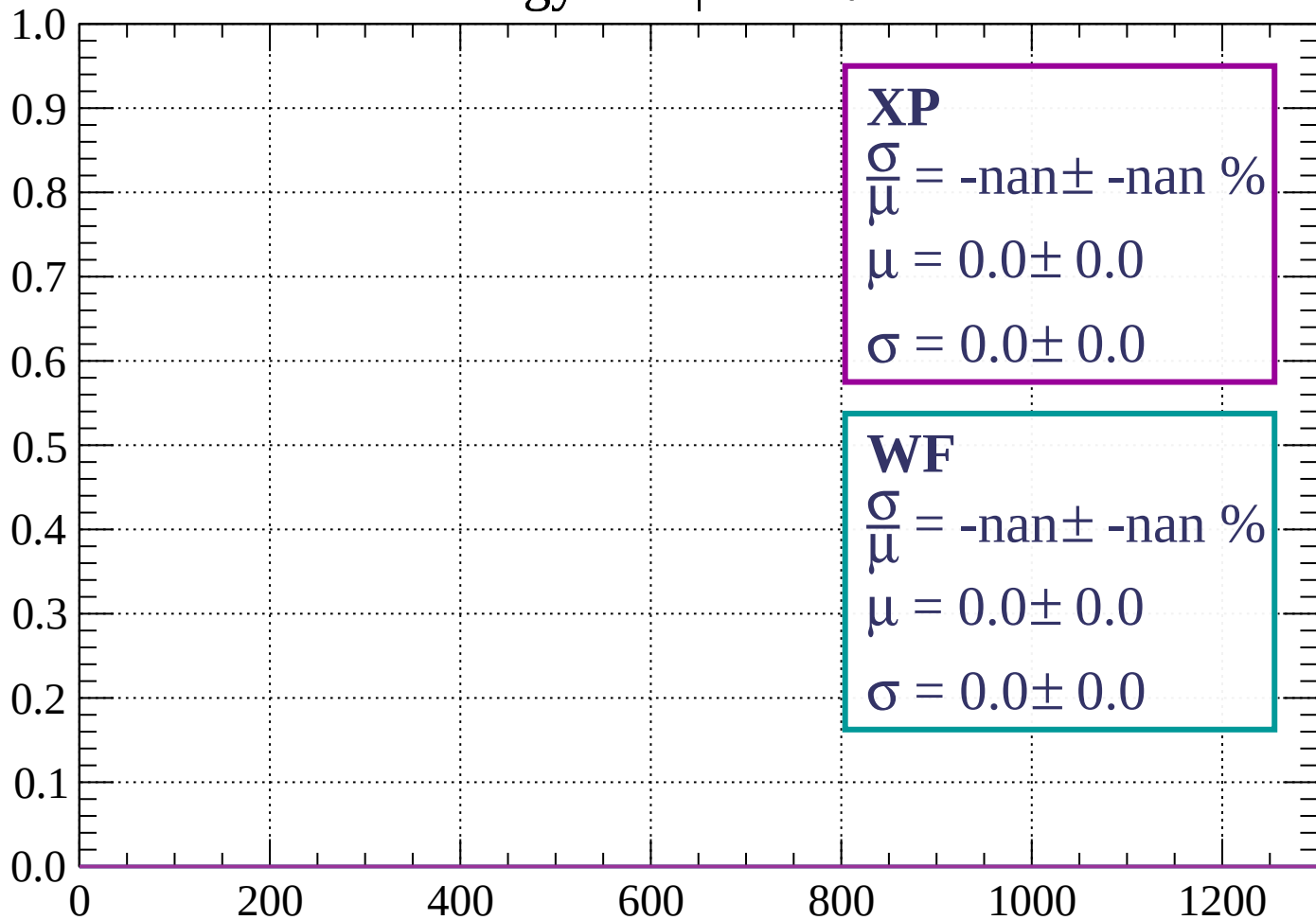
Count



dE/dx (ADC counts/cm)

Energy loss | $57 < \theta < 60$

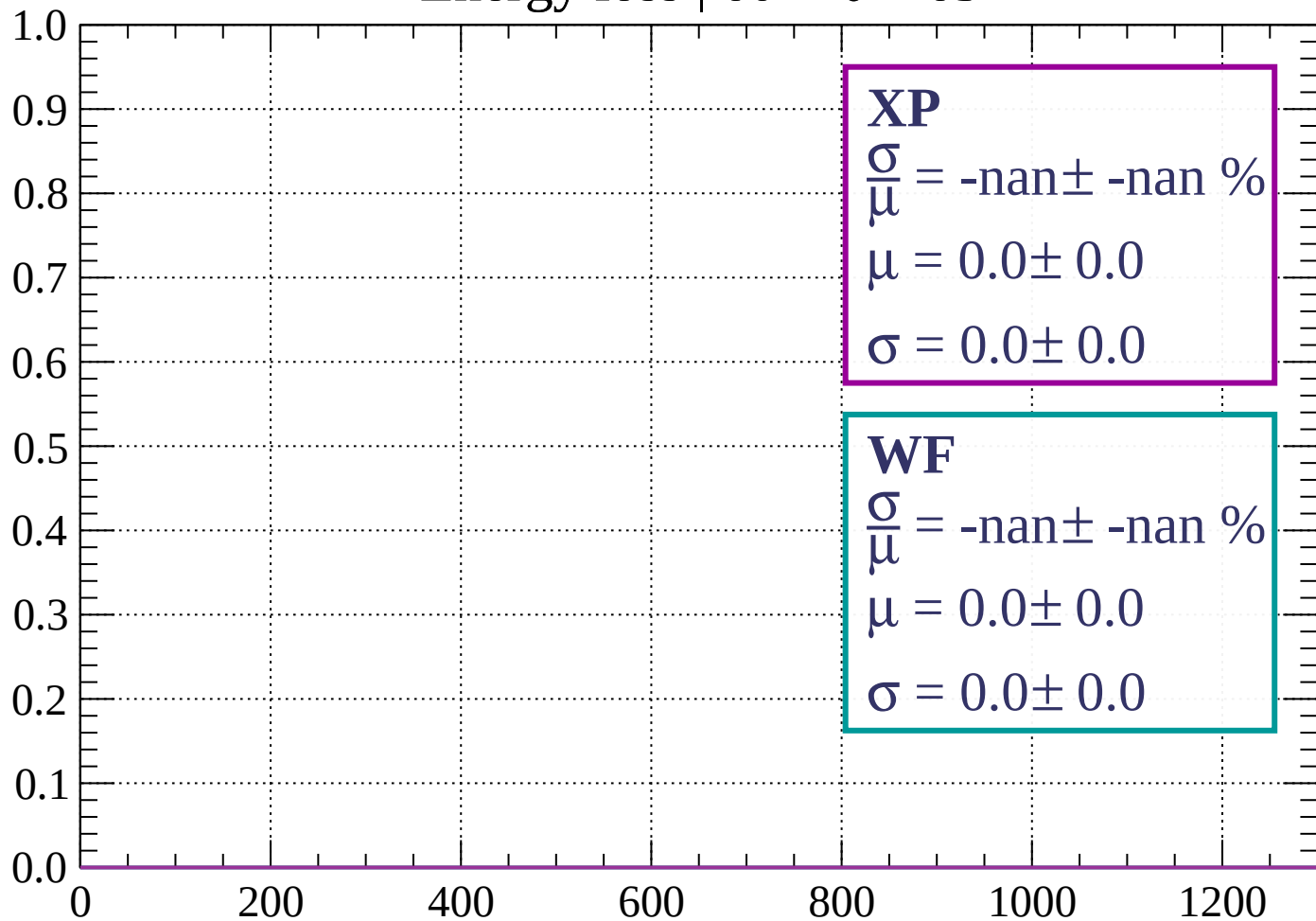
Count



dE/dx (ADC counts/cm)

Energy loss | $60 < \theta < 63$

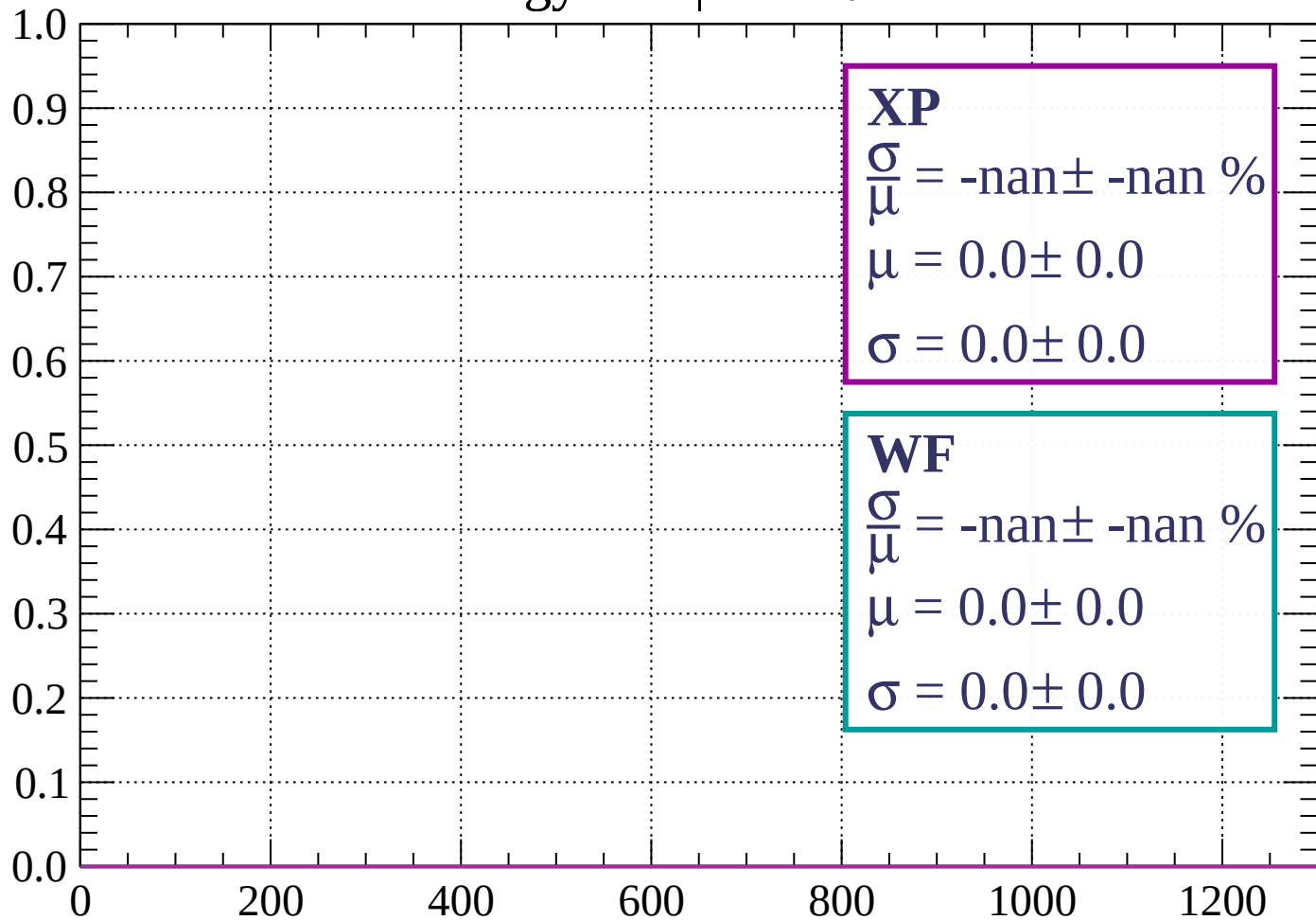
Count



dE/dx (ADC counts/cm)

Energy loss | $63 < \theta < 66$

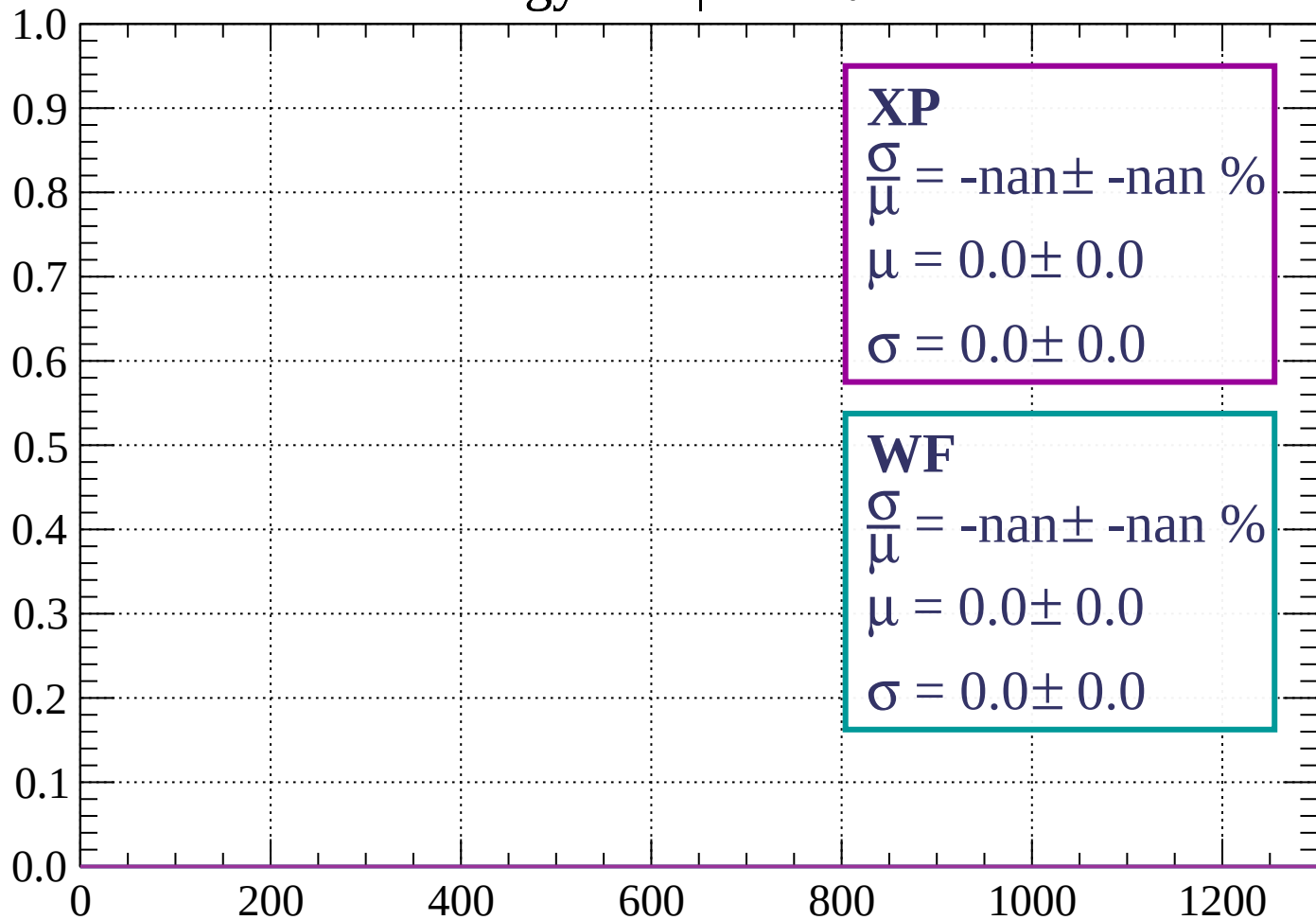
Count



dE/dx (ADC counts/cm)

Energy loss | $66 < \theta < 69$

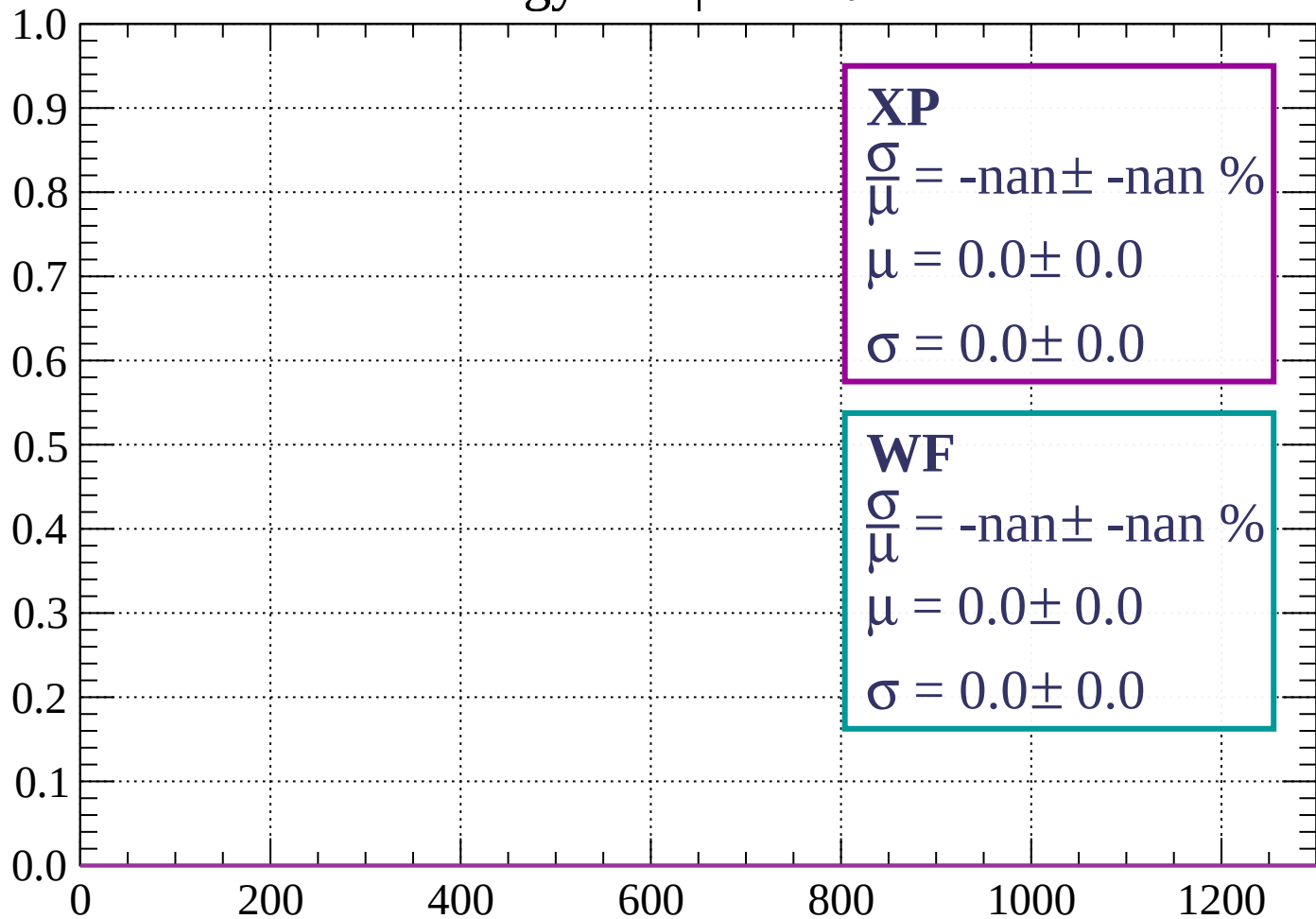
Count



dE/dx (ADC counts/cm)

Energy loss | $69 < \theta < 72$

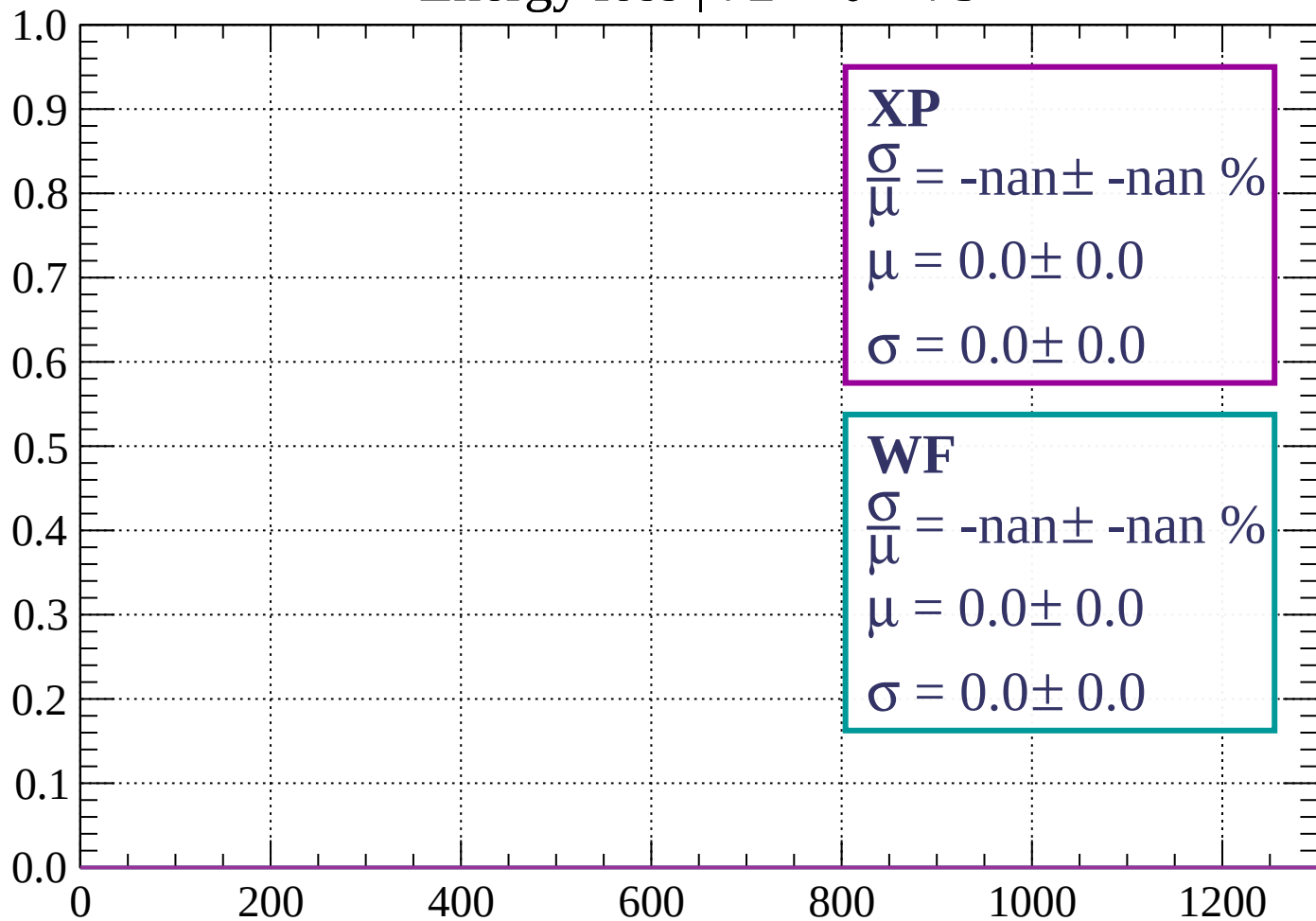
Count



dE/dx (ADC counts/cm)

Energy loss | $72 < \theta < 75$

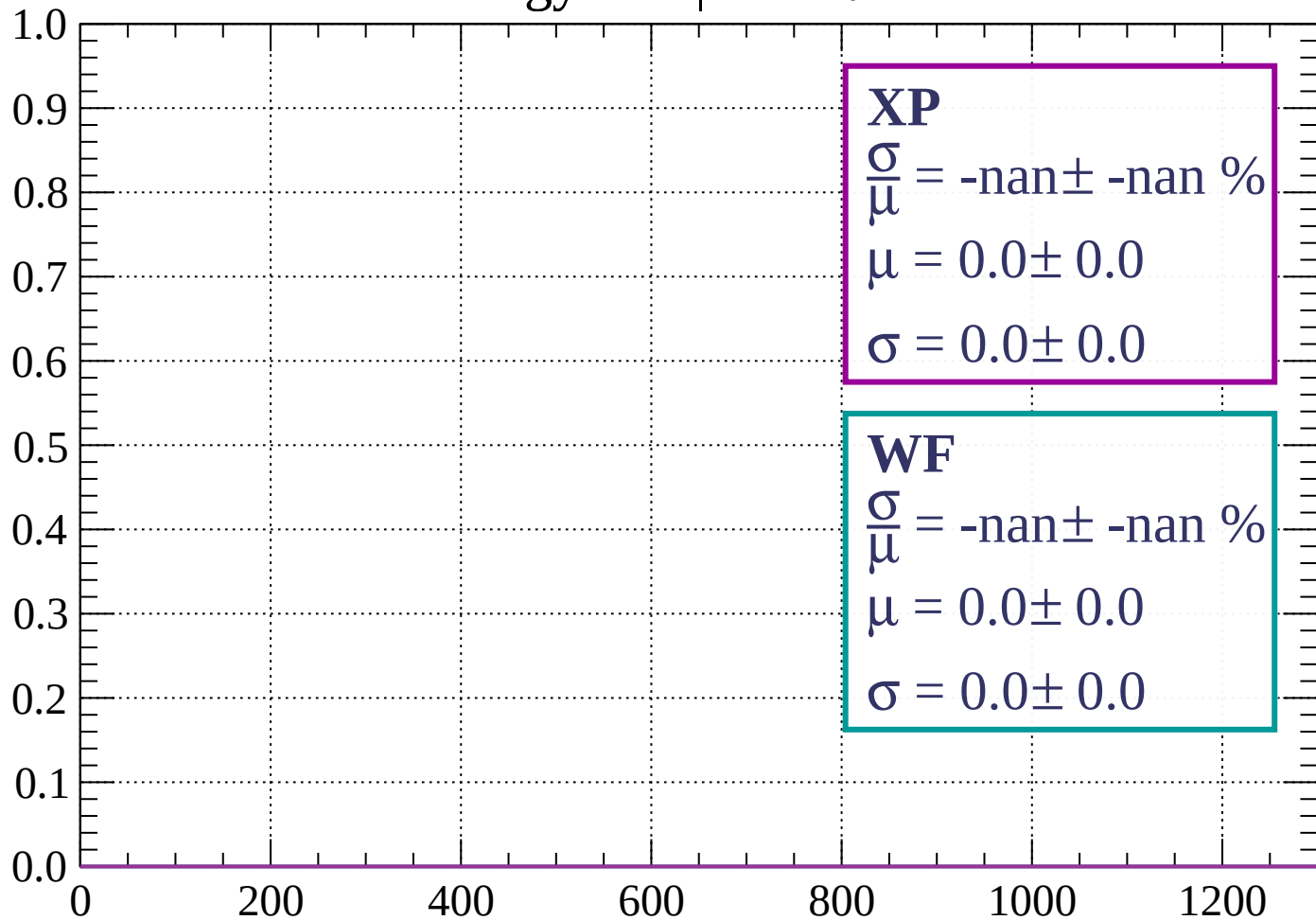
Count



dE/dx (ADC counts/cm)

Energy loss | $75 < \theta < 78$

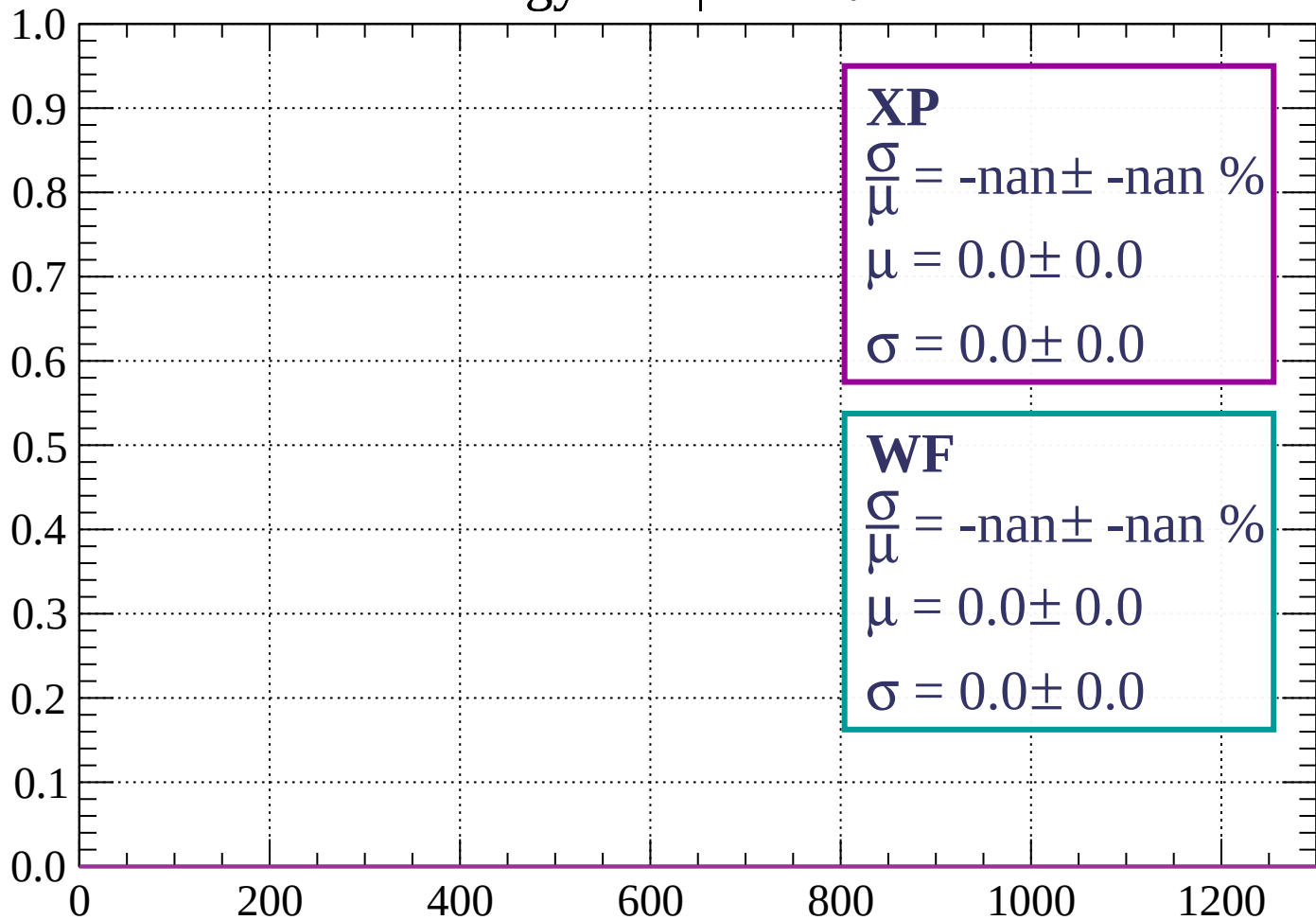
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 81$

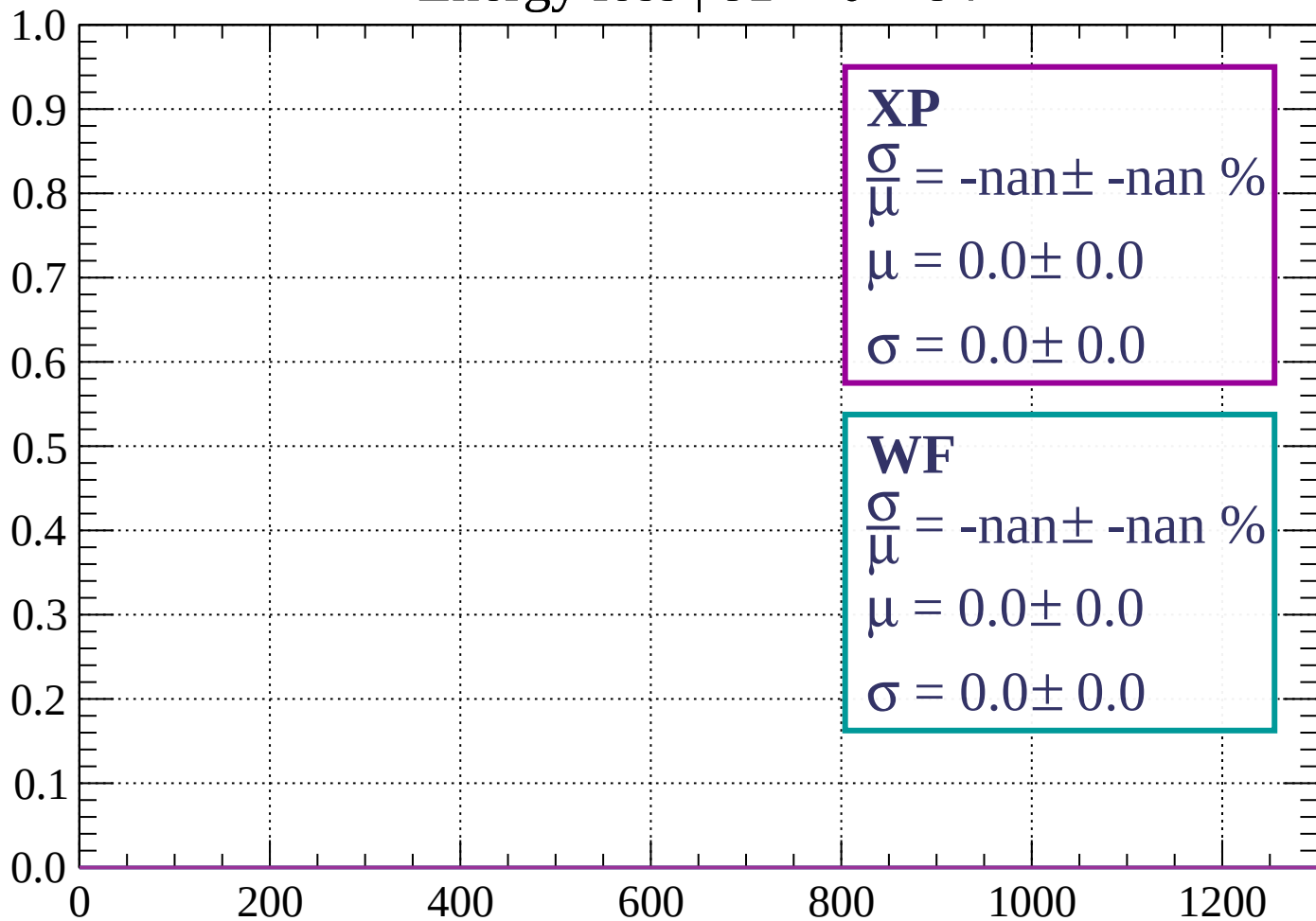
Count



dE/dx (ADC counts/cm)

Energy loss | $81 < \theta < 84$

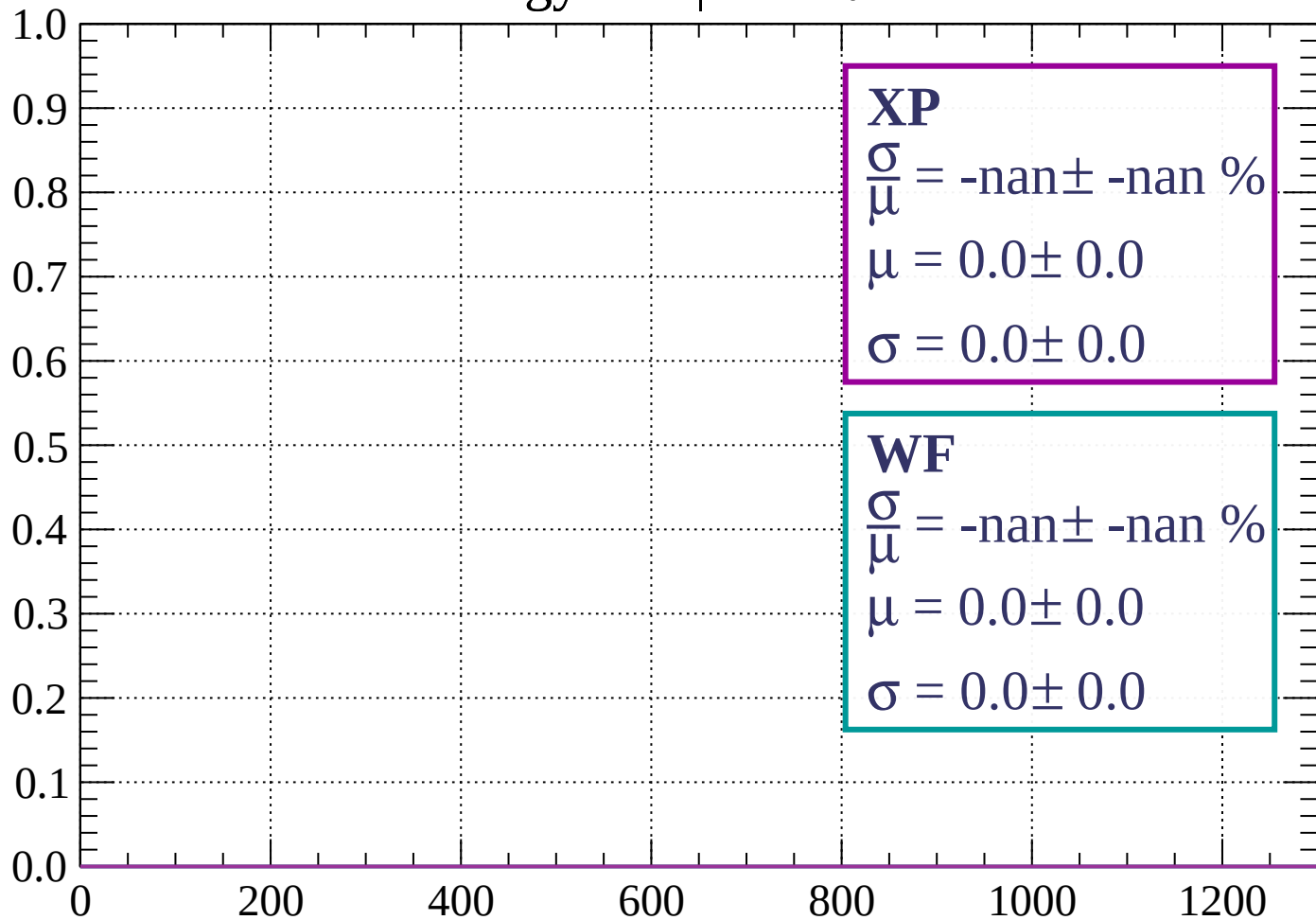
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 87$

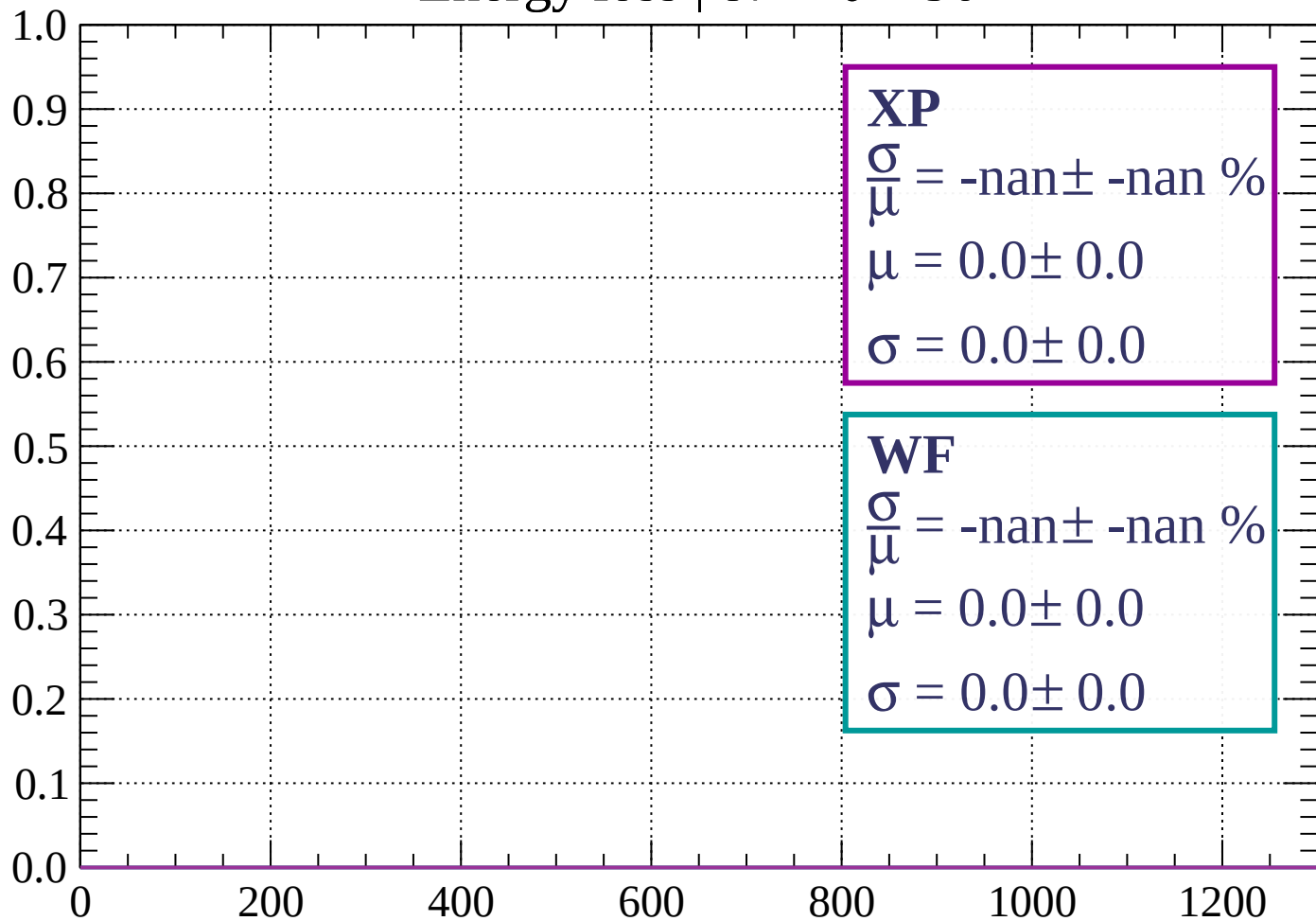
Count



dE/dx (ADC counts/cm)

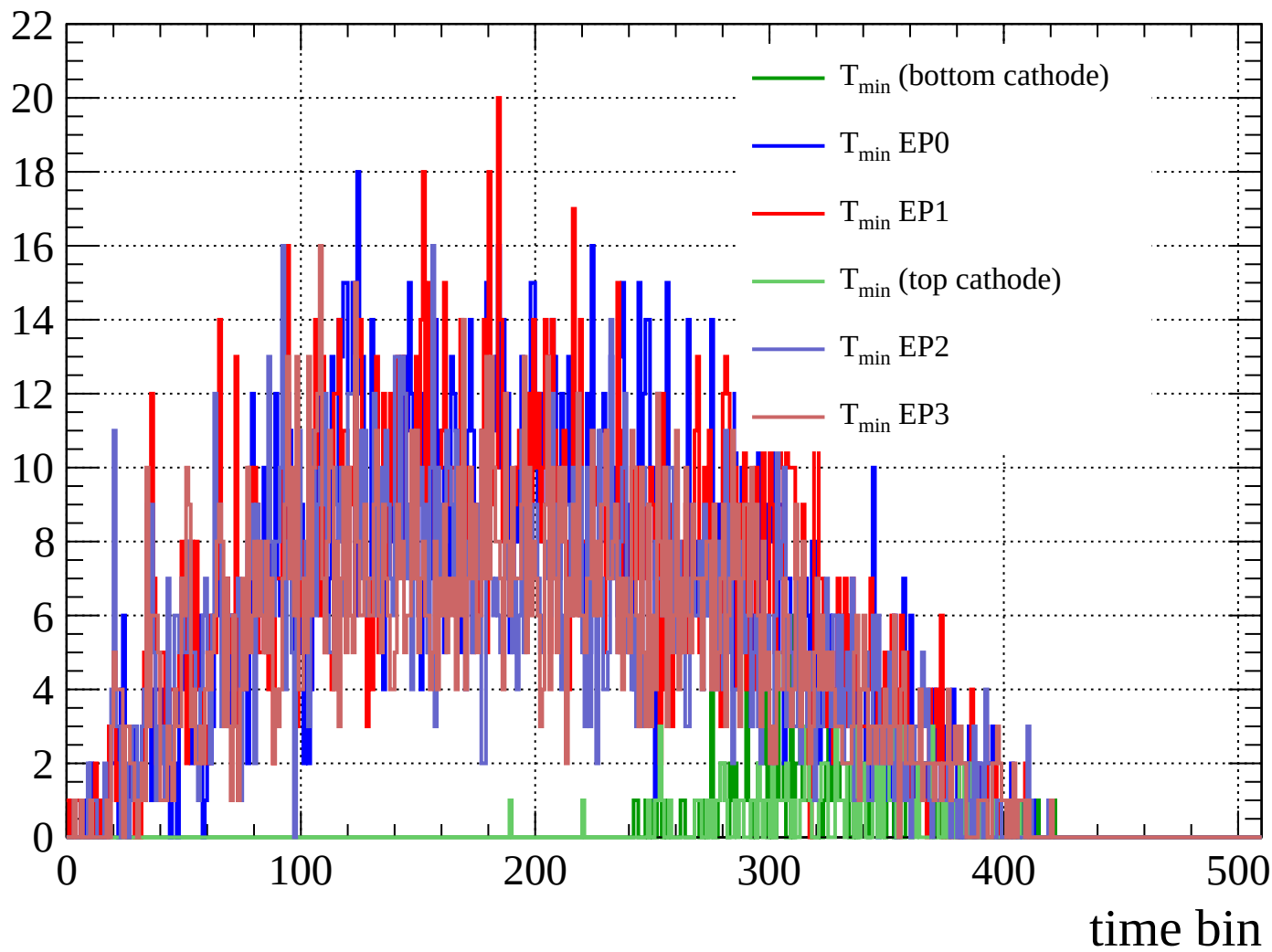
Energy loss | $87 < \theta < 90$

Count



dE/dx (ADC counts/cm)

Count



Count

